



**GREEN
CLIMATE
FUND**

Meeting of the Board
10 - 13 July 2023
Incheon, Republic of Korea
Provisional Agenda Item 12

GCF/B.36/02/Add.06

19 June 2023

Consideration of funding proposals – Addendum VI

Funding proposal package for FP211

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Hardest-to-Reach” submitted by Acumen Fund. Inc.;
- b) No-objection letters issued by the national designated authorities or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “Hardest-to-Reach” submitted by Acumen Fund. Inc. is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title:	Hardest-to-Reach
Country(ies):	16 sub-Saharan African countries: Benin, Burkina Faso, Burundi, Chad, DRC, Guinea, Guinea Bissau, Lesotho, Malawi, Mozambique, Niger, Sierra Leone, Somalia, Togo, Uganda, Zambia
Accredited Entity:	Acumen Fund
Date of first submission:	<u>[2023/01/09]</u>
Date of current submission	<u>[2023/06/16]</u>
Version number	V.3.0



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Private
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. <u>Not applicable</u>		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.		
		GCF contribution	Co-financers' contribution ¹
	Mitigation total	50 %	50 %
	☒ Energy generation and access	50 %	50 %
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %
	Adaptation total	50 %	50 %
	☒ Most vulnerable people and communities	25 %	25 %
	☒ Health and well-being, and food and water security	25 %	25 %
☐ Infrastructure and built environment	<u>Enter number</u> %	<u>Enter number</u> %	

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

	☐ Ecosystems and ecosystem services		Enter number %	Enter number %
A.5. Expected mitigation outcome <i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>	5,109,205 tCO ₂ eq <i>Indicate greenhouse gas (GHG) emission reductions or removals in tCO₂eq over total lifespan of the project/programme²</i>	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	54,331,312 (25,544,159 female, 28,787,153 male) <i>Indicate total number of direct and indirect beneficiaries</i>	
			9,944,285 (3,182,171 female, 6,762,114 male) <i>Indicate number of direct beneficiaries</i>	44,387,027 (22,361,988 female, 22,025,039 male) <i>Indicate number of indirect beneficiaries</i>
			3.0% <i>Indicate % of direct beneficiaries vis-à-vis total population</i>	13.5% <i>Indicate % of indirect beneficiaries vis-à-vis total population</i>

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

A.7. Total financing (GCF + co-finance³)	\$250 million USD	A.9. Project size	Medium (Up to USD 250M)
A.8. Total GCF funding requested	\$65 million USD		

A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.	
	☒ Grant \$15 million (Repayable Grant: \$10 million) ☐ Loan <u>Enter number</u> ☐ Guarantee <u>Enter number</u>	☒ Equity \$50 million ☐ Results-based payment <u>Enter number</u>

³ Refer to the Policy of Co-financing of the GCF.

A.11. Implementation period	12 years (10 years + 2 year extension) <i>Indicate the number of years and months the project/ programme is expected to be implemented.</i>	A.12. Total lifespan	12 years (see D6) <i>Indicate the maximum number of years over which the outcomes of the investment are expected to be effective, i.e. to lead to adaptation and/or mitigation results.</i>
A.13. Expected date of AE internal approval	<i>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.</i> Q4 2023	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☐ No ☒		

**A.20. Executing Entity
information**

Acumen Fund, Inc.
“Market Support Facility”
“Market Expansion Fund”
“MEF 2”
“Manager”
“General Partner”

If not the Accredited Entity, please indicate the full legal name of the Executing Entity(ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

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Provide an executive summary of the project/programme including:

1. Climate change problem 2. Proposed interventions 3. Climate results/benefits

Acumen is establishing a \$250m inclusive, innovative and multi-country investment initiative to deploy impact-focused capital and targeted technical assistance to off-grid-solar companies. The goal of 'Hardest-to-Reach' is to develop the off-grid solar (OGS) sector in countries where it is currently nascent or far below the capacity required to address the energy access challenge. The initiative will focus on least developed countries (LDCs) in sub-Saharan Africa, particularly with electrification rates at-or-below the sub-Saharan African average and high rates of poverty (45% or above), where investors have not been willing to enter at sufficient scale to meet the energy access and climate resilience needs of underserved segments.

Climate change problem

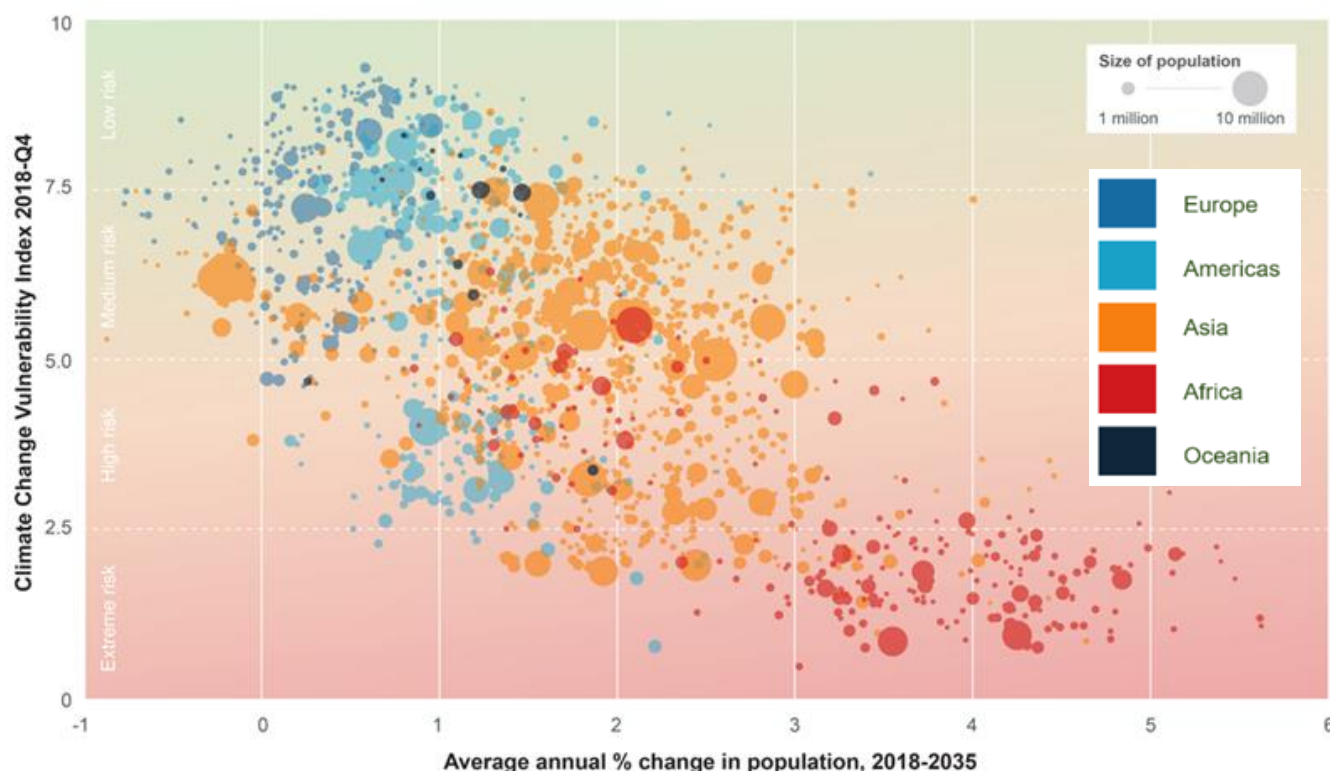
Across the 16 eligible Hardest-to-Reach (H2R) geographies, there are around 54 million African households that do not have access to modern electricity services. Even compared to their sub-Saharan African peers, these countries are being 'left behind' on energy access and are not on track to achieve Sustainable Development Goal 7 (SDG 7). On average, 78% of the population in these countries lives below the lower-middle-income poverty line,⁴ and these communities are both fast-growing sources of new energy demand and are residing in acutely climate vulnerable contexts – in contexts of worsening heat stress, more frequent droughts, more intense floods, and extreme weather events including tropical cyclones. Nearly half of the bottom-twenty countries listed in the ND-Gain Climate Vulnerability Index⁵ are Hardest-to-Reach countries, with H2R countries (for which there is available data) obtaining an average index score of 36.6, compared to a global average of 59.8. Maplecroft's Climate Vulnerability Index⁶ Fig 1 (below) displays the disproportionate incidence of climate vulnerability across Africa, where population growth is also at its highest.

Fig 1: Maplecroft Climate Vulnerability Index (2018)

⁴ World Bank [PovCal](#) (n.d)

⁵ The [ND-GAIN Country Index](#) summarizes a country's vulnerability to climate change and other global challenges in combination with its readiness to improve resilience. Higher index scores are given to less climate vulnerable countries.

⁶ The [Maplecroft Climate Change Vulnerability Index](#) provides a single figure rating for climate change exposure, sensitivity and adaptive capacity, helping companies calculate risks from future climate change. It combines exposure to climate extremes and change with the current human sensitivity to those climate stressors and the capacity of the country to adapt to the impacts of climate change. Higher index scores are given to less climate vulnerable countries.



In the face of escalating climate variability and change, the delivery of climate-resilient energy access infrastructure and services is now more critical and urgent. The most recent data from the IEA Africa Energy Outlook projects that, for every person in sub-Saharan Africa to have access to electricity by 2030, thus achieving SDG 7, 6% of the current total population or 90 million people, including a significant proportion from rural areas, would need to gain power access each year from 2022.⁷ Without energy that is both clean and climate resilient, households in Africa's LDCs will face the climate future with increasing uncertainty and risk to their livelihoods, health and wellbeing.

The interplay of several factors points to the need for an energy access solution that puts climate adaptation and resilience at the forefront.

Inadequacy of existing infrastructure

The past decade of climate trends and events has displayed the fragility of existing network infrastructure against extreme weather shocks, with the significant damage and long periods without repair being frequent, leading to severe and underappreciated consequences. Along with power outages come disruptions to household routines, diminished public safety, and halted economic activity, while the regularity of these incidents mean that households and small businesses are forced to invest in inefficient coping measures, ranging from the purchase of water tanks and diesel generators, to keeping expensive back-up inventories.⁸

Vulnerability of people without access

⁷ IEA [Africa Energy Outlook](#) (2022)

⁸ World Bank Policy Research - [Underutilized Potential: The business costs of unreliable infrastructure in developing countries](#) (2019)

Households without any electricity access, meanwhile, are trapped in climate vulnerability, as they lack the social, technological, and financial resources that are enabled and multiplied by energy access, such as early warning and post-disaster public notifications. For these households, energy access will likely be the springboard for climate resilient sustainable development. For example, addressing the energy access challenge through OGS products can have a transformational impact on economic activity, which can boost economic resilience to climate shocks. Around a third of the solar home systems (SHS) in East Africa were purchased with the intent to extend the working day or boost enterprise. Usage of SHS in East Africa boosted monthly earnings by US\$ 46 a month, around 14% increase on average income.⁹

Dependency on nature

Research from the Organisation for Economic Cooperation and Development (OECD)¹⁰ has pointed to the disproportionate dependence of the poorest and most marginalized on environmental resources to meet their basic needs, such as forests (for energy, materials and food), land (for farming) and water (for drinking, household use and farming). This is a reminder that the pathways of electrification that investors and developers deploy in low-income communities, should be the least nature-intensive, the most climate resilient, and the most supportive of community norms, even as the benefits of economic development unfold.

Inequitable distribution of climate burden

On these considerations alone, entry-level distributed residential solar compares favorably to other alternatives, separate from its mitigation and energy security benefits. And though the continent has contributed to less than 4% of global emissions,¹¹ the burden of climate change on least developed countries and livelihoods across the continent is disproportionately high.¹² Such climate injustice already threatens to undermine development gains and inflict additional economic costs and social disruption. Adaptation alone is projected to cost the continent at least \$500 billion annually by 2050.

The African Development Bank's African Economic Outlook summarizes several dynamics at play in the continent's potential low-carbon transition out of energy poverty:¹³

- Low-carbon transitions in Africa will vary by country, but they could be transformational. These variations reflect differing ecological zones, climates, settlement patterns, economic structures, resource bases, and governance systems;
- Africa is generously endowed with renewable energy resources, such solar irradiation, which present significant opportunities for the green transition;
- With Africa's limited lock-in to fossil-based energy technologies, renewable energy technology opportunities could help the continent build a climate-resilient and integrated sustainable energy sector;

⁹ GOGLA, [Powering Opportunity global report](#) (2020)

¹⁰ OECD, [Poverty and Climate Change Reducing the Vulnerability of the Poor through Adaptation](#) (2006). Referenced in DFID [Literature Review on Energy Access and Adaptation to Climate Change](#) (2015)

¹¹ Carbon Disclosure Project [CDP Africa Report](#) (2020)

¹² AfDB [Climate Resilience and a Just Energy Transition in Africa](#) (2022)

¹³ AfDB [African Economic Outlook](#) (2022)

- Interventions to achieve climate resilience and a just energy transition in the continent should be inclusive, “leaving no one behind.”

Africa needs a decade of highly adaptive and resilience-promoting infrastructure investment to deliver on the promise of sustainable development by 2030 and in this century. Yet, across the African continent, only a small number of countries have seen strong growth in electrification in the past decade, in particular Kenya, Senegal, and Ghana, as well as Ethiopia and Rwanda where there is still significant scope for development. A substantial scale-up of investment is needed throughout sub-Saharan Africa, as it totals two-thirds of the required additional global investment in electricity access. In terms of the emissions mitigation pathway, the lesson from across the globe, historically, is that in a business-as-usual environment, electricity and heat generation have been the greatest contributor to anthropogenic drivers of greenhouse gas emissions, sometimes twice as large as the second most-contributing economic sector. Still, today, power generation remains the largest GHG-emitting sector in developed economies across Europe¹⁴ and the second-largest in the US,¹⁵ suggesting that the carbon-intensity of power generation and economic development trajectories is highly material for mitigation outcomes, and showing that the established pathways of electrification are inconsistent with net zero objectives and climate targets of the United Nations SDGs and the Paris Accord. A similar fossil-fuelled trajectory in sub-Saharan Africa would be catastrophic for our climate; if African emissions per person were to rise to only the current global average, the continent would emit 12.5 billion tonnes of greenhouse gasses per year by 2050 — roughly equivalent to all emissions from China and India today.^{16,17}

Proposed intervention

Acumen’s Hardest-to-Reach energy access initiative seeks to mobilize a blend of institutional and donor finance to deliver much-needed investment into the region, to contribute to closing the energy access gap in sub-Saharan Africa towards achieving SDG 7 by providing more than 60 million people with clean, climate-resilient, off-grid solar energy products. Hardest-to-Reach (H2R) will establish two separate financing facilities with different investment risk profiles, and invest with different forms of capital to match the needs of off-grid solar companies and markets. This is designed to help off-grid solar companies to expand in H2R countries and extend their reach to underserved segments, thus achieving sustainable and inclusive scale with cross-cutting climate impacts. The first facility, the \$50M grant-funded Market Support Facility (MSF), intends to deploy debt, equity, and grants using a patient capital approach that is consistent with Acumen’s core investment experience. The principal work of the MSF is to prove an approach in a new market with a range of capital investment options and technical assistance accompaniment. A second vehicle, the \$200M blended finance Market Expansion Facility (MEF), intends to extend impact-linked debt to OGS players, with the objective of responsibly expanding the reach and impact of OGS technologies, while achieving high levels of inclusion and sustainability. To this end, H2R seeks to support companies with technical assistance and invest in ecosystem development to ensure the sustainability of the intervention and its impacts beyond the life of the fund.

¹⁴ European Environment Agency [CO2 Intensity of Electricity Generation](#) (2020)

¹⁵ United States Environmental Protection Agency [Sources of Greenhouse Gas Emissions](#) (2020)

¹⁶ Tony Blair Institute for Global Change [Africa Needs Support If It Is to Play Its Part in the Drive for Net Zero Carbon Emissions](#) (2021)

¹⁷ Evidence of decoupling pathways (decoupling of GHG emissions and economic development) demonstrate the ability of successive waves of investment to drive and prove cleaner pathways of sustainable development. C/f: Financial Times [Economics may take us to net zero all on its own](#) (2022)

The H2R initiative is designed to overcome a number of market barriers that currently hinder more widespread climate-resilient energy access in sub-Saharan Africa. These are outlined in the table below.

Market barrier	H2R response
Markets in H2R areas are too risky or nascent for many investors.	Perceived risk may differ from the real risk. At the portfolio level, H2R will leverage deeper market research to best assess the risk of each market. At the fund level, the H2R initiative's blended finance structure mitigates risk by crowding in investors with different risk appetites. The \$50M Market Support Facility aims to use philanthropically-backed patient capital to support market entry for OGS companies into H2R markets. The first loss tranche for the Market Expansion Facility can unlock more commercial capital from mainstream investors and build the case for subsequent investment flows, allowing companies to access greater amounts of funding for operations in H2R areas.
OGS companies experience capacity barriers to expand into H2R areas.	The targeted use of set-up grants, equity and debt from the Market Support Facility aims to address management, marketing, distribution and risk assessment barriers based on the country's difficulty and company's needs. The TA will seek to improve capacity and ensure responsible, sustainable growth (including through high standards of consumer protection).
There is a lack of flexible funding for OGS companies looking to achieve scale. Those that do receive funding aren't incentivized to serve the poorest and expand into H2R areas.	Acumen intends to raise a \$200M vehicle, specifically designed to invest via impact-linked finance in OGS companies. Acumen seeks to link the interest rate to impact-related KPIs to incentivize companies to reach low-income, first-time energy access customers, thereby avoiding mission drift. The grant capital as part of this vehicle is intended to permit interest rates adjustments to further incentivize first-time energy access.
People in H2R areas can't afford OGS products upfront, are unaware of their benefits, or live in areas with limited to no OGS company coverage. The majority of people do not have access to credit, which would	Acumen incentivizes and facilitates OGS companies to set up, market and sell solar energy products via PAYGo financing. The PAYGo model means that solar energy products are affordable for people in H2R areas, who can spread the cost of the solar product over a few months or years. Acumen, through Hardest-to-Reach, is also forming partnerships with the World Bank Group and donor countries to address the affordability gap through demand side subsidies to expand our reach to the lowest income level customers. H2R aims to accompany OGS companies to implement strong consumer protection practices, aligned with GOGRA Consumer

allow them to extend payments over time.

Protection Principles,¹⁸ and work with domain experts where required to deliver specialist technical assistance.

Climate results/benefits

By the end of the program, investments developed, financed, and accompanied by H2R will be able to provide:

- ~72.4M people with year-round access to clean, affordable, reliable and modern off-grid solar energy.
- ~120 MW of newly-installed, decentralized solar energy generating capacity across H2R geographies.
- ~54.3M people with first time energy access and improved climate resilience through decentralized solar power delivery.
- ~5.4M people with improved climate resilience through diversified income sources enabled by first-time access to a solar powered product.
- 5.1M tCO₂e in emissions reduced or avoided (excl. black carbon mitigation impact).
- In addition, some 13,922 clean energy jobs, with 4,641 as sources of employment for women.

¹⁸ [GOGLA Consumer Protection Code](#) consists of a set of principles and an assessment framework to enable companies to measure, demonstrate and improve their practices, and provide investors and other stakeholders with a framework to promote good consumer protection (CP) practice. The code establishes the de facto industry standard for CP that adds value to off-grid solar companies, investors and other stakeholders.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate change problem: Describe the climate change problem the proposal is expected to address. Describe the mitigation needs (GHG emissions profile) and/or adaptation needs (climate hazards and associated risks based on impacts, exposure, and vulnerabilities) that the proposed interventions are expected to address. Also describe the most likely scenario (prevailing conditions or other alternative) that would remain or continue in the absence of the proposed interventions. Include baseline information. The methodologies used to derive such information, including the mitigation and adaptation needs, should be included in the feasibility study.

Context: In describing the mitigation and/or adaptation needs, briefly describe the target region/area of the proposed interventions including information on the demographics, economy, topography, etc.

Related projects/interventions: Also describe any recent or ongoing projects/interventions that are related to the proposal from other domestic or international sources of funding, such as the Global Environment Facility, Adaptation Fund, Climate Investment Funds, etc., and how they will be complemented by this project/programme (e.g. scaling up, replication, etc.). Please identify current gaps and barriers regarding recent or ongoing projects and elaborate further how this project/programme complements or addresses these.

Africa is the most vulnerable continent to climate change impacts under all climate scenarios above 1.5 degrees Celsius.¹⁹ Exponential collateral damage in the face of climate change poses systemic risks to the continent's economies, infrastructure investments, water and food systems, public health, agriculture, and livelihoods, threatening to undo its modest development gains and slip into higher levels of extreme poverty. There are substantial disparities across Africa, with many countries in sub-Saharan Africa being the most vulnerable and exposed to the impacts of climate change. Many of these form the group of Hardest-to-Reach target countries. All are highly vulnerable, and least ready, under the NDI-GAIN indexation, with ranks reaching from 1st – 56th most vulnerable to, and 127th to 176th most ready to cope with, adaptation challenges.

Constraints to the supply and demand of electricity drive the energy access problem. Two drivers - household poverty, which limits a household's ability to pay for electricity, and the lack of physical access to electricity, due to costly, weak or non-existent coverage of grid electricity outside urban areas - are particularly harmful to climate vulnerable populations. Accordingly, the baseline context can be presented through two key indicators: poverty rate and electrification rates. At least half of all sub-Saharan Africans have no access to electricity at all,²⁰ and two-thirds of the global extreme poor population reside in this region,²¹ with more than three quarters living below the \$3.65/day poverty line across H2R program countries on average.²² Acumen's Hardest-to-Reach energy access initiative is seeking to contribute towards closing the energy access gap in sub-Saharan Africa's least developed countries (LDCs).

Fig 1: Percentage of population without electricity access and living below the poverty line for the 17 H2R countries and similar markets relative to the global average

¹⁹ AfDB [Climate Change in Africa](#) (2021)

²⁰ IEA [SDG7 Data and Projections: Access to Electricity](#) (2022)

²¹ World Bank [Where the extreme poor live](#) (2020)

²² World Bank [WB Data Bank: Poverty headcount ratio at \\$3.65 a day \(2017 PPP\)](#)

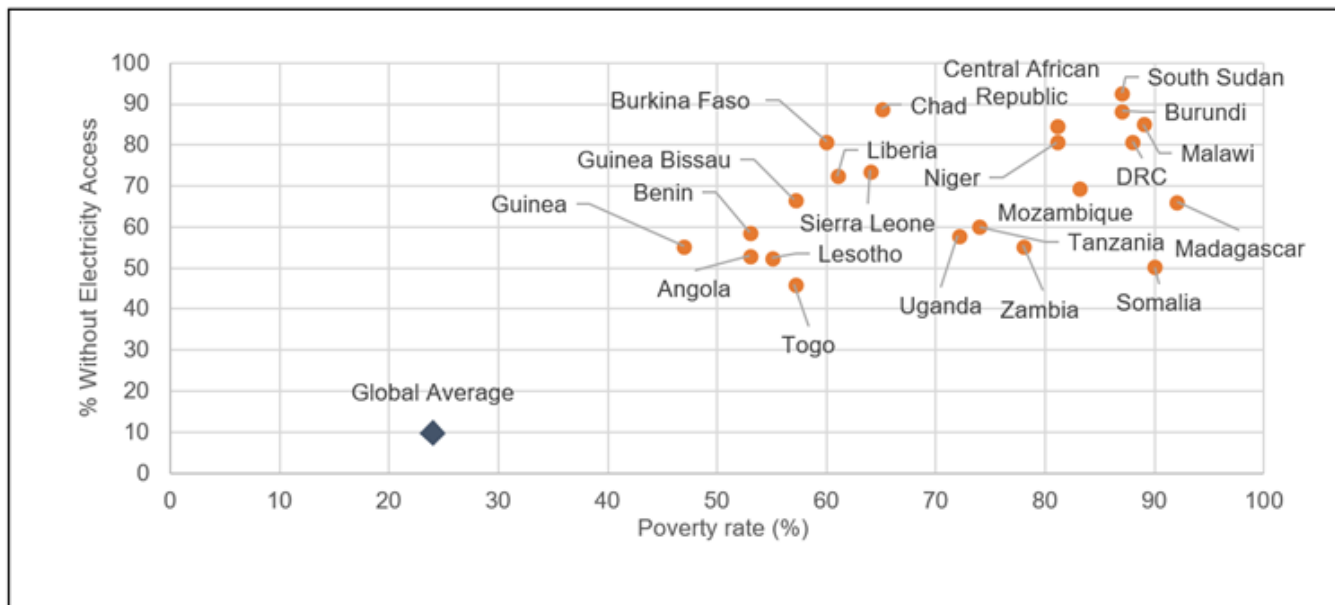


Table 1: Summary of Country Indicators for H2R: HDI and ND-GAIN Climate Vulnerability Ranking, Rural Energy Access and Poverty²³

Country	HDI	ND Gain Index Rank	Most Vulnerable	Most ready	Rural Electricity Access %	Rural Poverty (MPI) %
Benin	0.43	152	16	133	9%	79
Burkina Faso	0.45	161	27	159	2%	95
Burundi	0.43	165	22	173	2%	81
Chad	0.40	182	3	190	1%	90
DRC	0.48	178	10	189	1%	87
Guinea	0.47	146	31	146	24%	50

²³ UN HDI <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>, ND GAIN Country Index <https://gain.nd.edu/our-work/country-index/>, <https://ophi.org.uk/multidimensional-poverty-index/> for MPI and WEO 2019 for rural electricity access <https://www.iea.org/reports/world-energy-outlook-2019> .

Guinea-Bissau	0.48	180	3	171	7%	81
Lesotho	0.51	129	56	149	26%	28
Malawi	0.51	163	25	160	5%	55
Mozambique	0.45	156	48	171	22%	88
Niger	0.40	176	1	136	2%	97
Sierra Leone	0.48	155	19	144	6%	78
Somalia ²⁴	n.d	172	n.d	n.d	4%	81
Togo	0.54	129	42	127	19%	54
Uganda	0.53	166	13	160	17%	65
Zambia	0.57	138	42	143	6%	66
H2R Average	0.48	159	24	158	11%	73

This vulnerability has recently been confirmed by a large-scale climate survey of thousands of people across six sub-Saharan African countries. The survey found that 88% of people feel that climate change is already having an impact on their everyday lives, that 63% of people consider climate change and environmental degradation to affect their income and source of livelihood, and 57% of people have already incurred losses due to climate change.²⁵ These findings demonstrate that climate change is no longer only a future threat, but a reality that many vulnerable people in Africa are living with already.

Climate Adaptation Context

The IPCC sixth assessment report shows that most regions in Africa have experienced a gradual increase in temperatures in the past 40 years which has led to an increase in the frequency of annual heatwaves.²⁶ Rainfall trends in the continent have higher degrees of uncertainty and only a few regions show statistically significant historical rainfall trends. Projections indicate that both mean temperatures and temperature extremes will increase over most of the continent, while mean annual rainfall is likely to increase over the eastern Sahel, and eastern and central Africa.²⁷ The frequency and severity of heavy rainfall events are projected to increase throughout Africa, except in the northern and southwestern regions.²⁸ Conversely,

²⁴ n.d = no data. See Annex 2.2m for sub-national data points where this has been possible.

²⁵ EIB Climate Survey (2022)

²⁶ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

²⁷ Ibid.

²⁸ Ibid.

the frequency of drought conditions are projected to increase over southwestern Africa and coastal north Africa. Region-specific climate projection information is summarized in the table below. ²⁹

Table 2: Summary of key observed, and global warming level (GWL) projected climate variables for Central Africa, West Africa and Southern Africa IPCC AR6 WGII Chapter 9 (Africa)

		Central Africa	Eastern Africa
Temperature Change	Observed	• Mean annual and seasonal temperatures have increased by 0.75-1.2°C since 1960. • The number of hot days and heat waves has increased since 1979 while the number of Cold extremes has decreased.	• Mean and seasonal temperatures have increased by 1-3°C since the mid- 1970s. • The frequency, severity and duration of heat waves have increased relative to 1980-2000. • The frequency of tropical nights has increased by 4-13 nights each decade since 1961
	Mid-century Projection	• Mean annual temperatures are projected to increase by 0.6°C, 1.1°C and 2.1°C under 1.5°C, 2°C and 3 GWL respectively. • The number of annual very hot days whereby the maximum temperature will	• Mean annual temperatures are projected to increase by 0.6°C, 1.1°C and 2.1°C under 1.5°C, 2°C and 3°C GWL respectively. • Heatwave frequency and intensity is

²⁹ IPCC AR6 Assessment Reports, World Bank's Climate Change Knowledge Portal (CCKP), the Climate Analytics' Climate Impact Explorer and the GCF Climate Information Tool have been used to assimilate the observed historical and projected climates for the African continent, key H2R regions and target countries. SSP1-1.9, SSP2-4.5, and SSP5-8.5 climate scenario models have been used to assess climate projections predominantly over the period 2040 – 2059.

		exceed 35°C is projected to increase by 4-27 days at GWL 2 and 35-39 days at GWL 3°C. • Heatwaves frequency is projected to increase while the duration of extreme heatwave events may last longer than 180 days under 4.1°C GWL. • Lethal heat days will increase from <50 days to 50-70 days (1.6 GWL) and 100-150 days (2.5°C GWL)	projected to increase under all scenarios. • Heat-related mortality risks will become 6-9 times greater under 2°C GWL. • Lethal heat days will increase from 50 days to 50-150 days (1.6°C GWL) and 100-250 days (2.5°C GWL)
Precipitation Change	Observed	• A severe lack of weather station data has led to large uncertainties around observed Climate Change patterns for the region. • There is evidence of an increase in drought conditions since the mid-20th Century. Seasonal April-June precipitation levels have declined by 69mm annually in most regions except the northwest.	• The Gulf of Guinea and Sahel have experienced more severe precipitation events and the frequency of storm events has tripled. • Extreme precipitation has led to flooding in some areas. • Drought has increased in frequency since 1950.
	Mid-Century Projections	• There is low Confidence in mean rainfall Change under the low emission scenario, whereas	• Precipitation is projected to decrease in the west region and increase in the east

		mean annual rainfall will increase by 10-25% under GWL 3°C and GWL 4.4°C. - The intensity of extreme precipitation events is projected to increase under the high emission scenarios. This will increase the likelihood of widespread flood occurrences before, during and after the mature monsoon season. - There is low Confidence in the projected Change of drought frequency in the region.	region with medium confidence. - The severity, frequency and duration of drier and drought-prone Conditions is projected to increase. The duration of drought Conditions will double to four months under RCP 8.5°C. - Nevertheless, the frequency and intensity of extremely heavy precipitation events is projected to increase under mid-and-high emission scenarios
Southern Africa			
Temperature Change	Observed	- Mean annual and seasonal temperatures have increased by 1.04°C and 1.44°C respectively since 1961. - The number of hot days and increased heat stress has increased in the last four decades while the frequency of cold extremes and cold days has decreased.	
	Mid-century Projections	Mean annual temperatures are projected to increase by 1.2°C, 2.3°C and 3.3°C under 1.5°C, 2° and 3°C GWL respectively. - The annual number of heatwaves is projected to increase by between 2-4 (1.5°C GWL), 4-8 (2°C GWL) and 8-12 (3°C GWL)	

		• Hot and very hot days are projected to increase under 1.5°C and 2°C GWL.
Precipitation Change	Observed	• The frequency of dry spell and agricultural drought in the region has increased since 1961. • The frequency of meteorological drought increased by between 2.5-3 events per decade. • The number and intensity of extreme precipitation events have increased over the last century.
	Mid-century Projections	• Mean annual rainfall in the summer rainfall region is projected to decrease by 10-20% • The number of consecutive dry days during the rainy season are projected to increase under RCP8.5 • The frequency and duration of droughts is projected to increase over large parts of southern Africa at 1.5°C and extreme droughts at 2°C GWL.

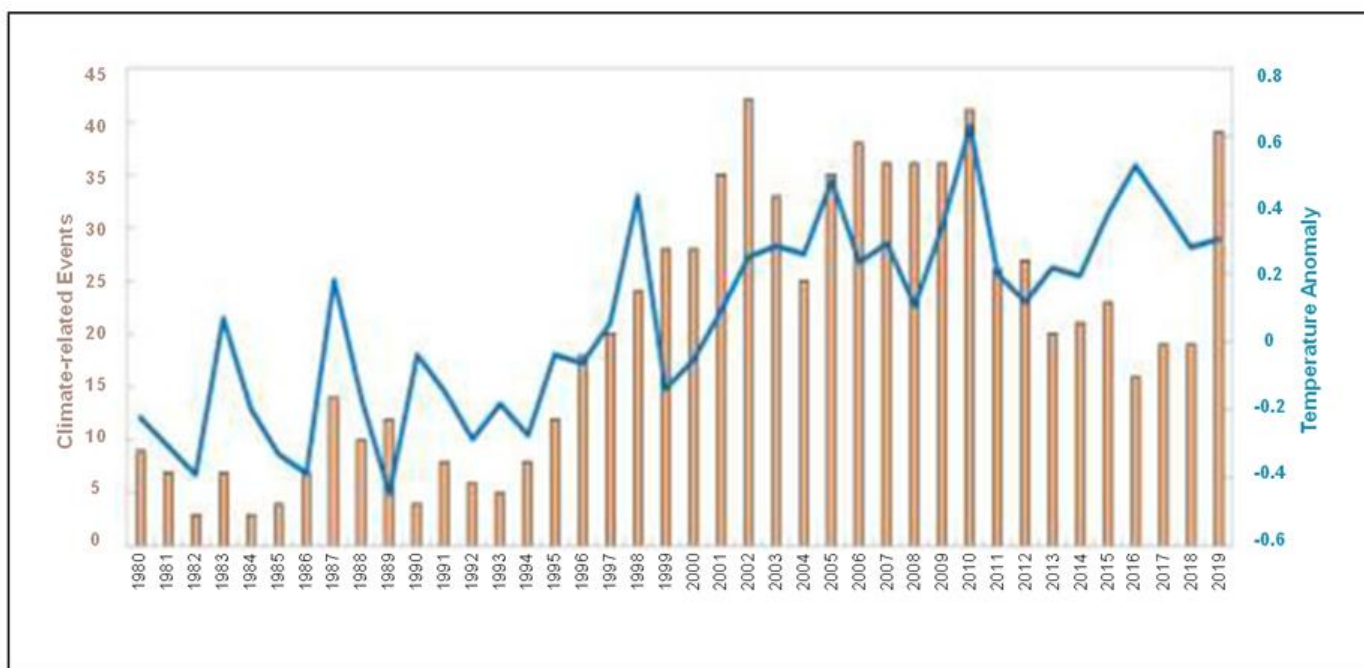
The number of observed climate-related hazard events in sub-Saharan Africa has also increased in recent years (Fig 2) and at a faster pace than the rest of the world.³⁰ Given the high poverty levels and dependence of most rural households on agriculture, climate events exacerbate the vulnerability of the region. As seen in Fig 3, climate-related epidemics, floods and storms have been the most common climate-related disasters faced by the continent's fragile states.³¹ An increase in the frequency and intensity of natural disasters has also been observed, which has been accompanied by an increase in the number of people being affected by these events. Six times more people were affected by droughts in 2010-2019 relative to 1970-1979, while flooding impacted eight times more people over the same period.³² Perhaps unsurprisingly, the Hardest-to-Reach countries rank as the top 15% most vulnerable countries in the world on the ND-Gain index.

³⁰ Africa's Pulse [An Analysis of Issues Shaping Africa's Economic Future](#) (2021)

³¹ IMF [Climate Change in Sub-Saharan Africa Fragile States: Evidence from Panel Estimations](#) (2022)

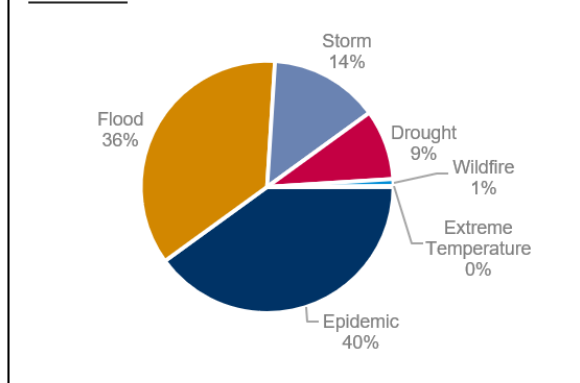
³² Africa's Pulse [An Analysis of Issues Shaping Africa's Economic Future](#) (2021)

Fig 2: Historical climate-related events and temperature anomalies for sub-Saharan African fragile states between 1980 and 2019



Regional vulnerability to the changing climate and increasing climate hazards is exacerbated by non-climate drivers such as high poverty levels, weak infrastructure systems, a lack of emergency response capabilities, and fragile grid electricity and telecommunications networks. In particular, a lack of access to modern energy sources heightens Africa's climate vulnerability. The region's energy systems have been compromised both by extreme and slow-onset climate events, with resulting economic losses, disruptions of services and impacts to wellbeing. Accordingly, the Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6) considers decentralized energy generation to be an important component of the systems transition of energy, where adaptation, mitigation and development solutions are pursued together to exploit synergies and reduce trade-offs within the sector.³³ Within energy system transitions, the most feasible adaptation options support infrastructure resilience, reliable power systems and efficient water use: "Energy generation diversification, including with renewable energy resources and generation that can be decentralized depending on context (e.g., wind, solar, small scale hydroelectric)...can reduce vulnerabilities to climate change, especially in rural populations (high confidence)".³⁴

Fig 3: Share of climate-related natural disasters in Sub-Saharan African fragile states between 1980 and 2019



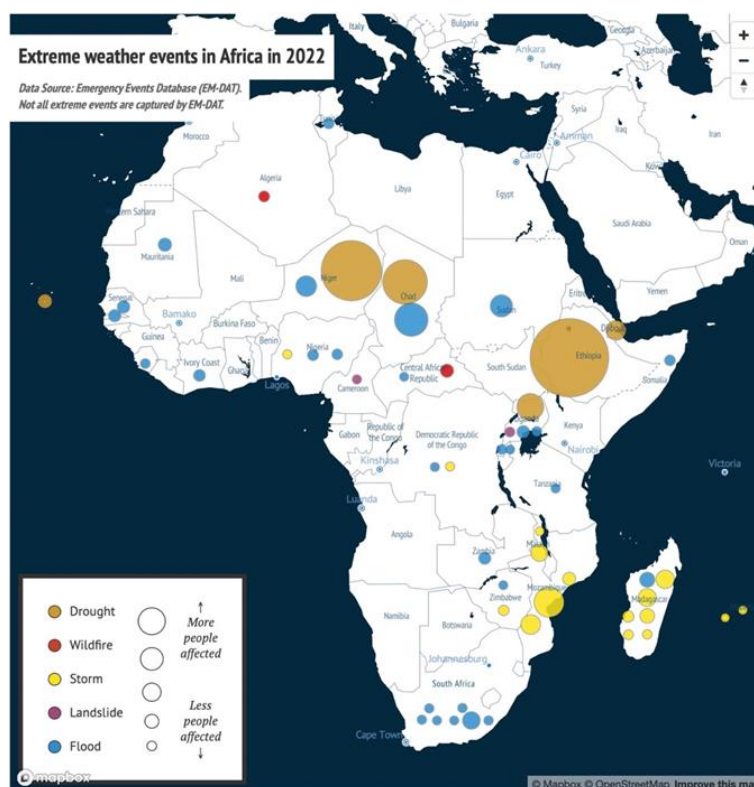
³³ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability](#) (2022)

³⁴ Ibid [Chapter 9: Africa](#) (2022)

It is now well understood that extreme weather events induced by climate change are a major source of vulnerability.³⁵ While these impacts are well tracked in developing countries, extreme weather events and their impacts tend to be underreported when they happen in Africa, meaning that we have a much lower level of understanding of their effects on peoples' lives.³⁶ This is closely linked to a lack of effective hydrometeorological systems in particular in the North and Central African regions.³⁷ Furthermore, even where events are registered internationally, they often are those affecting the more population-dense, and better connected urban locations, while similar events in rural regions simply go unnoticed.

With this being the case, it is likely that the vulnerability profile of African countries understates the levels of exposure of the rural poor, who are least able to implement measures to protect themselves from this. EM-DAT provides a detailed look at extreme weather events impacting Africa in 2022 (see Figure 4 below).

Figure 4: Extreme Weather Events in Africa, 2022³⁸



The energy access-climate-rural poverty context

Today's climate change impacts are felt most acutely in developing economies. In sub-Saharan Africa, where 70% of the world's LDCs are located, climate hazards such as floods, rising temperatures, aridity and droughts, storms, among others, are real and present threats to lives and livelihoods, inflicting tangible

³⁵ <https://unfccc.int/topics/resilience/resources/climate-related-risks-and-extreme-events>

³⁶ <https://www.preventionweb.net/news/analysis-africas-unreported-extreme-weather-2022-and-climate-change>

³⁷ <https://www.ctvnews.ca/climate-and-environment/africa-needs-better-weather-warning-systems-urge-experts-1.5934651>

³⁸ <https://www.preventionweb.net/news/analysis-africas-unreported-extreme-weather-2022-and-climate-change> data from <https://www.emdat.be>

collateral damage and physical risks to infrastructure investments, water and food systems, public health and agriculture.³⁹ The economies and livelihoods of most sub-Saharan African nations are driven by rain-fed agriculture and pastoralism. FAO reported that agriculture alone provided jobs for almost 50% of the total labor force of these countries.⁴⁰

At the same time, rural poverty is widespread in the target countries of H2R, and considerably exceeds the rate of urban poverty (see Figure 5 below). As mentioned, all H2R countries are LDCs with a high incidence of both poverty and climate vulnerability among the millions of smallholder farmers and the communities that dwell in rural, often informal, settlements with low access to the basic services sometimes available in cities. In addition, as the Global Center for Adaptation (GCA) 'State and Trends in Adaptation Report 2021' notes, large swathes of Africa, and especially the drylands areas that cover 60 percent of the continent, are warming at a rate twice the global average, putting half a billion people at risk.⁴¹

Rural poverty is also exacerbated by the lack of access to basic services provided by electricity. This explains why spatial and temporal variations in rainfall and recurrent drought episodes have severe impacts on the livelihood of people living throughout the continent.⁴²

Figure 5: Rural and Urban Poverty Rates in H2R target countries as share of the population⁴³

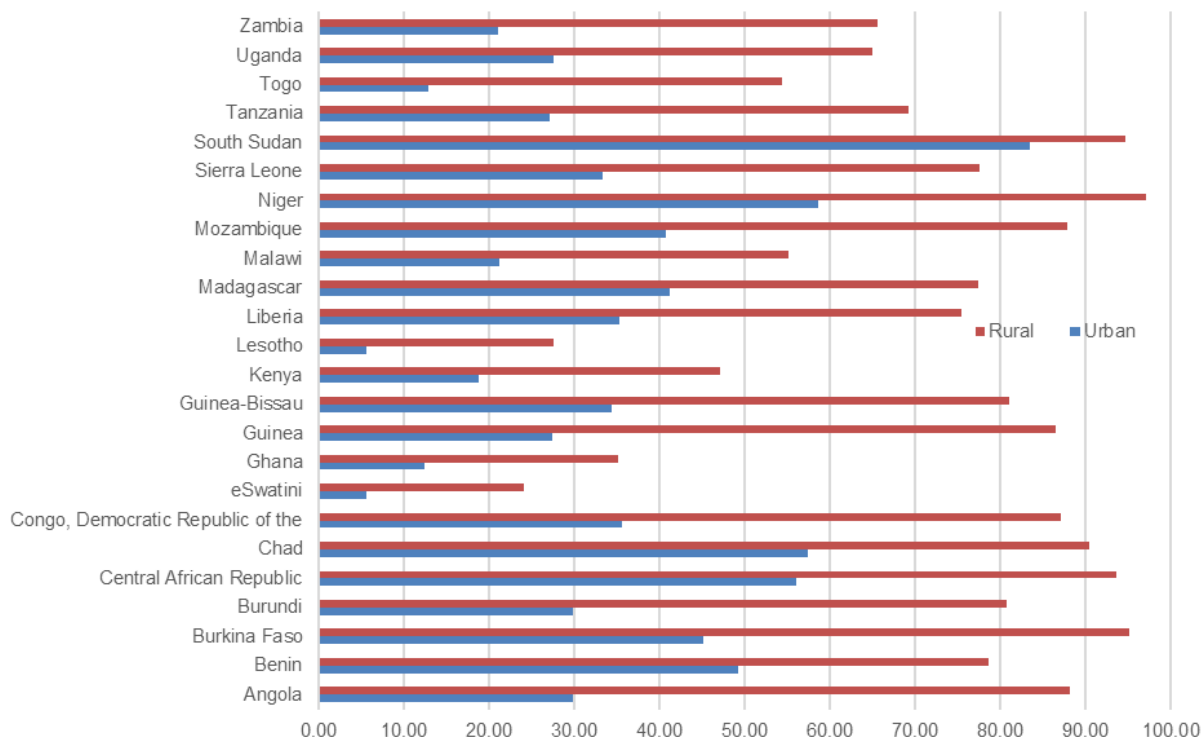
³⁹ GCA [State and Trends in Adaptation Report](#) (2021)

⁴⁰ <https://www.fao.org/3/bo092e/bo092e.pdf>

⁴¹ Ibid.

⁴² Vulnerability Assessment of West African Countries to Climate Change and Variability

⁴³ <https://ophi.org.uk/multidimensional-poverty-index/>



Up until now the continent has seen adaptation interventions that are primarily reactive rather than proactive, such as efforts focused on disaster relief or reconstruction and recovery following severe floods or a pandemic. The Africa-Led Consultative Dialogue Report on adaptation highlights the complementary need for a hazard and risk-oriented approach to adaptation, which considers it holistically across factors such as vulnerability, exposure, and adaptive capacity. The risk approach to adaptation can potentially “enable preparedness to not just address the climate risks but also the various shocks such as natural disasters...that are either exacerbated by climate change or exacerbate climatic impacts.”⁴⁴

The GAC 2021 report points to new research which considers interlinkages between climate, conflict and energy.⁴⁵ In the face of increasing climate stressors, there is the risk that certain energy systems are potentially maladaptive and could play a contributory role in increasing the risk of conflict. The disruption of energy flows at the local, national, regional and global scale can increase the risk of conflict, and energy sources can also become the target of sabotage or attacks, particularly by non-state armed groups. Additionally, there is a close link between energy and food security. Restricted energy access and increased energy prices can have significant impacts on overall food security, contributing to drivers of conflict. Though the research in this area is nascent, it points to the need to develop energy systems that are the most resilient and least vulnerable to the physical and human dimensions of climate risk. By many of these adaptation-based criteria, the case for decentralized solar power, particularly residential solar, as a prominent part of the electricity mix, has become a compelling one.

⁴⁴ ARIN & UKRI [Adaptation Research in Africa: Progress and Gaps Insights from the COP26 Africa-Led Consultative Workshops](#) (2021)

⁴⁵ Ibid.

The ultimate measure of the effects of energy access on climate resilience will be the extent to which off-grid residential solar electrification can be associated with reducing sensitivity and exposure, and increasing adaptive capacity during adverse effects of shocks and stresses related to climate (see Figure 2). Some examples of this impact pathway can be seen in Table 3 below. Energy access which, for example, supports early warning systems can prevent human and economic losses in case of extreme events, while extending rural electrification can enable adaptation and mitigate climate vulnerability. If based on a diversified and decentralized network of energy sources, it can reduce the vulnerability of energy supply.⁴⁶ As a review of NAPs and NDCs highlights⁴⁷, low levels of energy access are considered to reduce resilience, while measures to increase resilience and adaptability are often leading to higher energy needs, risking increased emissions. It is therefore important for energy access measures to capitalize on the opportunities of energy transition. For example, modern energy access in drylands reduces reliance on biomass fuels, leading to less wood extraction and charcoal use - which currently exceeds tree production, and increases resilience against the climatic trend of soil depletion and tree loss (roughly three million hectares of forest are lost annually⁴⁸). At the same time, lack of reliable access affects the ability of people to access important health, or early warning information in the context of disaster preparedness - both of these are areas highlighted for improvements in NDCs and NAPs⁴⁹, but the impact of this improvement will be limited without basic access. Per capita energy consumption is also strongly correlated to economic output. At higher levels of energy consumption and through productive uses of energy, we observe how clean energy can power adaptation through value-added agro-systems that unlock socioeconomic opportunities.⁵⁰ Adaptation impacts are also unlocked when we understand and capture the potential synergistic links among actions and technologies, for example household entry-level energy access alongside mobile telecommunications, TV or radio to enable early hazard warning.

Socio-economic development is one of the foremost strategies to reduce community vulnerability,⁵¹ and access to clean energy is a cornerstone of sustainable socio-economic development. Energy access as a whole is thus linked to community climate resilience and adaptation, and OGS have an important role to play in providing the stepping stone towards wider access. Therefore, delivering enhanced, even basic energy access has the potential to unlock a wide range of adaptation-linked socio-economic development outcomes that drive resilience of the population.^{52 53 54 55}

At a practical level, lighting, cooking, heating, cooling, communication and information access services are derived from energy. A community's basic needs, economic activities and institutions (schools, health centers, government buildings) are substantially enhanced through access to these energy services. For these reasons, long-term access to reliable energy builds self-resilience and adaptive capacity within

⁴⁶ Davide, M.; de Cian, E.; Bernigaud, A. Building a Framework to Understand the Energy Needs of Adaptation (2019)

⁴⁷ Davide, M.; de Cian, E.; Bernigaud, A. Building a Framework to Understand the Energy Needs of Adaptation (2019)

⁴⁸ N & C Reij (eds) [Dryland restoration successes in the Sahel and Greater Horn of Africa show how to increase scale and impact](#) (2020)

⁴⁹ Davide, M.; de Cian, E.; Bernigaud, A. Building a Framework to Understand the Energy Needs of Adaptation (2019)

⁵⁰ GCA [State and Trends in Adaptation Report](#) (2021)

⁵¹ Davide, M.; de Cian, E.; Bernigaud, A. Building a Framework to Understand the Energy Needs of Adaptation (2019)

⁵² GOGLA How to make climate finance a game-changer for the off-grid solar industry (2022)

⁵³ Mulugetta, Y.; Hagan, E., Kammen, D. Energy access for sustainable development (2019)

⁵⁴ Murphy, B. and Corbyn, D. Energy and Adaptation: Exploring how energy access can enable climate change adaptation (2013)

⁵⁵ Energy and Environment Partnership Trust Fund (EEP Africa) Energising Resilience: Climate Co-Benefits from Clean Energy Projects (2020)

households and local communities. Specific benefits of enhanced access include (see also Table 3 for examples):

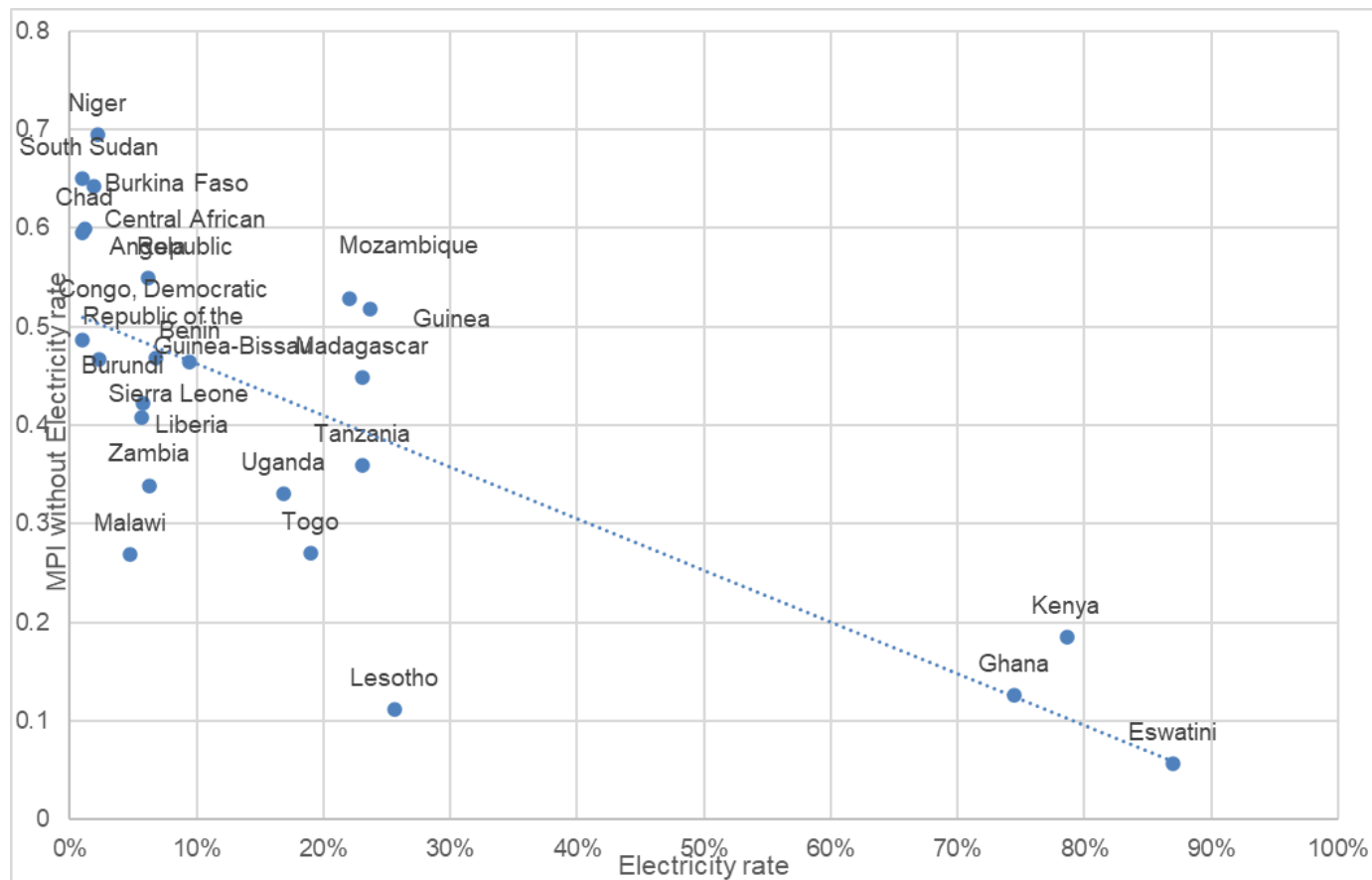
- Enhancing energy, food and water security by improving self-reliance in the face of increasing climate-induced risks;
- Providing easier and lower-cost access to information and digital services which may enhance disaster planning and response to climate-driven events such as EWE;
- Improving the health of households and communities by e.g. enabling access to health-information, providing clean lighting without accompanying indoor air quality degradation, thus reducing climate-induced health risks (e.g. vector borne diseases) while utilizing mitigation measures to improve health and living standards;
- Diversifying jobs, particularly for women who can expand their opportunities as a result of time-saving achieved through greater household productivity, thus increasing their resilience to slow-onset climate events; and
- Diversifying livelihoods and the local economy by boosting incomes, savings, and productivity of households and enterprises, also increasing their resilience to slow-onset events.

There is also a moderately strong link between rural poverty and access to electricity, with a correlation factor of 0.74.⁵⁶ While there are some countries that see lower levels of rural poverty on the MPI index, there is a clear clustering around combined high levels of rural MPI and low levels of rural electricity access, as shown in Figure 6 below.

Figure 6: Adjusted* rural MPI and Electricity Access in H2R Countries⁵⁷

⁵⁶ <https://www.statology.org/what-is-a-strong-correlation/>

⁵⁷ <https://ophi.org.uk/multidimensional-poverty-index/> for MPI and WEO 2019 for rural electricity access <https://www.iea.org/reports/world-energy-outlook-2019> .



*The MPI has been adjusted by replacing electricity access with a uniform dummy indicator in order to avoid creation of a feedback loop.

Through financing the delivery of affordable, reliable and modern off-grid solar products, Hardest-to-Reach will be able to contribute to climate resilience that is anticipatory, absorptive and adaptive, at the household and national level.

Table 3: Climate Hazards and Adaptation Impact of H2R

Climate hazard	SSA Region	Risks to people and energy sector	Adaptation/resilience impacts of off-grid solar	GCF results area	Example
Temperature rises - Temperatures in Africa are	- All	- Excess death rates from non-optimal temperature in Sub-Saharan Africa are estimated to be nearly double the global	- Off-grid solar helps to diversify the energy mix and improve energy security and resilience, as it reduces pressure on the grid. This happens because	ARA 1: Most vulnerable people and communities	South African Energy Crisis, 2007-present: Increased energy demand for cooling during heatwaves has contributed to the

projected to rise faster than the global average increase during the 21st century - It is expected that there will be an increase in the number of hot days and nights, and a decrease in the number of cold days and nights across Africa.⁵⁸		average, and are set to increase with global warming.⁵⁹ - Heatwaves lead to increased energy demand, as more people use air conditioning and fans. In fact, energy demand for cooling over Africa is expected to increase to 1.2% of total final energy demand by 2100 compared to 0.4% in 2005.⁶⁰ - Higher temperatures reduce the efficiency and capacity of thermal power plants and can reduce the availability of water for cooling because of increased evapotranspiration. - The co-occurrence of higher electricity demand and lower production efficiency of the grid exacerbates risk of life and health loss during a heatwave.	distributed renewable energy provides households with independent energy sources, and continues to provide energy to people even if the grid has been compromised because of surges in demand during high temperatures due to increased cooling needs, or damage caused by an extreme weather event. - Off-grid solar also makes people more able to adapt to heatwaves, as they are not reliant on the grid for electricity to power cooling appliances like fans. Sales of fans, which are often bundled with SHS systems, have been growing steadily in sub-Saharan Africa over the last 2 years, demonstrating that people have increasing demand for solar cooling products.⁶¹	<u>ARA 2:</u> Health, well-being, food and water security	ongoing energy crisis in South Africa. When demand exceeds supply there are energy blackouts, which prevents people from using cooling technologies and leads to heat-related illnesses and death.⁶² Sales of off-grid solar products in South Africa grew by 127% between 2021-21, despite the prevalence of the national grid. This demonstrates that off-grid solar products can provide energy security benefits to people with insecure grid connections, as well as people with no access to electricity.⁶³
Droughts - There will likely be decreases in mean annual precipitation in some regions of Sub-Saharan Africa. - It's expected that there will be a greater number of	- All: particularly southern and central Africa	- Water shortages caused by droughts reduce energy security through disruption to power generation. African countries that rely primarily on hydropower are particularly sensitive to climate variability. - There is projected to be up to a 20% decline in gross hydropower potential across Africa because of decreased water resources for energy generation.⁶⁵	- Many countries in sub-Saharan Africa are heavily reliant on hydropower, which is increasingly vulnerable to the effects of climate change. Policymakers across drought-prone and hydro dependent countries increasingly recognise the role of off-grid RE to strengthen the resilience of the country's energy provision and reduce the impact of power outages	<u>ARA 1:</u> Most vulnerable people and communities	<u>Southern Africa drought, 2015-2016:</u> Power generation reduced by 75% at Kariba dam (Zambia) in 2016, causing a national power deficit of around 985 megawatts, and the Cahora Bassa dam (Mozambique) reduced to 34% of its capacity with widespread impact

⁵⁸ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

⁵⁹ Ibid.

⁶⁰ Ibid.

⁶¹ GOGLA [Off-Grid Solar Market Trends Report 2022: State of the Sector](#) (2022)

⁶² IEA [Africa Energy Outlook](#) (2022)

⁶³ GOGLA [Global Off-Grid Solar Market Report Semi-Annual Sales and Impact Data](#) (multiple years)

⁶⁵ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

severe droughts across Sub-Saharan Africa. ⁶⁴		- There is also often a regional component to hydropower vulnerability, as grids in multiple countries rely on similar rainfall patterns for electricity generation, meaning they will also suffer the consequences of lack of water from droughts at the same time, causing concurrent climate-related electricity supply disruption. If all the large dams that are currently planned in eastern and southern Africa are constructed, by 2030 70% of total hydropower generating capacity in eastern Africa will be dependent on areas with similar rainfall patterns. In southern Africa 59% of hydropower generation will depend on areas with similar variability in rainfall.⁶⁶ - Droughts can also lead to a decline in thermopower potential because of decreased water resources for cooling.	on households. Off-grid solar products themselves do not need water to produce electricity. This means that people will therefore have access to electricity even throughout periods of drought. - E.g., 90% of Mozambique's grid electricity is generated by hydropower, making its energy supply vulnerable to any climate events that affect water availability or hydropower infrastructure. As a result, the Government of Mozambique has developed multiple strategies that prioritize diversifying its energy sources and increasing the share of on- and off-grid renewables in its energy mix. In fact, the country aims to connect 32% of its population to off-grid renewable energy sources by 2030.⁶⁷ - Decentralized, off-grid energy reduces pressure on the grid, thereby reducing pressure on water resources, and strengthens the resilience of the energy mix to climate events like droughts.		on electricity supplies across southern Africa.⁶⁸ Malawi, Tanzania, Zambia and Zimbabwe all experienced widespread and prolonged load shedding due to low rainfall. This lasted up to 8 hours at a time in Zambia, and up to 25 hours at a time in Malawi. The impact was felt throughout the economy and reflected in reduced GDP growth in Zambia.
Floods - Flooding is already a major hazard across Africa	- All	- Flooding can cause damage to energy infrastructure and disrupt energy generation and transmission.	- Decentralized, off-grid energy reduces pressure on the grid and strengthens the resilience of the energy mix to climate events like floods.	ARA 1: Most vulnerable people and communities	Malawi floods 2019: The total cost of damages to the energy sector was

⁶⁴ Ibid.

⁶⁶ CDKN [Hydropower plans in eastern and southern Africa could put electricity supply at risk](#) (2018)

⁶⁷ FUNAE Global off-grid solar forum & expo 2022 Opportunity for solar off-grid in Mozambique (2022)

⁶⁸ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

and is increasing. - It is likely that there will be an increase in the number of heavy rainfall events in all regions in Africa, causing more frequent and intense floods. - There are also projected to be increases in annual rainfall in some African regions.⁶⁹			This is because they are not dependent on power plants or grid infrastructure that can be damaged by floods, and prevented from supplying electricity to places that are potentially many miles away from the place that has been damaged. - People with off-grid solar products will therefore maintain access to energy even if the grid has stopped supplying electricity because of floods. - The portability of many OGS products means that if people are displaced because of a flood, they will be able to carry their OGS product with them, and will still have access to energy in their new accommodation.⁷⁰		\$3.1 million. 86,976 people were displaced.⁷¹
Extreme weather events - It is predicted that there will be more frequent extreme weather events such as tropical cyclones across Africa throughout the 21st century.⁷²	- All	- Extreme weather events can cause damage to energy infrastructure and disrupt energy generation and transmission.	- Decentralized, off-grid energy reduces pressure on the grid and strengthens the resilience of the energy mix to extreme weather events. - This is because OGS products are not dependent on power plants or grid infrastructure that can be damaged by extreme weather events and prevented from supplying electricity to places that are potentially many miles away from the place that has been damaged.	<u>ARA 1</u> : Most vulnerable people and communities	<u>Cyclone Idai, 2019</u>: the cyclone destroyed or damaged: • 1,345 km of transmission lines, • 10,216 km of distribution lines • two 90 MW generation plants • 30 sub-stations • 4,000 transformers

⁶⁹ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

⁷⁰ Overseas Development Institute [How solar household systems contribute to resilience](#) (2017)

⁷¹ Malawi Government [Floods Post Disaster Needs Assessment Report](#) (2019)

⁷² IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability, Chapter 9: Africa](#) (2022)

			- People will therefore maintain access to electricity even if there is damage to the grid because of extreme weather events. - Many entry-level, smaller OGS products are portable single units, and therefore easily carried. The portability of many OGS products means that people could store them inside their houses during extreme weather events, therefore protecting them from the impacts of climate change. Furthermore, if people are displaced because of a climate event, they will be able to carry their OGS product with them, and people will still have access to energy in their new accommodation. - People with OGS products may also have better access to information both prior to and following extreme climate events because of the ability to charge mobile phones. This will allow people to better plan for and recover from extreme weather events.		This resulted in an estimated loss of \$47.9 million in the energy sector in Mozambique.⁷³ <u>Uganda, 2012:</u> In 2012 the Government of Uganda implemented the Strengthening Climate Information and Early Warning Systems Project to warn smallholder farmers and allow them to respond to extreme climate events, including drought.⁷⁴ This project shared relevant weather alerts over cell phones giving farmers access to valuable information that helped them adapt to climatic changes. Farmers received up-to-the-minute weather information and improved crop forecasts that helped them know when to plant, when to harvest, and when to take their crops to market. The new weather monitoring systems translated into saving lives, building better livelihoods for smallholder farmers, and enabling climate-smart decisions. This boosted Uganda's agricultural
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⁷³ Ibid.

⁷⁴ UNDP [Uganda Annual Report](#) (2017)

					productivity, and drove economic development. OGS will allow smallholder farmers to charge their phones more easily and cheaply, thus giving them access to climate information and early warning systems.
Coastal erosion and flooding - Coastal systems across Sub-Saharan Africa will experience impacts through sea level rise - They will also experience impacts through high sea levels combined with storm swells.	- Eastern, western and southern Africa	- Coastal areas are undergoing significant environmental degradation, leading to deaths, ecosystem losses and damages to assets, including energy infrastructure.	- Decentralized, off-grid energy reduces pressure on the grid and strengthens the resilience of the energy mix to coastal erosion and flooding. - This is because OGS products are not dependent on power plants or grid infrastructure that can be damaged and prevented from supplying electricity to places that are potentially many miles away from the place that has been damaged. - People will therefore maintain access to electricity even if there is damage to the grid because of coastal erosion and coastal flooding. - The portability of many OGS products also means that if people are displaced because of a coastal flood, they will be able to carry their OGS product with them, and people will still have access to energy in their new accommodation.	<u>ARA 1: Most vulnerable people and communities</u>	<u>West African Coast:</u> Environmental degradation in the coastal areas of Benin, Côte d'Ivoire, Senegal, and Togo cost \$3.8 billion, or 5.3% of the four countries' GDP in 2017. Part of this cost was due to damage to coastal power plants.⁷⁵

⁷⁵ World Bank [The Cost of Coastal Degradation in West Africa](#) (2019)

Attention to principles for demonstrating the climate rationale of adaptation activities

Table 4: Response to H2R Submission to B33/05 requirements

Classification	B33/05 Requirement	Acumen H2R
(a) Identification:	Adaptation proposals should show how the activity addresses current or future projected climate change risk or impact, and why it is likely to be an effective response. Proposals should identify the systems at risk and the climate change hazard affecting them or expected to in the future. They should show how climate change has led, or will lead, to the specific risk or impact that the proposed activity addresses using the best available information. Where relevant, proposals should also consider any non- climatic factors that may be causing or exacerbating the risk or impact and describe the interactions between climate change and non-climatic drivers. Vulnerability assessments can be used to identify groups, sectors and sub regions most susceptible to the climate change impact and therefore will provide information to select and prioritize appropriate adaptation outcomes.	
(b) Response:	Proposals should explain how the activity will reduce the exposure and/or vulnerability (of people, systems, or ecosystems) and thus lessen the climate change risk or impact. Where relevant, a justification should be given of why the proposed activity was selected over alternatives. Proposals should consider barriers (e.g., technical, social, institutional, regulatory) to the implementation of the activity and describe how the project aims to overcome those barriers. Proposals should apply a methodological approach for the quantification of the number of beneficiaries expected to result from the activity.	N/A
(c) Alignment:	Proposals should confirm alignment of the proposed activity with the host country's national plans and climate strategies (including their NAPs, NAPAs, long-term climate strategies, and adaptation	

	communications including those submitted as components of NDCs, as applicable).	
(d) Monitoring and evaluation:	Projects with a well-designed theory of change are more likely to result in successful outcomes that can be measured and evaluated. Proposals should include a description of the monitoring and evaluation system that will be used to assess the climate impact of the proposed activity and quantify the adaptation beneficiaries.	

Emissions Context

Sub-Saharan Africa currently accounts for less than 4% of cumulative historical global greenhouse gas (GHG) emissions.⁷⁶ However, the region's economies and populations are growing rapidly and emissions have been rising steadily. Between 1950 to 2016, Africa's CO₂ emissions increased by a factor of 14. Today, Africa is home to 1.3 billion people; this number is projected to grow to 3 billion by 2060.⁷⁷ More than half of today's population currently has no access to electricity. For a significant share of peri-urban and rural households, access to electricity services is made possible through local kiosks which provide cell phone charging, refrigeration and other electricity-driven services. In off-grid areas, these kiosks and other local infrastructure such as clinics are primarily supplied by diesel generators or hybrid mini-grids and people's periodic access to these service points constrains their electricity usage. As a result, off-grid communities are experiencing suppressed energy demand, and current household energy consumption does not reflect the real demand for energy services.

Due to the low base of current energy use on the continent, and particularly in H2R program countries, and the rapidly growing and youthful populations, the continent's future energy trajectory has real implications for global net zero targets and the safety of our planet. In one scenario, an African transition out of energy poverty that relies on fossil fuels, could result in increased CO₂ emissions that would offset a 20 percent cut in the emissions of China, the United States, India, Russia, Japan, and Germany combined.⁷⁸

Currently, the majority of rural households rely on emission-intensive kerosene and biomass as their primary energy source for lighting, cooking, heating etc.⁷⁹ Across the region, the percentage of total wood harvested used as fuelwood varies from 90% in Africa, 62% in Asia, 50% in South America to less than 20% in Europe, North America and Oceania.⁸⁰ Under current projections, electrification can play a role in alleviating the pressure on native forests, while also reducing the share of agriculture, forestry and land

⁷⁶ Carbon Disclosure Project [CDP Africa Report](#) (2020)

⁷⁷ Wilson Center, [The Battle for Earth's Climate Will Be Fought in Africa](#) (2021)

⁷⁸ Wilson Center, [The Battle for Earth's Climate Will Be Fought in Africa](#) (2021)

⁷⁹ World Health Organization [Household air pollution fact sheet](#) (2022)

⁸⁰ IPCC [Agriculture, Forestry and Other Land Uses \(AFOLU\)](#) (2022)

use (AFOLU) activities in the region's emissions of dangerous greenhouse gases such as methane (CH₄) and nitrous oxide.⁸¹ Currently, sub-Saharan Africa's AFOLU activities account for 60% of CH₄ emissions.⁸²

To mitigate and adapt to an increasingly warming planet, it is critical that all countries transition to renewable energy willand, even where energy consumption is currently low, embark on clean energy pathways. The energy pathways and resource dependencies developed across the continent today will have profound climate impacts in the medium and long term. Data shows that developing economies (excluding China) now account for roughly 45% of carbon emissions and will propel almost all the growth going forward.⁸³ Providing first-time electricity access via a low carbon trajectory avoids approximately 268 million tonnes of CO₂eq emissions, with a technology mix that is 45% decentralized renewables.⁸⁴

54 countries in Africa signed the Paris Agreement and submitted ambitious Nationally Determined Contributions (NDCs). Most African NDCs incorporated two targets: an unconditional target that was designed to be met by African countries with their own resources (about 15%), and a conditional target, subject to financial support from the international community (about 85%). Each country's specific NDC adaptation and emission targets as well as climate policies have been assessed and documented in the accompanying country reports. Acumen's objectives through Hardest-to-Reach are directly aligned to National climate priorities, including National Adaptation Plans, National Climate Change Action Plans, and NDC's (see Section D.5).

Complementarity & Coherence

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Present the theory of change (ToC) that contains a goal statement and describes how the proposed project/programme will contribute towards the goal statement by using results chain links from activities, outputs, to outcomes. By referring to the sample ToC diagram template available in the guidance note, present a ToC diagram (approximately 1 page) which visually represents the same logic in the narrative description. The ToC diagram and narrative may include a wide range of co-benefits⁸⁵ as applicable in the context of the project/programme.

Note all co-benefits will need to be further elaborated in section D.3 (sustainable development potential) and correspondent co-benefit indicators should be provided under section E.5 (project/programme specific indicators).

The theory of change should also include any relevant barriers (social, gender, fiscal, regulatory, technological, financial, ecological, institutional, etc.) that need to be addressed as well as risks and assumptions. Note that the assumptions can be elaborated further in sections E.3 (GCF outcome level: reduced emissions and increased resilience) and E.5 (project/programme specific indicators) for each relevant indicator, as appropriate.

Theory of Change Narrative

⁸¹ Ibid.

⁸² Ibid.

⁸³ Center for Global Development [Developing Countries Are Responsible for 63 Percent of Current Carbon Emissions](#)

⁸⁴ Shell Foundation [Unlocking Climate Finance to Accelerate Energy Access in Africa](#) (2021)

⁸⁵ GCF categorizes co-benefits into six areas which are: environmental, social, economic, gender, adaptation (relevant for pure mitigation projects) and mitigation (relevant for pure adaptation projects). Further guidance is available in the funding proposal (FP) guidance note.

In 2020, 733 million people were still living without access to (affordable, clean) electricity, of which 298 million people were in nascent off-grid solar (OGS) markets where there is little commercial OGS activity.⁸⁶ Approximately 70% of this number have the ability to pay but live in areas deemed too remote or too risky to be served by traditional grid extensions and business as usual approaches in these markets. However, if the appropriate investment capital and technical assistance is provided to OGS companies, then they will likely be able to deliver clean energy access and adaptive capacity to low-income, underserved people and build sustainable markets. This will happen because they will be adequately incentivized and capacitated, through an enabling environment, to operate and grow in currently untested, 'Hardest-to-Reach' (H2R) areas.

There are however multiple barriers that prevent access to alternative forms of energy access, such as OGS. There is a funding gap, as the OGS market for the most deprived customers is considered highly risky, with the potential customers often the poorest populations in a country, and there are few track records due to market fragmentation. OGS companies themselves struggle to grow, or even survive, due to a lack of internal capacity and inability to access finance. Markets are often either unregulated, or the regulation is not appropriate for the types of products on offer. Taken together, these barriers lead to a lack of access to capital, in turn leading to OGS companies being underfunded, and to markets stagnating, rather than growing rapidly in response to the clear needs of the rural populations where energy access is lowest.

The Hardest-to-Reach energy access initiative aims to deliver energy access to 72.4M low-income customers in least developed and climate-vulnerable countries in sub-Saharan Africa, 54.3M of whom will be accessing energy for the first time. It will do so by engaging with stakeholders, including financing partners, governments and donors, to establish two new funds, H2R MSF and H2R MEF, which will provide early-stage and scaled-up funding to OGS companies. The engagement will also seek to lower barriers where these relate to e.g. regulation or a lack of shared knowledge. H2R MSF/MEF will then provide impact-linked capital and technical assistance to incentivize off-grid solar companies (OGS) to expand into new, under-reached markets, to build sustainable business platforms and to scale up, and for new OGS to be established. Investee OGS companies can then provide more off-grid solar energy products such as solar home systems to low-income customers via Pay-as-you-go mechanisms, providing many people with first-time access to modern energy.

H2R will be focused on learning how our interventions are able to drive impact at the household, enterprise and market level through a range of different measurement techniques, including feedback from customers (project beneficiaries) through lean data surveys. Furthermore, we will aim to disseminate the learnings to all stakeholders. The sharing of knowledge will be supported through our partnerships with industry, government, civil society and private sector, creating the long-lasting impact of a demonstration effect and the blueprint for other investors to follow.

Hardest-to-Reach focuses on creating impact at the sector, enterprise and household levels in fledgling markets, and thus will seek to deliver a number of outputs by scaling the OGS markets in the participating countries:

⁸⁶ [GOGLA Market Trends Report](#): State of the Sector (2022)

- Through the establishment of H2R MSF/MEF achieve greater provision of climate-resilient clean energy and greater installed renewable energy capacity across energy sectors of participating countries;
- Through engagement at AE/EE level deliver new partnerships to enable last-mile energy access for the poorest;
- Support the creation/growth of new and fledgling OGS businesses through H2R MSF finance and support to allow them to become sustainable, inclusive, and investable, and providing clean energy access to low-income people in hard-to-reach areas;
- Enable the scaling up of existing OGS businesses through H2R MEF finance and support to allow them to reach new clients to whom they can provide clean energy access to low-income people in hard-to-reach areas;
- Achieve sales of OGS systems to 13.3M households, benefiting 72.4M people;
- Deliver 120 MW of new renewable energy capacity.

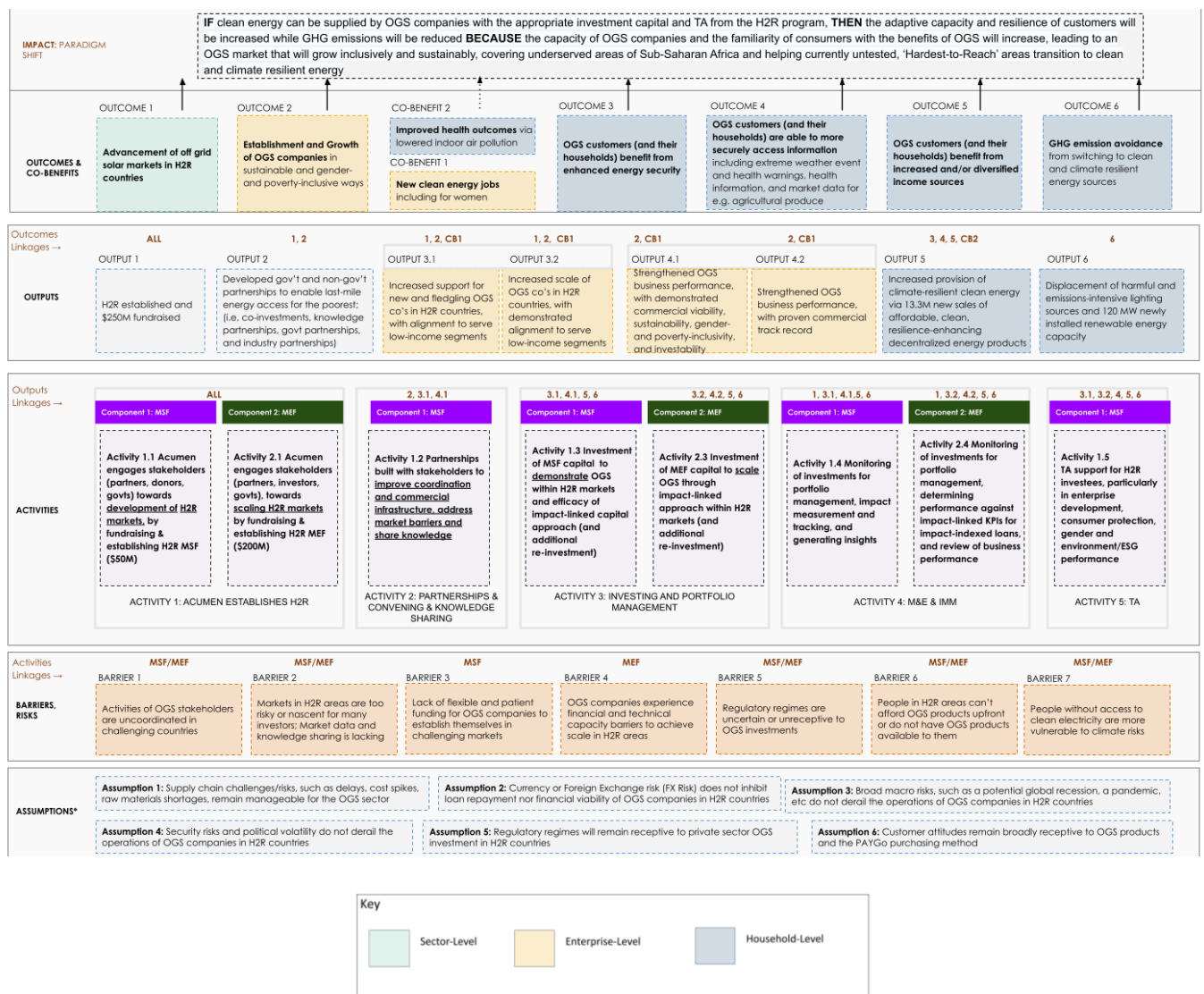
Taken together, these outputs will seek to deliver substantial adaptation, mitigation and enabling environment outcomes, specifically:

- The advancement of the OGS sector in hardest-to-reach countries
- The establishment and growth of OGS companies in sustainable and gender- and poverty-inclusive ways in H2R countries;
- Creation of new jobs in the green economy, including employment women;
- OGS customers and their households benefit from enhanced energy security;
- OGS customers and their households are able to more securely access information including extreme weather event and health warnings, health information, and market data for e.g. agricultural produce;
- OGS customers and their households benefit from reduced harmful emissions from fossil fuels through reductions in indoor air pollution;
- OGS customers and their households benefit from increased and/or diversified income sources;
- Emissions of 5.1MtCO₂eq over the lifetime of the fund are avoided.

Beyond adaptation and resilience benefits, H2R plans to deliver mitigation benefits. By reducing the need to rely on kerosene and other dirty fuels for lighting, through the provision of renewable energy technologies, the programme is expected to reduce over 5MtCo₂ over its lifetime, about equal to the total annual fossil-fuel driven emissions of Burkina Faso in 2022⁸⁷.

The approach is broken down by sector/enterprise/household level below. These impacts are summarized in the Theory of Change diagram below.

⁸⁷ [EDGAR](#), CO₂ emissions of all world countries - JRC/IEA/PBL 2022 Report (2022)



Sector-level: Advancement of the OGS section in H2R countries, and increasing provision of climate-resilient, clean energy across the energy sector of Program Host Nations through market creation, blended finance, partnerships

Challenge

There is a significant investment gap in the sector due to the perception of the high risk and low return of underdeveloped OGS markets. Current OGS stakeholder efforts are fragmented, resulting in overlapping and uncoordinated efforts and missed opportunities for greater impact through collaboration. The use of

blended finance to crowd in climate finance is constrained in part because the industry does not yet have a clearly defined narrative on how OGS products and services contribute to adaptation and resilience. In addition, the scale of many proposed initiatives in this space has been too small to attract commercial investors.

Activities and Outputs

Acumen will establish the H2R blended finance structure to crowd in investors with different risk appetites. Acumen also will develop partnerships that convene different financiers, TA providers, policy advocates, government, and OGS company investees in an effort to jumpstart the OGS market in H2R areas, support the creation of an enabling environment for other investors and businesses and share knowledge within the OGS sector. Specifically:

- Activities 1.1 and 2.1 comprise of fundraising and the establishment of the two funds that facilitates the investments, discussions and interest in the H2R regions.
- Activity 1.2 intends to drive partnership development designed to crowd in new investors, improve the level of knowledge sharing in the sector, coordinate the actions of various stakeholders, and make the initiative's impact sustainable.

Furthermore, the H2R initiative encourages more investment by improving the risk-return profile and track record of these markets. It does so by providing new market entrants with concessional finance and targeted technical assistance, and supporting the development of local networks and capacity. It intends to also increase the capacity of local distributors to become reliable partners to suppliers expanding in the region, and spur inclusive market development with a range of actors.

Outcome

H2R markets are developed and scaled with greater engagement towards an improved enabling environment and market conditions. Markets are shown to be viable, with the right approach, providing incentives and conditions to sustainably attract OGS companies, local participants and investors (philanthropic, development and private capital). The building blocks for OGS market advancement - such as the creation of market intelligence studies, capacity building for off-grid solar manufacturers and local distributors, consumer awareness efforts, policy engagement, access to finance and risk mitigation, and consumer subsidies, where needed - are developed in H2R countries.

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This leads to increased capital inflows (philanthropic, development, private) and to a better-coordinated OGS sector in H2R geographies through blended finance solutions. Investment in the OGS sector will likely lead to clean, climate-resilient energy access for vulnerable people and increased installed capacity across the energy sectors of Program Host Nations.

The impact of OGS on climate change (mitigation and adaptation), and demonstration effect of the intervention, including replicable aspects, will also be better established through the generation and sharing of expert insights and empirical evidence.

Enterprise-level: Fostering the sustainable and inclusive growth of OGS businesses

Challenge

OGS companies struggle to serve the poorest and most vulnerable in H2R areas due mainly to the lack of flexible funding and the concentration of investment in a small number of large players, operating in a handful of well-understood markets. Companies also have limited capacity to serve riskier customers in new markets that lack distribution channels.

Activities and Outputs

Acumen intends to manage the Hardest-to-Reach initiative combining two facilities with \$250M under management. \$50 million of market support capital is available via the philanthropically-backed Market Support Facility, to de-risk H2R markets by building local management and distribution capacity, as well as increasing awareness of OGS products. In addition, the Market Expansion Facility provides companies with \$200 million of impact-indexed loans, providing targeted concessional capital. This impact-linked capital intends to also incentivize companies to retain a focus on entry-level solar products for lower-income customers and prevent mission-drift.

The provision of TA alongside investment aims to also help companies develop capacity in the areas of consumer protection, business development, gender and ESG. Per activity 1.3, 2.3 and 1.5, H2R emphasizes the support (financial and via TA) of enterprises at different stages, in particular to ensure sustainability and impact of the investments.

Outcomes

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As a result of the market development activities, H2R countries benefit from the sustainable and inclusive growth of OGS businesses, able to reach and impact the Hardest-to-Reach. The H2R initiative will seek to help new companies to enter the OGS market, help fledgling companies to grow to critical mass to be able to survive and continue to grow, and finance existing companies to scale up sales of OGS systems.

Demonstration and validation of climate-positive business models in these challenging contexts, through the support of the mechanisms within the Market Support Facility, allows for replication and scale of impact achieved through the Market Expansion Facility. Strengthened local management capacity and distribution networks dynamize the sector, encouraging participation by other players in the market, while increased awareness of OGS solutions and positive customer experiences serve to build trust and stimulate demand. OGS solutions become more widespread, affordable, and cost effective.

In addition, jobs are created and sustained in the clean energy economy, with a significant portion of jobs being a source of employment for women.

Household level: Delivering 13.3M sales of off-grid solar projects and increasing the resilience and adaptive capacity of 54.3M people in Hardest-to-Reach areas and avoidance of 5.1M tCO₂.

Challenge

In H2R areas, many people are climate-vulnerable because they are located in areas that are highly exposed to the impacts of climate change and their sources of income are tied to activities which are weather dependent, such as agriculture, while access to basic energy services is both costly and time-consuming, impacting on household incomes. Their incomes can therefore be intermittent and they typically lack access to credit on reasonable terms.

Furthermore, these people are often living in unelectrified areas with no grid connection. As a result of their low asset bases, inconsistent incomes and lack of supporting infrastructure, people in H2R areas may be unable to adapt to the effects of climate change. The result is that many people living in H2R areas have to rely on fossil fuels to power their day-to-day activities, such as light, charging and cooking. Fossil fuels such as kerosene and diesel are harmful to those using them, harmful to the environment through the carbon emitted, and are notoriously expensive and unreliable. They are also made more vulnerable by not having assured access to communication tools to keep abreast of weather, health and market information.

Finally, women and children are likely to be disproportionately affected as they often spend more time at home with the harmful and emissions-intensive fossil fuel source which can lead to respiratory illnesses, or with candles which can lead to fires and burns.

Activities and Outputs

All fund activities contribute to seeing impact at the household level. Critically, products and services supplied through H2R investments intend to support household resilience/adaptation. (See B.1). Acumen incentivizes and facilitates OGS companies to set-up, and market and sell solar energy products via PAYGo financing to low-income people in H2R areas. This means that unelectrified people are able to access clean energy sources, many for the first time, and are able to afford them thanks to longer-term payment plans. Furthermore, after a payment period of 12-36 months, households should own their SHS outright for the remaining lifetime of 24-48 months. Companies are expected to monitor impact by collecting gender-disaggregated sales data as well as representative sampling on product accessibility, use and benefits.

Outcomes

Clean energy pathways are created for 72.4M people in H2R areas. This leads to increased resilience and adaptive capacity of the 54.3M beneficiaries accessing energy for the first time, including through improved energy security, income diversification and access to climate information. Data shows that a portion of customers will be able to diversify their sources of income, making their livelihoods more resilient to climate change. Beneficiaries will also be able to readily access climate-related information through the use of ICTs such as mobile phones and radios, allowing them to prepare for upcoming weather events. Furthermore, some people should benefit from a safer home and better health due to reduced use of lighting sources that can cause fires and burns or are harmful to respiratory health (co-benefit).

OGS use can also lead to reduced vulnerability in the context of climate change via a less climate-vulnerable energy supply. OGS is less vulnerable to weather events than grid-connected energy sources because they are decentralized. As the owners of a high quality OGS product, in the event of displacement due to a climate-related natural disaster, a person and their household are more able to retain access to electricity (with the benefit of it being clean and safe). Furthermore, OGS should take less time to repair or replace, meaning that people with SHS are able to access energy more quickly following a weather event.

Finally, displacement of emissions-intensive lighting sources is intended to lead to 5.1M tCO₂ being avoided.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1 Advancement of off-grid solar markets in H2R countries	☒	☐	☐	☐	☒	☐	☐	☐
Outcome 2 Establishment and Growth of OGS companies in sustainable and gender- and poverty-inclusive ways	☒	☐	☐	☐	☒	☐	☐	☐
Outcome 3 OGS customers (and their households) benefit from enhanced	☐	☐	☐	☐	☒	☐	☐	☐

energy security								
Outcome 4 OGS customers (and their households) are able to more securely access information including extreme weather event and health warnings, health information, and market data for e.g. agricultural produce	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 5 OGS customers (and their households) benefit from increased and/or diversified income sources	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 6 GHG emission avoidance from switching to clean and climate resilient energy sources	☒	☐	☐	☐	☐	☐	☐	☐

Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐
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If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Job Creation	☐	☒	☒	☒	☐	☐
Improved Health Outcomes	☐	☒	☐	☒	☐	☐
Co-benefit ...	☐	☐	☐	☐	☐	☐
Co-benefit ...	☐	☐	☐	☐	☐	☐

B.3. Programme description (max. 2500 words, approximately 5 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that will be undertaken by the project/programme to attain the intended outcomes elaborated in Section B.2 (a). Please also elaborate how the project/programme activities will address the barriers described in Section B.2.

This section of the funding proposal should be consistent with sections C.2 (financing by component), E.5 (project/programme results level) and E.6 (project/programme activities and deliverables).

Referring to the feasibility study, describe why this set of interventions was selected instead of alternative solutions and how the project/programme can help unlock the needed support in a sustainable manner. Also identify trade-offs of the selected interventions, if applicable.

For Enhanced Direct Access (EDA) proposals and projects/programmes with financial intermediation (loans or on-granting), describe the selection criteria of the sub-project and types.

Introduction to Hardest-to-Reach

Hardest-to-Reach (H2R) is an energy access initiative dedicated to reaching more than 70M of the 286M people who live in H2R countries without electricity. Using benchmarks from the GOGLA Market Trends Report 2022, approximately 190M people in H2R countries could afford an entry level off-grid solar connection, but are at risk of being left behind by business-as-usual market approaches to electrification.⁸⁸

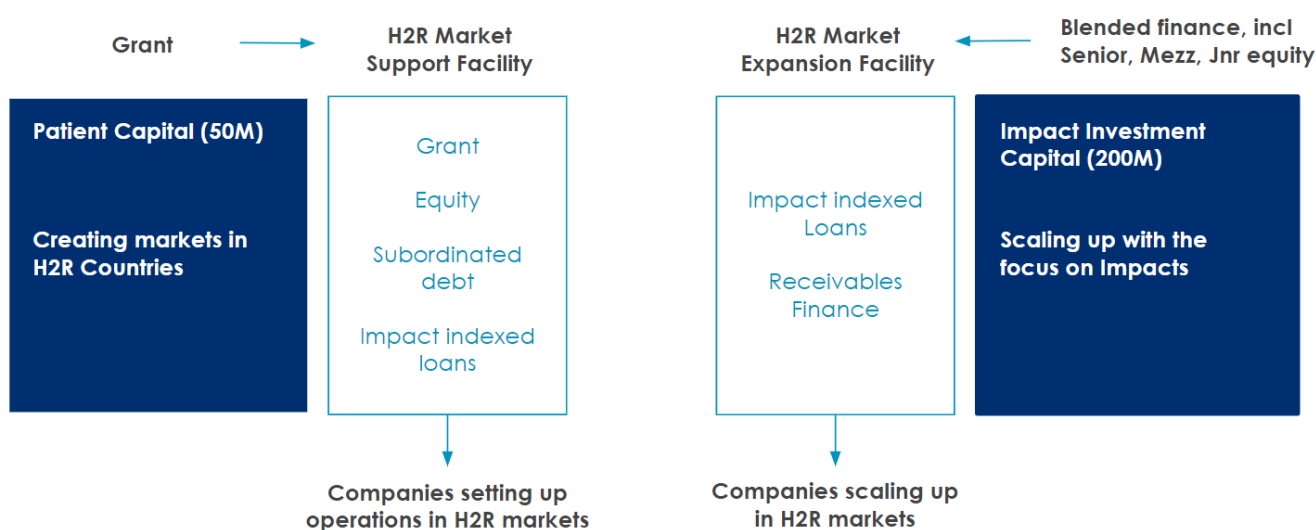
⁸⁸ [GOGLA Market Trends Report](#): State of the Sector (2022)

H2R aims to unlock first-time clean and resilient energy access for these communities. These are challenging markets with electrification rates at-or-below the sub-Saharan Africa average (48%) where investors have not been willing to enter to serve low-income segments. Our approach is to incentivize and accompany companies to enter into and grow in these challenging geographies to serve underserved segments by mobilizing capital at scale from various investor bases in off-grid solar. This approach is based on Acumen's deep knowledge of the energy access challenge, relationships with players in the off-grid solar sector with an ability to scale, and the experience of deploying innovative, patient capital to achieve impact.

H2R will target 16 sub-Saharan African countries with electrification rates of 48% or below and high poverty rates (above 45%). Out of this total, all eligible countries are considered least developed countries and they hold around 54 million African households that do not have access to modern electricity services. On average, 78% of the population living in these geographies lives below the lower-middle-income poverty line.

All 16 H2R countries are eligible countries for Acumen's H2R program. As part of H2R preparation, in depth country studies have been conducted across a number of H2R countries. In the first window of the Market Support Facility (MSF), explained in more detail later, country level research has identified barriers and guided the development of investment approaches and ensured policy alignment. Ongoing research and engagement in these countries will help to ensure that H2R achieves complementarity and coherence with other interventions and investments that donors, DFIs and private investors are doing.

At Acumen our approach is always to use the right kind of capital to tackle the right kind of problem. Hardest-to-Reach expects to use a diversified approach to inject capital and technical assistance to match the needs of off-grid energy companies at each stage of development and capacity. To build an off-grid solar market in these challenging H2R geographies, "start-up to scale-up" or "whole-of-life cycle" financing is needed to support entrepreneurs to set up in these markets and then scale up their operations, and thereby achieve their impact. This is why Acumen is structuring two independent financing mechanisms, tailored to the different stages of operators' evolution and the different risk appetites of investors.



Components, outcomes, and underlying activities within Hardest-to-Reach.

Component 1: Sustainable Market Creation in H2R countries (Market Support Facility)

A philanthropically-backed Market Support Facility aims to support the development of local players and regional companies' expansion into these new challenging markets. This \$50 million facility is expected to support companies with the flexibility to set up operations through grants, equity and debt, and the technical assistance to enter and navigate unfamiliar territory of high risk countries. The MSF is specifically designed to provide the right type of financing and accompaniment to enable companies to enter new markets. Set-up grants are expected to be provided to companies to address the high costs associated with expansion activities. Equity/equity-like capital, such as convertible notes, simple agreements for future equity (SAFEs) and recoverable grants, enables companies to innovate and scale up their operations. Concessional debt lowers the cost of operating in H2R markets compared to business-as-usual. Technical assistance aims to help reduce the real and perceived risks of entering new markets and build the capacity of companies in key areas such as management and strategy, credit risk analysis, consumer protection, gender equity and market studies. The facility aims to also improve the capacity of local distributors to become reliable partners to suppliers expanding in the region to build a local ecosystem for off-grid solar markets.

Component 2: Sustainable Market Expansion in H2R countries (Market Expansion Facility)

A subsequent blended finance-backed \$200 million Market Expansion Facility aims to provide impact-linked concessionary loans to scale up companies' operations and settle in H2R countries. A blended finance structure helps mobilize different types of investors with different risk appetites for the same purpose. The primary instrument used in the Market Expansion Facility is expected to be impact-indexed loans, an innovative RBF approach, where we intend to determine interest rates based on the impacts achieved in terms of first-time energy access and poverty level of the customers. The interest rates should range between a market rate and a zero percent, depending on the impact performance. The impact-linked incentives are designed to motivate the companies to provide first-time energy access to low-income customers and minimize the level of concessionality to the required level for delivering specific impacts. The objective of this approach is to integrate the impact into our financing structure, therefore into companies' operations, which will likely allow us to adapt, strengthen and maximize impacts along the way. Certain KPIs will also be used to ensure companies manage their growth sustainably and inclusively, disincentivizing sales growth at the cost of portfolio quality, for example.

MEF also seeks to invest in accounts receivable facilities, which should allow OGS companies to finance themselves rather than rely on third party investors, making them self-sustaining. Fundraising for the MEF is ongoing. We are not aware of any investors proposing exclusions or restrictions on their use of proceeds. Our intention is to deploy loans only to the borrowers for operations in NOL countries. Assuming there are no exclusions, our intention is to have a single set of criteria when selecting borrowers, and that all investors will receive returns on loans pro rata to their contributions to MEF. We also seek to ensure that all MEF investors will participate in each financing transaction entered into by MEF pro rata to their commitments to MEF.

See sections C.1 and C.2 for a summary of MSF/MEF breakdown by financing source and component.

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Component 1 Intended Activities:

Activity 1.1: Acumen engages stakeholders (partners, donors, govts) towards development of H2R markets, by fundraising & establishing H2R MSF (\$50M)

Acumen intends to set up the philanthropically-backed Market Support Facility (MSF). We seek to develop and manage the strategy, naming, branding and legal arrangement of the facility, ready for the implementation stage. Acumen raises \$50M for MSF across several donors, including a \$10M repayable grant from GCF. Acumen engages relevant stakeholders (partners, donors, governments) throughout. This includes engaging with nationally designated authorities to receive no-objection letters, workshops with government ministries and entities, town halls with project affected communities, and other engagement strategies with relevant public and private sector organizations.

In relation to the overall management of MSF, the MSF will have a sole member, the AE. The MSF will set up a management committee to oversee the activities and progress of the MSF, currently contemplated to be the Chief of Strategic Initiatives, CFO and GC of the AE. The MSF will have an investment committee to make grant and investment decisions, currently contemplated to be the President and CFO of the AE, together with a Board member of the AE. The MSF team is expected to consist of a director of investment, a director of operations, a manager of grants and impact, and support personnel. The intention is not to limit the term of the LLC or have it necessarily be required to unwind at the end of the program. Depending on the success and evolution of the vision for the H2R project, the entity may have continued utility, which is hard to predict at this time. However, we may choose to liquidate the entity if there is no further need for the entity at such time.

- Sub-Activity 1.1.1: H2R strategizes on the investment and impact thesis as well as investor mandate required in development of Market Support Facility (MSF)
- Sub-Activity 1.1.2: H2R names and brands the Market Support Facility
- Sub-Activity 1.1.3: H2R identifies and meets with aligned HNW donors and philanthropic institutional donors.
 - The intended result is \$50M raised for investment and project activities.
- Sub-Activity 1.1.4: H2R internal legal counsel reviews the appropriate legal structure of the MSF and seeks to develop the appropriate legal structure for the Market Support Facility.
- Sub-Activity 1.1.5: H2R engages with government entities and other relevant public and private sector organizations in H2R countries.

Activity 1.2: Partnerships built with stakeholders to improve coordination and commercial infrastructure, address market barriers and share knowledge

H2R aims to develop partnerships and engage with stakeholders (government, investors, opcos, capacity building programs) to improve coordination, address market barriers and share knowledge in H2R markets. Hardest-to-Reach seeks to work with and invest in established off-grid solar operators – manufacturers, distributors, and local distribution networks – and, through the Market Support Facility, expects to expand its reach to locally-rooted value chains that will serve these populations over the long term. By encouraging partnerships between established off-grid energy players and local distributors, H2R aims to build robust and coordinated off-grid energy markets in Hardest-to-Reach geographies. Across the range of H2R partnerships, coordinated knowledge sharing may help address market barriers to investment. We aspire to generate coordinated insights and opportunities to help reduce inefficient use of resources, create opportunities to leverage stakeholder experiences and competencies, and minimize distortive risks. H2R's frontier investments and coordination efforts serve to de-risk and prove the viability of OGS investments in these underserved markets. We aspire to support the REACH partnership with lean but impactful staffing engagement that will be part of our knowledge sharing ambitions. Hardest-to-Reach aims to inspire other activity by sectoral leaders, industry experts, and relevant stakeholders within the REACH partnership and other forums. The aspiration is that by sharing insights from our investment activity with stakeholders across the sector, including those in the REACH partnership, we can generate an enabling environment for others to participate in impactful ways in OGS sectors in H2R markets.

- Sub-Activity 1.2.1: Acumen seeks to support the development of a high level REACH Partnership between OGS leaders, practitioners, and stakeholders committed to sharing knowledge and insights.
- Sub-Activity 1.2.2: We intend to engage in REACH Partnership convenings at the country-level coordination committees to facilitate knowledge sharing and efficiently use resources.
 - These REACH partnership convening will likely provide the team the opportunity to align with and learn from national energy access activities.
- Sub-Activity 1.2.3: H2R seeks to share results and lessons learned, both from deployment of capital and from larger coalition-building efforts.

Activity 1.3: Investment of MSF capital to demonstrate OGS within H2R markets and efficacy of impact-linked capital approach (and additional re-investment)

The team will seek to identify a pipeline of potential H2R joint ventures and ensure their strategic fit with the project goals. Companies fitting the JV profile will be vertically integrated OGS companies already having successful set-up and expansion of OGS businesses in Africa. Acumen will seek to partner with the companies to replicate success cases in H2R countries. The team will conduct thorough due diligence on the company including a gender assessment and ESG diligence to be mission aligned and support responsible market development.

The MSF also will make equity-like investments and impact indexed loans to distributors as fit for the stage of the companies and needs. Similar to above, the team expects to identify a pipeline of distributors for OGS products that can or already do operate in H2R markets with a focus on locally owned and managed distributors to build a healthy local ecosystem.

H2R aims to provide impact indexed loans to OGS manufacturers to distribute OGS products to last mile distributors operating in H2R markets. The MSF team will provide inventory financing to OGS distributors through manufacturers to minimize the financial burden to smaller distributors while they build the capacity.

The investment and operations teams will source, diligence, negotiate and present investments to the investment committee. Diligence will include financial, operational, legal and ESG lens. MSF intends to disburse small last mile distributor loans, which may be supported by an advisor with local lending market knowledge, where appropriate. A third party advisor with local equity market knowledge may support the equity-like investments of H2R 's MSF. The investment committee will approve or reject investments.

- Sub-Activity 1.3.1: H2R intends to identify pipeline, perform due diligence on potential investments, create terms, develop JVs, and disburse funds to JV formed with a vertically integrated OGS company.
- Sub-Activity 1.3.2: H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and disburse equity/equity-like capital (including but not limited to convertible notes and recoverable grants) to last mile distributors.
- Sub-Activity 1.3.3: H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and disburse loans to last mile distributors via manufacturers or directly.⁸⁹
- Sub-Activity 1.3.4: H2R collects returned capital during the investment period and reinvests repaid capital into OGS companies
- Sub-Activity 1.3.5: H2R collects repayments and exits its equity-like investments after the investment period and disburses repaid capital to funders

Activity 1.4: Monitoring of investments for portfolio management, impact measurement and tracking, and generating insights

Hardest-to-Reach MSF will monitor and evaluate investments and follow them closely throughout the lifespan of the loan, equity investment, or joint venture. Baseline studies are conducted of potential investees sales, customer demographics, and climate resilience. Companies are expected to share financial and impact data on a quarterly, semiannual, and annual basis depending on the data and level of detail of the data. The MSF team is expected to monitor this data and investee activity for both compliance with the terms of investment and producing the intended outcome of the investments. The MSF team is expected to monitor for deviations from expectations and intervene if companies have an incident, are not complying with investment agreements, or are in financial distress. Monitoring may inform technical assistance and post-investment support.

⁸⁹ The intention is for the director of investment to monitor debt and equity investments including market opportunities for exits, where relevant. The director of operations will monitor joint ventures, with the intention that this person will sit on the board of directors of such ventures. The director of investments and the manager of impact will monitor technical assistance grants.

Companies are required to report to H2R on a quarterly, semi annual, and annual basis on impact, climate, ESG, gender, sales, financial, and other relevant data. MSF monitoring and reporting is expected to include impact, financial performance, and climate information to the donors. Finally, the MSF team expects to share community reports with relevant stakeholders, and will also host annual stakeholder town hall meetings.

- Sub-Activity 1.4.1: H2R monitors growth and impact progress of companies receiving Market Support capital.
 - Reporting mechanisms are developed, and data on growth, impact, and other key metrics is captured in a centralized database for monitoring and analysis. Qualitative inquiries are performed at key intervals into the drivers and barriers for MSF capital to build local management and distribution capacity. At agreed-upon intervals, data are summarized and analyzed in reports to funders.
- Sub-Activity 1.4.2: For companies eligible to receive growth capital, H2R measures baseline impact scenarios.
 - Baseline studies are conducted on potential investee sales, customer demographics, and climate resilience.
- Sub-Activity 1.4.3: H2R measures impact of borrower operations and calculates final repayment for impact-indexed loans.
 - Companies report on sales and operational data. Endline surveys are conducted. Data is synthesized and analyzed to determine final repayment structure through a transparent impact formula. H2R may seek third party verification to ensure the veracity of the data shared from companies.
- Sub-Activity 1.4.4: H2R submits final impact reports to all partners upon disbursement of repaid capital to funders.
- Sub-Activity 1.4.5: H2R intends to monitor loan conditions and covenants, gender, ESG, and climate and impact work.
- Sub-Activity 1.4.6: H2R intends to develop an annual MSF report to MSF donors. This report will include impact, financial, and other relevant data.
- Sub-Activity 1.4.7: H2R intends to develop an annual MSF community report to be shared with stakeholders and will hold annual stakeholder town halls.
- Sub-Activity 1.4.8: H2R intends to communicate findings from capacity-building work to larger energy access ecosystem.

Activity 1.5: TA support for H2R investees, particularly in enterprise development, consumer protection, gender and environment/ESG performance

Growing sustainable and inclusive commercial businesses in Hardest-to-Reach markets will demand more than patient and flexible investment capital. Through its Technical Assistance (TA) activities, Hardest-to-

Reach intends to support investees to navigate the business challenges, risks and barriers of delivering energy solutions to underserved segments; to grow sustainably and inclusively, with an eye on protecting consumers; to access capacity building and training for better business performance, to identify and mitigate environmental, social and governance risks, and understand and address the inequities women face in energy, as users, entrepreneurs and members of the workforce. TA activities are expected to ensure investees are able to access the right set of supports to achieve growth with maximum impact. Hardest-to-Reach supports investees across four areas of technical assistance; Consumer Protection, ESG, Gender and Enterprise Development.

TAF grants and expenditures may be approved by the H2R team and the MSF investment committee when appropriate.

- Sub-Activity 1.5.1: H2R intends to identify a group of appropriate and committed technical experts to advise H2R companies. Establishes formal and/or informal partnerships, as appropriate
- Sub-Activity 1.5.2: H2R intends to assess H2R company needs prior to investment, and secure commitment for improvement in key areas as part of conditions precedent.
 - A 'report card' for every H2R investment is developed and stored, then used to design appropriate support.
- Sub-Activity 1.5.3: H2R works with experts to support companies in the areas of consumer protection, enterprise development, gender and ESG through consultation, webinars, and other resources.
- Sub-Activity 1.5.4: H2R partners with market development and off-grid experts to research market development in H2R markets, and determine the market growth that can be attributed to H2R.

Component 2 Intended Activities:

Activity 2.1: Acumen engages stakeholders (partners, investors, governments), towards scaling H2R markets by fundraising & establishing H2R MEF (\$200M)

Acumen sets up the Market Expansion Facility. The strategy, naming, branding and legal arrangement of the facility are successfully developed and managed, ready for the implementation stage. Acumen raises \$200M for MEF across commercial investors and development finance institutions, anchored by a \$50M equity tranche from GCF. Acumen engages relevant stakeholders (partners, investors, governments) throughout.

The MEF is expected to be formed as a partnership. Under this expected structure, the General Partner of the MEF and the Manager will be an entity indirectly wholly owned by the AE. Investors will invest as limited partners in the MEF, such investments will be categorized as senior, mezzanine, or junior investments. Additional details are provided in Section B.4 below.

- Sub-Activity 2.1.1: H2R strategizes on investment and impact thesis, as well as investor mandate, for the Market Expansion Facility (MEF)
- Sub-Activity 2.1.2: H2R establishes debt fund manager.
- Sub-Activity 2.1.3: H2R names and brands the MEF.
- Sub-Activity 2.1.4: Acumen internal and external legal counsel review the appropriate legal structure of the MEF.
- Sub-Activity 2.1.5: H2R creates an indicative financial model and prospectus for review by potential partners.
- Sub-Activity 2.1.6: H2R identifies and engages commercial investors and subordinate capital providers.
 - Acumen raises \$200M in blended capital for MEF across several investors and concessionary capital providers, including a \$50M equity tranche from GCF and \$5M grant tranche. We intend to call from GCF an amount equal to 27.8 of the total equity commitments raised in each closing up to \$50M total. In one lump sum, \$5M of MEF grant will be requested in the first call (as will \$10M recoverable grant for the MSF).

Activity 2.3: Investment of MEF capital to scale OGS through impact-linked approach within H2R markets (and additional re-investment)

H2R aims to provide impact indexed loans to OGS companies seeking to expand in Hardest-to-Reach markets. The MEF team will identify vertically integrated businesses and last mile distributors in the pipeline that have the capacity and experience to expand in new markets. Some of these companies may have received investments in the MSF. If the company has received MSF financing, the MEF team will have procedures that help to ensure there is not a conflict of interest. The MEF team will diligence vertically integrated businesses and the last mile distributors including ESG diligence and gender assessments.

Hardest-to-Reach MEF seeks to invest in accounts receivable facilities for OGS distributors. The MEF team will work together with receivable financing managers to enable distributors together with vertically integrated companies to have access to receivable financing facilities. It will seek to incentivize the companies to generate quality receivables from proper customer management and support the companies as they move toward becoming self-sustaining without relying on external financing.

- Sub-2.3.1: H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and extend impact-indexed loans to OGS companies
- Sub-Activity 2.3.2: H2R intends to invest via accounts receivables financing
- Sub-Activity 2.3.3: H2R intends to incentivize borrowers to serve Hardest-to-Reach customers

- Sub-Activity 2.3.4: H2R collects repayments during the investment period and reinvests repaid capital into OGS companies and accounts receivables facilities
- Sub-Activity 2.3.5: H2R collects repayments after the investment period and disburses repaid capital to funders

Activity 2.4: Monitoring of investments for portfolio management, determining performance against impact-linked KPIs for impact-indexed loans, and review of business performance

Hardest-to-Reach MEF will monitor and evaluate investments and follow them closely throughout the lifespan of the impact indexed loans and receivables financing. Baseline studies are conducted of potential investees sales, customer demographics, and climate resilience. Companies are expected to share financial and impact data on a quarterly, semiannual, and annual basis depending on the data and level of detail of the data. The MEF team are expected to monitor this data and investee activity for both compliance with the terms of investment and producing the intended outcome of the investments. They will pay special attention to the KPIs for the impact indexed loans to determine interest rates. Managers are expected to monitor for deviations from expectations and intervene if companies have an incident, are not complying with investment agreements, or are in financial distress. Monitoring may inform technical assistance and post-investment support. The MEF team may seek third party verification of impact data with a special focus on data informing the impact indexed loans.

Companies are required to report to H2R on a quarterly, semi annual, and annual basis on impact, climate, ESG, gender, sales, financial, and other relevant data. In turn, the MEF team expects to report annual activities, financial statements, and impacts on an annual basis. The AE will also report MEF activities to GCF via an annual performance report. Finally, the MEF team intends to distribute community reports to relevant stakeholders, and will also host annual stakeholder town hall meetings.

- Sub-Activity 2.4.1: H2R monitors growth and impact progress of companies receiving Market Expansion capital
- Sub-Activity 2.4.2: H2R measures baseline impact scenario
- Sub-Activity 2.4.3: H2R measures impact of borrower operations and calculates final repayment for impact-indexed loans
 - Companies report on sales and operational data. Endline surveys are conducted. Data is synthesized and analyzed to determine final repayment structure through a transparent impact formula. H2R may seek third party verification to ensure the veracity of the data shared from companies
- Sub-Activity 2.4.4: H2R submits final impact reports to all partners upon disbursement of repaid capital to funders
- Sub-Activity 2.4.5: H2R monitors loan conditions and covenants, gender, ESG, and climate and impact work.

- Sub-Activity 2.4.6: H2R will develop a fulsome annual MEF report to MEF investors. This report will include impact, financial, and other relevant data.
- Sub-Activity 2.4.7: H2R will develop an annual MEF community report to be shared with stakeholders and will hold annual stakeholder town halls.
- Sub-Activity 2.4.8: H2R communicates findings from capacity-building work to larger energy access ecosystem

Outputs for Components 1 and 2

Output 1: H2R established and \$250M fundraised

- MSF development includes fundraising, forming the necessary MSF entity, project development, and stakeholder engagement. The intended end result is successfully fundraising a \$50 million facility to be passed through a limited liability company. By the end of the fundraising and project development period, the project is expected to be fully formed and ready for execution.
- MEF development includes fundraising, forming the necessary MEF entity, project development, and stakeholder engagement. The intended end result is successfully fundraising a \$200 million facility to be managed by the Manager. By the end of the fundraising and project development period, the project is expected to be fully formed and ready for execution.

Output 2: Developed gov't and non-gov't partnerships to enable last-mile energy access for the poorest; (i.e, co-investments, knowledge partnerships, govt partnerships, and industry partnerships)

- Through effective fundraising and blended finance structuring, H2R aims to invest in first-time energy access with a diversity of capital and co-investors.
- Across the range of H2R partnerships, coordinating learning and information sharing to help address other market barriers to investment, including developing the right VAT and duty structures for renewables products and narrowing the affordability gap for low-income customers. Coordinated information sharing efforts like these may help reduce inefficient use of resources, create opportunities to leverage stakeholder experiences and competencies, and minimize distortive risks.
- H2R's frontier investments and coordination efforts should serve to de-risk and prove the viability of expanding OGS investments in these underserved markets.

Output 3.1: Increased support for new and fledgling OGS co's in H2R countries, with alignment to serve low-income segments

- Concessional funding and set-up grants de-risk H2R areas for OGS companies, and allow them to enter and scale up in new markets.
- Flexible financing that matches the needs of the company in terms of their stage of growth and status of target markets is provided for OGS companies specifically in the form of equity for

distributors, joint ventures with vertically integrated SHS companies, distributor finance via OGS manufacturers, and lending to early-stage last mile distributors.

- Companies build capacity and start or improve operations in H2R markets, and are able to serve low-income and climate-vulnerable people with OGS products.

Output 3.2: Increased scale of OGS co's in H2R countries, with demonstrated alignment to serve low-income segments

- Impact-indexed loans incentivize OGS companies to retain a focus on entry-level solar products for lower-income customers and prevent mission-drift.

Output 4.1: Strengthened OGS business performance, with demonstrated commercial viability, sustainability, gender- and poverty-inclusivity, and investability

- H2R proves that companies selling OGS products to low-income people via PAYGo business models can be commercially viable, and creates a demonstration effect.
- Targeted technical assistance in the areas of consumer protection, enterprise development, gender and ESG builds the capacity of companies and enables them to address the significant challenges of H2R markets.
- In particular, companies improve their climate resilience, sustainability, consumer protection, ESG and gender equity practices to both mitigate risks, improve operations, and improve investability with sustainable investors.

Output 4.2: Strengthened OGS business performance, with proven commercial track record

- H2R's scale-up capital and targeted TA enables strong business performance and creates the runway for sustained growth at scale, which delivers a commercial track record. Companies have sufficient capacity to thrive in riskier markets in ways which manage risk and improve long-term viability, growth, customer retention, and market insights.

Output 5: Increased provision of climate-resilient clean energy via 13.3M new sales of affordable, clean, resilience-enhancing decentralized energy products

- As companies deliver OGS products to new geographies and underserved segments, low-income and climate-vulnerable people have access to clean energy sources, many for the first time.

Examples of household products^{*90}

⁹⁰ [GOGLA Market Trends Report](#): State of the Sector (2022)

Product Category	Definition	Power Range (Wp)	MTF Level
Multi-light systems	Multiple light & mobile charging	3-10.99	Enables Tier 1 Electricity Access for at least one person and up to a full household
Entry-level SHS	Three to four lights, phone charging and powering a radio	11-20.99	Enables Tier 1 Electricity Access for a household

*Non exhaustive

Output 6: Displacement of harmful and emissions-intensive lighting sources and 120 MW newly installed renewable energy capacity

- Low-income and climate-vulnerable people replace their traditional fossil fuel based energy sources with OGS products. This leads to reduced emissions and increased installed renewable energy capacity.

Outcomes of Component 1 and 2

Outcome 1: Advancement of off-grid solar markets in H2R countries

H2R intends to extend financing and technical assistance to OGS companies in sub-Saharan Africa, who will then sell off-grid solar products to customers, many of whom will be accessing energy for the first time. By generating investment in Hardest-to-Reach OGS markets, investing in a diverse portfolio of companies, creating new OGS customer bases, deploying intentional technical assistance, knowledge sharing with partners, and implementing consumer protection, ESG, and gender activities, we seek to activate and sustainably expand Hardest-to-Reach OGS markets, leading to more OGS companies and increased crowding in of public and private funding. To that end, we seek to partner with market development and off-grid experts to research market development in H2R markets, and determine the market growth that can be attributed to H2R. We will define the metrics that activate and expand markets and measure our contribution to the market activation indicators. We intend for our investments and program activities to activate and support the development of these H2R OGS markets. By defining these market activating “tipping points” and demonstrating our impact, we hope to create a blueprint for other impact investors seeking to make market level changes.

Indicative Metric: Hardest-to-Reach activates 9 tipping points across 3 H2R countries.

Outcome 2: Establishment and Growth of OGS companies in sustainable and gender- and poverty-inclusive ways

H2R seeds new ventures and backs existing players in challenging markets, supporting the establishment and growth of these nascent ventures and markets through partnerships and productive engagements with government and non-government stakeholders. New sales of OGS products help provide first-time connections and electrification through clean energy technologies. By supporting stronger and more viable businesses, creating a local investment history, and incentivizing impact, we hope to create sustainable and impactful opportunities for companies operating in or seeking to expand into H2R markets.

We seek to demonstrate that OGS companies operating in Hardest-to-Reach markets can build climate resilient, sustainable, and inclusive businesses with technical assistance. H2R companies will receive technical assistance across consumer protection, gender, ESG, and enterprise development activities. Consumer protection activities focus on enhancing consumer protection policies, procedures, and sales strategies for companies, and monitoring portfolio quality to manage credit risk and consumer protections. On gender, gender experts will support companies in developing and executing gender action plans, improving gender policies, and collecting sex disaggregated data. Additionally, technical assistance funds will be assigned to be used to improve gender equity in the OGS market by training women workers and raising female customer awareness of OGS products. ESG and climate TA focuses on ESG action plans for companies, support companies on enhancing e-waste policies and procedures, and building climate and ESG resources for OGS companies. Enterprise development work focuses on making companies more investable, supporting talent retention, improving operations, building customer and market knowledge, and supporting financially distressed companies. We expect companies to more strongly protect customers and mitigate credit risk, identify and engage with ESG risks, build more equitable and empowering workplaces and customer experiences, and develop and manage stronger, more talented and investable OGS companies because of targeted technical support.

Outcome 3: OGS customers (and their households) benefit from enhanced energy security

OGS products reduce the vulnerability of people's energy supply to climate change. OGS is less vulnerable to climate-change-related weather phenomena than grid-connected energy sources because they are decentralized. Furthermore, OGS should take less time to repair or replace, meaning that people with SHS are able to access energy more quickly following a weather event.

Indicative # number of beneficiaries who have been involuntary displaced due to climate-resilient natural disasters [OGS customers benefiting from enhanced energy security via portable, durable OGS products]:

Total Beneficiaries: 1,646,768

MSF: 265,873

MEF: 1,380,895

Outcome 4: OGS customers (and their households) are able to more securely access information including extreme weather event and health warnings, health information, and market data for e.g. agricultural produce

OGS products increase the resilience and adaptive capacity of 54.3M low-income and climate-vulnerable people in H2R areas. People may also have access to climate and weather information through the use of ICTs such as mobile phones and radios, allowing them to prepare for upcoming weather events. They might also have access to information on social safety nets following a climate event through those ICTs, making them better able to recover from weather events.

Indicative # of direct and indirect beneficiaries:

Total: 54,331,312

MSF: 8.8M lives reached

MEF: 45.5M lives reached

Outcome 5: OGS customers (and their households) benefit from increased and/or diversified income sources

OGS products reduce the vulnerability of people's energy supply to climate change. OGS is less vulnerable to climate-change-related weather phenomena than grid-connected energy sources because they are decentralized. Furthermore, OGS should take less time to repair or replace, meaning that people with SHS are able to access energy more quickly following a weather event. Moreover, research has shown that businesses in rural communities increase income with access to electrification and lighting by keeping stores open longer and being able to sell a broader assortment of products. Income generation is a proven climate resilience intervention that helps people rebound from extreme weather events and access other tools to adapt to climate change.

Indicative # of OGS customers that benefit from increased and/or diversified income sources:

Total: 5,424,647

MSF: 876K lives reached

MEF: 4.5M lives reached

Outcome 6: GHG emission avoidance from switching to clean and climate resilient energy sources

The displacement of emissions-intensive lighting sources such as kerosene, petrol & diesel leads to 5.1M tCO₂eq being avoided across the lifetime of the H2R initiative. People in H2R areas are able to embark on clean energy pathways and make the first step on the energy ladder.

Indicative GHG emissions reduced, avoided or removed/sequestered:

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Total: 5.1 MtCO₂e

MSF: 1 MtCO₂e

MEF: 4.1 MtCO₂e

Co-benefit 1: New clean energy jobs including for women

OGS companies starting, expanding, and scaling their enterprises in Hardest-to-Reach markets will create new job opportunities ranging from sales agents to country level operations management. H2R seeks to help companies create safe, impactful, and sustainable green jobs that can help staff increase income, improve economic stability, and improve health and well being outcomes for their families. H2R also seeks to support these companies adopt a gender lens when hiring and generate an impact on women workers in the OGS space through training, awareness building, and improved company-level policies and procedures.

Co-benefit 2: Improved health outcomes via lowered indoor air pollution

SHS products will be replacing a few forms of energy for some H2R customers. The primary energy replacement will likely be kerosene lanterns whose toxic fumes have been associated with negative health impacts. Switching to clean energy should improve health outcomes for customers using SHS products.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Provide a description of the project/programme implementation structure, outlining legal, contractual, institutional and financial arrangements from and between the GCF, the Accredited Entity (AE) and/or the Executing Entity(ies) (EE) or any third parties (if applicable) and beneficiaries.

- *Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function, particularly for Enhanced Direct Access (EDA) proposals.*
- *Provide information on the financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE or any third party and beneficiaries. For EEs that will administer GCF funds, indicate if a Capacity Assessment has been carried out. Where applicable, summarize the results of the assessment.*
- *Describe the experience and track record of the AE and EEs with respect to the activities (sector and country/region) that they are expected to undertake in the proposed project/programme.*

Provide a diagram(s) or organogram(s) that maps such arrangements including the governance structure, legal arrangements, and the flow and reflow of funds between entities.

MSF Structure

- MSF is intended to raise grant and repayable grant capital to build capacity for entities to establish and bolster a market for clean energy access for individuals in least developed countries. Target size is \$50 million, of which a majority has already been raised and for which several pilots are in planning.
- GCF is requested to provide \$10 million in the form of a partially repayable grant for MSF. Such repayable grant shall be used for all activities of the MSF, including expenses, grants and investments; of such GCF funding toward grants and investments is hereinafter referred to as “GCF Investment Proceeds”.
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- MSF is expected to provide technical assistance grants and potentially recoverable grants/convertible notes/equity, engage in joint ventures and provide impact indexed loans to incentivize established solar home systems manufacturers and distributors to enter LDC markets, and to bolster existing distributors in LDCs.
- Any joint ventures are expected to contain mechanisms for ownership to transfer to the counterparty once the ventures are financially self-sustaining.

Joint venture agreements may incorporate grants to cover set up and other initial costs.
- The operations of MSF are overseen by a management committee consisting of the Chief of Strategic Initiatives, Chief Financial Officer and General Counsel of the AE. The management committee seeks to ensure that the operations of the MSF are compliant with GCF requirements, applicable laws, regulations and policies. Grants and investments made by MSF are subject to review and approval of the global Investment Committee of the AE
- Members of the AE that will operationalize the MSF include the General Counsel, the Chief Strategic Officer, the Chief Financial Officer, a Director, an Associate Director, and the Legal Counsel of the AE.
- MSF is expected to build capacity within entities to be a viable pipeline for MEF borrowers. MSF will take the risk during the development stage of LDC operations for these potential borrowers with its high risk grant and repayable grant capital, reducing the risk for MEF.
- MSF expects to have an investment period of 7 years.
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MEF Structure

- MEF is intended to bring together senior, mezzanine and junior equity capital to incentivize investors with different risk profiles to provide clean energy access for individuals living off-grid in the least developed countries. MEF also intends to raise grant capital to enable further interest rate adjustments payable by borrowers as a reward for first-time energy access, among other metrics, which grant capital is intended to be swept to the operating account of MEF and made available for fees, expenses and deployment, as an offset for such adjustments.
- MEF is designed to make loans for working capital to borrowers providing clean energy access in H2R countries, with incentives to borrowers to enable first time clean energy access for their customers, with interest rate adjustments to the extent of such access.
- MEF is expected to use performance incentives within the loan structure to incentivize businesses to achieve specific impact targets, in the form of interest rate adjustments to the extent of achieving impact, to essentially provide a partial offset to interest.
- Borrowers may receive impact adjustments based on certain predetermined metrics. Such metrics will be collected and analyzed annually (using 3rd party verification of certain metrics, such as lean data studies), and result in an impact index score that will be aligned with an interest adjustment.
- In addition to impact adjusted loans, the MEF expects to lend to accounts receivable financing facilities, to support cash and inventory management of OGS companies operating in our target geographies. Current facilities under review have 4 year terms.
- Target MEF size is \$200 million, with indicative allocations across layers as follows: senior – \$60 million; mezzanine – \$60 million; junior – \$60 million; and grant – \$20 million. This proposal requests GCF to provide \$50 million for junior equity and \$5 million for non-repayable grant capital.
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- The above structure is designed to incentivize capital with different risk tolerance, noting that mission oriented and concessionary capital is expected to provide the mezzanine, junior and grant capital, while more commercially oriented capital is expected to provide senior capital.
- MEF is expected to have a 7 year investment period, which may be extended for up to two one-year periods
- Investment decisions for MEF will be made by an investment committee designated by the GP
- MSF is expected to provide pipeline borrowers to MEF. To mitigate conflicts of interest that may be perceived to bias MEF decisionmakers, 1) underwriting criteria has been developed to vet the viability of

the prospective borrowers, and 2) the investment committee and LPAC will be apprised of borrowers that have also received MSF funding for transparency and the opportunity to discuss any concerns.

- The non-reimbursable grant will flow from GCF to the AE via the FAA entered into between the AE and GCF. The money in turn will flow to the MEF via a subsidiary agreement to be entered into between the MEF and the AE. No additional money will flow to the borrowers; they benefit by virtue of an interest rate adjustment, so pay less to the MEF. The MEF retains the grant capital and is able to use it for operating purposes to the extent of the adjustment offset.

Track Record

Acumen is an AE that has three funding proposals in implementation. The first program is KawiSafi Ventures, a venture capital fund focused on creating clean energy access in East Africa, and is at the end of its investment period, having successfully deployed approximately \$42 million in investments that as of end of Q3 2022, impacted 110 million low income individuals and offset 24 million metric tons of carbon dioxide and kerosene. The second program is Acumen resilient Agriculture Fund, a venture capital fund focused on building climate resilience for smallholder farmers in West and East Africa, and is in the middle of its investment period, having successfully deployed \$20.2 million in investments that as of end of Q3 2022, impacted almost 1.6 million low income individuals. The third program is Energy Access Relief Fund, a debt fund focused on providing concessional capital to support businesses in the solar home system sector survive the impacts of COVID, and is in the middle of its investment period, having successfully deployed approximately \$49 million in loans. In addition, for over 20 years, Acumen has been directly investing in small and medium-sized enterprises that serve low-income communities in developing countries across South Asia, Sub-Saharan Africa, and Latin America, with the goal of changing the way the world tackles poverty. As of the end of 2021, Acumen had deployed approximately \$146 million into 151 companies, impacting approximately 443 million lives. A significant sector of focus for Acumen has been the off-grid energy sector, and Acumen contributes to many sectoral initiatives, including with GOGILA.

Capacity Assessment

Manager, as executing entity, is expected to be a wholly owned subsidiary of Acumen. Such an entity will be formed with policies and procedures directed by the AE, and with staff screened for experience in debt investing, the OGS sector and our target host country markets. We are in discussions with certain service providers to potentially engage for discrete functions, such as back office and support in risk assessments, but not to be decision makers for the Manager or MEF.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Explain why the project/programme requires GCF funding to address mitigation or adaptation measures, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.

Why H2R needs GCF

Addressing the unelectrified populations requires affordable solutions, which themselves require affordable, abundant capital. Yet the perceived risk and small size of these markets means that capital there is both scarce and expensive. Blended finance, anchored by the most influential concessional investor focused on climate change adaptation (GCF) is the only way to overcome this impasse and bring energy to people who are vulnerable to climate change.

GCF's involvement in this project is critical. GCF's concessional capital will play a catalytic role to mobilize capital at scale and channel it toward serving low income and climate vulnerable customers in LDCs. Public and private capital directed to the energy access sector has been concentrated in a few lower-risk markets in Africa and has avoided the regions with the lowest electrification rates. Between 2012 and 2019, western, central and southern Africa accounted for 71 percent of SSA's unelectrified population, yet received just 24 percent of cumulative off-grid solar investment.⁹¹ Research shows that even when investment dollars are directed to low-income regions, they are not always directed where the need is greatest. Most investors only operate in markets that are lower risk or proven to yield financial returns. For this reason, progress has been contained to a few investor-ready countries.

According to IEA research, the affordability of clean energy transitions will depend on reducing the cost and improving the availability of capital.⁹² Addressing the remaining unelectrified population demands a

⁹¹ IEA [SDG 7 Database – Access to Electricity](#) (n.d)

⁹² IEA [Financing clean energy transitions in emerging and developing economies](#) (2021)

shift towards more challenging market environments and thus keeping financing costs low will be vital to enabling clean energy transitions while keeping them inclusive and affordable. However, capital is substantially more costly in emerging and developing economies than in advanced economies. Nominal financing costs are up to seven times higher than in the United States and Europe, with higher levels in riskier segments.⁹³ This points to a relatively high bar for projects to raise debt finance and offer sufficient returns on equity.

OGS companies have therefore not been able to expand into these riskier markets, because their cost of financing is too high, financing doesn't exist, or they don't have the capacity to expand into nascent markets. Sales of off-grid solar products are very low or non-existent in many of these geographies, even though the technology has been proven as an effective means of providing energy access for low-income communities. For example, OGS sales in Burundi over the last 3 years were less than 0.1 percent of total sales in East Africa across the same period. Furthermore, sales growth in Sierra Leone over the last 3 years has been four times lower than the average across West Africa. What's more, sales of OGS products in DRC over the last 3 years were 1.6 percent of the total SSA sales, despite the country making up over 8 percent of the population of sub-Saharan Africa. The business-as-usual approach is not serving low-income communities in Hardest-to-Reach geographies, even though they have a significant need for energy access.

The GCF will enable H2R initiatives to channel much needed financing and technical assistance to OGS companies to support them to enter riskier markets and grow sustainably. The concessional component provided by GCF is vital because it means that Acumen can issue financing with lower interest rates and longer tenors. Companies will therefore be able to access affordable capital and should not suffer from over-indebtedness. Furthermore, they will be able to pass the concessionality onto customers in the form of cheaper products. Without a concessional component, OGS companies would have to charge potentially prohibitively high prices to their customers, many of whom will be earning below \$3.20 a day. If this were to happen, the H2R initiative would not be able to have an impact on the people who need energy access most.

GCF's financing package

H2R is creating a paradigm shift in African energy markets by unlocking finance from investors that would otherwise find these markets too risky. GCF's catalytic capital will be vital in providing companies with concessional financing and mobilizing different types of investors with different risk appetites at scale to tackle the aforementioned problem. Each type of financing will likely provide unique benefits:

- **Reimbursable grant:** GCF's \$10 million reimbursable grant to MSF, together with a target \$40 million grant from other partners, is needed for the Market Support Facility to help develop markets that are currently too underdeveloped or risky for most investors to enter. This concessional

⁹³ IEA [Financing clean energy transitions in emerging and developing economies](#) (2021)

financing will allow Acumen to provide a range of financial instruments to meet the needs of companies entering risky markets, including start-up grants and technical assistance.

- **Junior equity:** GCF's \$50 million investment of junior equity, along with a target \$10 junior equity from other financiers, is needed to create the Market Expansion Facility's blended finance structure, which is designed to mobilize an additional target \$120 million from investors with different risk appetites at scale. GCF's risk-taking capital in this structure will seek to mobilize financing from private sector investors, thereby crowding mainstream investors into a market that has mostly received financing from foundations, risk-supportive DFIs and impact-focused investors. This will allow the H2R initiative to reach significant scale and maximize its impact.
- **Grant:** GCF's \$5m in grant funding will be used to provide performance incentives to investee companies based on the performance of impact targets particularly targeting low-income customers with first-time energy access - sustainably and inclusively.

The 1:1:1 structure is intended to match the risk appetite of the investors with the incentive needed by companies to expand into H2R markets. Concessional financing is needed by companies to go to hardest-to-reach markets and thus the cost of capital has to be low enough to incentivize the companies to do so. At the same time Acumen would like to attract private investors into the sector. This tiered structure provides a de-facto guarantee for senior investors, despite high risk, low return investments.

Without GCF's \$10 million repayable grant to MSF, this intervention would not be able to innovate and bolster the LDC markets in which H2R seeks to operate. Without GCF's \$55 million anchor investment in the Market Expansion Facility, Hardest-to-Reach would not be able to crowd in other investors and would not be able to reach sufficient scale to have a significant impact in the region. With GCF's investment, Hardest-to-Reach aims to shift the paradigm and develop a new frontier for energy investment by creating new markets that have the potential to thrive beyond the lifetime of the initiative.

What GCF brings to H2R

The GCF is the world's largest dedicated climate fund, and as such brings significant credibility to the Hardest-to-Reach initiative. In addition to the direct investment into the initiative, GCF's involvement will have the following benefits:

- Provide a framework for mobilizing support from relevant governments, with nationally designated authorities as touch points for continued engagement and alignment
- Mobilize investors by enhancing both the risk profile and reputability of the H2R initiative
- Enhance stakeholder engagement with the initiative through its dedication to country ownership
- Strengthen the climate rationale of the initiative through its robust investment criteria and integrated results management framework
- Have strong signaling effects to the market of the importance of energy access in increasing the adaptive capacity of beneficiaries

Uniqueness of H2R

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There are a handful of other funds and facilities that invest in energy access in SSA, such as the Energy Access Fund, Access to Clean Power Fund, Facility for Energy Inclusion Off-Grid Energy Access Fund, and Africa Go Green Fund . However, none of these funds have a specific objective to support off-grid solar energy companies' expansion into riskier, hardest-to-reach markets with low electrification rates. Hardest-to-Reach plugs that funding gap by delivering capital to the markets most often overlooked by investors.

H2R is also unique in its two-stage approach. Through the Market Support Facility, H2R can provide a flexible blend of capital to meet companies' financing needs in these riskier markets. This includes grants for companies to cover the start-up costs of expanding into a new market, which is a differentiating feature. Through the Market Expansion Facility, H2R expects to provide impact-linked debt that incentivizes a focus on sales to first-time energy access customers. This distinguishes H2R's approach from others as particularly high-impact. H2R intends to furthermore provide TA support to companies as needed through both facilities in order to build capacity and strengthen markets for the long term.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Explain how the project/programme will successfully exit once implementation is completed, including how results and benefits will continue beyond the project/programme period and how the contribution to paradigm shift will be maintained.

Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through Initial Public Offerings, trade sales, etc.

Provide information on additional actions to be undertaken by public and private sector or civil society as part of the project/programme to ensure sustainability of the results attained.

By investing in nascent off-grid markets with few to no examples of off-grid solar players operating at scale, the Hardest-to-Reach initiative seeks to be a pioneer in untested markets, making operating companies more attractive to other investors and more sustainable beyond the life of the initiative. H2R will seek to achieve a paradigm shift by building OGS company capacity and scale through technical assistance and investment, demonstrating the viability of these markets and replicability of the investment approach, and ensuring the long-term interest of beneficiaries, companies, investors, and the public sector.

Sustainability

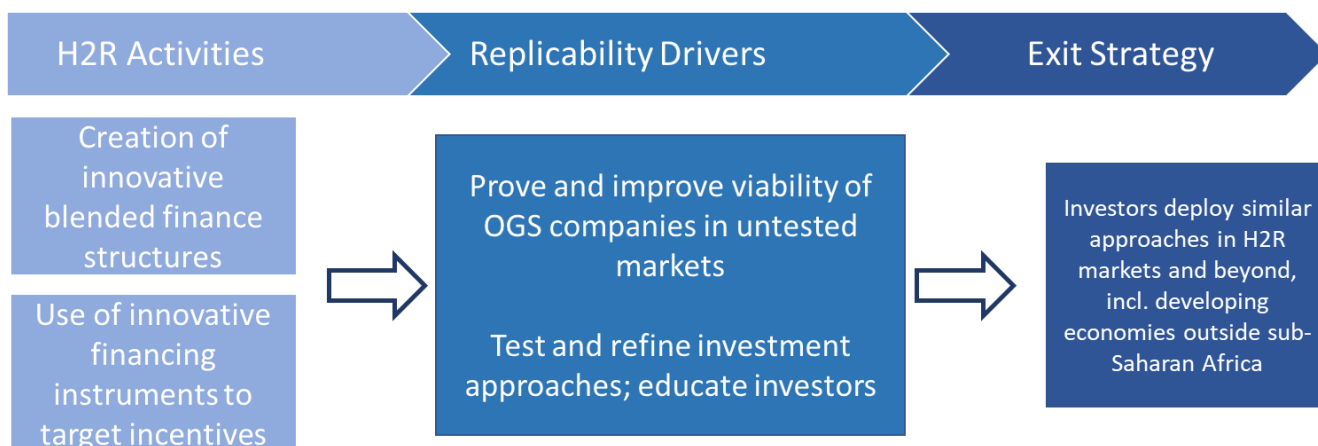
The initiative intends to provide investees with targeted accompaniment in the areas of consumer protection, enterprise development, gender and ESG. Technical assistance supports capacity building, can act as a risk management approach, and should enable companies to embed best practices that sustain responsible growth over the long term. Additionally, in order to develop a sustainable off-grid solar market in underserved geographies, H2R aims to boost the capacity of local distributors and companies and help establish local distribution networks.. H2R aims to source and invest in eligible local entities, with the objective being to ensure that local businesses also receive the concessionary capital and the technical assistance needed to operate in high-risk markets. By providing accompaniment to companies and local

distributors, the initiative aims to develop robust off-grid solar sectors in key H2R countries, which will enable markets to self-sustain their growth and become more attractive to investors.



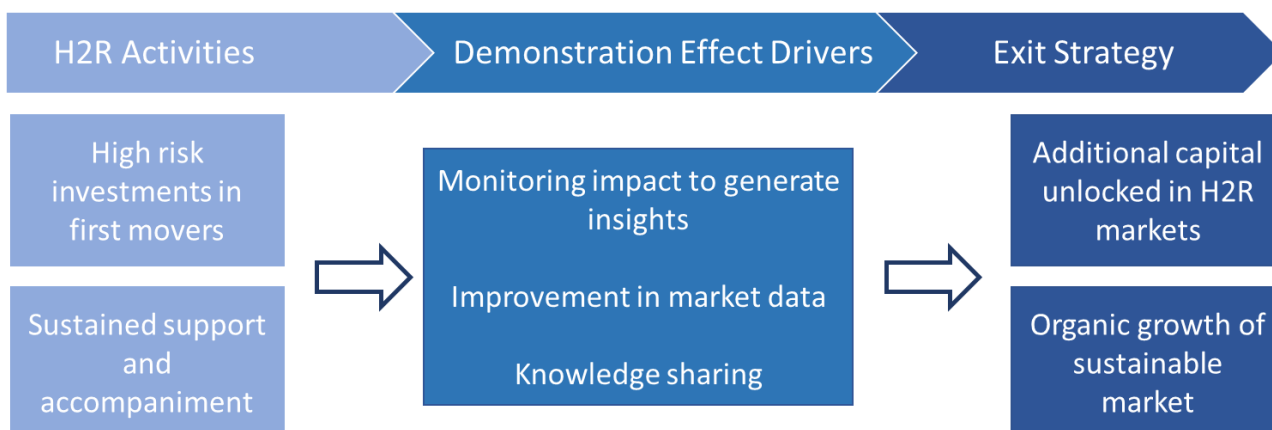
Replicability

The proposed approach has replicability in-built. Two separate financing facilities are proposed with different investment risk profiles, and will invest with different capital blends to match the needs of the company and market, as scale is achieved. The Market Support Facility (MSF) expects to extend debt, equity/equity-like capital, and grants using a patient capital approach that is consistent with Acumen's core investment experience. While the MSF seeks to prove an approach in a new market with a range of capital, the Market Expansion Facility (MEF) deploys impact-linked debt that aim to help responsibly expand the reach of investees in the long-term. As H2R will share lessons-learned and insights widely, successful expansions into new markets can be replicated by that company in another eligible geography or the playbook can be copied by another investor. For example, other investors can apply impact-indexed loans to target incentives or utilize flexible patient capital to support companies in the so-called 'pioneer gap' - the stage of enterprise development where businesses face real challenges raising capital to validate and prepare their business models for growth and build out the systems and infrastructure that are expected to enable them to scale effectively and responsibly.



Demonstration effect

High-risk investments made via the Market Support Facility provide the right balance of incentives and support to help companies weather the first-mover disadvantage, educate customers, test market and product approaches, and ultimately demonstrate the viability of these markets. By sharing insights and results from the initiative, valuable lessons can be passed on to other investors and OGS enterprises. The demonstration effect is key to unlocking other investments from broader sources of capital and sustaining the growth of these markets in the long term.



Policy alignment

Investments at the operational level will be aligned with long-term national policy, such as integrated energy planning and rural electrification strategies, to meet the needs of the recipient and prevent avoidable competing interests and inefficiencies, which are a threat to business sustainability. Section D5 specifies the context and policies of countries relevant to the initiative. These complementary efforts will help to ensure long-term interest in the project. The broader enabling environment is also critical to reducing market barriers and ensuring the long-term viability and attractiveness of Hardest-to-Reach markets and companies. By ensuring policy alignment, investees may also benefit from lowered duties and taxes, and the sharing of

public data, among other benefits, to cut the cost and timeline of business set-up and improve the accuracy of customer targeting. In addition, reporting and engagement with government entities will give us opportunities to share how they can support this work. Furthermore, H2R intends to only invest in countries in which it has received a no-objection letter from the National Designated Authority, thus ensuring country ownership of investments and supporting an enabling environment for off-grid solar.



C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	65M			USD		
GCF financial instrument	Amount		Tenor			Pricing
(i) Senior loans	0		-	-		
(ii) Subordinated loans	0		-	-		
(iii) Equity	50M		Up to 12 years	-		
(iv) Guarantees	0		-	-		
(v) Reimbursable grants	10M		Up to 12 years	-		
(vi) Grants	5M		-	-		
(vii) Results- based payments	0		-	-		
(b) Co- financing information	Total amount			Currency		
	185M target			USD		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing ⁹⁴	Seniority

⁹⁴ This is Expected Net IRR

Senior Equity Investors	<u>Senior Equity</u>	<u>60M</u>	<u>USD</u>	<u>Up to 12 years</u>		<u>Senior</u>									
Mez Equity Investors	<u>Mez Equity</u>	<u>60M</u>	<u>USD</u>	<u>Up to 12 years</u>		<u>Mezzanine</u>									
Junior Equity Investors.	<u>Junior Equity</u>	<u>10M</u>	<u>USD</u>	<u>Up to 12 years</u>		<u>Junior</u>									
Donors for MSF	<u>Grant</u>	<u>40M</u>	<u>USD</u>												
Donors for MEF (Performance incentives)	<u>Grant</u>	<u>15M</u>	<u>USD</u>												
(c) Total financing	Amount			Currency											
(c) = (a)+(b)	<u>250M</u>			<u>USD</u>											
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee). Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets. Please also include parallel financing associated with this project or programme (refer to the co-financing policy).														
C.2. Financing by component															
Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below .															
Component	Output		GCF financing	Co-financing											

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			Amount Options	Financial Instrument		Financial Instrument	Name of Institutions
Component 1: <u>Sustainable Market Creation in H2R countries (MSF)</u>	<u>Output 1:</u> <u>H2R</u> <u>established</u> <u>and \$250M</u> <u>fundraised;</u> <u>\$50M for</u> <u>Market</u> <u>Support</u> <u>Facility</u> <u>(MSF)</u>		<u>USD 10M</u>	<u>Repayable</u> <u>Grant</u>		<u>Grant</u>	<u>Philanthropic</u> <u>donors</u>
	<u>Output 2:</u> <u>Developed</u> <u>gov't and</u> <u>non-gov't</u> <u>partnerships</u> <u>to enable</u> <u>last-mile</u> <u>energy</u> <u>access for</u> <u>the poorest;</u> <u>(i.e. co-</u> <u>investments</u> <u>, knowledge</u> <u>partnerships</u> <u>, gov't</u> <u>partnerships</u> <u>, and</u> <u>industry</u> <u>partnerships</u> <u>)</u>						
	<u>Output 3.1:</u> <u>Increased</u> <u>support for</u> <u>new and</u> <u>fledgling</u> <u>OGS co's in</u> <u>H2R</u> <u>countries,</u> <u>with</u> <u>alignment to</u> <u>serve low-</u> <u>income</u> <u>segments</u>						
	<u>Output 5:</u> <u>Increased</u> <u>provision of</u> <u>climate-</u> <u>resilient</u>						

	<u>clean energy via 1.1M new sales of affordable, clean, resilience-enhancing decentralized energy products in MSE</u>						
	<u>Output 6: Displacement of harmful and emissions-intensive lighting sources and 6MW newly installed renewable energy capacity in MSE</u>						
	<u>Output 4.1: Strengthened OGS business performance, with demonstrated commercial viability, sustainability, gender- and poverty-inclusivity, and investability</u>						
Component 2: <u>Sustainable Market Expansion in H2R countries (MEF)</u>	<u>Output 1: H2R established and \$250M fundraised; \$200M for Market Expansion Facility (MEF)</u>		<u>USD 5M</u>	<u>Grant</u>		<u>Grant</u>	

	<u>Output 3.2: Increased scale of OGS co's in H2R countries, with demonstrated alignment to serve low-income segments</u> <u>Output 4.2: Strengthened OGS business performance, with proven commercial track record</u> <u>Output 5: Increased provision of climate-resilient clean energy via 12.2M new sales of affordable, clean, resilience-enhancing decentralized energy products in MEF</u> <u>Output 6: Displacement of harmful and emissions-intensive lighting sources and 114MW newly installed renewable energy</u>		USD 50M	Junior Equity		<u>Snr Equity</u> <u>Mezzanine</u> <u>Equity Jnr</u> <u>Equity</u>	
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	<u>capacity in MEF</u>						
Indicative total cost (USD)			<u>65M</u>				

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☐ No ☒

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

N/A

The H2R intervention seeks to provide capacity building activities in the form of technical assistance to investee companies and ecosystem development to strengthen the off-grid solar markets in Host Nations. A Technical Assistance Facility (TAF) is expected to be established as part of the Market Support Facility. By integrating capacity building into the initiative through the TAF, H2R aims to develop more resilient and impactful business models that can scale their operations sustainably in H2R markets. As a result, these markets should become stronger and more competitive, paving the way for other OGS businesses to thrive even beyond the life of the initiative.

TA will be provided for four main areas: Consumer Protection; Enterprise Development; ESG; and Gender. In particular, Hardest-to-Reach aims to support investees to navigate the business challenges, risks and barriers of delivering energy solutions to underserved segments; to grow sustainably and inclusively, with an eye on protecting consumers; to access capacity building for better financial performance; to enhance commercial and leadership capabilities for greater business sustainability, particularly for locally-owned and managed companies; to identify and mitigate environmental, social and governance risks; and understand and address the inequities women face in energy, as users, entrepreneurs and members of the workforce. H2R aims to also support the development of off-grid solar markets with a gender equity lens by providing training for female professionals and community members, and by uplifting women in OGS markets, as leaders, entrepreneurs, valued workforce and consumers. Similarly, H2R aims to strengthen ESG

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risk management and mitigation in these markets through training and awareness-raising, particularly as it relates to e-waste.

The specific activities for each TA area are outlined below.

Consumer Protection	Enterprise Development	ESG	Gender
• Consumer protection policies and procedures • Consumer protection practices • Consumer protection M&E	• Financial readiness • Financial development/ sustainability strategies • Financial management systems • Financial stability support • Commercial and leadership capacity-building	• ESG policies and procedures • ESG capacity building • E-waste support	• Gender-sensitive policies and procedures • Recruitment • Training, mentorship and career development • Ecosystem development

H2R will assess technical assistance needs during due diligence, through targeted TA due diligence questionnaires around each of the four areas. H2R will work with investees to develop TA action plans, and outline areas for support, alongside clear targets and a results framework. Investees will then implement their TA action plans with the appropriate level of support from TA expert partners, including H2R support. H2R will track TA implementation, and collect, monitor and evaluate impact data of TA interventions. The lifecycle of technical assistance interventions can be seen in the diagram below.



H2R intends to also analyze the effectiveness of TA interventions based on feedback gathered from companies and technical experts to guide future interventions. Furthermore, these findings and lessons-learned about the impact of capacity-building are expected to be shared with the OGS sector as part of H2R's thought leadership agenda.

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D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme benefits for mitigation and/or adaptation. Provide the intended outcomes for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the intended outcomes for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.3 reporting GCF's core indicators.

Hardest-to-Reach expects the Project outcomes as described in the Theory of Change to contribute to the following core indicators, consistent with GCF's result areas.

The following outlines impact potential targeted by the end of the intervention timeline.

MRA 1: Energy Generation & Access

- **Total project direct and indirect beneficiaries: 54.3M**
- Impact Potential (i): 5.1MtCO₂e emissions reduced or avoided
- Impact Potential (ii) : 120 MW installed capacity
- Impact Potential (iii): 72.4M lives impacted with clean energy
- Impact Potential (iv): # of H2R investments and countries invested
- Impact Potential (v): 13,922 jobs (4,641 women)
- Impact Potential (vi): Low-Emission Sustainable Development Pathways (see below)

Low-Emission Sustainable Development Pathways

The economies and populations of sub-Saharan Africa are growing rapidly and while they account for less than 4 percent of cumulative global greenhouse gas (GHG) emissions to date,⁹⁵ by 2050 a quarter of the global population will live in the region, and energy demand will rise significantly.⁹⁶ To mitigate and adapt to an increasingly warming planet, it is critical that all countries transition to low carbon-intensity and affordable modern energy and, even where energy consumption is currently low, embark on clean energy pathways. The energy pathways and resource dependencies developed across the continent today will have profound climate impacts in the medium and long term. Data shows that developing economies

⁹⁵ AfDB [African Economic Outlook Summary](#) (2022)

⁹⁶ The Guardian [By 2050, a quarter of the world's people will be African – this will shape our future](#) (2022)

(excluding China) now account for roughly 40 percent of carbon emissions,⁹⁷ and will propel almost all the growth going forward.

Sub-Saharan Africa produces, on average, only 0.8 tonnes of CO₂eq per person per year - compared to a global average of 4.8 tonnes.⁹⁸ However, highly developed and coal-dependent South Africa (85 percent electrified) produces nearly ten times that per person - an extreme example and a cause for some concern if large nations such as Mozambique (29% electrified), DRC (19%), and Uganda (29%) were to only partially follow suit. These are all countries with electrification rates below the regional average of just over 48 percent,⁹⁹ which is the primary selection criteria for Hardest-to-Reach target countries. Predictive modeling illustrates Africa's low-carbon SDG 7 scenarios and their correlation to SDG 13 as the region electrifies.¹⁰⁰ Providing first-time electricity access via a low carbon trajectory avoids approximately 268 million tonnes of CO₂eq emissions, with a technology mix that is 45 percent decentralized renewables.¹⁰¹ If African emissions per person were to rise even to the current global average, the continent would emit 12.5 billion tonnes of greenhouse gasses per year by 2050 — roughly equivalent to all emissions from China and India today.¹⁰²

The H2R initiative seeks to contribute to SSA's low-emissions sustainable development pathway. By providing people with first-time energy access through low-emissions energy sources, such as solar home systems, customers gain trust and familiarity in the technology and payment models and opt to use similar technologies in the future, even as their energy consumption grows. This aligns with the idea of low-carbon development pathways, where economic development is decoupled from increasing carbon emissions.

ARA 1: Most vulnerable people and communities

- **Total project direct and indirect beneficiaries: 54.3M**
- Impact Potential (i): 72.4M lives impacted with clean energy
- Impact Potential (ii): 54.3M people with first time energy access and improved climate resilience through decentralized solar infrastructure
- Impact Potential (iii): 5.4M people with improved climate resilience through diversified income sources enabled by first-time access to a solar powered product
- Impact Potential (iv) # of investments, including follow-on investments
- Impact Potential (v): 13,922 jobs (4,641 women)
- Impact Potential (vi): stronger more robust primary energy systems through decentralization of energy supply (see below)

⁹⁷ EnerData [GHG emissions in developing countries - issues and perspectives for COP-26](#) (2021)

⁹⁸ Wilson Centre [The Battle for Earth's Climate Will Be Fought in Africa](#) (2021)

⁹⁹ World Bank [Data Bank](#) (2020)

¹⁰⁰ Shell Foundation [Unlocking Climate Finance to Accelerate Energy Access in Africa](#) (2021)

¹⁰¹ Ibid.

¹⁰² Tony Blair Institute for Global Change [Africa Needs Support If It Is to Play Its Part in the Drive for Net Zero Carbon Emissions](#) (2021)

- Impact Potential (vii): anticipatory, absorptive and adaptive resilience for households via OGS characteristics (see below)

Energy System Resilience through decentralized solar

Access to electricity is considered a key part of adaptive capacity; one of the three principal factors which determine climate change vulnerability in a region, along with sensitivity and exposure to hazards and impacts. Adaptive capacity, the characteristic which allows people to cope with these hazards and impacts, includes elements such as social networks, livelihood diversity, asset diversity, innovation and access to financial services. Anticipatory capacity, the characteristic which allows people to plan for and invest and/or change behavior and habits in advance of events, includes disaster preparedness, including access and the ability to respond to weather forecasts and warnings of hazards. Electricity access is the thread linking all of these, as it plays a key role in enabling all these components. As such it plays a significant role in making livelihoods less vulnerable in H2R target countries.¹⁰³

Limited power access in turn threatens livelihoods which rely on agriculture for a number of reasons. First and foremost, the inability to access power means the inability to reliably access information and to communicate. This is a crucial weakness during natural disasters, which are expected to continue to increase in frequency and intensity as a result of climate change. In terms of enabling anticipatory capacity, energy access can contribute to increasing household income by increasing working hours through access to light and increasing opportunities for home based business, while direct access to power reduces the cost and time investment in keeping a mobile phone charged, or of keeping a kerosene lamp fueled.

Energy systems have been compromised both by extreme and slow-onset climate events, with resulting economic losses, disruptions of services and impacts to wellbeing. The threat to the reliable provision of energy is exacerbated by the centralization of power generation and where distribution is interconnected, including across regional power pools. Centralized energy systems across sub-Saharan Africa are already facing the strain of climate change. For example, following low rainfall and droughts in 2015-16 in southern Africa, power generation reduced by 75 percent at the Kariba dam in Zambia, and by 66 percent at the Cahora Bassa dam in Mozambique due to insufficient water resources, with widespread impact on electricity supplies across neighboring countries. As a result, Malawi, Zambia and Zimbabwe all experienced widespread and prolonged load shedding. The impact was felt throughout the regional economy and reflected in reduced GDP growth.¹⁰⁴ Furthermore, in 2019 Cyclone Idai caused significant damages to Mozambique's energy sector, and resulted in an estimated loss of \$47.9 million. The cyclone damaged or destroyed 1,345 km of transmission lines, 10,216 km of distribution lines, two 90 MW generation plants, 30 sub-stations, 4,000 transformers.¹⁰⁵ Similarly, Malawi lost 30% of its electricity to Tropical Storm Ana in February 2022 because of damage to the 130 MW Kapichira hydropower plant.¹⁰⁶ The World Bank approved a loan of \$45M to restore the power plant, which is set to reopen in May 2023, leaving the country at 2/3 of its energy generation capacity for over a year.¹⁰⁷ These examples

¹⁰³ ODI, [How solar household systems contribute to resilience](#) (2017)

¹⁰⁴ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability](#) (2022)

¹⁰⁵ Ibid.

¹⁰⁶ VOA News [Malawi Loses 30% of Its Electricity to Tropical Storm Ana](#) (2022)

¹⁰⁷ The Times Malawi [Blame game as Kapichira Dam rehabilitation delays](#) (2022)

demonstrate some of the ways in which traditional utility scale energy systems are vulnerable to the effects of climate change.

The IPCC has therefore recognised the role that decentralized renewable energy can play in increasing the resilience of energy systems, as it helps to diversify energy supplies and reduce pressure on the grid.¹⁰⁸ Decentralized energy also has a role in reducing the vulnerability of rural populations to climate change. In fact, evidence suggests that solar home system usage is considered as a coping mechanism against natural disaster-induced electricity disruption by households in climate vulnerable areas. The research points in particular to the attributes that make distributed renewable energy easier to be rebuilt (where necessary), reconnected, and fully functioning in a shorter period.¹⁰⁹ By diversifying energy sources, increasing the distribution of energy solutions and adopting technical improvements to reduce systems' climate vulnerability, we are able to see more adapted and resilient energy systems in H2R countries.¹¹⁰

Anticipatory, absorptive and adaptive capacity for households through characteristics of OGS

Energy access impacts the ability to anticipate, avoid, plan for, cope with, recover from and adapt to (climate related) shocks and stresses'.¹¹¹ Three dimensions or capacities of climate resilience can be improved through off-grid solar: adaptation (adaptive capacity), anticipation (anticipatory capacity), and absorption (absorptive capacity). By enabling the sale of climate resilient, affordable and reliable OGS products to underserved communities, Hardest-to-Reach contributes to all three. Firstly, OGS energy products are durable and portable, meaning that they are able to withstand climate-change related phenomena, and can be moved and safely stored in the event of extreme weather events.¹¹² They are also repairable, enabling a resumption of energy provision following product damage. Thirdly, they lead to a more diversified energy system and take pressure off the grid, which can reduce systemic climate vulnerability. As described, distributed renewable energy (DRE) reduces vulnerability to climate impacts by increasing the resilience of energy provision in the face of natural disasters and climate shocks, which are increasingly frequent and intense in these geographies.¹¹³ Each solar home system is an important independent energy source that can continue to provide a household with energy even if the grid has been compromised because of increased demand or damage caused by an extreme weather event. Recent data show that an estimated 775 million people worldwide are connected to weak grids, undermining the potential social and economic development of access to electricity. Off-grid solar can play an important role in improving the quality of electricity access here as well as deliver significant climate mitigation and adaptation benefits; Diesel generators used to provide grid back-up emit more than 100 megatons of CO₂

¹⁰⁸ IPCC [Sixth Assessment Report Climate Change 2022: Impacts, Adaptation and Vulnerability](#) (2022)

¹⁰⁹ Amin et al. [Solar energy and natural disasters: Exploring household coping mechanisms, capacity, and resilience in Bangladesh](#) (2021)

¹¹⁰ Practical Action [Understanding the Links Between Energy Access and Climate Change Adaptation](#) (2022)

¹¹¹ Bahadur A.V., Peters K., Wilkinson E., Pichon F., Gray K. and Tanner T; 'The 3as: Tracking resilience across BRACED' (2015). As quoted in Practical Action [Understanding the Links Between Energy Access and Climate Change Adaptation](#) (2022)

¹¹² ODI [How solar household systems contribute to resilience](#) (2017)

¹¹³ Brookings [Increasing droughts and floods on the African continent](#) (2021)

every year. In sub-Saharan Africa alone, replacing these generators with solar alternatives would avoid as much CO₂ as 20 percent of the region's vehicles being replaced with clean alternatives.¹¹⁴

ARA 2: Health, wellbeing, food and water security

- **Total project direct and indirect beneficiaries: 54.3M**
- Impact Potential (i): 54.3M people with first time energy access, displacing harmful and/or emissions intensive lighting sources
- Impact Potential (ii): 70% of customers (~50.6M) indicating quality of life improvement
- Impact Potential (iii): Improved health, wellbeing and quality of life through energy access (see below)

Improved health, wellbeing and quality of life through energy access

In least developed country contexts, in particular, energy access is an enabler for poverty reduction that is crucial for enhancing people's livelihoods and wellbeing. According to the IPCC and research by the UK's Foreign, Commonwealth & Development Office (formerly DFID) there is growing evidence of the role of DRE in reducing people's overall vulnerability, especially in underserved populations, through more well lit private land and assets, less hazardous household lighting, such as via open flames, and cleaner air in the absence of black carbon emitted from kerosene lanterns.¹¹⁵ Energy access can also reduce the impact of climate hazards. Without electricity, for example, households cannot receive SMS, radio or TV news to receive early warning messages or public safety announcements before and after a climate event. Thus, across sub-Saharan Africa, where 33 of the world's 46 LDCs are situated, delivering energy access to enable specific applications such as information communication technologies, and by driving socio-economic development (i.e. improving incomes, health outcomes and access to education), is highly impactful for quality of life perceptions, health and overall well being. The 60 Decibels Energy Benchmark, a Lean Data survey of off-grid energy customers in developing economies, found that 51% of people felt that their quality of life had improved significantly after buying an off-grid product in general, compared to 68% for solar-powered off-grid products in particular.¹¹⁶ Acumen's PEII portfolio companies scored high on this metric, where 60 Decibels surveys of PEII portfolio customers found that 64% of off-grid energy customers felt their quality of life had improved substantially. Whilst this measure is subjective and could incorporate a number of different factors, it indicates how significant an impact energy access has on people's lives. Finally, residential solar PV requires neither water nor additional land to generate electricity, meaning OGS products reduce stress on land use and water systems, with broader positive impacts on food security and biodiversity.

Even though it can be concluded that the impact of energy access through SHS on livelihood resilience in SSA is conditional and limited, solar lamps and SHS are nevertheless an important milestone on the route to initial last-mile electrification in rural areas of sub-Saharan Africa. In countries where grid expansion is

¹¹⁴ [GOGLA Market Trends Report](#): State of the Sector (2022)

¹¹⁵ DFID [Literature Review on Energy Access and Adaptation to Climate Change](#) (2015)

¹¹⁶ 60 Decibels [Online Energy Benchmark](#) (2022)

too costly and time-consuming, it is ultimately not realistic in many regions. Even the introduction of larger mini-grids is fraught with complexities in the H2R target regions, because the communities with a high share of rural poor simply lack the power demand to sustain a mini-grid.¹¹⁷ Therefore, at present, micro-solutions such as lanterns and SHS offer the best and only option to bring some form of electric power to these households.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Paradigm shift potential is defined as 'degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment'. In this section, elaborate on the contribution to paradigm shift and how the proposed project/programme aims to contribute towards it based on the theory of change described in section B2(a). Also describe how and to what extent the project/programme will be able to promote or contribute to paradigm shift through the below.

- Potential for scaling up and replication
- Potential for knowledge sharing and learning
- Contribution to the creation of an enabling environment
- Contribution to the regulatory framework and policies
- Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

Paradigm Shift

The H2R initiative will deliver a paradigm shift by:

- I. Nurturing and developing companies to serve underserved off-grid solar markets; and by
- II. Unlocking the provision of - and investment in - clean, climate-resilient decentralized solar, that will
- III. Power low-carbon economic development for the long-term.

One of the key aims of Hardest-to-Reach is to transform the African energy sector by building (and eventually scaling up) off-grid markets where there currently are none or they are fledgling. The initiative will focus on LDCs in sub-Saharan Africa, particularly with electrification rates at-or-below the regional average, where investors have not been willing to enter. H2R's blended finance structure is transformational in its design as it will crowd-in investors that would otherwise find the markets too nascent or risky. The initiative intends to de-risk these markets by providing new market entrants with impact-linked concessional finance and targeted technical assistance, and seeks to support the development of local networks and capacity. By proving the OGS business model in untested markets, companies can also crowd in additional capital to achieve scale in these markets. In the scale phase, H2R's impact-linked debt financing aspires to incentivize OGS companies to maintain their focus on low-income, first-time energy access customers.

A blend of capital is needed to achieve this. Data from Shell shows that approximately \$11 billion will be required in debt into SHS enterprises towards SDG 7 in non-established markets.¹¹⁸ A further \$2.8 billion of

¹¹⁷ GTM, [Recovering the Missing Millions Owed to Africa's Minigrids](#) (2020)

¹¹⁸ Shell Foundation [Achieving SDG 7: The Need to Disrupt Off-Grid Electricity Financing in Africa](#) (2018)

debt is required in established markets. Equity and equity-like capital is critical as fledgling and latent markets require substantial early-stage, risk tolerant capital in order to be unlocked. H2R's approach is based on allocating the right type and size of capital for the stage of the company and development of the market intending to create an enabling environment in these difficult to operate regions. Last-mile distributors (LMDs) have a crucial role to play in delivering OGS products to H2R consumers, but capital is not flowing into the space. Research from the GDC shows that all types of capital - grant, equity and debt - are needed to unlock the full potential of LMDs, depending on their growth trajectory.¹¹⁹ The MEF allows a targeted use of loans to established companies seeking to scale.

Technical assistance will play a fundamental role in delivering H2R innovation. By focusing on TA that overcomes enterprise and market barriers to achieving long-term and inclusive growth, there is a holistic, whole-of-company-lifecycle approach that represents a paradigm shift compared to current investment practices and sector interventions. Acumen will also develop partnerships that convene different financiers, TA providers, policy advocates, government, and OGS companies in an effort to jumpstart the OGS market in H2R areas. Through Hardest-to-Reach, Acumen is working to form a partnership with the World Bank Group, Power Africa, and other leading energy access practitioners and financiers to address the affordability gap through demand side subsidies to expand our reach to the lowest income level customers. Across the range of partners are those able to help address other market barriers, including developing the right VAT and duty structures for renewables products and narrowing the affordability gap for low-income customers. Coordinated efforts like these should reduce inefficient use of resources, create opportunities to leverage stakeholder experiences and competencies, and minimize distortive risks. This is expected to lead to a paradigm shift in the investing landscape in H2R areas, and help increase capital inflows and more coordinated investments on the ground, with multiple stakeholders and government buy-in for climate resilient off-grid solar. Through H2R's innovations - not only scaling up innovative business models and technologies, but also strengthening the entire OGS ecosystem through investing in local distribution networks, delivering consumer protection and other TA, fostering partnerships and conducting robust impact monitoring - Acumen intends to use GCF's much-needed concessional financing to create a paradigm shift by de-risking OGS markets, creating a commercial track record and crowding-in private finance to deploy new climate solutions at scale. It is a blueprint that can set a new course for decentralized and decarbonized electrification in these Hardest-to-Reach markets.

Acumen seeks to develop the knowledge base in the sector by generating and sharing insights and lessons so that other development partners can learn from the H2R initiative and replicate successful aspects in their own energy access initiatives. As outlined in the program activities (Sect E.6) and the program-specific indicators (Section E.5), H2R intends to develop as many as 10 independent thought pieces to be shared with the wider community of stakeholders, such as white papers, blog posts, insight studies, toolkits, or playbooks. Subjects may include lessons from our development of more robust consumer protection practices, studies to deepen the sector's understanding of household-level impacts of decentralized solar and how these can be maximized, and what role investors can play.

Acumen is playing a convening role in the sector and through sharing the knowledge and evidence-based insights gained through H2R, particularly on the impact of energy access and climate resilience, will be able to increase capital inflows and more coordinated investments on the ground, with the buy-in of multiple stakeholders and government toward climate resilient off-grid solar. H2R intends to coalesce all the relevant stakeholders (NDAs, government entities, academics, OGS businesses, industry associations,

¹¹⁹ Global Distributors Collective [The growth and fundraising journeys of last mile distributors \(LMDs\)](#) (2021)

indigenous communities, civil society organizations, and women's groups). Additionally, the team hopes to report on insights to GOGLA, investors, and other climate financiers as to crowd in climate adaptation financing for energy access projects. We believe that these efforts will help to bolster H2R's demonstration effect of "energy access as adaptation". This aspect should bring positive externalities in unlocking funding for powering resilience and building a resilient OGS ecosystem.

The 16 target countries for Acumen's Hardest-to-Reach initiative signed the Paris Agreement and submitted Nationally Determined Contributions (NDC). All 16 countries have pledged both conditional and unconditional targets to reduce their GHG emissions by 2030 under the Business-as-Usual (BAU) scenario. The unconditional targets are targets expected to be met by countries with their own resources without external support, whilst conditional targets are subject to financial support from the international community. Two of the 16 countries have set their conditional mitigation targets at over 50% emission reduction by 2030, namely: Malawi and Togo at (51% and 51%) respectively.¹²⁰ The lowest conditional emission reduction targets were from Burundi at 16%. The NDCs further incorporate mitigation and adaptation actions. For instance, Uganda's NDC provides 48 priority adaptation actions and 82 indications with adaptation targets for 2030, one of the targets is to achieve 75% electrification by 2030.¹²¹ Acumen's objectives through Hardest-to-Reach are directly aligned with national climate priorities, including National Adaptation Plans, National Climate Change Action Plans, and NDCs. Further details on country electrification, climate policies and climate adaptation overviews can be found in the country climate reports .

D.3. Sustainable development (max. 500 words, approximately 1 page)

Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide the potential in terms of:

- *Environmental co-benefits*
- *Social co-benefits including health co-benefit*
- *Economic co-benefits*
- *Gender-sensitive development benefit*

Sustainable Development

Modern society depends on reliable and affordable energy services to function smoothly and to develop equitably. Carbon intensive development patterns have historically relied on inexpensive and energy-dense fossil fuels, which are also the primary source of greenhouse gas emissions contributing to climate change. Hardest-to-Reach is aligned with SDG 7 (affordable, reliable, modern and sustainable energy for all) and through the power of energy access will enable broader sustainable development outcomes; good health and wellbeing (SDG 3), improved educational outcomes (SDG 4), economic savings and income generation (SDG 1 and SDG 8), and climate action (SDG 13). Acumen's intervention is also partnership-focused, envisioning a more coordinated investment ecosystem to transform the prospects for growth in the off-grid solar sector (consistent with SDG 17).

¹²⁰ United Nations Development Programme (UNDP). [Where We Work – Africa](#) (2022)

¹²¹ Ministry of Water and Environment Uganda. [Updated Nationally Determined Contributions - Uganda](#) (2021)

The H2R initiative has a number of broader sustainable development impacts. Firstly, off-grid solar products can improve people's health, safety and wellbeing. For those trapped in energy poverty, particularly the rural households in H2R target countries, fossil-fuel based energy sources such as kerosene may be the only available energy source. These traditional lanterns provide low-quality light, are harmful to people's health because of household pollutants, and can cause burns and fires. This disproportionately affects women and children who spend more time at home. Replacing fossil-fuel based energy sources with clean off-grid solar products can therefore have positive impacts on people's health and safety as customers will have access to better quality lighting,¹²² which can prevent falls at night, help with security, and allow for cleaner air at home. A 2018 study in Uganda cited the fact that single-wick lanterns were once responsible for 70 percent of fire incidents and 80 percent of burn injuries in energy poor households.¹²³ These data points are likely to be less stark in today's Uganda, thanks in part to some success of off-grid solar technologies. There is still a task to do to ensure no household has to rely on these fuels for lighting. Meanwhile, where households have replaced kerosene lanterns with single use dry cell battery LEDs, these batteries pose a health (and environmental) risk, as they contain heavy metals.¹²⁴ Additionally, fuel combustion releases hundreds of pollutants into the air, including carbon monoxide, formaldehyde and benzene, along with a myriad of other damaging particles. Secondly, OGS products can enable people to generate income through their use.¹²⁵ This can be directly (use in a business) or indirectly (task-shifting activities to the evening because of better quality light). This can help increase people's economic security. Thirdly, the PAYGo financing structure should enable many people to access mobile wallets and credit for the first time, leading to the financial inclusion of low-income people and helping to build their credit scores.¹²⁶ This is particularly important for women, who are often excluded from accessing credit because of land-based collateral requirements. Fourthly, the local distribution networks that will be created and after-sales services provided will be a source of livelihood options and income generation opportunities, as they should create employment for people in low-income areas.¹²⁷ The increased income they receive is expected to be an important contributor to their increased resilience and adaptive capacity. Lastly, H2R envisions a co-benefit of job creation that is consistent with SDG 8. As OGS companies scale, they will require additional human capital to support their growth and sustain their level of service. Jobs include customer service personnel, office and administrators, product sales staff, and technicians for installations and repair.

Environmental and Social

To ensure strong environmental and social co-benefits, Acumen developed an Environmental and Social Impact Assessment to better understand the positive impacts, environmental risks, and potential mitigants. The ESIA informed the development of the H2R Environmental and Social Management System. The H2R ESMS is also aligned with the GCF Environmental and Social Policy and the Acumen ESMS for GCF-funded Projects. The ESIA and ESMS are submitted as part of this proposal.

Our Project seeks to support the development of the OGS sector in H2R markets. An important co-benefit of supporting these nascent markets is building and sustaining green jobs. The H2R program targets the

¹²² 60 Decibels [Why Off-Grid Energy Matters](#) (2020)

¹²³ FINCA [Perceived Health Benefits of OGS Products](#) (2018)

¹²⁴ Bensch, Sievert & Peters [The lighting transition in rural Africa — From kerosene to battery-powered LED and the emerging disposal problem](#) (2017)

¹²⁵ GOGLA [Powering Opportunity The Economic Impact of Off-Grid Solar](#) (2018)

¹²⁶ CGAP [Digitally Financed Energy](#) (2016)

¹²⁷ GOGLA [Off-Grid Solar. A Growth Engine for Jobs](#) (2019)

creation of ~13,900 additional green jobs as a co-benefit of this intervention. An estimated 4,600 of these jobs could be a source of employment for women.

See [Annex 6](#).

Gender

Hardest-to-Reach intends to have significant benefits for women both by creating more opportunities for women to participate in the OGS industry and ensuring access to electricity for women. As mentioned previously, H2R expects to improve the health and safety of women and expects to also lead to their financial inclusion. The initiative will target the industry's benchmark of having 32% of its customers as women¹²⁸ and will aim to support women-led businesses. H2R also seeks to engage companies and the OGS ecosystem in developing more opportunities for women to work in the OGS system in H2R markets with the Project Technical Assistance Facility. Additionally, H2R intends to diligence and monitor Sexual Exploitation, Abuse and Harassment (SEAH) across Investees operations and engagement with customers. H2R commits to supporting equitable and empowering workplaces for women with Investees. Investees are expected to address opportunities for growth in their Gender Action Plans, and with the support of gender experts using technical assistance. Importantly, H2R expects to track gender-disaggregated data for certain indicators, which improves industry-wide knowledge on how women engage with energy products and services, and allows for the greater consideration of gender throughout the OGS industry.

The H2R team developed a gender assessment in accordance with industry best practices, guidance and support from gender experts, community input, and following the GCF Gender Policy. The gender assessment informed the Project's Gender Action Plan and Gender Lens Investing Strategy.

These documents can be found in [Annex 8](#).

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.).

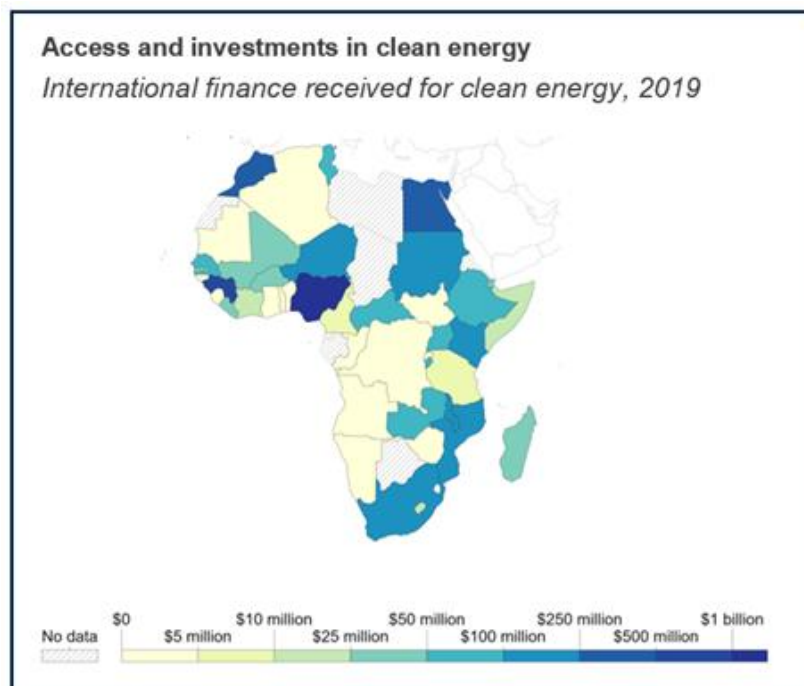
Describe how the project/programme addresses the following needs:

- *Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)*
- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity*

Country-level context

¹²⁸ 60 Decibels [Why Off-Grid Energy Matters](#) (2020)

The investment deficit is at the root of both the energy access gap and the lack of adaptive capacity faced by countries in sub-Saharan Africa. Despite the tremendous progress made by off-grid solar companies in improving energy access in sub-Saharan Africa, a substantial number of financially viable customers in difficult, hard-to-reach markets will not be catered for by current market-based interventions. OGS companies were estimated to have cumulatively raised US\$1.5 billion¹²⁹ as of 2019 from an increasingly diverse investor base as the industry has matured. However, there is a clear geographic bias in the flow of funding. 80 percent of the capital raised in the sector, between 2012-2019, has flowed to larger and more established, first-generation, companies whose primary location and focus is East Africa, with Kenya attracting the lion's share of the capital. Additionally, as OGS start-ups evolve towards profitability, they have increasingly focused on entrenching their market position and attracting higher margin customers in the most active markets at the expense of expanding into the less-active markets with a higher actual and perceived risk. As the two Figures show, the regional disparities in the size and capitalisation of the OGS sector are largely incongruent with the distribution of people without access to electricity. Thus, it is imperative to direct more resources, financial and otherwise, to where they are needed the most.



The overall level of investment has been adversely affected by the pandemic, with foreign private investment into Africa falling some 16% in 2020.¹³⁰ Significant financial commitments are required to close this gap and governments are unable to carry the fiscal burden alone. One estimate suggests that \$35 billion a year is required to reach SDG 7 by 2030. At the same time, the NDCs of 40 African countries cumulatively show a need for an estimated \$331 billion in investment for adaptation through 2030, with only 20% of this figure coming from public budgets.¹³¹ The map shows the dearth of international financing going to developing economies in support of clean energy, particularly given the size of developing-economy populations.¹³² Within sub-Saharan Africa a concentration of investment flows to now-proven markets risks leaving other countries and regions behind. However, barriers exist, including political instability, macroeconomic uncertainty (because of inflation and exchange rates), policy and regulatory issues, institutional weaknesses, and lack of transparency. All these make for a less favorable investment climate. Underinvested geographies need capital that is willing to bear the risk of proving and scaling off-grid business models in untested conditions, and bring with it practices that drive the growth of responsible

¹²⁹ Vivid Economics, [Off-Grid Solar: Market Trends Report 2020](#) (2020)

¹³⁰ World Bank [Driving Foreign Direct Investment \(FDI\) to Africa Video](#) (2021)

¹³¹ Global Centre for Adaptation [State and Trends in Adaptation Report 2021: Africa: Finance](#) (2021)

¹³² UN SDG Tracker [Measuring Progress Towards the SDGs](#) (2018)

companies. Acumen's approach to Hardest-to-Reach markets, with its focus on market development and incentivizing investment where it is needed most, addresses the urgent financing needs of the identified countries.

Benin Country Context

Benin is a low-income country, with a population of more than 13 million people and a population growth rate of 2.8%.¹³³ The rural electrification rate is 9%. Benin's current electrification rate stands at 33% which is below the sub-Saharan African average of 48%, leaving 7.9 million people without access to electricity.¹³⁴ This is disaggregated with 58% and 9% urban and rural electrification respectively.¹³⁵ As the country develops, the energy demand is anticipated to increase, thus there is a need to match the energy supply to the growing energy demand.

DRC Country Context

DRC is a low-income country, with a population of more than 96 million people and a population growth rate of 3.2% per year.¹³⁶ According to the World Bank Group, poverty in DRC is high and is among the five poorest nations in the world. The 2019 electrification rate for the DRC stood at 9% relative to the sub-Saharan African average of 48%, leaving 79 million people without access to electricity.¹³⁷ 19% of the urban population has access to electricity whereas this figure is less than 1% for the rural population.¹³⁸ In 2021, DRC had an installed electricity capacity of 2.8 GW comprising 99% hydropower and 1% non-renewable sources. The national electricity grid powers 19% of the population living in urban areas but only 1% for the population living in rural areas.¹³⁹

Burundi Country Context

Burundi is a low-income country, with a population of more than 12.55 million people and a population growth rate of 2.7%.¹⁴⁰ In 2019, Burundi's national electrification rate stood at 11%, relative to the sub-Saharan African average of 48%, leaving 11 million people without access to electricity.¹⁴¹ There is a 66% connection rate in urban areas, but only 2% in rural areas. As of 2019, the total energy supply in Burundi was sourced from 85% renewables, 14% oil, and 1% coal and others.¹⁴² Of the renewables, 98% is biomass, 0% solar, and 2% hydro. Hydroelectric power alone provides electricity to 9% of the population, which translates to 75% of those with access to electricity.

Mozambique Country Context

¹³³ World Bank. [Data: Total Population](#) (2021)

¹³⁴ IEA. [Access to Electricity](#) (2020)

¹³⁵ USAID. [Power Africa - Country Fact Sheet: Benin](#) (2022)

¹³⁶ World Bank. [Data: Total Population](#). (2021)

¹³⁷ IEA. [Access to Electricity](#) (2020)

¹³⁸ IRENA. [Energy profile DRC](#) (2022)

¹³⁹ USAID. [Off-Grid Solar Market Assessment Democratic Republic of the Congo](#) (2022)

¹⁴⁰ World Bank. [Data: Total Population](#) (2022)

¹⁴¹ International Energy Agency. [Access to Electricity, Burundi](#) (2019)

¹⁴² IRENA. [Energy Profile: Burundi](#) (2020)

Mozambique is a low-income country, with a population of more than 32 million people and a population growth rate of 2.9%.¹⁴³ Only 22% of the population in rural areas had access to electricity in 2019.¹⁴⁴ This is comparable to the urban electrification rate of 57%. Mozambique's national electrification rate stood at 35% in 2019, relative to the sub-Saharan African average of 48%, leaving 21 million people without access to electricity.¹⁴⁵ The country has an installed capacity of 2.9 GW, comprising 90% hydropower and 10% natural gas. However, only 36% of the produced power is used for domestic consumption, whilst the remaining 64% is exported to nearby countries.

Zambia Country Context

Zambia is a low-income country, with a population of more than 18 million people and a population growth rate of 2.9%.¹⁴⁶ Zambia's primary sources of energy include wood fuel, hydropower, coal and solar. In 2019, Zambia's national electrification rate stood at 37%, relative to the sub-Saharan African average of 48%, leaving 11 million people without access to energy. Its urban electrification rate was 76% and its rural electrification rate was 6%. Zambia has an installed capacity of 3.3 GW, comprising 80% hydropower, 11% coal, 4% diesel, 3% heavy fuel oil (HFO) and 3% solar.¹⁴⁷

Gender Context

The link between energy access and gender equality is compelling. Energy poverty has a strong gender dimension, as the lack of access to energy inhibits a woman's freedom from household drudgery, her freedom to allocate time, to access economic opportunities, and enjoy improved household living conditions. Off-grid energy technologies have a direct benefit to women's health, particularly in terms of reduced risks of pulmonary and respiratory diseases, caused by long-term proximity to fumes from kerosene lanterns.¹⁴⁸ At the enterprise level, women's participation and leadership in the OGS sector can help accelerate the transition to a low-carbon economy by promoting clean energy use and addressing energy poverty, while also advancing women's economic empowerment. However, multiple barriers limit women's ability to access OGS technologies and to fully participate in OGS value chains as business leaders, employees and entrepreneurs. These barriers include, but are not limited to, (i) a historically unequal distribution of economic assets which impacts women's ability to access OGS products; (ii) the limited collection and use of sex-disaggregated data to help better understand the needs of women as energy users, and more broadly as stakeholders in the sector; (iii) a lack of targeted training and mentorship opportunities for women to enter the workforce and thrive in the clean energy economy. Overall, evidence shows that women have less access to off-grid electricity than their male counterparts, and they mostly use pico-solar products and lanterns or other lower-tier technologies.¹⁴⁹ SHS customers are predominantly rural men living under the median poverty line of \$3.10.¹⁵⁰ In a survey conducted on 35,000 customers of 49 OGS companies in 16 countries, 60 Decibels found that only 32% of customers

¹⁴³ World Bank. [Data: Total Population](#) (2022)

¹⁴⁴ IEA. [Access to Electricity](#) (2022)

¹⁴⁵ IRENA. [Energy Profile: Mozambique](#) (2022)

¹⁴⁶ World Bank. [Data: Total Population](#) (2022)

¹⁴⁷ IRENA. [Energy Profile: Zambia](#) (2022)

¹⁴⁸ Lam et al. [Exposure reductions associated with introduction of solar lamps to kerosene lamp-using households in Busia County, Kenya](#) (2017)

¹⁴⁹ ESMAP (2022) Gender equality in the off-grid solar sector. A primer. The World Bank Group, Washington DC 2022

¹⁵⁰ GOGLA (2021) Powering opportunity: Energising work, enterprise and quality of life with off-grid solar.

are women.¹⁵¹ The Gender Country Assessments in [Annex 8](#) outline these national-level dynamics in detail.

Energy Access Companies

Equity investors push for profit margins, creating pressure on energy companies to shift in focus toward multi-purpose and higher-end SHS and away from entry-level products and first-time access as a result. Companies are often reluctant to pursue more first-time access customers in unfamiliar territories and have indicated that entering a market as a second or third mover is far more advantageous. This is due to the significant hurdles that need to be overcome in a completely new market such as finding good local management, establishing logistics lines, and understanding the nuances of the local finance and regulatory system. While the challenges faced by OGS are of a similar nature across Sub-Saharan Africa, they can be prohibitively more pronounced in some countries, substantially diminishing the attractiveness of a market as an expansion destination. OGS companies in these countries also face geographical barriers such as vast distances between the location of targeted customers and the capital city or warehousing locations, which is exacerbated by poor road networks. These challenges or viability gaps can be segmented into three broad categories-company-specific constraints, market constraints and funding constraints-as summarized in the table below. Company-specific viability gaps refer to constraints which are an outcome of the OGS company's strategic and market positioning and includes issues with existing capital structures and lack of management bandwidth. Market constraints on the other hand are specific to each country and are the result of structural and economic deficiencies such as high poverty rates, low mobile penetration rates, different fiscal systems and political risk. Finally, funding constraints refer to lack of bespoke financial products that can be leveraged by OGS to expand into more challenging markets. These viability gaps are presented in the table.

Table: Viability gaps summary

¹⁵¹ 60 Decibels (2020) Why Off Grid Energy Matters. <https://60decibels.com/insights/why-off-grid-energy-matters/>

Viability Gap	Nature of constraint
Company-specific	These include: ▪ Lack of management capacity to expand. ▪ High existing leverage. ▪ Conflicting organisational priorities.
Market-specific	These include: • Low population density which increases distribution costs. • Affordability gaps. • Low mobile money penetration. • Low levels of consumer product awareness. • Few suitable distribution partners which necessitate costly in-house distribution infrastructure. • Political instability. • Inadequate understanding of credit risk in new markets.
Funding-specific	These include: • Lack of local currency facilities and elevated foreign exchange risks. • Concentration of funding towards a small proportion of OGS companies.

Hardest-to-Reach's investment approach enables capacity building of OGS enterprises by addressing key viability gaps, through patient capital that meets the needs of companies, alongside targeted technical assistance. The investment approach has the built-in flexibility to support all business models in the OGS space including vertically integrated companies and pure manufacturers who leverage distribution channel partnerships, both of whom are set to be the principal beneficiaries of the program. The impact-linked structure of the Market Expansion Facility loans rewards first-time energy access sales, improving the economic incentive to serve often more remote and lower-margin customers. Local African founders and business owners in particular face structural inequities in respect of these challenges, with greater difficulty to access challenge being a challenge for the majority. Just 6 percent of startups that secured more than \$1 million in Kenya were led by local founders in 2019, while a survey of 70 predominantly local last mile distributors revealed that three quarters experience access to finance as a key barrier to growth, with 54% specifically seeking debt finance.¹⁵²

Energy Access Customers

Forecasting models for achieving universal energy access project that off-grid solar could reach as many as 624 million people by 2030, but 'game changing' interventions are needed.¹⁵³ Traditional market-based solutions, including grid extension and cash sales of off-grid solar products, are not likely to be sufficient and are most often available in favorable markets or near urban areas where access is less of a barrier. 1.1 billion people will require access to reliable, modern, and affordable energy by 2030 under the SDG7 scenario presented by GOGLA, and 89% of the 464 million to-be newly connected OGS users live in

¹⁵² Global Distributors' Collective, [How crowdfunding can be an option to raise debt finance](#) (2021)

¹⁵³ GOGLA [Off-Grid Solar Market Trends Report](#) (2022)

nascent and challenging emerging markets. In addition, there are those for whom commercial access to energy access will not be a viable option. Some \$4.5 billion is needed just to close the affordability gap to be able to reach SDG7, and the pandemic, inflation and the energy crisis have set back global progress on reaching the lowest income populations with electricity access.

Payment plan models, such as PAYGo, can enable lower upfront investment and payment over time, making systems affordable for poor customers. PAYGo increases by 40% the number of people who can afford an entry-level (Tier I) solar product.¹⁵⁴ Even in the short term, the PAYGo model enables customers living in off-grid regions to transition away from fossil fuel-based energy sources, such as kerosene or diesel generators, to solar energy, with material GHG emissions avoidance impacts.

In addition, a climate justice component is tied to the needs of communities living in energy poverty. Low-income people in Hardest-to-Reach regions have contributed the least to climate change, yet they stand to suffer the most from it and benefit the least from global capital allocated for renewable energy and climate-friendly investment. Governments and climate investors have bypassed them, particularly in the most remote and fragile regions, prioritizing the countries that have higher greenhouse gas emissions and offer better returns on investment. Through Hardest-to-Reach, Acumen will invest where others won't, to bring remote, low-income people access to clean, renewable solar energy, helping them adapt to a changing climate and build the resilience needed to face climate and economic shocks.

D.5. Country ownership (max. 500 words, approximately 1 page)

¹⁵⁴ GOGLA [Off-Grid Solar Market Trends Report](#) (2020)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- Existing national climate strategy
- Existing GCF country programme
- Relevance to and alignment with existing policies such as Nationally Determined Contributions (NDCs), Nationally Appropriate Mitigation Actions (NAMAs), and National Adaptation Plans (NAPs)
- NDCs, NAMAs, and NAPs
- Capacity of Accredited Entities or Executing Entities to deliver
- Role of National Designated Authority
- Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups

According to UNEP (2017), the energy sector is considered a focus area in 45% of the INDCs in all total Non-Annex 1 countries¹⁵⁵. Based on the data from UNEP, the energy sector is the most important sector for sub-Saharan Africa. The prioritization of the energy sector in most NDCs is likely to reflect both its relevance as a large emission source and the availability of various cost-effective mitigation measures. All developing countries include mitigation measures within the energy sector in their NDCs.

In the early part of the investment period, Hardest-to-Reach will seek to work with OGS companies with different business models (i.e. vertically integrated and pure-play distributors) to enter or expand their reach in a range of countries, some of which rank among the bottom markets across indicators. The countries below provide an example. They represent an addressable market of approximately 147 million people without electricity, roughly 25% of the total unelectrified population of sub-Saharan Africa. The alignment of these countries to the mission of H2R is strong across existing policies, such as Nationally Determined Contributions (NDCs), national electrification strategies, and National Adaptation Plans (NAPs):

- **Benin:** In Benin, the national electrification rate is 33%. In 2020, the country adopted a new electricity code that provides for the opening of the electricity market to private investors, who can now produce, transport, distribute and market electrical energy in the country in the same way as the national utility.¹⁵⁶ Benin contributes 0.06% to global GHG emissions and its Government has committed to reducing emissions by 20% by 2030 compared to the reference scenario, accounting for 48.75Mt CO₂eq.¹⁵⁷ Energy and agriculture are recognized in its NDC as the key reduction areas. Benin's National Adaptation Plan (NAP)¹⁵⁸ outlines specific measures for improving the climate resilience of the energy sector. Certain of these measures, including the improvement in the use and availability of early warning systems, are based on higher levels of energy availability. The plan also mentions the adoption of quality standards for energy infrastructure and equipment, which H2R is aligned with.
- **Democratic Republic of Congo (DRC):** The DRC has a national electrification rate of 9%. The government aims to reach a target of 65% electrification by 2025. Currently, the DRC contributes 1.39% of global GHG emissions, and its government is committed to reducing emissions by 21% by 2030 compared to the business-as-usual scenario. The sectors targeted for this reduction are forestry, agriculture, energy, and waste. The DRC's National Adaptation Plan¹⁵⁹ aims to increase

the climate resilience of priority thematic sectors, including energy. Low-carbon energy development, in particular via solar and other renewable sources, is outlined in the plan.

- Guinea:** The national electrification rate in Guinea is 46%. Guinea aims to achieve universal energy access by 2030.¹⁶⁰ There is the potential for off-grid solutions in the country, and one of Guinea's strategic priorities is to promote decentralized energy solutions involving local communities and the private sector at the rural level. In its NDC, Guinea outlined an unconditional emissions reduction target of 9.7% by 2030 relative to the business-as-usual scenario, and an emissions target of 17% by 2030 that is conditional on external support. Guinea relies on hydropower for 58% of total electricity generation, meaning its energy sector is vulnerable to climate change.¹⁶¹ Reduced rainfall combined with warming is projected to reduce flows in the Konkouré River in western Guinea by 30–50 percent by 2100.¹⁶² This is a major concern, given that several dams along the river provide much of the country's hydropower. Water has therefore been highlighted as a key adaptation area in Guinea's NDC.
- Sierra Leone:** Sierra Leone has a national electrification rate of 26%. The government has a target to reach a 92% electrification rate by 2030 and recognized that 42% of the population would be best reached by off-grid stand-alone systems by 2023.¹⁶³ Sierra Leone contributes to 0.02% of global GHG emissions and the government has committed in its NDC to reduce emissions by 25% by 2050 compared to the projected business-as-usual level.¹⁶⁴ There is strong alignment between Sierra Leone's NAP and H2R.¹⁶⁵ It aims at universal access to energy through the promotion of renewables, in particular promoting off-grid stand-alone solar investments through technical advice and knowledge sharing. The government has a dedicated Off-grid Solar Energy Strategy (OGSES) with a strong focus on mobilizing financial institutions to create investment packages and counterpart funding.
- Uganda:** Uganda has a national electrification rate of 29%. Uganda has made huge progress in closing the energy access gap, as electrification rates grew by 3% per year between 2010 and 2019. Uganda has made electrification a national priority and aims to achieve a national electrification rate of above 98% by 2030 under the Sustainability Energy For All (SE4All) Initiative.¹⁶⁶ The Government estimates there are potentially 2,906,525 rural households that could be electrified through the provision of solar home systems and up to 130,000 households that could be electrified through mini-grids. In its NDC Uganda outlined an emissions reduction target of 22% by 2030 relative to the business-as-usual scenario, conditional on external support of 70% of the required financial resources.¹⁶⁷ Energy is one of its key sectors of focus for emissions reduction. The country's energy sector is also highly vulnerable to climate change as it is dominated by hydropower. Energy supply is at risk of being disrupted because of extreme weather events, such as reduced river flows resulting from droughts, and damage to power plants and transmission lines from floods and landslides.¹⁶⁸ Energy is therefore an adaptation priority under Uganda's NDC.
- Zambia:** The national electrification rate in Zambia is 37%. The country is targeting universal energy access by 2030, which means connecting over 11M people. The Government also aims to connect 31.6% of the population to SHS by 2030.¹⁶⁹ Zambia's NDC outlined an ambitious unconditional commitment to reduce GHGs by 25% compared with BAU by 2030, and a commitment to reduce emissions by 47% compared with BAU based on conditional support.¹⁷⁰ Energy is one of its key areas of emissions reductions. The energy sector is also vulnerable to climate change, as 84% of its electricity generation is from hydropower.¹⁷¹ Energy generation has

been adversely impacted by droughts and floods over the past two decades, and the country suffers from power shortages due to reduced river flows.¹⁷² Energy was therefore identified as a priority sector in its NDC.

The list of countries in which we operate will naturally expand throughout the life of the Market Support Facility and Market Expansion Facility, and believe this model is also applicable to the following Hardest-to-Reach markets in sub-Saharan Africa: Burundi, Burkina Faso, Chad, Guinea Bissau, Lesotho, Malawi, Mozambique, Niger, Somalia and Togo.

In the project development phase, the AE has developed the Hardest-to-Reach Stakeholder Engagement Plan . The AE has conducted significant desktop research on civil society organizations, and other relevant stakeholders including NDAs, indigenous groups, women, and other vulnerable groups.

As of April 28, 2023, we received 16 NOLs from Benin, Burkina Faso, Burundi, Chad, DRC, Guinea, Guinea Bissau, Lesotho, Malawi, Mozambique, Niger, Sierra Leone, Somalia, Togo, Uganda, Zambia. The AE is in advanced discussions with several other countries, including (without limitation) Liberia, Madagascar and Tanzania. These countries all expressed significant interest in the project, but haven't been able to complete their internal sign-off processes for various reasons. Nevertheless, we expect to receive NOLs from most, if not all, of these countries before the publication deadline. Additional countries may be added to the programme after initial Board approval, provided they submit an NOL and the addition is approved by the Board.

Additionally, the AE has begun hosting stakeholder town halls with our first town hall in Sierra Leone. We engaged 10 different groups in that meeting, received project feedback, and strengthened relationships in Sierra Leone. We anticipate hosting in-person and virtual stakeholder town halls for stakeholders in each H2R country.

The AE has developed improved capacity to engage relevant stakeholders as reflected by the H2R Stakeholder Engagement Plan, Acumen's Stakeholder Engagement Plan for GCF-Funded Projects, and Acumen's recent re-accreditation with GCF.

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.

Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation outcome the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.

Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.

Please specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.

Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

Hardest-to-Reach's Market Expansion Facility is expected to be funded by capital from a range of investors, including private sector financiers, Development Finance Institutions (DFIs), governments, foundations, and corporations. The blended structure with the support of GCF should mobilize different types of investors with different risk appetites at scale for the same purpose, providing first-time energy access for low income customers in Hardest-to-Reach geographies. Generally, the initiative's goal is to substantially preserve investor capital, not to produce significant returns. The H2R initiative anticipates providing funding to energy access companies with the intermediate goal of providing sufficient and appropriate liquidity to firms that are in the pioneer gap. Lack of returns of invested capital (including some losses) that enable the sector's preservation or development would be deemed a successful outcome.

Hardest-to-Reach has set a fundraising target of \$200 million for the overall MEF, inclusive of the potential GCF investment. The targeted raise is meant to be sufficient to sustain the operations and impact of a significant number of energy access companies. Government agencies, DFIs, and even commercial investors have indicated an interest in layering their capital with first-loss grant capital. In fact, GCF's \$55 million investment into the facility will likely allow Acumen to crowd-in \$145 million of co-financing from other investors, including those from the private sector. GCF is mobilizing nearly three times as much capital into an LDC-focused initiative that investors would otherwise find too risky.

The Hardest-to-Reach initiative is also very efficient and effective in terms of its impact. The \$250m raised for both the Market Support Facility and Market Expansion Facility is targeting the provision of electricity to 72.4 million people, many of whom live below the poverty line. The initiative intends to provide electricity to millions of people at a very low-cost, especially when compared with the material impact that electricity access will have on people's lives.

As a further marker of efficiency, we note that the total investment amount well exceeds the total H2R project size through reinvestments. Without GCF's participation, Hardest-to-Reach may be able to proceed with an initial raise, but it will likely be unable to meet more than a fraction of the demonstrated demand for liquidity. If GCF is not able to contribute funding, there is a high risk that many companies will not maintain operations, as many investor entities consider such firms too risky. The resulting lack of

access would mean that current customers are therefore at risk of switching back from solar to kerosene or diesel, and the huge potential of the sector to drive sustainable development and reduce emissions is curtailed. Electricity access for 72.4 million people will also be put at significant risk.

An important aspect of the efficiency and effectiveness of the investment is the establishment, strengthening, and scaling up of OGS companies to achieve wider impacts. H2R will monitor impacts across the 12 year lifespan of the initiative, but it is assumed that the impact of the H2R funds will continue after the funds themselves cease investing and the programme ends.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	There is an insufficient number of OGS deployments in nascent markets due to the current geographic concentration of off-grid energy investment in lower	<u>MEDIUM</u>	The potential target scenario would be a situation in which existing companies can scale up their business models and the number of OGS companies in the market increases to reach sufficient scale to provide clean energy to a large population of unelectrified people.	Scale-up of existing players and establishment of new players: H2R's innovative and impact-focused financing will incentivize and support companies to expand into new challenging markets, to build solid operational platforms, and to grow in such markets, thereby creating a new level of scale in the sector. By proving the OGS business model in untested

	risk, more proven markets. A cadre of under-capacitated last mile OGS distributors with latent potential remain overlooked, under-invested and require expertise to develop and scale the PAYG business model and to help mature their businesses. There is an insufficient volume of flexible and patient capital to plug the financing gap required for the sector to contribute to universal electrification through clean and affordable electricity. Of the total \$23.3 billion in funding required by the sector, up to			markets, companies can also crowd in additional capital to achieve scale in these markets. Increased scale of appropriate capital: H2R will raise and invest capital that is able to respond to the gaps left by traditional finance, which has under-invested in the sector. In particular, the \$250M raised and deployed through H2R will reflect the very different needs of ‘slow and stable’ versus ‘high-growth’ entities along their life cycles, backing both last mile distributors and established vertically integrated operators and tapping new sources of funding to scale-up the sector. In addition, by adopting a blended finance approach, investors with different risk appetites and returns expectations are accommodated.
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	\$16.5 billion will be required just in nascent and emerging markets. ¹⁷³			
Replicability	The conventional design of development interventions, with finite donor windows and one-off sales drives via implementation agencies, has resulted in the cliff-edge effect, where there is little sign of a	<u>MEDIUM.</u>	The potential target scenario would be a situation in which H2R demonstrates the viability of blended financing solutions for the off-grid solar sector, and demonstrates the effectiveness of targeted TA in building companies' capacity to absorb investment, thus creating a blueprint for other financiers. This approach could then be replicated by other implementing agencies, thereby building companies' capacity to	• Line-of-sight funding: The proposed approach has replicability in-built. Two separate financing facilities are proposed with different investment risk profiles, and investing with different capital blends to match the needs of the company and market, as scale is achieved. The Market Support Facility (MSF) will extend debt, equity and

¹⁷³ GOGLA [Off-Grid Solar Market Trends Report 2022:Outlook](#) (2022)

	self-sustaining ecosystem or a replicable blueprint for the private sector. Other investment interventions have not foregrounded capacity building, leaving companies unable to fully absorb the impact of private investment and deliver sustainable growth.		absorb future investments, avoiding the cliff-edge effect, and mobilizing investment to address the \$23.3 billion shortfall.	equity-like capital, and grants using a patient capital approach that is consistent with Acumen's core investment experience. While the MSF seeks to prove an approach in a new market with a range of capital, the Market Expansion Facility (MEF) deploys impact-linked debt that will help responsibly expand the reach of investees in the long-term. Successful deployments with one market expansion can be replicated with that company in another eligible geography or the playbook can be copied by another company or investor. • Demonstration effect: High-risk investments made via the Market Support Facility provide the right balance of incentives and support to help companies weather the first-mover disadvantage, educate customers, test market and product approaches, and
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				ultimately demonstrate the viability of these markets. The demonstration effect is key to unlocking other investments from broader sources of capital and sustaining the growth of these markets in the long term.
Sustainability	Many OGS companies, particularly last mile distributors, have insufficient capacity to grow sustainably in riskier H2R markets. Many have business models that do not adequately address risks (consumer protection, credit risk, ESG), which can lead to unsustainable growth and undermine the impact of the intervention.	<u>LOW</u>	The target scenario would be a situation in which off-grid solar companies develop inclusive and sustainable business models and are operational and capable of growing beyond the life of the H2R initiative.	• Policy alignment: Investments at the operational level will be directed in best alignment with long-term national policy, such as integrated energy planning and rural electrification strategies, to benefit from complementary public investments and prevent avoidable competing interests and inefficiencies, which are a threat to business sustainability. The broader enabling environment is also critical to reducing market barriers and ensuring the long-term viability and

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				attractiveness of the Hardest-to-Reach markets and companies. By ensuring policy alignment, investees may also benefit from lowered duties and taxes, and the sharing of public data among other benefits, to cut the cost and timeline of business set-up and improve the accuracy of customer targeting. • Technical assistance: The intervention is intended to provide investees with targeted accompaniment including: ○ risk management (including credit assessment and customer protection), ○ market studies, ○ hiring, ○ legal, and ○ ESG assessment.
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				• Technical assistance supports capacity building, acts as a risk management approach, and will enable companies to embed best practices that sustain responsible growth over the long term. • Consumer protection & ESG monitoring: An important objective of the Hardest-to-Reach approach is that the consumers of off-grid solar products, who are typically low-income and for whom a solar home system constitutes a major investment, are not exposed to an unreasonable risk of financial loss or indebtedness. Hardest-to-Reach investees are expected to adhere to the principles of the GOGLA Consumer Protection Code. These include making sure that consumers can afford to pay for the product and /or service without becoming overburdened and that
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				consumers' characteristics are taken into account in the price, payment structure and fees of the product and/or service. Acumen is working with a coalition of stakeholders to ensure strong monitoring in this area and a more holistic approach to consumer protection. Additionally, Hardest-to-Reach aspires to be responsible and sustainable stewards of capital, and will have a robust ESG strategy to minimize potential environmental and social harms to the communities and investors we serve. • Localization: Acumen understands that in order to develop a sustainable off-grid solar market in underserved geographies, it is crucial to boost the capacity of local distributors and companies and help establish local distribution networks. Hardest-
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				to-Reach aims to source and invest in eligible locally-owned entities, with the objective being to ensure that local businesses also receive the concessionary capital and the technical assistance needed to operate in high-risk markets.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ¹⁷⁴	

¹⁷⁴ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

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<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Portfolio data (as provided in Annual Reports), informed by company sales data 3rd party user surveys to understand hours of usage of OGS products by customers¹⁷⁵	0	3.4MtCO₂e	5.1MtCO₂e	- Discount for unused product is 10% - Typical mitigation over product lifespan (entry SHS) 0.43tCO₂eq - Program-level Av.annual emission reduction 0.52MtcO₂e - Emissions factor per mitigation methodology Final target breakdown: - MSF: 1.0MtCO₂e - MEF: 4.1MtCO₂e
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¹⁷⁵ H2R expects to work with a consumer impact surveying company to conduct consumer-level end user surveys to get beneficiary feedback and data for certain indicators

<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	Portfolio data (as provided in Annual Reports), informed by company sales data	0	80 MW (off-grid)	120 MW (off-grid)	Final target breakdown: - MSF: 6 MW - MEF: 114 MW
<u>Total project beneficiaries</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Portfolio data (as provided in Annual Reports), informed by company sales data	0	36,220,875 (17,029,439 female, 19,191,435 male)	54,331,312 (25,544,159 female, 28,787,153 male)	- Final target breakdown: - MSF: 8.8M lives reached - MEF: 45.6M lives reached
	<u>Direct beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data	0	6,629,523 (2,121,448 female, 4,508,076 male)	9,944,285 (3,182,171 female, 6,762,114 male)	- Direct female beneficiaries are based on the 60dB benchmark for sales to women (32%) Final target breakdown: - MSF: 830K lives reached

						- MEF: 9.1M lives reached
	<u>Indirect beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data	0	29,591,351 (14,907,992 female, 14,683,359 male)	44,387,027 (22,361,988 female, 22,025,039 male)	- Indirect female beneficiaries are based on the weighted percent of women in H2R countries (50.3%). Final target breakdown: - MSF: 7.9M lives reached - MEF: 36.5M lives reached
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time	0	36,220,875 (17,029,439 female, 19,191,435 male)	54,331,312 (25,544,159 female, 28,787,153 male)	- Final target breakdown: - MSF: 8.8M lives reached

		energy access customers				- MEF: 45.6M lives reached
	<u>Direct beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time energy access customers	0	6,629,523 (2,121,448 female, 4,508,076 male)	9,944,285 (3,182,171 female, 6,762,114 male)	- Direct female beneficiaries are based on the 60dB benchmark for sales to women (32%) Final target breakdown: - MSF: 830K lives reached - MEF: 9.1M lives reached
	<u>Indirect beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time	0	29,591,351 (14,907,992 female, 14,683,359 male)	44,387,027 (22,361,988 female, 22,025,039 male)	- Indirect female beneficiaries are based on the weighted percent of women in H2R countries (50.3%).

		energy access customers				Final target breakdown: - MSF: 7.9M lives reached - MEF: 36.5M lives reached
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	Portfolio data (as provided in Annual Reports), informed by company sales data. User surveys to determine beneficiaries with improved and/or new climate-resilient livelihood options	0	3,622,087 (1,702,944 female; 1,919,144 male)	5,433,131 (2,554,416 female, 2,878,715 male)	- Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options are people who will be using their product for income-generating activities - Final target breakdown: - MSF: 876K lives reached - MEF: 4.6M lives reached

	<u>Direct beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data. User surveys to determine beneficiaries with improved and/or new climate-resilient livelihood options	0	662,952 (212,145 female, 450,808 male)	994,429 (318,217 female, 676,211 male)	- Direct female beneficiaries are based on the 60dB benchmark for sales to women (32%) Final target breakdown: - MSF: 83K lives reached - MEF: 911K lives reached
	<u>Indirect beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data. User surveys to determine beneficiaries with improved and/or new climate-resilient livelihood options	0	2,959,135 (1,490,799 female, 1,468,336 male)	4,438,703 (2,236,199 female, 2,202,504 male)	- Indirect female beneficiaries are based on the weighted percent of women in H2R countries (50.3%). Final target breakdown: - MSF: 793K lives reached

						- MEF: 3.6M lives reached
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time energy access customers	0	36,220,875 (17,029,439 female, 19,191,435 male)	54,331,312 (25,544,159 female, 28,787,153 male)	Final target breakdown: - MSF: 8.8M lives reached - MEF: 45.6M lives reached
	<u>Direct beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time energy access customers	0	6,629,523 (2,121,448 female, 4,508,076 male)	9,944,285 (3,182,171 female, 6,762,114 male)	- Direct female beneficiaries are based on the 60dB benchmark for sales to women (32%) Final target breakdown: - MSF: 830K lives reached

						- MEF: 9.1M lives reached
	<u>Indirect beneficiaries</u>	Portfolio data (as provided in Annual Reports), informed by company sales data User surveys to determine first time energy access customers	0	29,591,351 (14,907,992 female, 14,683,359 male)	44,387,027 (22,361,988 female, 22,025,039 male)	- Indirect female beneficiaries are based on the weighted percent of women in H2R countries (50.3%). Final target breakdown: - MSF: 7.9M lives reached - MEF: 36.5M lives reached

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Hardest-to-Reach countries have limited investment directed at an energy access or a just transition. The H2R regions are inherently high risk and lack appropriate financial mechanisms or facilities to facilitate adequate investment and strengthen regulatory frameworks	<u>Low.</u>	H2R countries support the regulatory drive to develop an enabling environment for new investors and new businesses H2R countries support a just energy transition	The H2R initiative will work alongside NDAs and inter-regional initiatives to drive increased and <i>just</i> energy access. H2R intends to provide demonstration for governments that businesses can succeed in difficult environments	<u>Multi-countries</u>
<u>Core indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or</u>	Potential beneficiaries in rural areas of the H2R target countries have limited access to OGS technology like SHS. OGS penetration in H2R target countries is only 4.7%	<u>Low</u>	New market entrants and scaled-up activity by existing market players contributes to enhanced access to OGS technologies in rural areas. New, innovative business	The H2R program will deliver investment in new and existing companies to allow them to scale up sales both in terms of depth and reach, reaching new clients both in	<u>Multi-countries.</u>

<u>transfer and innovation.</u>	compared to 15% in Rwanda and 24% in Kenya.		models are developed. The penetration rate of OGS in H2R countries increases from its currently low level.	existing coverage areas and new geographies. This will increase the provision of OGS energy access for H2R communities.	
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	H2R areas lack sufficient success stories to install confidence in the markets Lack of adaptive capacity within national markets	<u>Low.</u>	New investors and businesses crowd into H2R markets such that capital market conditions are created Financial vehicles are developed to support the high-risk nature of H2R countries	The H2R initiative creates sufficient demonstration for new businesses to crowd into markets and to develop new markets	<u>Multi-countries</u>
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices.</u>	H2R target countries lack sufficient data and market understanding to meaningfully reduce investment risk, as no perceptible track record of OGS	<u>Low.</u>	<i>Information on market penetration and success/failure is provided, with a focus on providing statistics and data that can be used to undertake DD activities by investors,</i>	The H2R initiative creates sufficient knowledge for new businesses and investors to crowd into	<u>Multi-countries</u>

<u>methodologies and standards.</u>	markets in rural locations exist.		<i>and good practice for replication by OGS companies.</i>	markets and to develop new markets	
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E.5. Programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Advancement of off-grid solar markets in H2R countries	<i># of tipping points</i>	<i>Portfolio Company Data</i> <i>3rd Party Data Collection</i> <i>Third Party evaluation of data</i>	0	3	9	<i>Tipping points to be defined, related to investment capital, business activity, and other factors that are being determined by consultants with market research expertise.</i>

Outcome 2: Establishment and Growth of OGS companies in sustainable and gender- and poverty-inclusive ways	<i># of OGS manufacturers reporting sales in H2R countries</i>	<i>GOGLA data</i>	116	126	136	<i>GOGLA collects sales data from affiliate¹⁷⁶ OGS manufacturers semi-annually.</i>
	<i># of non-manufacturing OGS distributors reporting sales in H2R countries</i>	<i>GOGLA data</i>	28	34	41	<i>GOGLA collects sales data from affiliate non-manufacturing OGS distributors semi-annually.</i>
Outcome 3: OGS customers (and their households) benefit from enhanced energy security	<i># of people (and their households) with new or existing access to an OGS product</i>	<i>Portfolio reviews</i>	0	48,294,500 people (9,658,900 households)	72,441,750 people (14,488,350 households)	Average household size in H2R countries is 5. Final target breakdown: MSF: 11.7M people (2.3M households) MEF: 60.8M people (12.2M households)

¹⁷⁶ Affiliates are companies connected to the partner organizations involved in the reporting process. Companies include GOGLA members, companies selling products that meet VeraSol Quality Standards, and appliance companies that participated in the Global LEAP Awards or are engaging with the Low Energy Inclusive Appliances (LEIA) program.

Outcome 4: OGS customers (and their households) are able to more securely access information including extreme weather event and health warnings, health information, and market data for e.g. agricultural produce	<i># of people who will use OGS products for early warning systems and access to information</i>	<i>Portfolio reviews</i> <i>User surveys for people reporting use of OGS products for early warning systems and access to information</i>	0	36,220,875 people (7,244,175 households)	54,331,312 people (10,866,262 households)	75% of beneficiaries will be accessing an OGS product for the first time All first-time energy users benefit from more secure access to critical information on e.g. extreme weather events, agriculture, market prices, disaster risks, public health. Average household size in H2R countries is 5. Final target breakdown: MSF: 8.8M people (1.8M households) MEF: 45.5M people (9.1M households)
Outcome 5: OGS customers (and their households)	N/A	N/A	N/A	N/A	N/A	N/A

benefit from increased and/or diversified income sources						
Outcome 6: GHG emission avoidance from switching to clean and climate resilient energy sources	N/A	N/A	N/A	N/A	N/A	N/A
Output 1: H2R established and \$250M fundraised	Amount fundraised	Funding agreements	0	\$250M	\$250M	
Output 2: Developed gov't and non-gov't partnerships to enable last-mile energy access for the poorest; (i.e., co-investments, knowledge partnerships, govt partnerships, and industry partnerships)	# of annual government engagements per year	NDA annual reports and town halls	0	16	16	
	# of H2R countries with REACH Partnership Activities	Annual reports	0	3	6	E.g. countries where policies developed, stakeholder engagements, convenings
	# of thought pieces produced	Portfolio reportsIndustry reports / Government data	0	3	10	Thought pieces include white papers, blog posts, or insight studies shared with the wider community

Output 3.1: Increased support for new and fledgling OGS co's in H2R countries, with alignment to serve low-income segments						
	Additional sales by newly created portfolio companies	Market reports	0	370K	555K	
Output 3.2: Increased scale of OGS co's in H2R countries, with demonstrated alignment to serve low-income segments						
	Poverty focus (%<\$3.65pd)	User surveys	0	50%	60%	Country PPI benchmarks are up to date and conversions to new poverty level of \$3.65, where necessary, follow best practice

Output 4.1: Strengthened OGS business performance, with demonstrated commercial viability, sustainability, gender- and poverty-inclusivity, and investability						
	Number additional OGS products sold by investee companies (MSF)	Portfolio reports	0	737,473	1,106,210	Targets set using the fund model's MSF investments
	% of portfolio Local entrepreneurs	Portfolio reports	0	25%	33%	Local entrepreneur means an individual with either a passport or direct heritage to the country where the investment/enterprise is headquartered
	% of portfolio 2X gender compliant	Portfolio reports	0	30%	30%	% of companies that comply with one 2X metric is considered 2X compliant
	# of women reached through OGS awareness outreach	SMS 'messages to potential customers	0	500	1000	Allocation to specific TA intervention
	# of women trained	Portfolio report	0	275	275	Assumed that training will be delivered through the TA, following the H2R Gender Action Plan .
Output 4.2 Strengthened OGS business performance,						

with proven commercial track record	Number additional OGS products sold by investee companies (MEF)	Portfolio reports	0	8,101,891	12,152,837	Targets set using the fund model's MEF investments
Output 5: Increased provision of climate-resilient clean energy via 13.3M new sales of affordable, clean, resilience-enhancing decentralized energy products	# of sales of OGS products	Portfolio reviews	0	8,839,365	13,259,047	Annual review company reporting company sales data Final target breakdown: MSF: 1.1M sales MEF: 12.2M sales
	% reporting improved quality of life	User surveys	0	60%	70%	Improved quality of life is a proxy for a reduction of vulnerability and increased dignity; will track across both MSF and MEF with similar goals for both.
Output 6: Displacement of harmful and emissions-intensive lighting sources and 120MW newly installed renewable energy capacity	Increased MW of renewable energy installed capacity	Portfolio data (as provided in Annual Reports), informed by company sales data	0	80 MW (off-grid)	120 MW (off-grid)	Final target breakdown: MSF: 6 MW MEF: 114 MW
	% of SHS that displaced traditional energy source (incl wood, kerosene etc)	User survey/Insights piece	0	33%	33%	60dB found that 38% of SHS and lantern customers were using kerosene before their purchase, of which

						87% totally eliminate usage. This comes to a total displacement rate of 33%.¹⁷⁷ Insights study is comparable across regions and can be extrapolated across portfolio; will track across both MSF and MEF with similar goals for both.
Project/programme co-benefit indicators						
Co-benefit 1: New clean energy jobs including for women	Jobs created	Portfolio reports	0	9,281 (3,094 female, 6,187 male)	13,922 (4,641 female, 9,281 male)	Assume 1.05 full time jobs are created for every additional 1,000¹⁷⁸ units sold by companies. Jobs created include sales managers, product technicians and administrative staff. Additional jobs from TAF
Co-benefit 2: Improved health outcomes via lowered indoor air pollution	% reporting health improvements	User surveys	0	48%	48%	Reduced indoor air pollution from burning wood, dung, and kerosene leads to better respiratory health, particularly for women and children who spend more time at home.

¹⁷⁷ 60 Decibels Why Off-Grid Energy Matters (2020)

¹⁷⁸ Based on actual portfolio company data

						60dB found that 48% of OGS customers reported health improvements ¹⁷⁹
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E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1: <u>Acumen engages stakeholders (partners, donors, govts) towards development of H2R markets, by fundraising & establishing H2R MSF (\$50M)</u>	Acumen sets up the Market Support Facility (MSF). The strategy, naming, branding and legal arrangement of the facility are successfully developed and managed, ready for the implementation stage. Acumen raises \$50M for MSF across several donors, including a \$10M repayable grant from GCF. Acumen engages relevant stakeholders (partners, donors, governments) throughout.		
		Sub-Activity 1.1.1	

¹⁷⁹ 60 Decibels Why Off-Grid Energy Matters (2020)

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		H2R strategizes on the investment and impact thesis as well as investor mandate required in development of Market Support Facility (MSF)	The strategy of Market Support Facility is developed
		Sub-Activity 1.1.2 H2R names and brands the Market Support Facility	The naming and branding of the facility is successfully developed
		Sub-Activity 1.1.3 H2R identifies and meets with aligned HNW donors and philanthropic institutional donors	\$50M for MSF is raised across several donors, including a \$10M repayable grant from GCF
		Sub-Activity 1.1.4	The legal arrangement of the facility is successfully developed

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		H2R internal legal counsel reviews the appropriate legal structure of the MSF and seeks to develop the appropriate legal structure for the Market Support Facility	
		Sub-Activity 1.1.5 H2R engages with government entities and other relevant public and private sector organizations in H2R countries	16 NOLs obtained from H2R countries. Workshops, town halls and other engagements take place with government ministries, public and private sector organizations, and affected communities
Activity 1.2: <u>Partnerships built with stakeholders to improve coordination and commercial infrastructure, address market barriers and share knowledge</u>	H2R develops partnerships and engages with stakeholders (government, investors, opcos, capacity building programs) to improve coordination, address market barriers and share knowledge in H2R markets.		Across the range of H2R partnerships, coordination helps address market barriers to investment. Coordinated efforts help reduce inefficient use of resources, create opportunities to leverage stakeholder experiences and competencies, and minimize distortive risks. H2R's frontier investments and coordination efforts serve to derisk and prove

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			the viability of OGS investments in these underserved markets.
		Sub-Activity 1.2.1 Acumen seeks to support the development of high level REACH Partnership between OGS leaders, practitioners, and stakeholders committed to sharing knowledge and insights.	The partnership structure, goals, and coordination mechanism are agreed on by all parties
		Sub-Activity 1.2.2 We intend to engage in REACH Partnership convenings at the country-level coordination committees to facilitate knowledge sharing and efficiently use resources.	Country-level stakeholders participate in regular convenings and are given a comprehensive and regularly-updated overview of national energy access activities. H2R activities are aligned with country-level policies and priorities.
		Sub-Activity 1.2.3	The insights and impact of the H2R initiative are widely shared to interested audiences

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		H2R seeks to share results and lessons learned, both from deployment of capital and from larger coalition-building efforts	
Activity 1.3: <u>Investment of MSF capital to demonstrate OGS within H2R markets and efficacy of impact-linked capital approach (and additional re-investment)</u>	H2R invests capital into OGS companies through a variety of investment structures, depending on the needs of the company and the situation in the market.		
	MSF Joint Venture investing, in the appropriate markets with large off-grid populations	Sub-Activity 1.3.1 H2R intends to identify pipeline, perform due diligence on potential investments, create terms, develop JVs, and disburse funds to JV formed with a vertically integrated OGS company.	Funding disbursed to joint ventures
	MSF convertible note, equity investments or recoverable grants in last mile distributors which may be supported by an advisor with	Sub-Activity 1.3.2 H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and	Equity/equity-like capital to last mile distributors

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	local equity market knowledge, where appropriate	disburse equity/equity-like capital (including but not limited to convertible notes and recoverable grants) to last mile distributors	
	MSF distributor finance via manufacturers to distribute solar home systems and OGS products to last mile distributors. MSF lending to early-stage last mile distributors which may be supported by an advisor with local lending market knowledge, where appropriate	Sub-Activity 1.3.3 H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and disburse loans to last mile distributors via manufacturers or directly.	loans for distributor finance via manufacturers and loans to last mile distributors
		Sub-Activity 1.3.4 H2R collects returned capital during the investment period and	Returned principal and interest payments are collected and reinvested in OGS companies.

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		reinvests repaid capital into OGS companies	
		Sub-Activity 1.3.5 H2R collects repayments and exits its equity-like investments after the investment period and disburses repaid capital to funders	Principal and interest payments are collected and remitted to funders.
Activity 1.4: <u>Monitoring of investments for portfolio management, impact measurement and tracking, and generating insights</u>	H2R monitors MSF investments for portfolio management, impact measurement and tracking, and generating insights		
		Sub-Activity 1.4.1 H2R monitors growth and impact progress of companies receiving Market Support capital	Reporting mechanisms are developed, and data on growth, impact, and other key metrics is captured in a centralized database for monitoring and analysis. Qualitative inquiries are performed at key intervals into the drivers and barriers for MSF capital to build local management and distribution capacity. At agreed-upon intervals,

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			data are summarized and analyzed in reports to funders.
		Sub-Activity 1.4.2 For companies eligible to receive growth capital, H2R measures baseline impact scenarios	Baseline studies are conducted on potential investee sales, customer demographics, and climate resilience
		Sub-Activity 1.4.3 H2R measures impact of borrower operations and calculates final repayment for impact-indexed loans	Companies report on sales and operational data. Endline surveys are conducted. Data is synthesized and analyzed to determine final repayment structure through transparent impact formulae.
		Sub-Activity 1.4.4 H2R submits final impact reports to all partners upon disbursement of repaid capital to funders	Summary reports are submitted to all funders.

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		Sub-Activity 1.4.5 H2R intends to monitor loan conditions and covenants, gender, ESG, and climate and impact work.	Ongoing company monitoring.
		Sub-Activity 1.4.6 H2R intends to develop an annual MSF report to MSF donors. This report will include impact, financial, and other relevant data.	Annual MSF donor reports.
		Sub-Activity 1.4.7 H2R intends to develop an annual MSF community report to be shared with stakeholders and will hold annual stakeholder town halls.	Annual community report and annual town halls
		Sub-Activity 1.4.8	At the end of the project, insights and impact of various interventions

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		H2R intends to communicate findings from capacity-building work to larger energy access ecosystem	are collected and shared with appropriate audiences.
Activity 1.5: <u>TA support for H2R investees, particularly in enterprise development, consumer protection, gender and environment/ESG performance</u>	H2R team diligences companies to understand risk and opportunities for capacity building. Technical assistance action plans outline the capacity building roadmap and help ensure alignment between investor and investee on milestones and KPIs. H2R works with a range of partners and technical experts to support delivery of the action plans.		The real and perceived risks of entering new markets are reduced, and the capacity of companies is built in key areas such as management, credit risk analysis and market research/strategy. Local distributors have increased capacity to become reliable partners to suppliers expanding in the region.
		Sub-Activity 1.5.1 H2R intends to identify a group of appropriate and committed technical experts to advise H2R companies. Establishes formal	A roster of qualified and committed technical experts is developed, and partnerships established with key firms.

		and/or informal partnerships, as appropriate	
		Sub-Activity 1.5.2 H2R intends to assess H2R company needs prior to investment, and secure commitment for improvement in key areas as part of conditions precedent	A 'report card' for every H2R investment is developed and stored, then used to design appropriate support.
		Sub-Activity 1.5.3 H2R works with experts to support companies in the areas of consumer protection, enterprise development, gender and ESG through consultation, webinars, and other resources.	investee companies supported with technical assistance as needed
		Sub-Activity 1.5.4	Market activation research is carried out and a blueprint is created. As part of this, metrics are defined that activate and expand

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		H2R partners with market development and off-grid experts to research market development in H2R markets, and determine the market growth that can be attributed to H2R.	markets and measure H2R's contribution to the market activation indicators. Midline and Endline Reports on market activation are carried out.
Activity 2.1: <u>Acumen engages stakeholders (partners, investors, govts), towards scaling H2R markets by fundraising & establishing H2R MEF (\$200M)</u>	Acumen sets up the Market Expansion Facility. The strategy, naming, branding and legal arrangement of the facility are successfully developed and managed, ready for the implementation stage. Acumen raises \$200M for MEF across commercial investors and development finance institutions, anchored by a \$50M equity tranche from GCF. Acumen engages relevant stakeholders (partners, investors, governments) throughout.		
		Sub-Activity 2.1.1	The strategy of the Market Expansion Facility is developed

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		H2R strategizes on investment and impact thesis, as well as investor mandate, for the Market Expansion Fund (MEF)	
		Sub-Activity 2.1.2 H2R establishes debt fund manager.	H2R debt fund manager is established.
		Sub-Activity 2.1.3 H2R names and brands the MEF	The naming and branding of the facility is successfully developed
		Sub-Activity 2.1.4 Acumen internal and external legal counsel review the appropriate legal structure of the MEF	The legal arrangement of the facility is successfully developed
		Sub-Activity 2.1.5	The financial model and prospectus are developed and shared with interested parties

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		H2R creates an indicative financial model and prospectus for review by potential partners	
		Sub-Activity 2.1.6 H2R identifies and engages commercial investors and subordinate capital providers	\$200M in blended capital for MEF is raised across several investors and concessionary capital providers, including a \$50M equity tranche from GCF and \$5M grant tranche. We intend to call from GCF an amount equal to 27.8% of the total equity commitments raised in each closing up to \$50M total. In one lump sum, \$5M of MEF grant will be requested in the first call (as will \$10M recoverable grant for the MSF).
Activity 2.3: <u>Investment of MEF capital to scale OGS through impact-linked approach within H2R markets (and additional re-investment)</u>	H2R invests impact-indexed debt into OGS companies and invests in accounts receivables facilities.		

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	MEF extends impact-indexed loans to OGS companies in target countries.	Sub-Activity 2.3.1 H2R intends to identify pipeline, perform due diligence on potential investments, create terms, and extend impact-indexed loans to OGS companies.	impact-indexed loans extended to OGS companies
	MEF invests in accounts receivable facilities.	Sub-Activity 2.3.2 H2R intends to invest via accounts receivables financing.	H2R invests in receivables financing facilities
		Sub-Activity 2.3.3 H2R intends to incentivize borrowers to serve Hardest-to-Reach customers	Performance grants allocated to investee companies based on achievement of impact KPI targets

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		Sub-Activity 2.3.4 H2R collects repayments during the investment period and reinvests repaid capital into OGS companies and accounts receivables facilities	Returned principal and interest payments are collected and reinvested in OGS companies.
		Sub-Activity 2.3.5 H2R collects repayments after the investment period and disburses repaid capital to funders	Principal and interest payments are collected and remitted to investors.
Activity 2.4: <u>Monitoring of investments for portfolio management, determining performance against impact-linked KPIs for impact-indexed loans, and review of business performance</u>	H2R monitors investments for portfolio management, determining performance against impact-linked KPIs for impact-indexed loans, and review of business performance.		
		Sub-Activity 2.4.1	Reporting mechanisms are developed, and data on growth, impact, and other key metrics is

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		H2R monitors growth and impact progress of companies receiving Market Expansion capital	captured in a centralized database for monitoring and analysis. Qualitative inquiries are performed at key intervals into the drivers and barriers for MEF capital to build local management and distribution capacity. At agreed-upon intervals, data are summarized and analyzed in reports to funders.
		Sub-Activity 2.4.2 H2R measures baseline impact scenario	Baseline studies are conducted on potential investee sales, customer demographics, and climate resilience.
		Sub-Activity 2.4.3 H2R measures impact of borrower operations and calculates final repayment for impact-indexed loans	Companies report on sales and operational data. Endline surveys are conducted. Data are synthesized and analyzed to determine final repayment structure through transparent impact formulae.
		Sub-Activity 2.4.4	Summary reports are submitted to all funders.

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		H2R submits final impact reports to all partners upon disbursement of repaid capital to funders	
		Sub-Activity 2.4.5 H2R monitors loan conditions and covenants, gender, ESG, and climate and impact work.	Ongoing company monitoring.
		Sub-Activity 2.4.6 H2R will develop a fulsome annual MEF report to MEF investors. This report will include impact, financial, and other relevant data.	Annual investor reports.
		Sub-Activity 2.4.7 H2R will develop an annual MEF community report to be shared with stakeholders and will hold annual stakeholder town halls.	Annual community reports and stakeholder town halls

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		Sub-Activity 2.4.8 H2R communicates findings from capacity-building work to larger energy access ecosystem	At the end of the project, insights and impact of various interventions are collected and shared with appropriate audiences.
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements (e.g. annual performance reports) laid out in Accreditation Master Agreement (AMA), please give a summary of the project/programme specific arrangements for monitoring, reporting and evaluation including a description of the monitoring and reporting system that will be used to assess the climate results of the proposed activity. Please also summarize the types of interim and final evaluations. Describe Accredited Entity (AE) project reporting relationships, including to the National Designated Authority (NDA)/Focal Point and between AE and Executing Entity (EE) as relevant, identifying reporting obligations from the EE to the AE. This should relate to the frequency of reporting on project indicators, implementation challenges and financial status. Please note that interim and final evaluations are expected to embed an assessment of project/programme's contributions to a paradigm shift and enabling environment using a three-point scale rating. Refer to the guidance note for the summary requirements and factor in additional evaluation /assessment activities under this section accordingly.

H2R expects to engage in thoughtful and extensive monitoring, reporting, and evaluation of the Project.

Monitoring

The H2R team intends to monitor the H2R Project in accordance with the project Funded Activity Agreement and the Accreditation Master Agreement. The Management Team has developed a curated list of Key Performance Indicators to monitor throughout the Project to ensure it is accomplishing the outcomes shared in the Funding Proposal.

The Project team expects to assign a relationship manager to each Investee to ensure that Portfolio Companies are being fully and appropriately monitored throughout the duration of the loan. The RM seeks to monitor the Portfolio Company for compliance with the conditions and covenants

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of the investment obligation, company-level KPIs impacting the performance of the Project, climate results and impact performance. The Investees should be expected to monitor legally binding conditions and covenants of the term sheet, company-level KPIs, climate results, and the impact performance of the investment. H2R expects the ongoing company-level monitoring to inform adjustments to project design, implementation, and potential changes to monitoring and evaluation tools.

The Project team intends to also utilize several other sources for ongoing Project monitoring. First, the team expects to work with a consumer impact surveying company to conduct consumer-level surveys to get beneficiary feedback and data on a number of topics. H2R hopes to learn about impact, poverty, consumer protection, gender, ESG, climate resilience and adaptation, and other topics. This data is expected to inform Project implementation and strategy.

Finally, the H2R team expects to monitor the Project through extensive stakeholder engagement. The H2R team wants to share Project results with stakeholders to get their feedback on an ongoing basis to ensure that their voice informs our monitoring practice and understanding of the Project impact.

Reporting

The Project expects to report financial, impact, and climate results to a number of stakeholders on an annual basis. Investors are expected to receive reporting on a quarterly and annual basis detailing the Project activities, financial statements, climate results, and impact data. GCF will also receive reporting via the APR. Reporting requirements are expected to be established in legal agreements between donors, investors, and the project team. H2R expects to provide investors with an annual MEF report and an annual impact and ESG report. MEF investors will also receive an annual audited account of the MEF within 120 calendar days of the end of the relevant accounting period. Donors will receive an annual MSF report that includes impact and ESG reporting.

H2R expects to engage NDAs on an annual basis. The team expects to share a Community Report with NDAs, and seeks to engage with them on an annual basis to review project activities and seek feedback. NDA feedback will inform project activities and potentially strategy.

H2R also expects to share an annual Community Report with relevant stakeholders including industry associations, NGOs, Civil Society organizations, academia, and other institutions representing underserved communities.

Where H2R MSF works with third party advisors with local market knowledge on deals they will follow the Project-level reporting requirements and have the proper arrangements to collect and verify Portfolio Company reporting, financial, ESG, gender, and impact data, and climate results. Third party consultants supporting technical assistance may also have a reporting expectation.

Portfolio Company reporting requirements are expected to be detailed in the term sheet and shared through the diligence process. Portfolio companies are expected to report financial and impact data on a quarterly basis. The H2R team expects to send Investees reporting templates that need to be filled out and sent with evidence to H2R prior to the annual reporting deadline. The team intends to then verify and compile company-level reporting to inform Project-level reporting.

Portfolio companies are expected to report on financial performance on a quarterly and annual basis. H2R will pay special attention to financial metrics like cash flows, cash balance, working capital needs, sales data, collection rate, write-off ratio, and PAR rates. Portfolio companies are also expected to report on ESG Action Plans, Gender Action Plans, consumer protection, and other relevant data. H2R companies are also expected to report on impact KPIs related to sales, sex-disaggregated beneficiaries, emission reductions, and other relevant data.

Evaluation

H2R expects to implement independent interim and final evaluations aligned with the GCF Monitoring and Evaluation Framework and the Evaluation Policy.

Acumen's evaluation team intends to hire an independent evaluator to implement each evaluation to ensure our commitment to evaluation's best practices: impartial, fair, and objective, robustness and credibility, and relevance and participation. The team will develop an evaluation project scope ahead of seeking a partner. Acumen's evaluation team is expected to evaluate the scope of the project. Acumen will use the AE fee to pay for the evaluation. The team expects to use a mixed methodology approach to learn about the project. H2R intends to disclose the interim and final independent evaluations to GCF.

Additionally, the team is exploring the development of long term evaluation projects for H2R. As this project has a multitude of opportunities for developing unique insights to share with the broader climate and impact investing community, H2R is developing a strategy to determine what types of long term evaluation projects are feasible, appropriate, and important to conduct for the betterment of these markets, to demonstrate the paradigm shift of the project, to ensure the replicability of H2R, and to reach an interested audience to scale the project impact.

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F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1: Execution

Category	Probability	Impact
<u>Technical and Operational</u>	<u>High</u>	<u>Medium</u>

Description

Energy access companies that are newly entering a volatile market, may not be able to execute on first-time energy access goals, and may struggle to generate high volumes of quality sales. Last mile distributors may face difficulties with expanding their operations in H2R markets with immature business operations, and external market, political, currency, and macro level risks.

Additionally, Acumen is expanding into new markets that the firm has not invested in to date. Inexperience in the market could expose the project to unexpected political, legal, and other forms of risk. Acumen also may have some gaps in navigating the market landscape in H2R markets leading to longer due diligence or missed investment opportunities.

Mitigation Measure(s)

The chief mitigant to execution risk for Investees will be to ensure that companies have the funds and accompaniment they need to build capacity. The team engaged with many off-grid solar companies about the right investment instruments, tenors, and loan terms needed to incentivize companies into entering or scaling in H2R markets. Following the information gathering phase, creating realistic, data-based projections will be essential, as will combining funding with needed resources such as Technical Assistance (TA) and market research. The Market Support Facility, with its focus on building capacity and stable operational foundations, ahead of scale-up, offers calibration to the ambition of the project, demonstrating an acknowledgement that any expectation of immediately high sales would be unrealistic and hazardous. That being said, this project is ambitious and high risk (and in our view, necessary for the world's poor).

Acumen engaged in extensive market research, conducting market studies and reviewing the socio-political climate for each potential H2R country. The team also conducted studies on the climate and gender context of H2R countries.

Additionally, Acumen has experience investing in new markets and has built an accomplished investing team with operational expertise and experience in several of the H2R markets. Moreover, Acumen has successfully fundraised for and invested in funds operating in H2R markets. Finally, the Acumen team has developed an extensive, appropriately scoped, and modular due diligence process that incorporates market research, Acumen experience, and best practices.

Selected Risk Factor 2: Macro

Category	Probability	Impact
<u>Other</u>	<u>High</u>	<u>High</u>

Description

Hardest-to-Reach markets often face significant security risks, political volatility, regulatory inconsistency and legislative unpredictability. These risks do not exist in isolation, some even have the potential to be correlated across a given region. A major crisis or series of interlocking risk events could derail progress on energy access.

Moreover, global trends have been very impactful to volatile markets and unstable political environments. Inflation, a potential global recession, and the ongoing effects of the Covid-19 pandemic have created even more instability in H2R markets.

Macro and global risks have had significant impacts on potential offgrid solar customers. Rising inflation has increased the costs of necessary goods and services making other expenditures less affordable for people in poverty. Covid-19 created lock downs and supply chain issues that contributed to unemployment in some markets and harmful shipping delays for OGS companies.

Mitigation Measure(s)

Targeted technical assistance may support legal evaluation to understand risks. Political risk insurance can be a valuable tool to hedge some of these risks. The team has conducted multiple political risk studies on H2R markets. The team intends to create political risk assessments for each market. Macro risks for each investment will be summarized in investment memos and will be shared with the Investment Committee along with strategies to address the risk aligned to the context of the specific investment. It's also worth mentioning that H2R will have a broad portfolio, diversifying risk across a range of geographies.

Selected Risk Factor 3: Mission Drift

Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>

Description		
Energy companies that feel under pressure to repay loans and show growth may opt to sell more to higher-income segments, negating some or all of the company's proposed additional impact.		
Mitigation Measure(s)		
The use of impact-linked grants and loans, which will incorporate a measure of the depth of poverty of a customer base, will be used to incentivize companies to target lower-income segments. The impact-indexed loans will have KPIs to ensure customer retention is achieved responsibly and sustainably. Additionally, Acumen will conduct thorough due diligence on each Hardest-to-Reach investment to ensure that companies are capable of properly utilizing H2R funds. The IC will be instructed on ensuring that the mission is a core component of each investment. Term sheets will stipulate the use of funds for each loan or equity investment.		
Selected Risk Factor 4: Distortion		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Low</u>
Description		
The act of choosing a subset of companies to fund, some of which may operate in existing sub-Saharan African markets, could distort the future market for off-grid solar, and crowd out less well-connected companies. Relatedly, the volume of concessionary debt going towards energy companies could distort the overall energy access debt market.		
Mitigation Measure(s)		
While there are existing companies operating in Hardest-to-Reach markets, there are not many of them, and few if any that have reached a significant scale serving the most challenging reaches of the country. By incentivizing companies to reach first-time customers, H2R is creating and nurturing new markets. within weak-grid and off-grid areas, thus the risk of distorting an existing market remains low.. As <u>evidence from ODI suggests</u> , the risk of crowding out other possible (private) sources of finance is less relevant in low-income countries where financial markets are insufficiently developed, and investors are unlikely to have access to international capital markets at affordable rates in the absence of the concessional intervention. In addition, the concessionality provided to the companies is linked to the impacts achieved in terms of first time energy access and poverty level of customers, which will likely ensure that the benefits of concessionality pass down to the low income, first time energy access customers (in terms of increased supply of entry-level products) and minimize a potential for market distortion.		
Selected Risk Factor 5: Continuity		

Category	Probability	Impact
<u>Operational and Technical</u>	<u>Medium</u>	<u>High</u>
Description		
When the program ends, the end of concessionary financing could mean that firms find Hardest-to-Reach markets unsustainable to operate in or cease serving low-income segments within those markets.		
Mitigation Measure(s)		
Hardest-to-Reach is an integrated life-cycle financing to accompany companies to expand into new, challenging markets and to support their sustainable growth over the longer term through supportive patient capital and line of sight working capital financing which is linked to impact outcomes, which will contribute to building robust markets in Hardest-to-Reach geographies. Over time, the success of H2R investees can derisk these previously untested markets while also demonstrating the success pathway - including through the use of blended finance. Combined this should make it easier for these companies to access financing. In addition, H2R seeks to address the viability gaps faced by companies in terms of capacity, strategy and enabling environment. Over the life of the two vehicles, two pools of technical assistance capital will support ecosystem development to ensure that, through training, local capacity building, and stakeholder coordination efforts - among other things - the sustainability of the intervention is enhanced and impacts beyond the life of the fund continue in a self-sustaining way.		
Selected Risk Factor 6: Due Diligence		
Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>
Description		
Operating in certain countries and through certain on-lending structures (particularly onto smaller distributors) poses a DD challenge, Where good market information is scarce, this increases the complexity of conducting satisfactory due diligence and thus the risk of financing operationally, financially and/or reputationally unsuitable companies.		
Mitigation Measure(s)		
In addition to careful due diligence and investment structuring, in-depth market studies inform the H2R strategy at the country-level, as well as uncover macro risks. In the case of distributor finance via manufacturers, the investment team will work closely with the manufacturer borrowers/on-lenders to screen-out distributors that are not suitable for this facility for financial or ESG reasons.		
Selected Risk Factor 7: Currency		

Category	Probability	Impact
<u>Forex</u>	<u>Medium</u>	<u>Medium</u>
Description		
Currency or Foreign Exchange risk (FX Risk) is a key inhibitor of capital inflows and deal progress in the off-grid energy sector, particularly in least developed countries such as the ones that Hardest-to-Reach aims to impact. Local companies face challenges when receiving loans in 'hard currency' while their operations and sales are managed with local currency. Significant depreciation in local currency value could make it very challenging for distributors to service their H2R debt.		
Mitigation Measure(s)		
The due diligence process will incorporate an assessment of country risk, including foreign exchange and hard currency liquidity risk. Acumen is exploring the feasibility of local currency lending alongside hedging partners.		
Selected Risk Factor 8: Forced Labor		
Category	Probability	Impact
<u>Reputational</u>	<u>Low</u>	<u>Medium</u>
Description		
Raw materials for solar products and some solar products may come from geographies currently under international scrutiny for forced labor practices. Several solar product manufacturers also operate in areas under international scrutiny for forced labor practices. <u>It is also estimated</u> that there are 3.7 million forced laborers in sub-Saharan Africa, largely in the informal economy (particularly in informal, artisanal cobalt mining industries within several H2R target countries). There is therefore some risk of forced labor in the value chain.		
Mitigation Measure(s)		
Acumen will include forced labor in its ESMS and will integrate it into its company due diligence, including requesting disclosure of suppliers from potential portfolio companies, and conducting ongoing monitoring. Acumen will mitigate forced labor through the use of screening tools, in its deal terms and as part of its monitoring and evaluation process.		
Selected Risk Factor 8: Governance		
Category	Probability	Impact

<u>Governance</u>	<u>Medium</u>	<u>Low</u>
Description		
Some companies that receive H2R investment are early-stage startups, and may have inefficient management or governance structure which may result in ineffective decision making and the funds allowed to them may not be utilized properly. This risk could materialize due to separation of key-executives or change in group share-holding structures. The materialization of this risk could result in ineffective decision making at company's level and the company might not achieve desired outcome from the availed funds which will ultimately impact on their repayments. Projected H2R impact, including GHG emissions, households accessing renewable energy, and retained employment, may fail to materialize. This failure could result from a number of factors, including, but not limited to, ongoing effects of the COVID-19 pandemic, customers' inability to pay, and/or mismanagement of companies receiving H2R investments.		
Mitigation Measure(s)		
The facility manager has a well-designed credit due diligence process which involves detailed review of ownership, governance, management and board structure/effectiveness along-with interviews of senior management. In case any weakness is identified, at the assessment level the borrowers are advised to resolve these issues appropriately along-with these strong covenants are also incorporated to improve the governance structure. Furthermore, any major change in ownership structure is also restricted in the legal documentation. The screening process described in the ESMS is meant to ensure that H2R investments are directed to companies with a viable, impactful business model. These measures, combined with the monitoring and oversight described in Section E, should mitigate the risk of attenuated impact H2R expects to have limited exposure to governance risks. Locally owned and managed early stage distributors, who may have more governance risks than other investees in the H2R portfolio could be invested with the support of relevant advisors with local market expertise. If H2R finds that a company has particular governance risks, the manager may decide to create conditions subsequent to address these risks.		
Selected Risk Factor 10: Sanctions and AML/KYC		
Category	Probability	Impact
<u>Sanctions</u>	<u>Low</u>	<u>Medium</u>
Description		
Hardest-to-Reach is investing in countries that are on the UN Sanctions List. Hardest-to-Reach seeks to make investments in the Democratic Republic of Congo and Somalia. Investing in these countries increases the likelihood of investing in companies or individuals that are on UN Sanctions Lists or other sanctions lists. The materialization of this risk can have serious consequences such as blockage of funds from investors, regulatory or criminal proceedings against the funds/investors, sanctions, and may also cause monetary losses.		

Mitigation Measure(s)

The Hardest-to-Reach team has conducted extensive market and political risk research across H2R markets. The team has developed a strong level of knowledge of country-specific risks across H2R markets.

As a result of this research, Acumen has developed a project-specific AML/KYC plan that details the AML/KYC policies and procedure.

Each potential investee that has been IC approved is expected to undergo a thorough background check. The diligence team will also seek to diligence key persons including their board, investors, senior management, founders, and the C-suite. If any company is linked to money laundering or financing of terrorism then they will not receive H2R funds.

In the DRC and Somalia, prior to investment, the H2R team will check UN Sanctions Lists, Terrorist Exclusion Lists, The Office of Foreign Assets Control watchlist, and others to ensure that no key persons are on the +100 watchlists. If people are identified on the lists, H2R will not proceed with the investment.

Selected Risk Factor 11: Excluded Activities

Category	Probability	Impact
<u>Prohibited practices</u>	<u>Low</u>	<u>Medium</u>

Description

Hardest-to-Reach is reaching new markets and investing in early-stage companies with limited track records who may engage in prohibited practices. If regulatory entities or investors find that a company engages in prohibited practices, the company or individual could face criminal charges, shutting down of operations, or blockage of funds from investors. If H2R is found to invest in such a company, the materialization of this risk can have serious consequences such as blockage of funds from investors, regulatory or criminal proceedings against the funds/investors, sanctions and may also cause monetary losses.

Mitigation Measure(s)

Hardest-to-Reach has developed an Exclusion List to screen out potential Investees engaging in any prohibited practices. Potential Investees will be required to declare that they do not engage in any activity, do not fund any activity, and do not support any activity on the Exclusion List. If a company does engage in activity on the Exclusion List then they cannot receive H2R financing. H2R investees will also agree to a binding legal agreement that prohibits engagement in Exclusion List activities.

Selected Risk Factor 12: Portfolio quality risk

Category	Probability	Impact
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<u>Credit</u>	<u>Medium</u>	<u>High</u>
Description		
The imposition of ambitious, time-bound impact goals could create incentives to push the number of loans at the expense of quality. This could have a deleterious effect on the credit risk management of energy companies, reducing the stability of their cashflows and viability in the medium term. Companies could also respond by adopting aggressive collections techniques at the risk of breaching relevant consumer rights and legal protections. This could create overindebtedness among low-income households, and other consumer protection-related concerns.		
Mitigation Measure(s)		
For PAYGo companies, business viability is dependent on more than sales volume. Cash flow only comes from a pool of receivables if they are of sufficient quality. Assessing risk management capacity, monitoring portfolio health, and incorporating collections rate into the index for impact-linked loans will likely be essential. TA support is available for investees with portfolios showing signs of deterioration and the due diligence process is designed to flag the types of weaknesses in governance and business processes that make poorly managed growth more likely. Ensuring that companies subscribe and adhere to the GOGLA Consumer Protection Principles is one practical mitigant. H2R due diligence process will seek to identify any CP weakness/challenges faced by the potential investee. Investees will be supported to fulfill and grow beyond the minimum standards of practice in their treatment of customers, in order to safeguard impacts and respect the rights of consumers		

Selected Risk Factor 13: Finance		
Category	Probability	Impact
<u>Credit</u>	<u>High</u>	<u>High</u>
Description		
H2R debt investees may fail to partially or fully repay their obligations, either on-time or at all. This risk could materialize because of lower-than-expected customer repayments, political events, the occurrence of natural disaster, or myriad other possibilities. Accretive exits from H2R equity investments may prove difficult or impossible within the Program life. The MSF may be forced to exit from equity investments when market conditions are not favorable or when a particular asset has not matured, contrary to the plans set out in the initial investment strategy. Operating in H2R contexts is high risk and materialization of this risk could result in investee defaults and lower fund-level returns.		
Mitigation Measure(s)		
H2R investments will undergo thorough due diligence, assessing enterprise and macro level risks to recouping projected base case returns. Ongoing monitoring of investments will enable our teams to		

intervene with relevant accompaniment, including restructuring advisory, to protect the financial and impact outcomes of the investment alongside other finance-specific capacity building, such as financial readiness and consumer protection TA. For the Market Expansion Facility, a blended capital structure with a first loss layer is adopted to better insulate senior tranche investors from any fund losses. Fund documents include a 2x 1-year extension at the fund manager's discretion, to create flexibility around the timing of equity exits and possible debt restructuring.

Selected Risk Factor 13: Pipeline

Category	Probability	Impact
<u>Execution</u>	<u>Medium</u>	<u>High</u>

Description

H2R faces the risk of not being able to source investments and build pipelines of the target size. Local distributors may not be investable and H2R could spend time, money, and energy on due diligence for companies that do not meet H2R expectations. As H2R is a time bound program, any delays in deploying capital would negatively impact potential returns and scaling impact.

Mitigation Measure(s) The goals of the MSF are to create strength in OGS distributors and to support expansion of OGS manufacturers and distributors in LDCs. It is intended that recipients of MSF funds will be viable MEF pipeline borrowers. In addition, part of the MSF strategy is to establish new entities, through a JV structure, to initiate the growth of markets with few or no investable opportunities. H2R has developed strong screening and eligibility criteria for a variety of the investing activities to ensure quality of investment proposals.

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G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, Environment and Social Impact Assessment (ESIA), Environment and Social Management Framework (ESMF), Environment and Social Management Plan (ESMP), Environment and Social Management System (ESMS), environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's Environmental and Social Safeguard (ESS) standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF [Information Disclosure Policy](#) and [Environmental and Social Policy](#).

Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF [Indigenous Peoples Policy](#).

Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.

Acumen and the Hardest-to-Reach team will be leading this project. Using the Green Climate Fund Revised Environmental and Social Policy,¹⁸⁰ the Environmental and Social Impact Assessment, project fundamentals, and executing entity capacity indicate that the appropriate ESS Category for the Hardest-to-Reach program is **environmental and social safeguard category I-2** with predominantly E&S category C investments and activities.

Industry associations¹⁸¹ and similar projects¹⁸² investing in OGS have rated their projects as primarily Category C with the occasional Category B investment. Most of the AE investments in OGS have been categorized as E&S risk level of C. Investees will be ESS Category B or lower and engaged in energy access – primarily solar home systems (<1kw), to lesser extent biodigesters, clean cooking or micro-grids. These various applicants will be assessed according to the following E&S factors: impact on key biodiversity areas; human rights; labor; health & safety; potentially irreversible impacts; land use; extent of

¹⁸⁰ Green Climate Fund (2021) Revised Environmental and Social Policy

¹⁸¹ Ecowas Center for Renewable Energy and Energy Efficiency (ECREEE) (2019) ESRM Sector Guide on Off Grid Standalone Solar for Financial Service Providers

¹⁸² Trade and Development Bank (2020) E&S Risk Management Procedure for Off-Grid Solar Energy Projects

environmental/social issue. The context of inherent sector risks, scale, continued operations and locations will also be considered.

While we assume that most investments in off-grid solar will be E&S Category C, we acknowledge the potential for a very small amount of investments to have Category B risks contained within the scope of the investment. H2R expects to have a more rigorous investment process that may include an environmental and social impact assessment if the team perceives an investment may be Category B. Category B investments may be engaged in purchasing land or have a demonstrated pattern of site-specific and reversible negative environmental and social impacts.

Environmental and Social Risk Categories: General Characteristics (determined on a case-by-case basis)

Category B	Category C
• OGS investment ● Must be larger investment: \$5M+ ● Due diligence finding any of the following outcomes: ○ Funds used for land acquisition outside of commercial zones ○ Pattern of OHS incidents ○ Pattern of gender or SEAH incidents ○ Pattern of reversible environmental and social impacts	● OGS investment ● Investment smaller than \$5M ● Due diligence finds the following: ○ Funds used for working capital, accounts receivable, operational expenditures ○ Minimal environmental and social impacts ○ May not have sufficient policies and procedures

Scope of E&S assessment

Acumen seeks to do no harm with the H2R Project activities. Acumen used industry best practices, tools, metrics, methodologies, and engagements to develop a broad and detailed assessment of the project environmental and social risks and impacts. Hardest-to-Reach also extensively engaged with stakeholders to get their input into project design, market research, project management and implementation, climate, gender, and poverty strategies, assessing the viability of the initiative, and measuring interest in continued engagement. Hardest-to-Reach engaged with potential investors and donors, pipeline companies, potential partners, industry leaders and experts, industry associations, government entities, regulators, expert consultants, and beneficiaries in designing and developing the project. The team has held one on one meetings, group discussions, town halls, webinars, and surveys. Hardest-to-Reach intends to continue this

engagement with stakeholders throughout project development and implementation. The H2R project team also utilized extensive desktop research to understand specific risks for the project, investees, the country context, and relevant mitigants.

Key Project Level Environmental and Social Risks

The project has four primary environmental and social risk concerns:

- E-waste
- Consumer Protection
- Forced Labor
- OGS Company ESG Readiness

E-waste

While there are many strong environmental benefits to Hardest-To-Reach, the project team recognizes both environmental and climate risks for the project. One of Hardest-To-Reach's primary environmental concerns is solar e-waste. Off-grid businesses will be selling millions of products to first time buyers in the H2R markets. These markets have weak and nascent waste management sectors. Moreover, many African countries have weak regulatory requirements for e-waste. Last mile distributors may have weak or immature e-waste policies, procedures, and procurement. Moreover, the e-waste markets in Hardest-to-Reach countries are in a nascent stage with few players and little coverage.

Consumer Protection

There are two primary social risks for Hardest-To-Reach. First, pay-as-you-go (PAYGo), products sold on credit, create risks for customers who can face issues such as over-indebtedness, poor credit, and defunct products if they fall behind on their payments. PAYGo also allows many customers who could not afford to pay for a solar home system up front the opportunity to access electricity with smaller recurring payments.

Additionally, the team has concerns around consumer privacy and protections for PAYGo and other solar home system customers. While many companies are now GOGLA consumer protection signatories, customers still face the risk of customer data being compromised, improperly stored, stolen, or sold to malicious third parties. Infringing on customer data protections may open off-grid solar customers to stolen identity, fraud, and other issues surrounding identity theft.

Forced Labor

Second, the H2R management team is concerned about forced labor in the solar supply chain. Human rights investigators have found evidence of forced labor in some regions where there are solar manufacturers. Advocacy groups warn that marginalized people may be facing mass detention in certain

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regions. Moreover, these marginalized people face dangerous working conditions that may be exacerbated by ongoing global trends. Advocacy groups and human rights officials expect these global factors to create downward pressure on manufacturers to increase forced labor practices under worsening conditions.

OGS Company ESG Readiness

The project team expects that Hardest-to-Reach will invest in companies with varying degrees of ESG readiness. Larger and more established off-grid solar companies are expected to demonstrate maturity across a number of ESG issues. Several larger pipeline companies have indicated that they have strong ESMSs, consumer protection policies, and other environmental and social mitigants in place. Acumen's experience has demonstrated that early-stage SHS companies, especially last mile distributors, may not have robust ESG policies and procedures, nor would they have strong organizational capacity to mitigate ESG risks. These companies have a small footprint so their risks are marginal to the project success and impact on beneficiaries.

The Hardest-to-Reach ESMS includes the ESIA, which has a more extensive report on project and company level environmental and social risks and impacts.

Environmental and Social Risk Mitigants

To ensure the Hardest-to-Reach project achieves its mission and impact, the team has designed a thorough ESMS aligned with GCF Policy, Acumen Policy, and industry best practices.

The ESMS shares the project development, management, and monitoring and reporting policies and procedures. All investments will be screened against the project Exclusion List before being accepted for due diligence.

The project team will diligence companies with a variety of ESG diligence tools including base-line questionnaires, site visits, company evidence in the form of policies and procedures, desktop research, a climate resilience assessment tool, and supplementary questionnaires.

The ESG findings will be memorialized in an ESG due diligence report and the investment memo. The Investment Committee is expected to consider ESG factors in their deliberations. The ESG due diligence report will also provide the information needed for developing the term sheet binding, time bound, ESG Action Plan.

The project will offer Portfolio Companies technical assistance tailored to supporting company-specific ESG, consumer protection, gender, and business readiness needs. This will include: needs diagnostics, webinars, direct support, and policy and procedure guidance and resources.

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The team will monitor and report on ESG strategy across the portfolio and the project to relevant teams.

Hardest-to-Reach will also ensure that the team has the organizational capacity to identify and mitigate risks across the project and investments. The team will hire a Climate, ESG, and stakeholder engagement officer who will be responsible for ensuring the implementation of the ESMS.

The team will also ensure that they listen to stakeholder voices so that communities affected by the project can continue to shape and improve the project design and implementation.

Indigenous Peoples

The program intends to have a landing page on the AE website that will share information about the program including the ESMS. We seek to provide guidance on how indigenous communities can request support from the GCF independent Redress Mechanism and/or the GCF Secretariat's indigenous peoples focal point at any stage, including before a claim has been made. Additionally, the team seeks for H2R investees to have indigenous peoples' policies that will communicate how GCF can be reached. This is especially important for investees that directly engage with indigenous communities.

Acumen, as the AE, and the H2R team has reached out to indigenous organizations to learn from, share the project design, and ask for input. We have also asked NDAs to connect our team with indigenous communities. We seek meaningful engagement with indigenous populations who may be impacted by the project prior to first close. We intend to report to indigenous organizations about project activity and direct them to OGS expanding in areas that have indigenous populations so that they can benefit from access to affordable, clean, electricity opportunities.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Provide a summary of the gender assessment and project/programme-level gender action plan that is aligned with the objectives of GCF's [Gender Policy](#). Confirm a gender assessment and action plan exists describing the process used to develop both documents. Provide information on the key findings (who is vulnerable and why) and key recommendations (how to address the vulnerability identified) of the gender assessment. Indicate if stakeholder consultations have taken place and describe the key inputs integrated into the action plan, including: how addressing the vulnerability will ensure equal participation and benefits from funds investment; key gender-related results to be expected from the project/programme with targets; implementation arrangements that the AE has put in place to ensure activities are implemented and expected outcomes will be achieved, monitored and evaluated.

Provide the full gender assessment and project-level gender action plan as annex 8.

Executive Summary

Acumen has conducted a thorough and robust gender assessment for the Hardest-to-Reach project aligned with the objectives of GCF's Gender Policy, Acumen's Gender Sensitivity Policy for GCF-funded projects. The gender assessment informed the development of the Hardest-to-Reach Gender Lens Investing Strategy and the Hardest-to-Reach Gender Action Plan. The H2R gender assessment and GAP are detailed in [Annex 8](#).

As part of a H2R pilot project, Acumen hired a lead consumer impact surveying business to survey beneficiaries of the pilot project. The H2R team will need to learn from more beneficiaries, but the initial findings demonstrated similar impacts for men and women. Both groups felt the need and importance of using SHS products to attain electricity.

The team also conducted extensive stakeholder engagement across the Hardest-to-Reach markets including speaking with companies, government entities, industry associations, NGOs, and civil society organizations. The H2R team spoke with local gender experts and civil society organizations from multiple H2R markets on local gender challenges and opportunities that informed our gender lens investing strategy and gender work.

The H2R team worked to do project-level gender assessment that included 14 country assessments. The team generated two country assessments for the remaining Hardest-to-Reach countries. The team relied on desktop research, industry experts, gender experts, OGS companies, and survey data to inform the gender assessments, GLI strategy, and Gender Action Plan.

Findings

Barriers to gender equality in OGS sector

The link between energy access and gender equality is well recognized. Energy poverty has a strong gender dimension as the lack of access to energy inhibits women's well-being, and their access to economic opportunities, and it strongly affects women's living conditions and time-use.

One of the key barriers in the market relates to lack of sex-disaggregated data for the OGS sector both at the consumer and workforce levels.

Gender and social norms limit women's access to technologies and their abilities to enter the workforce.

Companies have a limited understanding of the business case for gender inclusion in their businesses, and of how to make their practices more inclusive. Several companies we spoke with shared that they are beginning to build institutional capacity on gender and want to build inclusive policies and procedures into their operations.

--

Serving women with OGS technologies

Overall, evidence shows that women have less access to off-grid electricity than their male counterparts, and they mostly use pico-solar products and lanterns or other lower-tier technologies.¹⁸³ An analysis conducted by Value for Women on 59 companies working in the energy sector found that only 23% of companies collect sex-disaggregated data on customers, and half of them reported that they serve 46% of customers who are women. In H2R country contexts, prevalent gender norms are likely to affect access to off-grid solar technologies, influencing companies' abilities to reach women. Discriminatory gender norms inhibit women's rights to own land and other assets, making it challenging for women to access credit and purchase renewable energy technologies. PAYGo companies may reach a lower percentage of women customers due to additional barriers related to mobile ownership and mobile money usage among women.

Women's participation in OGS companies' workforce

ESMAP and GOGLA recently surveyed 37 OGS companies from 20 different countries and found that on average women constitute 41.7% of workers across all levels. In contrast, a previous GOGLA study indicated that only 27% of women work in the OGS sector. An analysis completed by Value for Women found that on average women represent 33% of the workforce of energy companies.

The country assessments indicate that several barriers limit women's participation in the workforce and in the economy more generally, with prevailing gender norms, gaps in education levels and discriminatory policies and practices affecting women's productive activities and economic empowerment in several countries.

Hardest-to-Reach Gender Work

Hardest-to-Reach seeks to build OGS markets with a gender lens, and strong commitment to gender equity. H2R is also committed to supporting companies to build their gender capacity to support their operations and their customers. Additionally, H2R is committed to mitigating risks of sexual harassment, abuse, and exploitation across the project, investees, and beneficiaries.

All companies will undergo a gender assessment to determine their gender strengths and weaknesses. Opportunities for growth will be captured in a binding Gender Action Plan. All companies in the H2R portfolio will have to create a GAP that will be monitored over the company engagement with H2R. Companies must commit to strong anti-sexual harassment policies, non-discrimination policies, and their best effort to collect sex-disaggregated data.

¹⁸³ ESMAP (2022) Gender equality in the off-grid solar sector. A primer. The World Bank Group, Washington DC 2022

H2R intends to provide companies with technical assistance to support building and executing GAPs, creating strategies to hire women, and to retain female customers. H2R will also have technical assistance to train women for jobs in the offgrid solar market. Additionally, H2R intends to build women in H2R markets' awareness of OGS products and local climate hazards by partnering with gender experts on improving inclusive marketing and customer retention activities.

H2R will monitor and report on the gender assessments, GAPs, sex-disaggregated data, and other KPIs in the GAP and logic framework.

Lastly, H2R will continue to engage female stakeholders and stakeholders from marginalized or unheard communities to ensure that their insights inform the project design, implementation, and management.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Describe the project/programme's financial management including the financial monitoring systems, financial accounting, auditing, and disbursement structure and methods. Refer to section B.4 on implementation arrangements as necessary.

Articulate any procurement issues that may require attention, e.g. procurement implementation arrangements and the role of the AE under the respective proposal, articulation of procurement risk assessment undertaken and how that will be managed by the AE or the implementing agency.

1. Accounts Payable / Purchases / Expenses

1.1. *Initiating and Authorizing:*

a. When purchasing supplies, property, equipment, and services from an external source, the Program:

i) Will alert the AE's General Counsel for potential conflict of interests.

ii) Shall neither solicit nor accept gratuities, favors, or anything of monetary value from contractors, potential contractors, or similar service providers.

iii) Will follow the Procurement Policy including consulting the AE for any goods or services ≥ \$100k USD, awarding the winning bidder, documenting the decision justification, and the evaluation of each vendor following the contract term.

** The Program's Market Support Facility (MSF) will seek to engage service providers with track records of delivery in alignment with the above policies and the

AE's procurement standards. Service providers must demonstrate capacity and execute around the scope of work developed by the MSF team and shared with the AE for review, where relevant . All procurement decisions will be reviewed by the MSF Director.

**The Program's Market Expansion Facility (MEF) will seek to engage service providers with track records of delivery in alignment with the above policies and the AE's procurement standards. Service providers must demonstrate capacity and execute around the scope of work developed by the MEF team and shared with the AE for review, where relevant . All procurement decisions will be reviewed by the MEF Director.

iv) When required by investors, the Program may publish an RFP for a specific project.

b. Vendor selection is dependent on:

i) The results of a standardized background check. Each vendor and its related entities and owners are subject to Patriot Act searches . Each search reviews international sanctions lists, enforcement sources (i.e., civil, criminal offenses, indictments, and convictions), and financial history (i.e., foreclosures, bankruptcies, etc.).

ii) The different investing strategies the the Program will implore (Market Support Facility and Market Expansion Facility), and requisite sectoral expertise and operational infrastructure necessary for the distinct approaches.

1.2. **Recording and Processing:** Accounts payable will be reviewed and processed at the end of each week. Each invoice is submitted to the Program's accounting platform for review and approval and must have an associated vendor record in the system. Before approving, the appropriate finance employee will review all invoices, wires, checks and account coding.

1.3. **Reconciling:** The Program reconciles all bank, petty cash, and investment accounts monthly, as follows:

- a. Once available, the Finance Associate (or designate) reviews all relevant bank statements
- b. The Finance Associate (or designate) will complete the reconciliation within seven days
- c. The following is the general outline to be used to reconcile accounts:
 - i) Comparison of bank statements and cash receipts, verifying dates and totals deposited.
 - ii) Comparison of bank and/or wire transfer dates received against dates sent.

- iii) Investigations of items rejected by the bank (i.e., returned checks or deposits).
- iv) Comparison of canceled checks and the disbursement journal (check numbers, payee, and amount).
- v) Examination of canceled checks for authorized signatories, irregular endorsements, and alterations.
- vi) Accounting for the sequence of checks from month-to-month.
- vii) A review and proper mutilation of void checks.
- viii) Investigate, write-off or reissue/stop-pay checks outstanding for an extended period.

2. Expense Reporting Procedures

2.1. *Principles and Procedures:*

- a. The Finance Associate Director will be responsible for compiling monthly and year-to-date expense reports. Expense reports are reconciled to the general ledger and accounting records.
- b. Quarterly financial reports, analyzing the Program's financial position and the effectiveness of its management, will be presented to its Limited Partners as required.
- c. Records to be maintained include copies of contracts, invoices, changes to contracts or budgets, proof of payments, and allocation tracking.

3. Expected Contractors

The Program may contract consultants for several purposes. Technical Assistance consultants will be hired for expertise in ESG; Consumer Protection; Gender Equity; Business development (as it relates to fundraising and financial systems management) for companies; Legal and Accountants/auditors.

The Program's Market Support Facility (MSF) will focus on early-stage investments, employing equity, equity-like, impact index loans, grants, and joint venture financing options. Launched in January 2023, the MSF will run through to 2033. Although negotiations remain on-going, the Program has preliminarily selected both a local market advisor and a separate small-ticket debt advisor. In each case, the firms distinguished themselves from other providers through their unique value (i.e., their extensive experience in both debt and equity lending to early stage off-grid energy companies across the Program's target geographies).

Conversely, the Program's Market Expansion Facility (MEF) has both a separate investment vehicle structure, strategy, and instrument size. The MEF will target later stage, more established companies with larger ticket sizes, utilizing impact index loans (working capital), and receivables financing, and will be structured as a pooled investment vehicle. With an expected launch in January 2024, the MEF will run through to 2034. Given the difference in approach, in addition to the types of providers noted above, the Program is considering contracting with certain service providers to support fund management, such as an administrator and back office and support in risk assessments. While

discussions remain on-going, the Program intends to identify institutions with existing systems and processes in place (examples could include: interest tracking and invoicing, record maintenance, tax and regulatory compliance, accounting, and IT services related to debt investing).

Lastly, there may be other types of contracts with other service providers that the Program continues to explore including legal services and fund administration [see capacity assessment in B.4].

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

A. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
- ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, etc.)
- ☒
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)

H.2. Other annexes as applicable

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** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

¹⁸⁴ Annex 22 is mandatory for mitigation and cross-cutting projects.



N° 686 /DGEC/MCVDD/SD

To: Mr. Yannick Glemarec
Executive Director
Green Climate Fund Secretariat
G-Tower, Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Cotonou, 11 November 2022

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Mr. Yannick Glemarec,

We refer to the programme titled "*Hardest-to-Reach*" in Angola, Benin, Burkina Faso, Burundi, Central African Republic, Chad, Democratic Republic of the Congo, Guinea, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mozambique, Niger, Sierra Leone, Somalia, South Sudan, Togo, Uganda, United Republic of Tanzania, and Zambia as included in the funding proposal submitted by Acumen to us on 20 September 2022.

The undersigned is the duly authorized Director General of Environment and Climate, the National Designated Authority of Benin.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Benin has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Benin;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Martin Pépin AINA
National Designated Authority, Benin
General Director of Environment and Climate

PRIMATURE

SECRETARIAT GENERAL

**SECRETARIAT EXECUTIF DU FONDS VERT
POUR LE CLIMAT AU BURKINA FASO**

BURKINA FASO
Unité-Progrès-Justice

Ouagadougou, February 15 2023

N° 2023-001 /PM/SG/SE-FVC/BF

National Designated Authority

**To
The Green Climate Fund
("GCF")**

Republic of Korea

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Burkina Faso, as included in the funding proposal submitted by Acumen to us on 14 February 2023.

The undersigned is the duly authorized representative of the Prime Ministry, the National Designated Authority of Burkina Faso.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Burkina Faso has no-objection to the programme as included in the funding proposal;

- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Burkina Faso;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Nebyida Lamech KABORE

*Inspector of Waters and Forests
Policy Officer*

REPUBLIC OF BURUNDI

Gitega, le 18 / 01 / 2023

MINISTRY OF ENVIRONMENT,
AGRICULTURE AND LIVESTOCK

NATIONAL DESIGNATED AUTHORITY

N/Réf: AND-PF/FVC- BURUNDI/01/2023

To: Mr. Yannick Glemarec
Executive Director
Green Climate Fund Secretariat
G-Tower, Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Mr. Yannick Glemarec,

We refer to the programme titled "*Hardest-to-Reach*" in Angola, Benin, Burkina Faso, Burundi, Central African Republic, Chad, Democratic Republic of the Congo, Guinea, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mozambique, Niger, Sierra Leone, Somalia, South Sudan, Togo, Uganda, United Republic of Tanzania, and Zambia as included in the funding proposal submitted by Acumen to us on 24 October 2022.

The undersigned is the duly authorized representative of the Ministry of Environment, Agriculture, and Livestock, the National Designated Authority of Burundi.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

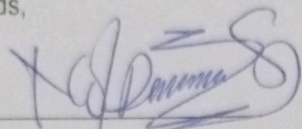
- (a) The government of Burundi has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Burundi;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

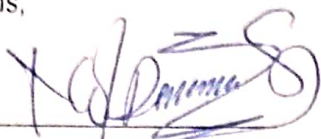
A handwritten signature in blue ink, appearing to read 'Diomède Ndayirukiye', is written over a horizontal line.

Mr. Diomède Ndayirukiye
Ministry of Environment, Agriculture, and Livestock
National Designated Authority
Burundi

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

A handwritten signature in blue ink, appearing to read 'Diomède Ndayirukiye', is written over a horizontal line.

Mr. Diomède Ndayirukiye
Ministry of Environment, Agriculture, and Livestock
National Designated Authority
Burundi

RÉPUBLIQUE DU TCHAD

PRESIDENCE DE TRANSITION

PRIMATURE

MINISTÈRE DE L'ENVIRONNEMENT, DE LA PÊCHE
ET DU DÉVELOPPEMENT DURABLE

DIRECTION DE CABINET

CELLULE DE COORDINATION

POINT FOCAL DU FONDS VERT POUR LE CLIMAT

N° 001/PT/PM/MEPDD/DC/CC/PFFVC/23

UNITE – TRAVAIL – PROGRES

N'Djamena, 10 March 2023

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Chad, as included in the funding proposal submitted by Acumen to us on 10 March 2023.

The undersigned is the duly authorized representative of the Ministry of Environment, Water, and Fisheries, the National Designated Authority of Chad.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Chad has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Chad;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Ibrahim Charfadine Abdelkerim
Ministry of Environment, Water, and Fisheries
National Designated Authority - Chad

No-objection letter template for programmes





To: The Green Climate Fund ("GCF")

Kinshasa, 18 October 2022

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Mr. Yannick Glemarec,

We refer to the programme titled "*Hardest-to-Reach*" in Angola, Benin, Burkina Faso, Burundi, Central African Republic, Chad, Democratic Republic of the Congo, Guinea, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mozambique, Niger, Sierra Leone, Somalia, South Sudan, Togo, Uganda, United Republic of Tanzania, and Zambia as included in the funding proposal submitted by Acumen to us on 18 October 2022.

The undersigned is the duly authorized representative of the National Designated Authority of the Democratic Republic of the Congo.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Democratic Republic of the Congo has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies, and plans of the Democratic Republic of the Congo;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Dr. Hans André Lohayo Djamba, Ph.D.
National Designated Authority
Democratic Republic of the Congo

MISTERE DE L'ENVIRONNEMENT ET DU
DEVELOPPEMENT DURABLE

SECRETARIAT EXECUTIF DE L'AUTORITE
NATIONALE DESIGNEE DU FONDS VERT
POUR LE CLIMAT



REPUBLIQUE DE GUINEE
Travail-Justice- Solidarité

000097

Réf...../MEDD/CAB/SE-AND-FVC/2022

Conakry, le 16 DEC 2022

Le Secrétaire Exécutif

A

Mrs Caroline FUENTES, Director of the Division of Country
Programming Green Climate Fund
Songd, Incheon, Seoul, Korea

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Mrs Caroline FUENTES,

We refer to the programme titled "Hardest-to-Reach" in Angola, Benin, Burkina Faso, Burundi, Central African Republic, Chad, the Democratic Republic of Congo, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mozambique, Niger, Sierra Leone, Somalia, South Sudan, Togo, Uganda, United Republic of Tanzania, and Zambia as included in the funding proposal submitted by Acumen to us on 1 November 2022.

The undersigned is the duly authorized representative of the Executive Secretary of the National Designated Authority for Climate Fund of Guinea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.
By communicating our no-objection, it is implied that:

- (a) The government of Guinea has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Guinea;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Pierre LAMAH





Bissau, 17 Avril 2023

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Guinea-Bissau, as included in the funding proposal submitted by Acumen to us on 16 March 2023.

The undersigned is the duly authorized representative of the Ministry of the Environment and Biodiversity/ General Direction of the Environment, the National Designated Authority of Guinea-Bissau.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Guinea-Bissau has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Guinea-Bissau;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Viriato Luís Soares Cassamá
National Designated Authority
Ministry of the Environment and Biodiversity / General
Direction of the Environment Guinea-Bissau

Avenida João Bernardo Vieira, Palácio do Governo, Bissau, República da Guiné-Bissau



**Ministry of Natural Resources
Africa House, Old High Court Road Maseru
Private Bag A91, Maseru 100, Lesotho
Tel: +266 22 311100**

Maseru, 12/04/2023

To: The Green Climate Fund ("GCF")
Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by Acumen Fund regarding "Hardest-to-Reach" Programme

Dear Madam, Sir,

We refer to the programme "**Hardest-to-Reach**" in Lesotho as included in the funding proposal submitted by Acumen Fund to us on 13/03/2013

The undersigned is the duly authorized representative of the Ministry of Energy and Meteorology, the National Designated Authority of Lesotho.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The government of Lesotho has no-objection to the programme as included in the funding proposal;
- b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Lesotho;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme. We acknowledge that this letter will be made publicly available on the GCF website.

For further comments or questions, please do not hesitate to write to the NDA team's official e-mail address: kabelo.lebohang@gov.ls.

Yours Faithfully,

A handwritten signature in black ink, appearing to read 'H. Makope', is positioned above a horizontal line.

Mr. Habofane MAKOPELA
Deputy Principal Secretary
GCF National Designated Authority for Lesotho

Telephone: +265 1 771 111
Telefax: +265 1 773 379

Our Reference: EAD/99/06/04N
Your Reference:.....



Environmental Affairs Department
Private Bag 394
Capital City
Lilongwe 3

24th March, 2023

The Green Climate Fund ("GCF")
Executive Director
Green Climate Fund
Song Do Business District
175 Art Center-Daero
Yeonsu-Gu, Incheon 22004
Republic of Korea

Dear Madam, Sir,

Re: Funding proposal for the GCF by ACUMEN Fund: Hardest-to-Reach

We refer to the programme titled Hardest-to Reach as included in the funding proposal submitted by ACUMEN Fund on 3rd March, 2023.

The undersigned is the duly authorized representative of the Environmental Affairs Department, the National Designated Authority of Malawi.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Malawi has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Malawi;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mrs. Tawonga Mbale-Luka
Director of Environmental Affairs
GCF Nationally Designated Authority for Malawi



**REPUBLIC OF MOZAMBIQUE
MINISTRY OF ECONOMY AND FINANCE
NATIONAL MONITORING AND EVALUATION DIRECTORATE
NATIONAL DESIGNATED AUTHORITY**

To: The Green Climate Fund ("GCF")

Maputo, 7 March 2023

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Mozambique, as included in the funding proposal submitted by Acumen to us on 6 February 2023.

The undersigned is the duly authorized representative of Ministry of Economy and Finance, the National Designated Authority of Mozambique.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mozambique has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mozambique specifically in the increment of access to modern and high quality energy for all national communities by 2030, without compromising the environment through the use of cleaner and more efficient energy sources ;
- (c) The interest in working with local national companies and ACUMEN's approach to ensuring long-term technical assistance is proving to be an opportunity for the development and strengthening of local companies.
- (d) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.



REPUBLIQUE DU NIGER



Fraternité-Travail-Progrès

CABINET DU PREMIER MINISTRE

Conseil National de l'Environnement pour un Développement Durable
Secrétariat Exécutif

To: The Green Climate Fund ("GCF")

Niamey, le 20 AVR 2023

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Niger, as included in the funding proposal submitted by Acumen to us on 14 February 2023.

The undersigned is the duly authorized representative of the National Council of the Environment for Sustainable Development, the National Designated Authority of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Niger has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Niger;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

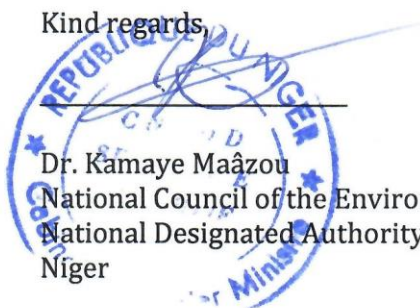
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Dr. Kamaye Maâzou
National Council of the Environment for Sustainable Development
National Designated Authority
Niger





GOVERNMENT OF SIERRA LEONE
Environment Protection Agency
Ministry of the Environment
92 Dundas Street, Freetown



To: The Green Climate Fund (“GCF”)

Freetown, 13 March 2023

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled “*Hardest-to-Reach*” in key Sub-Saharan African countries including Sierra Leone, as included in the funding proposal submitted by Acumen to us on 13 March 2023.

The undersigned is the duly authorized representative of the Environment Protection Agency, the National Designated Authority of Sierra Leone.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Sierra Leone has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Sierra Leone;
- (c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Dr. Bondi Gevaio
 National Designated Authority
 Environment Protection Agency
 Sierra Leone



FEDERAL REPUBLIC OF Somalia
MINISTRY OF ENVIRONMENT & CLIMATE CHANGE
Office of the Minister

Ref: MoECC/124/2023

Date: April 15, 2023

To: The Green Climate Fund

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled “*Hardest-to-Reach*” in key Sub-Saharan African countries including Somalia, as included in the funding proposal submitted by Acumen to us on 5 April 2023.

The undersigned is the duly authorized representative of the Ministry of Environment and Climate Change, the Choose an item of Somalia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of Somalia has no-objection to the programme as included in the funding proposal;
- The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Somalia;
- In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

H.E Amb. Khadija Mohamed Alмахzoumi
Minister, Ministry of Environment and Climate Change
Federal Republic of Somalia



Address: Villa Somalia, Mogadishu - Somalia

Website: moecc.gov.so | Email: minister@moecc.gov.so



MINISTERE DE L'ENVIRONNEMENT
ET DES RESSOURCES FORESTIERES



REPUBLIQUE TOGOLAISE
Travail – Liberté – Patrie

SECRETARIAT GENERAL

DIRECTION DE L'ENVIRONNEMENT

Point focal national du Fonds vert pour le climat

N° 001 /DE/FVC

To: The Green Climate Fund ("GCF")

Lomé, 30 January 2023

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in Angola, Benin, Burkina Faso, Burundi, Central African Republic, Chad, Democratic Republic of the Congo, Guinea, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mozambique, Niger, Sierra Leone, Somalia, South Sudan, Togo, Uganda, United Republic of Tanzania, and Zambia as included in the funding proposal submitted by Acumen to us on 30 January 2023.

The undersigned is the duly authorized representative of the Directorate of Environment, Ministry of Environment, the National Designated Authority of Togo.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Togo has no-objection to the programme as included in the funding proposal ;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Togo ;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Ms. Méry Yaou
Directorate of Environment, Ministry of Environment
National Designated Authority
Togo

Telephone : 256 41 4232095/341286
 Fax : 256 41 4233524
 Email : finance@finance.go.ug
 : treasury@finance.go.ug
 Website : www.finance.go.ug
 Plot No. 2-8 Apollo Kaggwa Road
 In any correspondence on
 This subject please quote No. ALD 79/251/03



Ministry of Finance, Planning &
 Economic Development,
 P.O Box 8147
 Kampala, Uganda

3rd November, 2022

The Executive Director
 Green Climate Fund
 Songdo Business District
REPUBLIC OF KOREA

LETTER OF NO OBJECTION FOR ENERGY ACCESS FOR HARDEST TO REACH

I refer to the above programme, which has been prepared and submitted to you by ACUMEN for funding by the Green Climate Fund (GCF).

The undersigned is the duly authorized representative of the Ministry of Finance, Planning and Economic Development, the National Designated Authority focal point of Uganda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme activities as included in the proposal.

By communicating our no-objection, it is implied that:

- a) The Government of the Republic of Uganda has no-objection to the programme as included in the funding proposal;
- b) The programme is in conformity with Uganda's national priorities, strategies and plans;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the proposal has been duly followed and that it applies to all projects or activities to be implemented within the scope of the programme.

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

We acknowledge that this letter will be made publicly available on the GCF website.



Ramathan Ggoobi

**PERMANENT SECRETARY/ SECRETARY TO THE
TREASURY/NATIONAL DESIGNATED AUTHORITY FOCAL POINT
FOR THE GREEN CLIMATE FUND**

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

All communication should be addressed to the
Permanent Secretary
Telephone (260 211) 252395
(260 211) 252394
(260 211) 252391



REPUBLIC OF ZAMBIA

In reply please quote
NDA/71/21/4
No.

MINISTRY OF GREEN ECONOMY AND ENVIRONMENT

OFFICE OF THE PERMANENT SECRETARY
CORNER OF JOHN MBITA & NATIONALIST ROADS
P.O. BOX 30147
LUSAKA, ZAMBIA

To: The Green Climate Fund ("GCF")

Lusaka, 3 February 2023

Re: Funding proposal for the GCF by Acumen regarding Hardest-to-Reach

Dear Madam, Sir,

We refer to the programme titled "*Hardest-to-Reach*" in key Sub-Saharan African countries including Zambia, as included in the funding proposal submitted by Acumen to us on 3 February 2023.

The undersigned is the duly authorized representative of Ministry of Green Economy and Environment, the National Designated Authority of Zambia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Zambia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Zambia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Francis Mpampi
National Designated Authority
Ministry of Green Economy and Environment
Zambia

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Hardest-to-Reach
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Acumen Fund, Inc
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Benin, Burkina Faso, Burundi, Chad, Democratic Republic of the Congo, Guinea, Guinea-Bissau, Lesotho, Malawi, Mozambique, Niger, Sierra Leone, Somalia, Togo, Uganda, and Zambia
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, June 7, 2023
Language(s) of disclosure	English, French, Portuguese, and Arabic
Explanation on language	These are an/the official languages in each of the target countries.
Link to disclosure	English: https://drive.google.com/file/d/1r5_CTjBqCpot_EG-MgtzFuOLN1FNKvyc/view?usp=drive_link French: https://drive.google.com/file/d/1rPGcuslbz-dtKMNIvkuZZpfmMqPP6C_B/view?usp=drive_link Portuguese: https://drive.google.com/file/d/1rGL28q1Lvy6o5IT-XSgRxpLkHBFNF34Q/view?usp=drive_link

	Arabic: https://drive.google.com/file/d/1rk_4G94nJKF4yIdVoyJOL9StLLrRpqPF/view
Other link(s)	https://acumencapitalpartners.com/about/
Remarks	An ESMS consistent with the requirements for a Category I-2 programme is contained in the “Environmental and Social Management System”.
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	Land Acquisition and Resettlement Policy, Guidance on Indigenous Peoples / Wednesday, June 7, 2023
Language(s) of disclosure	English, French, Portuguese, and Arabic
Explanation on language	These are the official languages in each of the target countries.
Link to disclosure	English: https://drive.google.com/file/d/1r5_CTjBqCpot_EG-MgtzFuOLN1FNKvte/view?usp=drive_link French: https://drive.google.com/file/d/1rPGcuslbz-dtKMNIvkuZZpfmMqPP6C_B/view?usp=drive_link Portuguese: https://drive.google.com/file/d/1rGL28q1Lvy6o5IT-XSgRxpLkHBFNF34Q/view?usp=drive_link Arabic: https://drive.google.com/file/d/1rk_4G94nJKF4yIdVoyJOL9StLLrRpqPF/view
Other link(s)	https://acumencapitalpartners.com/about/
Remarks	The Land Acquisition and Resettlement Policy and the Guidance on Indigenous Peoples are included in the Hardest-to-Reach Environmental and Social Management System.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Wednesday, June 7, 2023
Place	The Hardest-to-Reach Environmental and Social Management System is available at the Acumen East Africa and West Africa offices located at: 3rd Floor, ABC Towers Waiyaki Way, Westlands, Nairobi, Kenya Plot 18 Sikiru Alade Oloko Crescent Lekki Phase 1, Lagos Nigeria

	The Hardest-to-Reach Environmental and Social Management System can also be found at the Nationally Designated Authority offices in the beneficiary countries. These addresses are listed below. Benin 01 BP 3502 - 01 BP 3621, Cotonou, Benin Burkina Faso Prime Ministry, 9FFR+C8F, Avenue de l'Indépendance, Koulouba, Ouagadougou, Burkina Faso Burundi Ministry of Environment, Agriculture and Livestock Avenue Triomphe / Q Musinzira, Gitega Bujumbura, - 1850. Burundi Chad Avenue Idriss Miskine, N'djamena, Chad Democratic Republic of the Congo National Coordinator and GCF Focal Point for the Democratic Republic of the Congo Environment renewable natural resources in Kinshasa, Democratic Republic of the Congo 15, Av : Papa ILEO (ex-des cliniques), C : Gombe, Congo - Kinshasa Guinea Coléah Lansébounyi, Commune de Matam, Conakry, Guinea Guinea Bissau Avenida dos Combatentes da Liberdade da Pátria Palácio do Governo, Bissau, Guinée-Bissau, Bissau, Guinea-Bissau Lesotho Ministry of Energy and Meteorology Private Bag A91, Maseru, Lesotho Malawi Environmental Affairs Department Private Bag 394 Lilongwe 3, Malawi, Lilongwe, Malawi Mozambique National Directorate for Monitoring and Evaluation, Ministry of Economy and Finance, Av. Ahmed Sekou Touré n° 21 Maputo, Maputo, +258 Mozambique Niger Cabinet du Premier Ministre, PL 17 Rue du Gouverneur Jules Brévié, BP 10193, Niamey, Niger, Niamey, Niger
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	Sierra Leone Environment Protection Agency, 92 Dundas Street, Freetown, Sierra Leone Somalia Ministry of Environment and Climate Change, Villa Somalia, Mogadishu, Somalia Togo Directorate of Environment, Ministry of Environment, 240 rue des Nîmes Lomé - Togo -, Lomé, Togo Uganda Ministry of Finance, Planning and Economic Development, Plot 2/8 Apollo Kaggwa Road, P.O.Box 8147 Shimoni Rd 1, Kampala, Uganda Zambia Ministry of Green Economy and Environment P.O. Box 50555, Corner of John Mbita and National, Lusaka, Zambia
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 10, 2023

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP211

Proposal name:	Hardest-to-Reach
Accredited entity:	Acumen Fund
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. The Hardest-to-Reach (H2R) programme is designed to supply off-grid solar (OGS) energy to low-income populations in 16 least developed countries (LDCs) in sub-Saharan Africa. It aims to develop an OGS ecosystem in countries with low electrification and high poverty rates. The initiative supplies capital to OGS companies entering and expanding in these nascent markets. Although the OGS market is thriving in many parts of Africa, there is reluctance among companies to enter LDC markets and to engage with the poorest. These markets are considered too risky by OGS companies, which prefer to focus their limited capacities on more profitable and low-risk markets. The programme seeks to deliver clean first-time electricity access to over 54 million beneficiaries with high rates of poverty, where investors have not been willing to enter at sufficient scale to meet the energy access needs of underserved segments.

2. The USD 250 million programme plans to establish two separate financing facilities with different investment risk profiles. This is designed to help OGS companies expand in H2R countries and extend their reach. The USD 50 million Market Support Facility (MSF) shall be grant funded by philanthropies and donors and includes a USD 10 million reimbursable grant from GCF. The MSF aims to support and improve the capacity of local players and regional companies for market entry through grants, equity, debt, and technical assistance. For the USD 200 million Market Expansion Facility (MEF), Acumen intends to raise USD 145 million across commercial investors and development finance institutions (DFIs), anchored by a USD 50 million junior equity tranche and a USD 5 million non-repayable grant capital from GCF. The MEF aims to provide impact-linked concessionary loans to scale up companies' operations and settle in H2R countries. Interest rates shall be adjusted based on results achieved in terms of first-time energy access and poverty level of the customers. The USD 65 million concessional financing from GCF in the form of grant and junior equity is expected to mobilize private finance to support OGS companies and build a sustainable OGS ecosystem in the challenging markets for just energy systems transitions.

1.2 Impact potential

Scale: High

3. The programme has both mitigation and adaptation impacts. The mitigation impact is estimated at 5.1 megatonne of carbon dioxide equivalent (Mt CO₂ eq) reduced over the lifespan of the programme and, as adaptation outcome, the project expects to reach 54 million beneficiaries of which 9 million are direct beneficiaries.

4. The mitigation outcome is achieved by providing OGS products to people to achieve clean energy access (tier 1+). 120 MW of decentralized OGS energy generation capacity shall be newly installed. Currently, the majority of rural households rely on kerosene and biomass as

their primary energy sources for lighting, cooking, heating, etc. In the baseline scenario, a large portion of households do not have access to electricity due to lack of grids and low-income levels, which limit a household's ability to pay for electricity. The programme reduces the need to rely on fossil fuels and biomass by replacing them with solar energy.

5. The greenhouse gas (GHG) mitigation impact was calculated using the United Nations Framework Convention on Climate Change registered clean development mechanism methodologies AMS-III.AR. ("Substituting fossil fuel-based lighting with LED/CFL lighting systems") and AMS-I.L. ("Electrification of rural communities using renewable energy"). Assumptions for distribution of funds between solar home systems and lanterns, the number of units to be financed per area, as well as the expected lifetime of products are presented by the programme in a transparent manner and are considered to be reasonable. Black carbon emissions have not been included by the programme in the calculations, which is conservative especially as kerosene lamps have high potential black carbon emissions. The achievement of the expected mitigation outcome is reliant on reaching the expected fund size and product placement. Overall, the projected GHG emission reductions are deemed to have been calculated in a conservative manner.

6. The expected adaptation outcome is 54 million direct and indirect beneficiaries. Indirect beneficiaries are calculated by multiplying OGS sales by factor 5, which corresponds to the average number of people per household in H2R countries. The 54 million beneficiaries are people with first-time access to electricity. Rural electricity access in programme countries is 11 per cent on average.

7. The increase in resilience to climate change through the programme is based on the following expected outcomes: (i) most vulnerable people and communities would still have access to energy in the aftermath of displacement-inducing natural disasters caused by climate change due to owning OGS products. It was calculated that 1.6 million people could benefit from enhanced energy security through the programme based on the expected number of people to be displaced due to climate hazards within the total projected number of H2R beneficiaries; (ii) health, well-being, food and water security could be enhanced as OGS customers would be able to more securely access information including extreme weather event and health warnings, health information, and market data, such as for agricultural produce. OGS products linked with additional devices, such as radios or mobile phones, allow access to climate-related information. Seventy-five per cent of the total 72 million beneficiaries (of which 54 million are beneficiaries with first-time electricity access and therefore counted as beneficiaries in the adaptation outcome) are expected to access electricity power for the first time with an OGS product. This results in a total of 54 million beneficiaries with improved access to information; and (iii) most vulnerable people and communities' benefit from increased and/or diversified income sources. Energy access can contribute to increasing household income by increasing working hours through access to light and increasing opportunities for home-based business. A 60 Decibels energy benchmark study revealed that 10 per cent of OGS clients could diversify and increase their income base.¹ Based on the 54 million first-time customers of OGS, this results in 5.4 million direct beneficiaries profiting from increased and more diversified income sources. The total number of 54 million beneficiaries is based on the maximum number of beneficiaries of the three aforementioned impact areas. The programme thus has a potential impact on increasing the climate resilience of around 15 per cent of the total population in the participating countries.

8. The three core adaptation outcomes of the programme, while expected to be significant, may be challenging to measure or report to GCF. While the programme's logical framework includes "User surveys to determine beneficiaries with improved and/or new climate-resilient livelihood options", no mention is made of tracking enhanced access to climate-relevant

¹ 60 Decibels. (2020). Why Off-Grid Energy Matters.

information through OGS-powered communication media. In terms of the assumptions made that the OGS products installed will provide resilient energy access, through durability in the face of extreme events and portability in the case of displacement, no mechanism has been built in to monitor performance of the OGS products.

9. Overall, the independent Technical Advisory Panel (iTAP) considers that the programme has a high adaptation and medium to high mitigation impact. Cumulatively, the impact potential is considered to be high.

1.3 Paradigm shift potential

Scale: Medium

10. The programme promotes the entrance of companies in the African energy sector into off-grid markets, where investors have not been willing to enter up to the present. The blended finance structure has the ability to crowd in investors that would otherwise find the markets too nascent or risky through initiatives to de-risk these markets by providing new market entrants with impact-linked concessional finance and targeted technical assistance. Last-mile distributors have a crucial role to play in delivering OGS products, but capital is currently not flowing into the space. The programme can thus achieve a transformational impact in reaching clients and markets currently only marginally covered by private sector approaches.

11. Companies that thrive in these risky and nascent markets would be willing to increase their business in these markets, thus resulting in positive multiplication effects. The programme supports companies to build solid operational platforms in challenging markets, and to grow in such markets, thereby scaling up their ventures and providing clean energy to a larger population of unelectrified people.

12. Acumen intends to share the knowledge and evidence-based insights gained through the programme. This can result in increased capital inflows and additional investments on the ground. The programme's demonstration of linking energy access and adaptation, as well as showcasing that companies can also thrive in such difficult surroundings, should also unlock funding for OGS ecosystems in more difficult markets. Demonstrating the viability of challenging markets can unlock other investments from broader sources of capital, thereby sustaining the growth of these markets in the long term and adding to a replication effect.

13. Scaling up the programme would depend to a large extent on the sustainability of market efforts. Experiences with COVID-19 have shown that OGS companies struggled with high levels of default among their customers. The Acumen project Participation in Energy Access Relief Facility (FP148) outlined that energy access companies struggled with dramatically reduced sales and collections, coupled with disrupted supply chains, limited cash reserves, and tightening funding sources, which are crippling companies' ability to fund operations. This shows the fragile base of most companies in this market. While COVID-19 in conjunction with other external factors can be considered extraordinary, the exposure of OGS companies to external shocks remains high and the risk of defaults from their customers permanent. The targeted poorer segments of society are especially exposed to numerous external shocks, including climate-related ones, which can negatively affect their payment ability. For people who are close to or below the poverty threshold, the payments for OGS equipment might be one of the first to discontinue in case of existential needs.

14. The experience of many development activities is that successful interventions do not replicate automatically. On-ground demonstration of successful intervention is a positive first step but not sufficient for a high level of replication and transformational change. The programme does require activities to include all relevant stakeholders and to actively manage and spread the knowledge generated. This might not coincide with the goal of involved private enterprises to avoid inflow of competition. Overall, the planned programme activities in the area of knowledge management and outreach are not considered to be comprehensive enough

to result in a massive replication effect. However, this might well be taken up by international development agencies and entities more familiar and experienced with outreach activities, which could build upon the programme's on-the-ground activities.

15. The programme's focus is on working with private enterprises to make commercial ventures in this market successful. The programme recognizes the importance of creating an ecospace that is positive towards OGS, reducing barriers and creating incentives. This is largely related to government policies and activities. While the programme does relate with the involved governments, there is no clear and comprehensive engagement plan with the national governments to improve the framework conditions.

16. Overall, iTAP considers that the programme has the potential to increase the appetite of private companies and capital to also invest in more challenging markets and thus create a paradigm shift. To have an even larger transformational impact, additional measures in the realm of outreach, capacity-building, knowledge management, and stronger engagement with governments will be required. However, this is not necessarily the core expertise of the parties involved in the programme and it is highly probable that positive programme experiences would be taken up by the international development community to "spread the word". The paradigm shift potential is considered to be medium.

1.4 Sustainable development potential

Scale: Medium to High

17. The programme is aligned with Sustainable Development Goal (SDG) 7 (affordable, reliable, modern and sustainable energy for all), SDG 3 (good health and well-being), SDG 4 (improved educational outcomes), SDGs 1 and 8 (economic savings and income generation), and SDG 13 (climate action).

18. One factor that could limit sustainable development impacts, or even cause negative impacts, is the local currency rate risk from MEF on-lending to OGS companies. The iTAP is concerned about the financial sustainability of an MEF USD loan both from an OGS company and an end-customer perspective. In the funding proposal, it is written that "Acumen is exploring the feasibility of local currency lending alongside hedging partners" but, in discussions with iTAP, Acumen wrote that "In the meantime, the H2R MEF would make loans in USD. The target borrowers for H2R MEF would be expected to be sophisticated OGS companies with multinational operations, aware of the impact of currency fluctuations on their borrowings." This essentially means that the OGS companies will pass those currency risks to the ultimate hardest-to-reach consumers. Given current volatility in currency markets, such a pass-on can potentially have devastating impact on the ability of those consumers to pay. Alternatively, if the OGS companies are "forced" to keep that risk on their own balance sheet, it may jeopardize the companies' financial viability. For the very sophisticated OGS companies who truly understand the local currency risks in some of these markets, the MEF funding may not be even worth it.

19. In terms of benefits, OGS products can have positive impacts on people's health and safety. Traditional kerosene lanterns are harmful to people's respiratory health because of household pollutants, and can cause burns and fires. Indoor air pollution from kerosene lanterns is related to particulate matter and carbon monoxide and disproportionately affects women and children, who spend more time at home. According to a 60 Decibels study conducted in 2020,² 48 per cent of OGS customers reported health improvements after acquiring OGS products.

20. OGS systems can potentially help to preserve ecosystems and reduce deforestation by substituting unsustainable biomass harvesting for energy usage with OGS systems.

² 60 Decibels. (2020). Why Off-Grid Energy Matters.

21. OGS products can enable people to generate income through their use, either directly (e.g. expanding business hours or offering new services such as cell phone charging) or indirectly (e.g. shifting tasks or activities to the evening because of better quality light). In the case of biomass harvesting used in absence of OGS systems, it can also reduce workloads, primarily of women and children, allowing more time to be dedicated to other activities. The programme can also contribute towards increased financial inclusion and access to the formal banking system through usage of the PAYGo financing structure to provide OGS products, allowing people to build their credit histories.
22. The programme targets the creation of around 14,000 additional green jobs as a co-benefit. An estimated 4,600 of these jobs could be a source of employment for women. However, at least some of these jobs might not be additional ones but substitutes for jobs held before (e.g. vendors of kerosene lanterns).
23. The programme will target the industry's benchmark of having 32 per cent of its customers as women and will aim to support women-led businesses. The programme also intends to engage companies and the OGS ecosystem in developing more opportunities for women to work. The programme will track gender-disaggregated data for multiple indicators, allowing for the generation of knowledge on the engagement of women with energy products and services useful throughout the OGS industry.
24. Overall, the sustainable development potential of the programme is considered medium to high. It can make critical steps to improve the livelihoods of the poorest in a sustainable manner. Tackling the currency risk could strongly improve the sustainability potential of the programme.

1.5 Needs of the recipient

Scale: High

25. Across the 16 African countries included, there are around 54 million beneficiaries that do not have access to modern electricity services – representing roughly just under 10 per cent of the total unelectrified population of sub-Saharan Africa. On average, 78 per cent of the population in these countries lives below the lower-middle-income poverty line. They are residing in acutely climate vulnerable contexts in terms of worsening heat stress, more frequent droughts, more intense floods, and extreme weather events. Nearly half of the bottom twenty countries listed in the Notre Dame Global Adaptation Initiative (ND-GAIN) climate vulnerability index³ are programme countries. Regional vulnerability to the increasing climate hazards is exacerbated by non-climate drivers such as high poverty levels, weak infrastructure systems, lack of emergency response capabilities, and fragile grid electricity and telecommunications networks.
26. At country and company level, limited investment can currently be attracted to the market targeted by the programme. Eighty per cent of the capital raised by OGS companies between 2012–2019 has flowed to larger and more established, first-generation companies whose primary location and focus is East Africa, with Kenya attracting the lion's share of the capital. OGS start-ups also focused on entrenching their market position and attracting higher margin customers, leaving out the rural, poorer markets of LDCs with a higher actual and perceived risk.
27. International financing going to developing economies in support of clean energy focuses within sub-Saharan Africa on now-proven markets, and risks leaving other countries and regions behind. Political instability, macroeconomic uncertainty, policy and regulatory issues, institutional weaknesses, and lack of transparency contribute to a less favourable

³ The [ND-GAIN Country Index](#) summarizes a country's vulnerability to climate change and other global challenges in combination with its readiness to improve resilience. Higher index scores are given to less climate vulnerable countries.

investment climate in targeted countries. Underinvested geographies are in need of capital willing to take risks in untested OGS markets. The International Energy Agency states that the affordability of clean energy transitions will depend on reducing the cost and improving the availability of capital.⁴

28. The programme's market focus and instruments used incentivize investment in the areas and countries with the most urgent financing needs. GCF funding is critical to enable and mobilize private sector investment. GCF investment is in line with its core purpose of enabling and catalysing investment in critical and risky areas with a high need and demand where private capital would otherwise not flow. The iTAP therefore considers the needs of the recipient as high.

1.6 Country ownership

Scale: Medium

29. The prioritization of the energy sector in most nationally determined contributions (NDCs) is likely to reflect both its relevance as a large emission source and the availability of various cost-effective mitigation measures. All developing countries included mitigation measures within the energy sector in their NDCs. The programme is also aligned with national electrification strategies and national adaptation plans.

30. No-objection letters have been received from the 16 programme countries: Benin, Burkina Faso, Burundi, Chad, the Democratic Republic of the Congo, Guinea, Guinea Bissau, Lesotho, Malawi, Mozambique, the Niger, Sierra Leone, Somalia, Togo, Uganda, and Zambia.

31. The programme has developed a stakeholder engagement plan and has also realized initial stakeholder engagement meetings. The programme plans to engage with stakeholders, especially national designated authorities, at regular intervals though iTAP notes that a range of other public actors would also be important to engage, including ministries related to energy and trade and industry, and regulatory agencies. To date, the programme has engaged with over 90 different stakeholders across 28 countries, representing a variety of stakeholder groups. In early 2023, Open Capital Advisors, on behalf of the programme, carried out five in-country visits to five key programme markets to meet with relevant stakeholders and gain market insights. Senior members of the programme team joined Open Capital Advisors for three of the trips.

32. The country ownership is assessed by iTAP as medium. The programme aligns with the countries' NDCs and national adaptation plans. However, on-ground stakeholder engagement in the 16 countries has so far been limited. Stakeholder and country participation and involvement is thus to a significant extent a planned and hitherto unrealized activity.

1.7 Efficiency and effectiveness

Scale: High

33. The programme intends to leverage additional capital of factor 3 relative to the GCF contribution for the overall MEF, and factor 4 including the MSF and the MEF. Government agencies, foundations, DFIs, governments and even commercial investors have indicated interest in layering their capital with first-loss grant capital. If primarily government agencies, foundations and DFIs invest, the co-investment from the private sector will, however, presumably be fairly limited.

34. The total investment amount shall exceed by factor 3 the initial project investment amount, based on anticipated reinvestments.

⁴ International Energy Agency. (2021). Financing Clean Energy Transitions in Emerging and Developing Economies. Paris: International Energy Agency.

35. The USD 250 million programme (for the MSF and the MEF) targets the provision of entry level electricity access to 72 million people (of which 54 million access for the first-time electricity) at a cost of around USD 3.5 per person, which is considered to be low-cost.
36. The estimated GCF cost per t CO₂ eq avoided, considering the 50 per cent assigned GCF financing to mitigation, is around USD 6, which is considered low. The GCF funding generates significant adaptation benefits by providing first-time energy access and strengthening the climate resilience of 54 million people.
37. Acumen has successful and broad experience with similar initiatives, also with the GCF (i.e. KawiSafi Ventures Fund FP005 and Participation in Energy Access Relief Facility FP148). The capacity of Acumen to set up, oversee and manage such funds is deemed as high. The MSF funds shall be transferred to a newly formed limited liability company, which will be a pass-through entity wholly owned by the accredited entity (AE). The MEF is expected to be formed as a general partnership with the AE which is expected to indirectly wholly own the general partner.
38. The overall iTAP assessment concerning efficiency and effectiveness is high as the programme can achieve significant mitigation and adaptation impact at a low cost while crowding in potentially some, albeit limited, additional private finance.

II. Overall remarks from the independent Technical Advisory Panel

39. Based on the assessment above, iTAP endorses this funding proposal.
40. The iTAP notes that the programme is entering a potentially high-risk market and that, in the learning stage, various ventures will probably fail. However, the approach proposed, and the measures taken, as well as the experience of the AE in this type of venture, ensure that risks are managed well while still striving to reach people underserved by traditional private sector approaches. The role of GCF is to provide risk finance to push the borders beyond business as usual and to reach people not covered by traditional approaches through an effective approach with a high impact and replication potential. Therefore, the programme ticks all the boxes of the investment criteria set forward by GCF.
41. The iTAP recommends that the AE strengthen the logical framework to be able to track progress towards adaptation outcomes 3 and 4. The user surveys planned to track the number of beneficiaries with enhanced climate-resilient livelihood options (outcome 5) can also include questions to track enhanced access to climate-relevant information, or to communication media as a proxy for that (outcome 4), as well as the durability/portability of the OGS systems in the face of any extreme events experienced (outcome 3).
42. The iTAP recommends that the GCF concessionary capital be utilized to take local currency basket options or other hedging products at the MEF fund level rather than push such a risk onto OGS companies, or worse, onto end customers. The hedging products may be expensive for the MEF, but they will certainly be expensive for the OGS companies and their end customers who will bear that volatility (which must be borne by someone). Such a hedge may ultimately also address concerns from potential limited partners on the financial sustainability of MEF on-lending. Either way, the main goal of blended finance is to crowd in commercial capital, but it is equally important to utilize concessional capital to pay for “insurance” products, which minimize risk for the companies and customers operating in hardest-to-reach markets.

Reply to the Independent Technical Advisory Panel assessment findings

Proposal name: Hardest-to-Reach

Accredited entity: Acumen Fund

Impact potential

Acumen is pleased to note that the impact potential is considered 'high'. The commentary captures the cross-cutting scope and we note the emphasis on the adaptation and resilience impacts (#6 and #7), for which there is a pressing need in low income communities where this project will reach. As referenced, the 54 million adaptation beneficiaries are people achieving first-time access to electricity (around 15 per cent of the total population in the participating countries), and many will receive entry level solar home systems, via a pay as you go plan, to match their ability to pay.

Monitoring the adaptation outcomes of the program is a core consideration of the program design. Acumen's experience of incorporating user surveys has shown their effectiveness in capturing a range of impact dimensions, including climate resilience and adaptation metrics, and gender disaggregated impact. Survey questions can reveal patterns of household access to climate-relevant information through OGS-powered communication media, at baseline and endline, in terms of both objective and/or quantitative [e.g., point(s) of access, frequency of access, information subject matter] and subjective and/or qualitative [e.g., perceived ease of access, perceived usefulness of access] metrics. Tracking the frequency of access to climate information, for example, is a specific and measurable indicator to inform outcome 4, and we are considering additional monitoring methodologies to add depth to this picture, as well as verification capacity. In terms of Outcome 3, resilient energy access, through durability in the face of extreme events and portability in the case of displacement, in addition to event-responsive customer surveys, in the proximate aftermath of extreme climate events, our discussions are ongoing with providers of technologies able to capture and track location and usage data of off grid solar products that will enable accurate monitoring of energy consumption patterns during and after a climate event. Acumen recognizes the opportunity to apply innovative methods to monitor and verify climate resilience impact, with a possible demonstration effect for the other energy programs, with broad positive impact across the sector.

Paradigm shift potential

Acumen notes the iTAP's balanced and helpful assessment of the program's paradigm shift potential. As highlighted in the commentary, H2R has the potential to play a pivotal role in attracting investment into hard-to-reach markets where capital has not traditionally flowed, with the help of a blended finance structure which effectively mitigates the real and perceived risks associated with investing in these nascent markets. Overall, we are also positive about the multiplication effects that can be achieved through this approach, including through the knowledge sharing component.

Acumen is paying close attention to the various risks which could undermine the impact of the program. The sector's lessons from the shock of the COVID pandemic should be seen as a source of resilience for the sector in the medium and long-term. Off-grid solar companies faced challenges such as higher costs of raw materials and decreased incomes among their predominantly rural and economically disadvantaged customer base. As noted, there is still a need to address the fragility of OGS companies and the risks they face from external shocks, particularly to protect customers, as well as H2R investments, and our ability to achieve the desired volume of reinvestment. As mentioned, beyond H2R's investments, a dedicated technical assistance facility will support companies to build up their capacity and long-term viability. TA areas include consumer protection, gender, enterprise development and ESG.

Acumen is aligned with the view that broad stakeholder engagement is needed to enhance the potential for transformational impact. Acumen is a member of the REACH partnership, a newly formed group of leading energy access practitioners and a knowledge sharing platform, with expertise in areas including capacity building – including technical assistance – of all stakeholders (companies, financiers, and governments) and policy and regulatory support to create enabling environment.

Sustainable development potential

The iTAP's 'medium-to-high' rating of the program's sustainable development impact is noted. As the commentary suggests, there are interlinkages between clean energy access, economic development (income generation/diversification, savings, job creation), educational outcomes, climate goals, and health and wellbeing. The guidance to address currency risk is noted. Looking across the sector, we also note that the viability of different local currency lending approaches is being explored and the H2R approach will benefit from any lessons being shared.

Needs of the recipient

Acumen welcomes the iTAP's positive assessment of the alignment between the H2R intervention and the needs of the recipient. As outlined in the funding proposal, H2R has been designed in response to the clear needs of the rural populations where energy access is lowest, addressing the urgent financing needs of the identified countries, and enhancing climate resilience through energy access in least developed countries and to some of the most climate vulnerable communities in the world.

Country ownership

Acumen notes the 'moderate' rating for the level of country ownership and in particular the guidance for broad and in-country engagement. The program aligns with the countries' NDCs and national adaptation plans, and there is alignment with government electrification targets. In-country engagement is ongoing and Senior members of the program team are regularly in the region, enabling engagement across the spectrum of stakeholders.

Efficiency and effectiveness

Acumen notes the iTAP's positive assessment of the program's efficiency and effectiveness, and in particular Acumen's track record establishing and overseeing programs of this nature. As highlighted, with the support of the GCF funding, H2R can generate significant adaptation benefits by providing first-time energy access and strengthening the climate resilience of 54 million people.

Overall remarks from the independent Technical Advisory Panel:

Acumen notes the overall positive remarks from the iTAP on this program, along with recognition of Acumen's track record, in particular to deliver the impact while managing the risks of the proposed program.

Acumen also notes the clear mission alignment between H2R and GCF, to deliver a break from a business-as-usual electrification trajectory and to reach people not covered by traditional approaches of investment.

The guidance on the logframe is noted and aligned with the approach we are developing, as is the feedback to address currency risk, potentially by taking local currency basket options or other hedging products.

Hardest-to-Reach Gender Assessment

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Introduction

This document provides an assessment of the barriers and opportunities for advancing gender inclusion in Hardest to Reach (“H2R”) initiative countries. The assessment includes an overview of the evidence on women’s access to clean energy technologies, women’s participation in the off-grid solar (“OGS”) sector workforce, and the key barriers to advancing equality in the sector. This document outlines key entry points for the H2R initiative, followed by detailed in-country assessments for the sixteen target H2R countries.

Barriers to gender equality in the OGS sector

The link between energy access and gender equality is well recognized. Energy poverty has a strong gender dimension as the lack of access to energy inhibits women’s well-being, and their access to economic opportunities, and it strongly affects women’s living conditions and time-use. Off-grid energy technologies have a direct benefit to women’s health, particularly in terms of reduced risks of pulmonary and respiratory diseases. Women’s participation and leadership in the OGS sector can help accelerate the transition to a low-carbon economy by promoting clean energy use and addressing energy poverty, while also advancing women’s economic empowerment. However, multiple barriers limit women’s ability to access OGS technologies and to fully participate in OGS value chains as business leaders, employees and entrepreneurs.

One of the key barriers in the market relates to **lack of sex-disaggregated data for the OGS sector both at the consumer and workforce levels**. The limited sex-disaggregated data available about customer segments and demand for products leads to missed opportunities to reach women customers and meet their needs. In addition, the lack of data on women-owned enterprises and the potential for productive uses of energy prevents increasing the number of women customers and the chance to design business strategies and projects that could transform their economic opportunities.

Historical inequities of distribution and control of assets impact women’s ability to access OGS products, and affordability is a well-recognized barrier to accessing OGS technologies (for both women and men). In many countries, women face discriminatory property and tenure laws, resulting in women’s lower access to financing, to mobile phones and mobile money.

Gender and social norms limit women’s access to technologies and their abilities to enter the workforce. Intrahousehold decision-making and mobility constraints inhibit women’s ability to be aware of energy products and to purchase them. Women may be discouraged from seeking employment opportunities as sales agents and technicians because in some contexts these may not be considered suitable opportunities for them, or because they are discouraged from interacting with men due to social and gender norms.

Companies have a limited understanding of the business case for gender inclusion in their businesses, and of how to make their practices more inclusive. OGS companies may not strive to reach women customers or to hire women because they are unaware of the positive

impacts on their business performance, and they may see this as an extra cost rather than a strategy to meet profitability and business goals. Further, companies' policies often do not recognize gender equality as a priority and few companies have equal opportunities policies or are intentional in addressing gender equality in their strategies to attract, recruit, and promote women at all levels. Even when companies have an explicit gender strategy or they intentionally strive to make their practices more inclusive, they face challenges in attracting women workers, designing products and services for female customers and women entrepreneurs, and ultimately attracting gender-lens investments.¹

The sector lacks appropriate and dedicated training and mentorship opportunities for women to enter and thrive in the workforce. Because of prevailing gender norms, women may be excluded from educational opportunities and may not be considered suitable for employment opportunities in technical roles, as well as sales and distribution roles. Further, male-dominated workplaces are not attractive to women and pose additional risks to gender-based violence and sexual harassment.

Serving women with OGS technologies

Overall, evidence shows that women have less access to off-grid electricity than their male counterparts, and they mostly use pico-solar products and lanterns or other lower-tier technologies.² SHS customers are predominantly rural men living under the median poverty line of \$3.10.³ In a survey conducted on 35,000 customers of 49 OGS companies in 17 countries, 60 Decibels found that only 32% of customers are women.⁴

An analysis conducted by Value for Women on 59 companies working in the energy sector found that only 23% of companies collect sex-disaggregated data on customers, and half of them reported that they serve 46% of customers who are women. A small group of companies (6) also collect data on end-users, and they reported serving 60% women among their end-users.⁵ In another study, GOGLA (the global association for the off-grid solar energy industry) found that women buy solar lanterns and smaller products mostly on a cash basis. In a survey conducted with 37 OGS companies, ESMAP and GOGLA found that about 40% of companies collect sex-disaggregated data on customers, and those who do report that 45% of their customers are women.⁶ Another report indicates that the share of women in PAYGo solar customers is as low as 25%.⁷ Based on the limited and fragmented available data, it can be concluded that, on

¹ Value for Women analysis of survey with 56 clean energy sector companies

² ESMAP (2022) Gender equality in the off-grid solar sector. A primer. The World Bank Group, Washington DC 2022

³ GOGLA (2021) Powering opportunity: Energising work, enterprise and quality of life with off-grid solar.

⁴ 60 Decibels (2020) Why Off Grid Energy Matters. <https://60decibels.com/insights/why-off-grid-energy-matters/>

⁵ Value for Women sectoral analysis of energy companies who completed Nexus gender inclusion self-assessment report

⁶ ESMAP and GOGLA (2022). Gender Equality in the Off-Grid Solar Sector. Gender Survey early findings. <https://www.lightingglobal.org/wp-content/uploads/2022/10/FINAL-Gender-Survey-Early-Findings-29062022.pdf>

⁷ FinEquity: Applying a gender lens to PAYGo solar.

https://www.findevgateway.org/sites/default/files/users/user1456/Webinar%20Deck_PAYGo%20%26%20Gender_To%20Present.pdf

average, OGS and clean energy companies serve about 37% of women in their customer base (see Table 1).

Percentage of off-grid energy customers who are women	Source
32% ⁸	60 Decibels (2020)
46%	Value for Women analysis (2022)
45%	GOGLA (2022)
25% (for PAYGo solar)	FinEquity (2022)

Table 1: Available data on women served by clean energy and off-grid solar companies

These studies indicate that the sector overall lacks sex-disaggregated data, resulting in a limited understanding of gender-related barriers to access and limited knowledge on the impact that these technologies have on women's lives. The limited knowledge about customer segments and women's demand for OGS products leads to missed opportunities to reach women's customers. In addition, men are the main purchasers of energy products, even when women are the primary end-users; this results in skewed consumer data and products that do not always reflect the needs or preferences of women.

In H2R country contexts, prevalent gender norms are likely to affect access to off-grid solar technologies, influencing companies' abilities to reach women. Discriminatory gender norms inhibit women's rights to own land and other assets in most countries, making it challenging for women to access credit and purchase renewable energy technologies. PAYGo companies may reach a lower percentage of women customers due to additional barriers related to mobile ownership and mobile money usage among women (see Table 2). Overall, in Sub-Saharan Africa, the gender gap in mobile ownership is 30%. This gap has grown from 22% in 2017 since ownership of smartphones among men in the region has steadily outpaced that of women. Further, the gender gap in mobile internet usage in the region stands at 37%.⁹ For the countries for which sex-disaggregated data is available (e.g. Uganda and Zambia), the low rates of access to electricity in H2R contexts are even more pronounced in female-headed households.

Country	Women's mobile ownership	Women (age 15+) with mobile money accounts ¹⁰
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⁸60 Decibels benchmark is calculated from interviews with 35,000 customers served by 49 companies. Vfw, GOGLA and FinEquity benchmarks are instead calculated based on companies' reported rates of women customers. It can be argued that companies that collect sex-disaggregated data on customers may be able to serve more women customers, hence the higher % reported.

⁹ [GSMA \(2022\). The Mobile Gender Gap Report 2022](#)

¹⁰ [The World Bank \(2021\) The Global Findex Database](#)

Democratic Republic of Congo	NA	14% ¹¹
Uganda	64.6% ¹²	53%
Benin	51.1% ¹³	30%
Zambia	53% ¹⁴	39%
Mozambique	35% ¹⁵	21%
Sierra Leone	43% ¹⁶	17%
Guinea	68.9% ¹⁷	17%
Burundi	23.6% ¹⁸	1% ¹⁹
Togo	NA	32%
Malawi	33% ²⁰	30%
Lesotho	86.9% ²¹	31% ²²
Burkina Faso	33% ²³	19%
Guinea-Bissau	60.7%	NA
Niger	NA	5% ²⁴
Chad	NA	15% ²⁵

¹¹Findex Data from the 2017 round

¹²[Uganda Bureau of Statistics \(UBOS\) \(2021\). Uganda National Household Survey 2019/2020. Kampala, Uganda; UBOS](#)

¹³[Institut National de la Statistique et de l'Analyse Économique \(INSAE\) et ICF \(2019\). Enquête Démographique et de Santé au Bénin, 2017-2018. Cotonou, Bénin et Rockville, Maryland, USA : INSAE et ICF](#)

¹⁴[Zambia Statistics Agency, Ministry of Health \(MOH\) Zambia, and ICF \(2019\). Zambia Demographic and Health Survey 2018. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, Ministry of Health, and ICF](#)

¹⁵[INSTAT Madagascar \(2021\) Enquête Démographique de Santé](#)

¹⁶[Statistics Sierra Leone \(Stats SL\) and ICF. 2020. Sierra Leone Demographic and Health Survey 2019. Freetown, Sierra Leone, and Rockville, Maryland, USA: Stats SL and ICF.](#)

¹⁷[Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

¹⁸[Ministère à la Présidence chargé de la Bonne Gouvernance et du Plan \[Burundi\] \(MPBGP\), Ministère de la Santé Publique et de la Lutte contre le Sida \[Burundi\] \(MSPLS\), Institut de Statistiques et d'Études Économiques du Burundi \(ISTEEBU\), et ICF. 2017. Troisième Enquête Démographique et de Santé. Bujumbura, Burundi : ISTEEBU, MSPLS, et ICF.](#)

¹⁹Findex Data from the 2014 round

²⁰[National Statistical Office \(NSO\) \[Malawi\] and ICF. 2017. Malawi Demographic and Health Survey 2015-16. Zomba, Malawi, and Rockville, Maryland, USA. NSO and ICF.](#)

²¹[Bureau of Statistics \(2019\). Lesotho Multiple Indicator Cluster Survey 2018: Survey Findings Report. Maseru, Lesotho: Bureau of Statistics](#)

²²Findex Data from the 2017 round

²³[GSMA \(2017\). Accelerating affordable smartphone ownership in emerging markets](#)

²⁴Findex Data from the 2017 round

²⁵Findex Data from the 2017 round

Somalia	84.5% ²⁶	34% ²⁷
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Table 2: Country-specific mobile ownership data

African Indigenous People, which can be defined as “nomadic and semi-nomadic pastoralists and hunter/gatherers who live in situations of marginalization and discrimination,”²⁸ are faced with additional challenges in accessing resources and opportunities. In H2R countries, indigenous communities are hunter-gatherers, pastoralists and nomadic populations often without state recognition. They heavily rely on nature and land for their livelihood and are often located in forests and conservations areas, facing eviction from their land. Further, African indigenous people are generally poorer and lack access to education and health. Indigenous women face additional challenges such as sexual violence (e.g. Pygmies in DRC). In West Africa, political instability and conflicts disproportionately impact women and female-headed households. Women and girls make up a large proportion of internally displaced populations, they are victims of systemic attacks and kidnappings, and are forced into slavery.²⁹

In the context of a changing climate, indigenous communities and minority groups whose lives depend on the health of their natural environments are increasingly more vulnerable to shocks, while their specific energy needs are often overlooked. Without addressing their needs, the transition to clean energy is hindered in many SSA countries. Empowering Indigenous populations, women, youth, and other traditionally marginalized people is an essential part of protecting rights and creating jobs. In the context of H2R, OGS technologies can provide first access to clean and reliable energy for communities that are usually not included in traditional business models and energy programs. Their additional vulnerabilities (i.e. geographic remoteness, lower access to information and resources, higher poverty rates, discrimination) should be considered when assessing strategies for OGS product dissemination and when they consider hiring in the communities they serve.

In conclusion, it is safe to assume that product reach to women and other vulnerable communities in H2R markets will be slightly lower than average, or compared to countries where the OGS sector is well established (e.g. Kenya). The H2R initiative provides an important opportunity to fill some of the knowledge and data gaps on reaching women as OGS customers by requesting investee companies to collect and track sex-disaggregated data on sales.

Women’s participation in OGS companies’ workforce

Data on women’s participation as employees and leaders in the sector is scarce and fragmented. ESMAP and GOGLA recently surveyed 37 OGS companies from 20 different countries and found that on average women constitute 41.7% of workers across all levels.³⁰ In contrast, a previous

²⁶ [World Bank Gender Data on Somali Women’s Digital Access, 2023](#)

²⁷ Findex Data from the 2014 round

²⁸ [AfDB \(2016\) Development and Indigenous People in Africa](#)

²⁹ [OECD - Women and conflict in West Africa and beyond](#)

³⁰ ESMAP and GOGLA (2022). Gender Equality in the Off-Grid Solar Sector. Gender Survey early findings.

GOGLA study indicated that only 27% of women work in the OGS sector.³¹ An analysis completed by Value for Women found that on average women represent 33% of the workforce of energy companies. Similarly, in the renewable energy sector in general women hold roughly 32% of jobs.³² Another recent study based on 64 surveyed companies operating in SSA found that women represent 37% of the decentralised RE workforce, and that they are more likely to be employed in corporate support positions (administration, communications and sales).³³ Only 13% of women had higher-paying science, technology, engineering, and mathematics (STEM)-related jobs, and 16% held non-STEM, technical positions.

Based on the available market data, it can be concluded that globally women represent about 33% of the clean energy workforce across all levels and this figure can be used as a benchmark for female workforce participation.

Women funder, CEO or director	Women in senior leadership positions or senior management	Women in middle management	Women in other positions, including field staff and agents	Average across all levels	Source
49%	38%	41%	39%	42%	GOGLA survey (2022)
-	-	-	-	27%	GOGLA (2019)
28%	27%	33%	34%	33%	Value for Women analysis (2022)
-	-	31%	45% of women in admin positions	32%	IRENA (2019)
30%	16% (vice presidents and directors)	26%	-	37% for DRE companies	IFC (2022)

Table 3: Available data on women employed by clean energy and off-grid solar companies

Considerable work remains before OGS companies will reach a gender balanced workforce, and this is especially true with respect to increasing the participation of women in leadership and

³¹ [GOGLA \(2019\) Off Grid Solar. A growth engine for jobs](#)

³² IRENA (2019). Renewable Energy: A gender perspective.

³³ [IFC and Energy2Equal \(2022\) Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying barriers and opportunities for women leaders and employees.](#)

management positions.³⁴ The key challenges affecting companies operating in these contexts relate to workplace policies and practices, social norms and gender stereotypes and legal barriers.

The country assessments indicate that several barriers limit women's participation in the workforce and in the economy more generally, with prevailing gender norms, gaps in education levels and discriminatory policies and practices affecting women's productive activities and economic empowerment in several countries. Among H2R countries, only Uganda presents data for Distributed Renewable Energy ("DRE") labour force participation of women, with 27% of women being employed in the sector. Lesotho reports an overall high representation of women in private sector energy companies compared to regional and global averages (42%), however these women are reportedly concentrated in administrative roles rather than technical or managerial roles. For other countries, this data is not available but it is likely to be considerably lower, especially in markets such as Togo, Guinea or DRC where gender norms prevent women from undertaking many productive activities and where the OGS sector is still in its nascent stages. The rates of secondary school enrolment remain low across SSA (36% on average), and of the few women who do attend university, only a small segment completes a STEM degree, which is needed for higher-paid and more senior jobs in OGS companies. Once women enter the job market, they encounter discrimination, including the widespread misconception that women are not suited for technical roles. Companies that intentionally recruit women for these positions face pipeline-related challenges and need to address such gender and social norms.

Women entrepreneurship is also hindered by gender norms and discriminatory policy frameworks. In most of the assessed countries, women are predominantly represented in the informal sector, in some cases they do not have the same rights to register a business or open bank accounts, and they largely lack access to formal financing.

Country	Women's Labour Force Participation Rate (age 15 to 64)	Source
Democratic Republic of Congo	62%	World Bank, 2019
Uganda	68%	World Bank, 2019
Benin	70%	World Bank, 2019
Zambia	71%	World Bank, 2019
Mozambique	78%	World Bank, 2019
Sierra Leone	59%	World Bank, 2019
Guinea	65%	World Bank, 2019

³⁴ Lighting Global/ESMAP, GOGLA, Efficiency For Access, Open Capital Advisors (2022), Off-Grid Solar Market Trends Report 2022: State of the Sector. Washington, DC: World Bank.

Burundi	82%	World Bank, 2019
Togo	56%	World Bank, 2019
Malawi	74%	World Bank, 2019
Lesotho	63%	World Bank, 2019
Burkina Faso	60%	World Bank, 2019
Guinea-Bissau	67%	World Bank, 2019
Niger	62%	World Bank, 2019
Chad	64%	World Bank, 2019
Somalia	23%	World Bank, 2019

Table 4: Women's Labour Force Participation Rate data

The H2R initiative brings opportunities to mandate that companies collect and annually report a sex-disaggregated staff footprint, with data on female versus male employment at all levels. Such data can be also used to showcase the business case for the inclusion of women across all levels of the OGS workforce.

Stakeholder Engagement

As part of the process of understanding the needs and perspectives of women in Hardest-to-Reach geographies, we have engaged with over 25 individuals, notably via a Stakeholder Consultation Forum with GCF Gender Monitors. The session had representation from over 11 sub-Saharan African countries, as well as from specialist gender lens off-grid solar organisations. An example of the latter is Smiling Through Light (STL), a Sierra Leone-based social enterprise which focuses on clean energy access by working with a network of local women to distribute and sell solar products. STL focuses on the last mile which creates employment and income opportunities in underserved areas.

We also launched a customer survey for a pilot investment in Benin where we listened to and learned from 85 women about their experience with OGS products. These women shared their opinions on the value of the products, barriers to purchase, and challenges with the products. Their challenges and hopes mirrored what we heard from gender experts, NGO leaders, and other important stakeholders.

The following insights were gained from our consultation and studies:

- The sector needs more support of women leaders and entrepreneurs:** The view was expressed that, in order to address the challenges and inequities women face in energy access, it would be impactful to have more women in leadership within the sector. This could be achieved by investors backing more women-led businesses and by the provision of training to uplift women in the workforce into positions of influence and leadership. Our gender lens strategy is aligned with the 2X challenge to increase women participation in the OGS sector. Additionally, we expect to provide training for women to be sales agents and technicians in OGS as a part of our technical assistance.
- Awareness of off grid solar among women in some areas is still low:** This is particularly the case in very remote areas, where there are no distributors serving the locale. According to our survey, women were slightly more likely to gain first time access to electricity with OGS and have fewer OGS alternatives than men. As a result of our consultations, conversations with gender experts and surveying of female OGS customers, we are supporting several companies with technical assistance generating gender-inclusive marketing strategies.
- Off grid solar products are very important as entry level electricity in rural areas:** Hardest-to-Reach initiative is delivering a product which women in rural areas will benefit from, and will value if they are made aware of the various products and uses. 80% of women in our survey shared their lives improved with access to electricity. 50% said that their lives very much improved with their OGS product.
- There are grassroots women-led initiatives which are seeking NGO and donor support:** The consultation shone a light on some worthy initiatives which women are engaged in at the grassroots level. While direct support of these is outside the scope of the H2R initiative, we are hopeful the H2R's uplift of the OGS sector can have a positive knock-on effect, in particular by raising awareness of OGS.

Centre for 21st Century Issues

ACUMEN FUND

GREEN CLIMATE FUND

STAKEHOLDERS' CONSULTATION WITH AFRICA GCF GENDER MONITORS BY ACUMEN FUND INC.

THEME: **ENERGY ACCESS FOR THE HARDEST-TO-REACH (H2R)**
A NEW PROJECT BY ACUMEN FUND TO BE CONSIDERED FOR APPROVAL BY THE GCF BOARD

Tuesday 18th April, 2023 | 12pm GMT | Virtual Meeting

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Key entry points for the H2R initiative

1) Elevate women in the OGS sector as customers, employees and entrepreneurs. The country assessments indicate gender gaps in access to energy and employment opportunities, and a comprehensive programmatic approach is necessary to create opportunities for women at all levels.

2) Do no harm from a gender perspective. This entails evaluating potential investee companies' risk identification and mitigation with respect to negative impacts on women as part of its Environmental, Social and Governance standards. H2R's Environmental and Social Risk Management Plan will establish gender-specific screening criteria.

3) Provide clean and affordable energy to women customers and women-headed households. The country assessments and the global benchmarks indicate that women are predominantly affected by energy poverty and that poor and vulnerable households are mostly women-headed.³⁵ The H2R initiative can impact women's lives by requiring investees to collect and track sex-disaggregated data on customers reached and by supporting companies to implement strategies to better serve women, ultimately reducing gender gaps in access to clean energy technologies.

4) Track investments in women-led businesses and in businesses that have gender-forward business practices. Investors have a clear role to play in applying a gender lens to the investment process and to inform investment decisions by intentionally examining the gender balance of companies, their commitment to equality, and by measuring the progress of investees in reaching women customers. The H2R initiative can encourage companies to recruit, retain and advance women at all levels of the OGS value chain by integrating a gender lens in due diligence, deal structuring and in post-investment support. Companies that are intentional in addressing gender gaps in access to OGS technologies and opportunities can apply business practices that build customer trust and loyalty, strengthen women's capacity to enter and advance in the sector, and at the same time reach women customers.

5) Positively impact organisational culture by supporting portfolio companies to adopt and/or expand gender-inclusive policies and business practices. To increase the attraction, retention, and advancement of women in the OGS workforce, companies need to have in place inclusive workplace policies and practices. At a minimum, companies should be expected to have anti-harassment, anti-discrimination, and wage parity policies, which are considered necessary requirements to operate in H2R contexts where wage disparities persist and where gender norms and gender-based violence ("GBV") are inhibiting women's economic empowerment. Other policies and practices such as flexible working arrangements, parental leave policies, flexible return to work after parental leave, etc. are necessary to create women-friendly workplaces and to attract female talent. Further, companies can apply a gender lens to other areas of their business, particularly in product design and market research and customer engagement areas (sales, distribution, after-sales), as well as their supply chains. Such practices will enable

³⁵ [Energy Sector Management Assistance Program. 2022. Gender Equality in the Off-Grid Solar Sector. Washington, DC](#)

companies to understand women's needs and to better serve women customers, ultimately meeting sales targets and contributing to H2R's program impact.

The H2R initiative will mandate companies to assess the level of gender inclusion in their business, and support them in developing actions and strategies to become more inclusive. The gender inclusion assessment carried out during due diligence and the GAP developed within 90 days of investment (see Gender Action Plan for details) will provide detailed information about which policies companies are expected to develop or update, as well as the steps to increase women's participation in the workforce and strengthen the pipeline of women candidates. Evidence shows that companies that undertake actions to close gender gaps and that integrate intentionality in their strategies and business processes have been successful in attracting further financing.³⁶ Such an approach will contribute to H2R's gender equality objectives and it will help mobilize finance to enter these challenging contexts and to reach underserved markets.

6) Provide dedicated training and employment opportunities. At the ecosystem level, the H2R initiative will address the lack of training and mentorship opportunities by supporting women-dedicated training programs that aim at closing skills gaps. The H2R initiative should also focus on establishing partnerships between training institutions, the private sector and recruitment actors to increase awareness of job opportunities and to foster the creation of internships and mentorship programs. The initiative can also enhance collaboration by convening stakeholders such as gender lens investors, women's associations, women entrepreneurs, and other key local actors in engagement workshops to work together to improve the operating environment for women in OGS markets.

7) Facilitate OGS sector transformation. Acumen has a track record of building ecosystems with a gender lens. In H2R, Acumen will use insights to build the case for the benefits of investing in hardest-to-reach markets and taking a gender lens to OGS sector facilitation. There is a tremendous opportunity to gather insights on the strategies companies will adopt to close gender gaps in the OGS sector, from reaching customers to hiring women in their workforce, and disseminating lessons learned to the wider sector.

Country-level assessments

The assessment of the sixteen H2R countries identified context-specific challenges for advancing women's participation in the OGS sector. These countries present some of the lowest electricity access rates and some of the highest poverty rates. Disparities between men and women persist in the areas of economic opportunities, health, education, political participation, and empowerment. Gender norms and customary laws shape women's lives and their ability to access assets and resources and to participate in economic activities.

³⁶ ESMAP (2022) Gender Equality in the Off-Grid Solar Sector. A primer. The World Bank
 USAID (2019) Power Africa Case Study Ghana: Advancing Gender Equality in Africa Off Grid Energy Sector
https://www.usaid.gov/sites/default/files/documents/1860/PA_Case_Study.2019.04.09.508.pdf

Country	Gender Development Index	Gender Inequality Index	Electrification rate (men and women) ³⁷	% Population living below poverty line (\$2.15 a day) (World Bank)
DRC	0.885	0.601	19%	69.7% (2012)
Uganda	0.927	0.530	56%	42.2% (2019)
Benin	0.880	0.602	41%	19.9% (2018)
Zambia	0.965	0.540	32.8%	61.4% (2015)
Mozambique	0.922	0.537	30.6%	64.6% (2014)
Sierra Leone	0.893	0.633	26.2%	26% (2018)
Guinea	0.850	0.621	44%	13.8% (2018)
Burundi	0.935	0.505	11.7%	65.1% (2013)
Togo	0.849	0.580	54%	28% (2018)
Malawi	0.968	0.554	14.9%	70% (2019)
Lesotho	0.985	0.557	43.8%	32.4% (2017)
Burkina Faso	0.903	0.621	19%	30.5% (2018)
Guinea-Bissau	0.867	0.627	35.7%	21.7% (2018)
Chad	0.77	0.652	11%	30.9% (2019)
Somalia	n.a.	n.a.	18%	70.7% (2017)

Table 5: Overview of gender equality and energy access situation in H2R countries

Countries are at different stages of OGS market development, and in some contexts, like Uganda and Mozambique, several programs have been implemented over the last years with the intent of reaching women customers and of increasing participation of women as sales agents, technicians and employees in the sector.

Country	Programme description ³⁸
Uganda	Uganda has a 'Women Renewable Energy Association of Uganda' which aims to build the technical capacity of women in the field of solar, biomass, and hydro energies, organizing mentorship programs, and engaging with communities on the

³⁷ Citations are mentioned in detailed country-level assessments

³⁸ Citations are mentioned in detailed country-level assessments

	adoption of renewable energy. • The country currently has a program to Operationalize the National Content policy to enhance local Content and participation of nationals including women and youth in the oil and gas sector and is planning to launch a project on Climate Resilient Livelihood Opportunities for Women's Economic Empowerment (CRWEE) in Karamoja and West Nile Regions of Uganda (October 2018 - 30 September 2030)
Benin	ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE) is implementing a project entitled: Promoting Agricultural Competitiveness through Solar Energy (PAC-SOLAR) in Benin with the aim to enhance women's access to renewable energy. As part of this project, women have also been trained and charged with the job of cleaning the solar panels regularly, and maintain solar PV installations
Zambia	The Ministry of Energy conducted a Gender Assessment of the Energy Sector in Zambia in 2018, and is in the process of developing a Gender Mainstreaming Strategy and Action Plan. Zambia's Energy Regulatory Board (ERB) has also committed to participate in the Women in Energy Regulation Internship Program, organized by the National Association of Regulatory Utility Commissioners (NARUC) with funding support from the U.S. Agency for International Development (USAID) and Power Africa to provide education and training opportunities and foster a gender-equitable workforce in the energy sector
Mozambique	• The Mozambique Women Of Energy that is a platform for women in the energy sector who identify areas of action for gender equality in this sector • BRILHO programme, an SNV - Netherlands Development Organisation program in partnership with TechnoServe that aims to increase the number of women with access to modern energy solutions in off-grid communities and enhance the inclusion and empowerment of women in the sector as collaborators and leading businesses. They do this through technical training of companies on gender-forward company policies and practices
Sierra Leone	Sierra Leone is a part of ECOWAS and is implementing their policies for gender mainstreaming in Energy Access. Under ECOWAS, The Ministry of Energy is also responsible for supervising the Barefoot women's solar training. The Barefoot Solar Engineering Training Centre has been established to undertake rural electrification projects and to provide solar training to illiterate women. Between 2014 to 2017, 50 solar women technicians were trained under this program and solar PV systems were installed in 3,500 Households. The Barefoot women have also provided solar connections to 60 community clinics in northern Sierra Leone, which earlier lacked continuous electricity, putting lives at risk, especially making childbirth risky at these clinics.
Guinea	• Women's Fund for Gender Equality (FGE) developed a training program on renewable energy technologies to promote sustainable livelihoods for women and in partnership with a local grantee organization Partenariat - Recherches - Environnement - Médias (PREM), trained 300 women in Guinea's Tristao Islands how to plant, grow, harvest, and sell products made from moringa plants using solar

	polytunnel dryers. This resulted in 25,000 moringa trees being planted and greenhouse gas emissions being reduced by 40 tonnes as a result of solar drying technology, between 2013 to 2016 • Currently, there is an ongoing Research Project on Energy Transition for Women's Economic Empowerment through the Horticultural Value Chain in Guinea, to understand and address mechanisms for use of solar irrigation systems in the horticultural value chain, for irrigation, processing, and preservation. The project which aims to improve the profitability and working time of rural women by 2025, is also building the capacity of women to use solar technologies and access finance. The program is being implemented by the <i>Initiative prospective agricole et rurale</i> (IPAR) in collaboration with the <i>Centre d'étude et de coopération internationale</i> (CECI) and is funded by the International Development Research Centre (IDRC)
Burundi	• The Lumière program, implemented by local NGO FVS-Amade in partnership with UNICEF since 2015, works with Village Savings and Loan Associations (VSLAs) to serve the last mile. UNICEF subsidizes around 60% to 70% of the purchase price, the purchase of solar lanterns by the NGO FVSAmade, which sells them either directly to households or to VSLAs. The program operates in the provinces of Makamba and Rutana. • In 2020, the Republic of Burundi received a grant amounting to USD 100 million from the World Bank for the Solar Energy in Local Communities (SOLEIL) project. The project's Technical Assistance facility has a significant gender focus ○ <u>Scholarship and Mentorship for women's capacity development in the energy sector</u>: Establish a 100% scholarship program for 40 women studying STEM-related courses, supported by the General Director for Energy, who wish to join the energy workforce after graduation. Further, the program aims to provide leadership training and targeted mentorship programs for female talent in the sector ○ <u>Employment</u>: Enabling 30% employment of women in overall jobs through mini-grid development by supporting private sector companies in mapping out relevant employment opportunities; such as during construction, and O&M (i.e. solar panel cleaning and small-scale gardening activities) and also upstream employment across mini-grid company staff footprint through local partners. ○ <u>Entrepreneurship</u>: To foster female entrepreneurship in the sector, 20% of start-up grants are to be earmarked for female-headed or majority female-owned companies. Further, information sessions targeting female business associations (such as Association des Femmes d'Affaires au Burundi) would be hosted and business skills development support would be provided to women entrepreneurs in the sector ○ <u>Women customers</u>: Differentiated subsidies will be provided to companies in order to target female consumers and increase the purchase of off-grid systems (asset ownership) by vulnerable households. Further, women's preferences and affordability constraints will be mapped at the household and enterprise level, followed by an assessment of the positive impact on women as a result of adopting off-grid products (in terms of time savings, drudgery reduction, and livelihoods).

Togo	Entrepreneurs du Monde (EdM) is an initiative supported by the Economic Community of West African States (ECOWAS) and the government of France, created in 1998, that works with people living in fuel poverty by facilitating access to essential goods and services for their families. Based in Togo since 2012, EdM runs the Mivo Energie project that aims to combat global warming and deforestation in Togo by facilitating access to clean, modern, and affordable energy products for vulnerable populations, especially women.
Malawi	FINCOOP is a financial cooperative that provides affordable financial services, serving micro, small and medium entrepreneurs, farmers, and salaried employees. The company's Green Loan Programme targets low-income households across Malawi using multiple channels such as Village Savings and Loan Associations (VSLAs), civil services, and technology partners. FINCOOP has tailored its payment terms to fit women's requirements through the deployment of payment methods such as Pay As You Go models, cash, and group loans, and by offering longer repayment terms. From 2018 to 2020, more than 1212 female-headed households were able to access solar products through this program. ³⁹
Lesotho	Lesotho has launched a Women In Sustainable Energy Lesotho (WISE-L) initiative that aims to promote gender equality and equity in the energy sector and create opportunities for women. ⁴⁰
Burkina Faso	• The GEF-funded Africa Minigrids Program implemented by UNDP seeks to demonstrate productive uses of electricity to reduce costs, increase the viability and enable mini-grid development at scale. The primary targets and actors of the pilot projects will be small-scale farmers (women and youth, mostly) who will benefit directly from the electricity supplied by the solar mini-grids. The project started in 2022 and it is planned to end in 2026.⁴¹ • The GCF-funded Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries⁴² program, is implemented by the International Fund for Agricultural Development (IFAD) and local organizations from the participant countries, including Burkina Faso. The program seeks to provide concessional loans to foster the best adaptation and mitigation practices for green businesses within the selected agricultural value chains in each target country. Operated by local, national agricultural banks (LNABs) – the Agricultural Bank of Burkina Faso– the facility will offer special lines of financing for green business projects prepared by farmer's organizations (FOs), women and youth organizations, cooperatives, and MSMEs (including agribusiness dealers and solar operators). Besides providing loans, the program includes a Technical Assistance Facility (TAF) to address capacity, knowledge, and policy

³⁹ [UK Aid, AECF \(2021\). The impact of financing business models on access to solar home systems and women's financial inclusion: Insights from FINCOOP Savings and Credit Cooperative in Malawi and MoneyMart Finance in Zimbabwe. Retrieved January 4, 2023, from https://www.aecfafrica.org/wp-content/uploads/2021/12/AECF-Womens-financial-inclusion-Summary-Final-Draft-1.pdf](https://www.aecfafrica.org/wp-content/uploads/2021/12/AECF-Womens-financial-inclusion-Summary-Final-Draft-1.pdf)

⁴⁰ [Lesotho: Energy. WISE-L Celebrate Women's Month. \(2022, August 30\). APO Group: Africa Newsroom. Retrieved February 7, 2023, from https://www.africa-newsroom.com/press/lesotho-energy-wisel-celebrate-womens-month?lang=en](https://www.africa-newsroom.com/press/lesotho-energy-wisel-celebrate-womens-month?lang=en)

⁴¹ "Off-Grid Solar Market Assessment Burkina Faso Report" in the framework of the Regional Offgrid Electrification Project (ROGEP) cited in [UNDP & Government of Burkina Faso \(2021\) project document: National Child Project under the GEF Africa Minigrids Program – Burkina Faso](#)

⁴² The countries participating in this program are Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan.

	gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks, and clients (FOs, MSMEs, and cooperatives), with a special emphasis on women and youth. ⁴³
Chad	World Bank project for \$295M - Chad Energy Access Scale Up Project (PAAET) - aims to increase access to electricity and clean cooking solutions via expansion of the main power grid and mini-grids, standalone solar systems, deployment of improved stoves, and natural resource management. ⁴⁴ The project will include targeted interventions to improve access to electricity for female-headed households, including in refugee camps, and female-led enterprises, promote women entrepreneurship, and build awareness relevant to women and girls around energy services. ⁴⁵
Somalia	The Sustainable Charcoal Reduction and Alternative Livelihoods (SCREAL) project aims to reduce dependency on charcoal by promoting sustainable energy alternatives, including fuel-efficient stoves, and creating alternative livelihood opportunities, particularly for women and youth. ⁴⁶ The Africa Minigrids Program (AMP) aims to bring new development opportunities to rural communities and to increase access to electricity for 67K people, half of whom will be women. ⁴⁷

Table 6: Overview of relevant programmes and initiatives dedicated to training women, targeting women entrepreneurs or enhancing access to clean energy

Disaggregated data on women's participation in the renewable and green energy sector is scarce and available only for Uganda. Zambia, on the other hand, is the sole country to present sex-disaggregated household data on Multi-Tier Framework energy access.

The following sections provide an in-depth assessment of each country.

⁴³ [The Inclusive Green Financing Initiative \(IGREENFIN I\): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall \(GGW\) countries](#)

⁴⁴ [World Bank Chad Scales Up Its Access to Energy, 2022](#)

⁴⁵ [World Bank Project Information Document, 2020](#)

⁴⁶ [UNDP Somalia Projects, n.d.](#)

⁴⁷ [UNDP Sustainable Energy Hub, 2023](#)

Democratic Republic of Congo

Overview of gender equality and gender gaps in the country

The Democratic Republic of Congo is one of the biggest countries in Africa, with a population of around 92 million people – 50.1% of which are women – an annual population growth of 3.1%, with 33% of the total population under the age of 15.

The Gender Development Index for the DRC is 0.885, placing women's human development at about 83% of that of men. According to the GDI scoring, the DRC belongs to the group of countries with the lowest human development (countries with low equality in HDI achievements between women and men and absolute deviation from gender parity of more than 10 percent). On the Gender Inequality Index (GII) which measures disparities between men and women in the areas of economic opportunities, reproductive health, and empowerment, the country is ranked among the highest in the world (151).

The DRC also scores low on human development, with women being in worse conditions compared to men (0.449 HDI for women versus men 0.507), and DRC ranks 179th out of 191 countries on the Human Development Index (2019).

Indicator	Result	Source and date
Life expectancy at birth	61.5	UNDP (2022) Gender Development Index
Maternal mortality ratio	473	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	66	World Bank, 2016
Women's Labour Force Participation Rate (age 15 to 64)	62%	World Bank, 2019
Mean years of schooling ⁴⁸	5.6	UNDP (2022) Gender Development Index
Estimated gross income per capita	896 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

The DRC has ratified many UN Conventions relating to equity and inclusion, including the 1969 Convention on Eradication of Racial Discrimination, the 1966 Convention on Civil and Political Rights, and the 1966 Convention on Economic, Social and Cultural Rights (ICESCR). Further, DRC has ratified the Convention on the Elimination of Discrimination Against Women (CEDAW). The 2006 Constitution prohibits discrimination against women in its many forms and emphasizes gender parity in public institutions (Article 14). It also provides through Articles 5, 14, and 15 the foundations of legitimization of any policy of equality and equity in the country. Under the Constitution, women and men have equal rights to work, education, vote, justice, and political participation.

⁴⁸ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

The DRC developed a National Gender Policy (2017-2021), which aims to achieve gender equality and social empowerment through five strategic axes: (i) the consolidation of gender equality and the empowerment of women; (ii) strengthening the role and place of women and girls in the economy and employment; (iii) increased access of women and girls to decision-making spheres; (iv) fight against forms of sexual violence; and (v) strengthening the institutional mechanism for implementation of the National Gender Policy. Despite an overall favorable policy framework for gender equality, implementation of policies remains a challenge, with a lack of appropriate budget, limited awareness, poor coordination, a limited capacity of implementing agencies, as well as constraining societal gender norms.⁴⁹

Political participation

The Constitution forbids discrimination against women (Article 14), but there is no law or policy that ensures women's just representation. The DRC demonstrates only 13% female representation in the National Assembly, 10% of women elected at the provincial level, no female governors in the provinces, and just under 14% women representatives as decision-making authorities for public administration.⁵⁰ Gender norms impact women's political participation, and there is a strong notion that politics are for men and that women often need to follow their husband's opinions.

Social and cultural context

Men dominate both the political and social spheres, and they are the culturally and legally appointed heads of the household, presenting them with decision-making power over most aspects of life. Women are defined by their household or reproductive responsibilities, which include maintaining the physical and moral well-being of their families. Men are perceived to be responsible for the financial health of their families. According to the law, women are required to obey their husbands, and they cannot be head of the household in the same way as men.⁵¹ Further, they do not have the same rights to remarry. Traditional gender roles and social norms exist in both urban and rural areas; however, they appear to be more rigid in rural settings.

Sexual and gender-based violence (SGBV) has been a defining feature of the ongoing conflict in the country and the DRC has one of the highest incidences of rape in the world, and there is no legislation addressing domestic violence. Domestic violence is widespread and is generally accepted as a response to women's alleged failings in the household.⁵² In fact, sexual violence against women and girls has been prevalent, especially by a partner or household member. UN Women reports that 51% of Congolese women experience lifetime physical and/or sexual intimate partner violence, with 37% reporting violence in the last 12 months.⁵³

Household decisions are dominated by men, and the lack of agency within the home and within communities has a direct influence on women's productive activities and their ability to access and control resources. A survey conducted by the World Bank found that men take decisions over a woman's income, health, major household investments, and social visits. Gender norms mean that there is an expectation for men to be the primary household breadwinner, and consequently they are expected to pay for major

⁴⁹ [Agapitova, N., Braunmiller, J., & Dry, M. \(2022, May 26\). *Reforms to enhance gender equality in the Democratic Republic of Congo: From advocacy to implementation*. World Bank Blogs. Retrieved December 1, 2022, from https://blogs.worldbank.org/developmenttalk/reforms-enhance-gender-equality-democratic-republic-congo-advocacy-implementation](https://blogs.worldbank.org/developmenttalk/reforms-enhance-gender-equality-democratic-republic-congo-advocacy-implementation)

⁵⁰ [Women's International League for Peace and Freedom. \(2019\). \(rep.\). Review of the Democratic Republic of the Congo: Submission to the Committee on the Elimination of Discrimination Against Women. Retrieved from https://www.wilpf.org/wp-content/uploads/2019/11/CEDAW_DRC-EN_Web.pdf](https://www.wilpf.org/wp-content/uploads/2019/11/CEDAW_DRC-EN_Web.pdf)

⁵¹ [World Bank Group. \(2022\). \(rep.\). Women, Business and the Law: Congo, Dem. Rep.](https://www.worldbank.org/en/publications/women-business-law/congo-dem-rep)

⁵² [Davis, L., Fabbri, P., & Alphonse, I. M. \(2014\). \(rep.\). *Gender Country Profile: Democratic Republic of Congo - DRC*. Retrieved December 2, 2022, from https://www.lauradavis.eu/wp-content/uploads/2014/07/Gender-Country-Profile-DRC-2014.pdf](https://www.lauradavis.eu/wp-content/uploads/2014/07/Gender-Country-Profile-DRC-2014.pdf)

⁵³ [UN Women, Global Database on Violence against Women](https://www.unwomen.org/en/digital-library/publications/2019/05/global-database-on-violence-against-women)

household expenses. In fact, only 60% of women report participating in decisions about major household purchases.

Indigenous women face additional challenges in DRC. Indigenous Pygmies are traditional hunter gatherers, and thus remaining poor and marginalised with limited access to land rights. Women from these tribes are often subjected to sexual violence by Bantu people, due to being considered 'subhuman'.⁵⁴ With DRC's forest cover being a vast 62% of the country's land, conservation efforts have been made to protect this "largest carbon sink of the world", which include repossession of lands from them under the garb of protecting these forests, which threatens the traditional way of life for these Pygmies and makes indigenous women even more vulnerable due to displacement. Recently laws have been passed toward Promotion and Protection of the Rights of the Indigenous Pygmy Peoples, granting them the right and access to free, prior and informed consent before their land is exploited by government or industries.⁵⁵

Health

A high maternal mortality rate is a major health issue in the DRC (740 per 100,000 live births); it is caused by several factors such as lack of access to medical treatment, early marriage, and high birth rates. Throughout the country, the average age at first birth is 19.9 years, and 37% of women aged 20-24 were married before their eighteenth birthday, as compared to 6% of men in the same age group. Congolese women suffer from poorer health compared to men, with women's health being undermined by men's control over women's bodies, low access to resources, and women's lack of freedom to move.⁵⁶

Education

One of the key drivers of inequality in the country is linked to gender disparities in education, directly impacting women's participation in the economy and wage earnings. Articles 43 and 44 of the Constitution stipulate free obligatory education and the eradication of illiteracy. However, girls often drop out of school due to early marriage and pregnancy, which leads to a gender gap in education.

The average Congolese woman will benefit from 5.3 years of formal schooling compared to 8.4 years for men; as such, only 40% of women have attended secondary school compared to 69 % of men.⁵⁷ Nationally, an estimated 66% of Congolese women are literate compared to 89% of men.⁵⁸ Among young women, literacy is increasing; an estimated 80% of young women are literate compared to 91% of young men. Traditionally, education for women and girls is less valued and households often prioritize education for boys compared to girls.

Women's economic participation and entrepreneurship

Sixty percent of the country's population is under the age of 20, and this percentage will only expand, as DRC is projected to attain a population of 120 million by 2030. More than 60% of women live under the

⁵⁴ [IWGIA \(2012\). Country Technical Notes on Indigenous Peoples' Issues: Democratic Republic Of The Congo. Retrieved 2 March, 2023, from https://www.ifad.org/documents/38714170/40224547/conqodr_ctn.pdf/2c80e90b-b0a9-4171-a621-3f0df1925ff9](https://www.ifad.org/documents/38714170/40224547/conqodr_ctn.pdf/2c80e90b-b0a9-4171-a621-3f0df1925ff9)

⁵⁵ [Hemedi, P. S. \(2022, November 16\). After 14 years of advocacy, the Drc president finally signs new Indigenous peoples law \(commentary\). Mongabay. Retrieved March 2, 2023, from https://news.mongabay.com/2022/11/after-14-years-of-advocacy-the-drc-president-signs-new-indigenous-peoples-law-commentary/](https://news.mongabay.com/2022/11/after-14-years-of-advocacy-the-drc-president-signs-new-indigenous-peoples-law-commentary/)

⁵⁶ [Davis, L., Fabbri, P., & Alphonse, I. M. \(2014\). \(rep.\). Gender Country Profile: Democratic Republic of Congo - DRC. Retrieved December 2, 2022, from https://www.lauradavis.eu/wp-content/uploads/2014/07/Gender-Country-Profile-DRC-2014.pdf](https://www.lauradavis.eu/wp-content/uploads/2014/07/Gender-Country-Profile-DRC-2014.pdf)

⁵⁷ [UNDP \(2022\) Human Development Insights](#)

⁵⁸ [World Bank, 2016](#)

poverty threshold as compared to 51% of men.⁵⁹ Women are mostly engaged in agriculture, predominantly subsistence farming, and small businesses in the informal sector.

The labour force participation rate for women is 61.2% compared to 69.1% of men, and only 6.4% of women work in wage employment, compared to 23.9% of men.⁶⁰ Gender gaps persist across different economic sectors, with large discrepancies in agricultural production (17.8% gap between men and women), in entrepreneurship profits (66.5% gap) and in wage earnings (77.3% gap).⁶¹ Less than 10% of women are wage or salaried workers compared to 25% of men.⁶² In the private sector, the wage gap is nearly 80%. The underlying causes of such gaps include women's higher burden of care work, which is the primary factor behind women's lower labour force participation. Lower educational attainment, skills, and concentration in lower-paying sectors are instead the primary drivers of women's lower-wage earnings. The entrepreneurship gender gap indicates that businesswomen have lower levels of capital and lower numbers of workers. The female share of employment in senior and middle management is 6% while only 11% of firms have a female top manager (2013).

Further, across all sectors, lack of control over land, lack of agency in the household, prevalent gender-based violence, and marriage have been found to be consistent drivers of gaps in economic empowerment.⁶³

The first Family Code (1987) included legal restrictions to women's mobility, workplace, marriage, and assets, but the 2016 reform of the Code included the elimination of marital authorization for women's access to employment, bank accounts, and loans, directly impacting women's ability to explore new economic opportunities. While family laws have been updated, many have yet to be ratified. Specifically, Article 444 of the DRC Family Code (1987) provides that a husband is head of the household and the revised Family Code (2016) still maintains this. However, of note in the revised version, is the elimination of the need for a married woman to obtain her husband's permission to open an individual bank account, or access financial services such as credits or loans. As such, most married women's decisions remain under the discretion of their husbands, who retain control of all household, financial, and medical decisions in both statutory and cultural law. Widowed, divorced, or single women face ostracization and most often must submit to the will and authority of male family members.

The Women Business and the Law Index Score for the DRCF is 78.8 (2021). The overall score for the Democratic Republic of the Congo is higher than the regional average observed across Sub-Saharan Africa (71.5).

Gender and energy in DRC

Only 19% of the population have access to electricity. Gender norms restrict women's access to resources and assets, which directly impacts energy poverty. Because women often do not make decisions alone on household spending, they are likely to have less access to energy products and technologies. Sex-disaggregated data on access to energy as well as women's participation in the sector is absent.

⁵⁹ [Women's International League for Peace and Freedom. \(n.d.\). Gender Inequality And Social Institutions In The Drc. Peace Women. Retrieved December 14, 2022, from https://www.peacewomen.org/content/gender-inequality-and-social-institutions-dr-congo](https://www.peacewomen.org/content/gender-inequality-and-social-institutions-dr-congo)

⁶⁰ [Donald, A. A., Hwang, H., & Phipps, V. \(2022, June 2\). Obstacles and opportunities for women's economic empowerment in the DRC. Retrieved December 2, 2022, from https://blogs.worldbank.org/african/obstacles-and-opportunities-womens-economic-empowerment-drc](https://blogs.worldbank.org/african/obstacles-and-opportunities-womens-economic-empowerment-drc)

⁶¹ [Donald, A. A., Hwang, H., & Phipps, V. \(2022, June 2\). Obstacles and opportunities for women's economic empowerment in the DRC. Retrieved December 2, 2022, from https://blogs.worldbank.org/african/obstacles-and-opportunities-womens-economic-empowerment-drc](https://blogs.worldbank.org/african/obstacles-and-opportunities-womens-economic-empowerment-drc)

⁶² [Ayivi-Guedehoussou, N., Dervisevic, E., Donald, A., Hwang, H., Khonde, L. M., Lewis, C., Morrison, L., Phipps, V., Pierotti, R., & Vaillant, J. \(n.d.\). \(rep.\). Women's Economic Empowerment In The Democratic Republic Of The Congo: Obstacles And Opportunities. Retrieved December 2, 2022, from https://openknowledge.worldbank.org/bitstream/handle/10986/36703/P17518604b63b809d09a9e0ae31320c85e7.pdf?sequence=7&isAllowed=y](https://openknowledge.worldbank.org/bitstream/handle/10986/36703/P17518604b63b809d09a9e0ae31320c85e7.pdf?sequence=7&isAllowed=y)

⁶³ Ibid.

Women can play a key role in off-grid energy value chains, but challenges remain particularly related to access to finance, intra-household decision-making, and education levels. Social and gender norms and pervasive GBV affect women's ability to engage in entrepreneurial activities and inactively participate in energy value chains as consumers, entrepreneurs, and agents of change.

Uganda

Overview of gender equality and gender gaps in the country

Uganda has a total population of 34.6 million, of which 51.21% are women. The country has one of the world's youngest and fastest-growing populations with 55% of people under 18 years of age and a very high fertility rate of 5.4.⁶⁴

The Gender Development Index in Uganda is 0.927 and is ranked 166 out of 191 in Human Development Index, placing the country among other 'Group 3' countries with a high absolute deviation from gender parity, of 5–7.5%. This is largely driven by a gender gap in mean years of schooling and gross national income per capita.⁶⁵ Uganda is ranked 131 in the Gender Inequality Index which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.⁶⁶

Indicator	Result	Source and date
Life expectancy at birth	64.9	UNDP (2022) Gender Development Index
Maternal mortality ratio	375	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	71%	World Bank, 2018
Women's Labour Force Participation Rate (age 15 to 64)	68%	World Bank, 2019
Mean years of schooling ⁶⁷	4.9	UNDP (2022) Gender Development Index
Estimated gross income per capita	1,877 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Uganda has ratified various UN conventions demonstrating a commitment to equity and inclusion, including the Elimination of Discrimination Against Women (CEDAW) in 1985, the International Convention on Civil and Political Rights (ICCPR) in 1995, and the International Convention on Social Economic and Cultural Rights (ICSECR) in 1987. Further, for the protection of children, Uganda has also ratified the Convention on the Rights of the Child (CRC) in 1990. Regionally, Uganda has also ratified the Maputo protocol, i.e. the African Union (AU) Protocol to the African Charter on Human and People's Rights on the Rights of Women

⁶⁴ [Uganda Bureau of Statistics, 2014](#)

⁶⁵ [UNDP \(2022\) Gender Development Index](#)

⁶⁶ [UNDP \(2022\) Gender Inequality Index](#)

⁶⁷ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

in Africa. This protocol specifies provisions for civil, political, economic, social and cultural rights as well as the rights to development and peace, as well as reproductive and sexual rights for women.^{68,69}

Uganda also compiled a Gender Policy in 1997 (revising it in 2007), The National Policy on the Elimination of Gender-based Violence (2016) and the National Male Engagement Strategy, developed respectively (2017) with a commitment to eliminate all forms of gender inequalities in the long-run.^{70,71} The policy is meant to serve as a guiding framework for all levels of planning, resource allocation and implementation of development programmes with a gender perspective. Under the National Gender Policy, Uganda has identified 4 priority areas of focus including:

1. **Gender and livelihoods**; with the aim to redress three areas of inequality including gender differences in livelihoods, time poverty, and access to and control over productive resources.
2. **Promotion and protection of rights**; by committing to eliminating gender differences in women's access to justice through the Judiciary Gender Policy and Strategy released in 2012.
3. **Participation in decision-making and governance**; wherein the country aims to overcome obstacles to women's engagement in elective processes, limited capacity to participate effectively in leadership and decision-making, low presence of women in technical positions.
4. **Gender and Macro-economic Management**; The government commits to women's economic empowerment by laying down strategies like gender budgeting, reforms in investment regulations and domestic/local revenue generation systems and building the overall capacity of the national government to include gender in macroeconomic policies.

The Constitution of the Republic of Uganda, under Article 21, prohibits gender discrimination generally and under Article 31 sets the minimum age for marriage at 18 and provides for equal rights between men and women during marriage and divorce. The Article 33 pertains specifically to the rights of women and requires that the government provide opportunities to enhance the welfare of women and enable them to reach their full potential, provide women equal rights to men in political, economic, and social activities, and. Article 33 also prohibits laws, customs, traditions, and cultures that are "against the dignity of women".⁷²

While the above measures create a conducive policy environment in Uganda to address gender inequality, restrictive socio-cultural practices continue to curtail women's potential in the country. This has been further exacerbated by COVID-19 during which increased incidences of violence against women have been reported.⁷³

Political participation

With women representing 34.9% of parliamentary seats in Uganda, the country ranks high in Africa with regard to women's political participation⁷⁴. To ensure the representation of women in local governments (LGs) (*that have the powers enshrined under the LG Act, 1997 to make and implement development plans based on locally determined priorities*) the Ugandan government under the 1997 LG Act stipulates that women should constitute a third of any local council. This affirmative action has led to women constituting

⁶⁸ [UN Human rights treaty bodies, Ratification status for Uganda](#)

⁶⁹ [The Judiciary of the Republic of Uganda \(2012\) Judiciary Gender policy and strategy](#)

⁷⁰ [The Uganda Gender Policy, 2007](#)

⁷¹ [UN Human Rights Office of the High Commissioner, 2022](#)

⁷² [Legal Information Institute on the Constitution of the Republic of Uganda](#)

⁷³ [Kiiza, W. \(2022, February 24\). Brookings. Retrieved December 3, 2022, from https://www.brookings.edu/blog/africa-in-focus/2022/02/24/it-is-not-yet-uhuru-for-the-women-of-uganda/](#)

⁷⁴ [National Planning Authority. \(2020\). National Development Plan III 2020/21 to 2024/25, Uganda](#)

about 46% of total elected representatives in LGs.⁷⁵ However, women are underrepresented in executive committees where decision-making at the LG level takes place, and elected women representatives are unable to effectively fulfill their roles. This is found to be due to an overall patriarchal society in Uganda, wherein there exists misperception associated with women commanding leadership positions, and where women's caregiving responsibilities leave them time-poor to effectively engage with the demands of public life. Further, limited access to income and low education attainment levels among women limit their ability to navigate the commercialized electoral process and technical processes associated with political decision-making.⁷⁶ The high costs of political campaigning have also been highlighted by the UN Committee on the Elimination of Discrimination against Women.⁷⁷

Social and cultural context

In Uganda, Women's rights on land have been recognized in the Land Act (Cap 227) and Land Acquisition Act (Cap 226) and the country aims to promote tenure security in women's access to land under their IIIrd National Development Plan. These have led to a sharp increase in both land and home ownership by women and in joint ownership between the period 2013 to 2017, compared to male dominance in ownership prior to 2012-13. However, the country continues to be a patriarchal and patrilinear society with women's domestic and household responsibilities impacting their effective participation in productive work. Further, research has found that in Uganda, as women attain increased access to social networks and empowerment, especially at the age group 30 to 39, they are often subjected to violence by their current or former partners.⁷⁸ Almost 50% of women report experiencing some form of physical and/or sexual intimate partner violence in their lifetime and about 30% reported experiencing it in the last 12 months.⁷⁹ Uganda has developed a male involvement strategy, finalized in 2017 with the aim of engaging men and boys in prevention and response to GBV.⁸⁰

Health

Uganda has a maternal mortality ratio (MMR) of 375 per 100,000 live births (as of 2017)⁸¹ which is extremely high compared to the Sustainable Development Goal target of achieving no more than an MMR of 70 by 2030. Even with the introduction of 14 different policy instruments over 15 years by the government, to reduce MMR, the failures in referral policies, inadequacies in healthcare facilities and trained personnel and transportation issues remained.⁸² Unsafe abortions account for about 5% of maternal deaths in Uganda. Abortion is illegal in Uganda and termination is allowed only in cases the pregnancy threatens the

⁷⁵ [Uganda Bureau of Statistics. \(2017\). Women in Local Government: Baseline Statistics for Local Council Levels III-V in Uganda](#)

⁷⁶ [The Economic Policy Research Centre \(EPRC\). \(2021\). \(rep.\). Women's Participation In Politics At The Local Government Level In Uganda. Retrieved December 3, 2022, from \[https://africa.unwomen.org/sites/default/files/2022-06/Women%20Participation%20in%20Politics%20at%20the%20LG%20in%20Uganda_Occasional%20Paper%2053.pdf\]\(https://africa.unwomen.org/sites/default/files/2022-06/Women%20Participation%20in%20Politics%20at%20the%20LG%20in%20Uganda_Occasional%20Paper%2053.pdf\)](#)

⁷⁷ [UN Human Rights Office of the High Commissioner, 2022](#)

⁷⁸ [Uganda Bureau of Statistics. \(2019\). \(rep.\). Gender Issues in Uganda. Retrieved December 3, 2022, from \[https://www.ubos.org/wp-content/uploads/publications/03_2019UBOS_Gender_Issues_Report_2019.pdf\]\(https://www.ubos.org/wp-content/uploads/publications/03_2019UBOS_Gender_Issues_Report_2019.pdf\)](#)

⁷⁹ [UN Women, Global Database on Violence against Women](#)

⁸⁰ [The Ministry of Gender Labour and Social Development. \(2017\). \(rep.\). National Male Involvement Strategy for the prevention and response to Gender Based Violence in Uganda . Retrieved December 3, 2022, from \[https://uganda.unfpa.org/sites/default/files/pub-pdf/15_03_18_%20MALE%20INVOLVEMENT%20STRATEGY%2024%20JULY%202017.pdf\]\(https://uganda.unfpa.org/sites/default/files/pub-pdf/15_03_18_%20MALE%20INVOLVEMENT%20STRATEGY%2024%20JULY%202017.pdf\)](#)

⁸¹ [WHO, 2017](#)

⁸² [Mukuru, M., Gorro, J., Kiwanuka, S. N., Gibson, L., Musoke, D., & Ssenkooba, F. \(2021\). Designed to fail? revisiting Uganda's maternal health policies to understand policy design issues underpinning missed targets for reduction of maternal mortality ratio \(MMR\): 2000-2015. International Journal of Health Policy and Management. <https://doi.org/10.34172/ijhpm.2021.127>](#)

life of the mother, including in cases of cervical cancer and HIV-positive status. Further, pregnancies from rape, incest, and defilement can also be terminated.⁸³

Education

In Uganda, men are expected to complete 6.7 mean years of schooling compared to 4.9 years completed by women.⁸⁴ As of 2018, 83% of adult males are literate and 71% of adult females are literate. Interestingly, however, among the young population, there is gender parity in literacy levels with 90% of women in the age group of 15-24 being literate and 89% of men attaining literacy which shows an improving trend for the next generation of the Ugandan population.⁸⁵ However, the lower secondary completion rate is low among both men (27.7%) and women (25.1%) in Uganda.⁸⁶

Under its National Strategy for Girls Education (2015-2019) the country has created a policy implementation Framework to promote girls' education by aiming to harmonize education programs on Girls' Education for effective coordination, monitoring and evaluation, ensure adequate commitment to requisite human and financial resources to promote girls' education, institutionalize research on Girls' Education to inform policy development and build the capacity of education actors to address gender inequalities in education.⁸⁷ Uganda has also been commended internationally by the Committee on the Elimination of Discrimination against Women for the strides made in the education sector, including a particular focus on building infrastructure for the education of girls with special needs.⁸⁸ Notably, Uganda has a Science Education Policy (2004) under which physics, chemistry, and biology have been made compulsory in lower secondary schools,⁸⁹ and 75% of government sponsorships in tertiary institutions are for science-based courses. However, only 25% of students opting for science-based courses at the university level are women, hence they are unable to enjoy state-funded science education at this level.⁹⁰

Although the country has introduced key legislation to ensure improved education access for women the country is still falling behind in effectively transitioning women into the world of work. There has been a steep drop in labour force participation of women in the ages of 15-24, from 74.6% in 2013 to 54.2% in 2017.⁹¹ While some of this decline can be explained by improved access to schooling, the share of inactivity is also increasing among women with them being more likely than men to be both out of school and out of a job due to familial responsibilities and lack of job opportunities.⁹²

Women's economic participation and entrepreneurship

⁸³ [Reproductive Health Division Department of Community Health Ministry of Health. \(2006\). \(rep.\). The National Policy Guidelines and Service Standards for Sexual and Reproductive Health and Rights. Retrieved December 3, 2022, from https://www.prb.org/wp-content/uploads/2018/05/National-Policy-Guidelines-and-Service-Standards-for-Sexual-and-Reproductive-Health-and-Rights-2006.Uganda.pdf](https://www.prb.org/wp-content/uploads/2018/05/National-Policy-Guidelines-and-Service-Standards-for-Sexual-and-Reproductive-Health-and-Rights-2006.Uganda.pdf).

⁸⁴ [UNDP \(2022\) Gender Development Index](#)

⁸⁵ [World Bank, 2018](#)

⁸⁶ [World Bank, 2017](#)

⁸⁷ [Ministry of Education and Sports \(MoES\). \(2013\). \(rep.\). National Strategy for Girls' Education \(NSGE\) in Uganda \(2015 – 2019\) . Retrieved December 3, 2022, from https://scorecard.prb.org/wp-content/uploads/2018/05/National-Strategy-for-Girls%E2%80%99Education-in-Uganda-2015-2019.pdf](https://scorecard.prb.org/wp-content/uploads/2018/05/National-Strategy-for-Girls%E2%80%99Education-in-Uganda-2015-2019.pdf).

⁸⁸ [UN Human Rights Office of the High Commissioner, 2022](#)

⁸⁹ [Uganda National Commission for UNESCO. \(2017\). \(rep.\). Popularization of Science Using Champion Teachers. Retrieved December 3, 2022, from https://unesco-uganda.ug/wp-content/uploads/2019/02/REPORT-ON-THE-POPULARISATION-OF-SCIENCES.pdf](https://unesco-uganda.ug/wp-content/uploads/2019/02/REPORT-ON-THE-POPULARISATION-OF-SCIENCES.pdf).

⁹⁰ [Odaga, G. \(2020\). Gender in Uganda's Tertiary Educational Distribution. Social Sciences & Humanities Open, 2\(1\), 100023. https://doi.org/10.1016/j.ssaho.2020.100023](https://doi.org/10.1016/j.ssaho.2020.100023)

⁹¹ [World Bank](#)

⁹² [Merotto, D. \(2019\). \(rep.\). Uganda: Jobs Strategy For Inclusive Growth. World Bank Group](#)

As of 2019, the female Labour force participation rate for women in the age group 15 to 64 in Uganda is 68%.⁹³ As per the Mastercard Global Index of Women Entrepreneurs report (2021), of all the business leaders in Uganda, 25.3 % are women. Further, women outperform men in undertaking 'entrepreneurial activity' in 2021 was 39.9% for women compared to 36.1% for men. However, it is crucial to note that 21.9% of women entrepreneurship in the country is necessity driven compared to 15.2% of male entrepreneurship that is necessity driven. In The report also found that there has been a slight decrease in the entrepreneurial activity of women in the country between 2020 to 2021 from 42% to 39.9% which may be an impact of COVID-19.⁹⁴

In salaried employment, there are evident gender gaps in median monthly nominal wages earned by men and women, with women earning 32.41 USD while men earning more than double the income, i.e. 67.52 USD.⁹⁵ Further, there are glaring differences in women's representation at the senior management level, across sectors. Women senior managers are concentrated in sectors like non-profits, education and nursing, but have limited representation in financial service industries and IT sectors.⁹⁶

About 83% of Ugandan women are also engaged in unpaid care work, the most common forms being cleaning, cooking, and other household tasks. While the least percentage of persons report engaging in unpaid work for firewood collection, a disproportionately higher percentage of women report spending time on this activity (42%) compared to men (22%).⁹⁷

Gender and energy in Uganda

It is widely recognized that consistent and reliable access to energy is critical for both economic growth and social transformation. Further, access to clean energy can improve health & well-being as well as the status of time poverty among women by reducing the burden of collecting firewood. Over the past decade, Uganda has increased its electricity generation capacity from 601MW in 2010 to 1839MW in 2020.⁹⁸ However, in Uganda, biomass continues to be the predominant source of cooking fuel used by 94% of the population with 72.8 % using firewood and 21.4% using charcoal. Further, firewood use is 3% higher among female heads of households than men.⁹⁹ Overall 56% population has access to electricity, with the majority of the households (27%) using solar kits for lighting, 19% using grid electricity, and 11% using solar home systems. While 51.6% of women-headed households have electricity access in the country, 59% men headed households have electricity access.¹⁰⁰

There is high potential for women to be employed in the clean energy sector, as last mile agents generating demand for electricity consumption, which continues to be a challenge in Uganda. At 100 kWh per capita, Uganda currently has one of the lowest rates of electricity consumption per capita in the world.¹⁰¹ However, currently, the participation rate of women in Distributed Renewable Energy is 28% of the total workforce in

⁹³ [World Bank](#)

⁹⁴ [Mastercard. \(2022\). \(rep.\). The Mastercard Index of Women Entrepreneurs How targeted support for women-led business can unlock sustainable economic growth. Retrieved December 3, 2022, from https://www.mastercard.com/news/media/phwevxc/the-mastercard-index-of-women-entrepreneurs.pdf.](#)

⁹⁵ [Uganda Bureau of Statistics \(UBOS\) \(2021\). Uganda National Household Survey 2019/2020. Kampala, Uganda: UBOS](#)

⁹⁶ [Mastercard. \(2022\). \(rep.\). The Mastercard Index of Women Entrepreneurs How targeted support for women-led business can unlock sustainable economic growth. Retrieved December 3, 2022, from https://www.mastercard.com/news/media/phwevxc/the-mastercard-index-of-women-entrepreneurs.pdf.](#)

⁹⁷ [Uganda Bureau of Statistics \(UBOS\) \(2021\). Uganda National Household Survey 2019/2020. Kampala, Uganda: UBOS](#)

⁹⁸ [National Planning Authority. \(2020\). National Development Plan III 2020/21 to 2024/25. Uganda](#)

⁹⁹ [Uganda Bureau of Statistics \(UBOS\) \(2021\). Uganda National Household Survey 2019/2020. Kampala, Uganda: UBOS](#)

¹⁰⁰ [Uganda Bureau of Statistics \(UBOS\) \(2021\). Uganda National Household Survey 2019/2020. Kampala, Uganda: UBOS](#)

¹⁰¹ [National Planning Authority. \(2020\). National Development Plan III 2020/21 to 2024/25. Uganda](#)

the sector. The highest share of jobs held in the DRE sector by women is in mini-grid technology, and women continue to be concentrated in unskilled work, making up 35% of unskilled mini-grid workers. Evidently, women face myriad socio-cultural barriers as workers in the DRE sector, reporting workplace discrimination and stating that household responsibilities discourage their effective participation in the sector. On the supply side, among employers, misconceptions exist regarding women's technical abilities as installation agents and engineers. While the DRE sector in Uganda is not yet an equal opportunity employer, there is positive progress made towards achieving pay parity, where women are remunerated at 96% of what their male counterparts earn.¹⁰² It is encouraging to see that Uganda has a '[Women Renewable Energy Association of Uganda](#)' which aims to build the technical capacity of women in the field of solar, biomass, and hydro energies, organizing mentorship programs, and engaging with communities on the adoption of renewable energy.

The country currently has a program to Operationalize the National Content policy to enhance local Content and participation of nationals including women and youth in the oil and gas sector and is planning to launch a project on Climate Resilient Livelihood Opportunities for Women's Economic Empowerment (CRWEE) in Karamoja and West Nile Regions of Uganda (October 2018 - 30 September 2030)¹⁰³

¹⁰² [Power for All. \(2022\). \(rep.\). Powering Jobs Census 2022: Focus on Uganda. Retrieved December 3, 2022, from https://unreeca.org/wp-content/uploads/2022/10/Powering-Jobs-Census-2022-Uganda-1.pdf.](#)

¹⁰³ [National Planning Authority. \(2020\). National Development Plan III 2020/21 to 2024/25, Uganda](#)

Benin

Overview of gender equality and gender gaps in the country

Benin has a population of 12.8 million people, with a young population of 42% below the age of 14. The fertility rate in the country stands at 4.6 which is significantly high compared to other countries in the region.

¹⁰⁴

The Gender Development Index in Benin is 0.880 and the country is placed with 'Group 5' countries with a high absolute deviation from gender parity, of more than 10%. This is largely driven by a gender gap in expected and mean years of schooling.¹⁰⁵ Benin has a Gender Inequality Index of 0.602 and is ranked 152, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.¹⁰⁶

Indicator	Result	Source and date
Life expectancy at birth	61.4	UNDP (2022) Gender Development Index
Maternal mortality ratio	397	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	31%	World Bank, 2018
Women's Labour Force Participation Rate (age 15 to 64)	70%	World Bank, 2019
Mean years of schooling ¹⁰⁷	3.3	UNDP (2022) Gender Development Index
Estimated gross income per capita	2,998 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Benin has ratified various UN conventions which aim towards ensuring equity and inclusion, including the Elimination of Discrimination Against Women (CEDAW) in 1982, and the Convention on the Rights of the Child (CRC) in 1990.¹⁰⁸ In the African region, Benin has also ratified the Solemn Declaration of the Heads of State on Gender Equality in Africa and committed to action under the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa.¹⁰⁹

The Constitution of Benin accords equal rights to all without distinction of origin, race, gender, religion, political opinion, or social position under Article 26. Under its Code of Persons and the Family in the Republic of Benin, the country also sets the legal age of marriage at 18 for both men and women. It

¹⁰⁴ [United Nations Population Fund, World Population Dashboard, Benin](#)

¹⁰⁵ [UNDP \(2022\) Gender Development Index](#)

¹⁰⁶ [UNDP \(2022\) Gender Inequality Index](#)

¹⁰⁷ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

¹⁰⁸ [United Nations Human Rights Treaty Bodies, Benin](#)

¹⁰⁹ [World Bank \(2021\). Gender Assessment in Benin : Women's Participation in Economic Opportunities and Decision-Making. World Bank, Washington, DC. © World Bank. Retrieved from https://openknowledge.worldbank.org/handle/10986/38039](#)

promulgates that men and women are guaranteed the same inheritance rights under the law. Further, Benin has also established a policy framework to translate the legal commitment to action through its National Gender Promotion Policy, National Policy for the Promotion of Women in Agriculture, and the National Action Plan to Combat Gender-Based Violence. Benin, under its National Gender Promotion Policy (consolidated in 2009), aims to achieve gender equality by 2025, however, the action plan to implement strategies laid down under the policy was only articulated in 2015 and is yet to be reviewed and assessed which raises questions on how successful the country's progress is towards the envisioned goal of the policy.¹¹⁰ Other gender promotion systems in Benin include the establishment of Social Promotion Centers, a National Commission for the Promotion of Women, a National Institute for the Promotion of Women, and the Ministry of Social Affairs and Microfinance.¹¹¹

Although the legal and policy institutionalization in Benin is robust, policy implementation is replete with enforcement issues in the backdrop of heavily patriarchal social structures still prevalent in the country.

Political participation

Women in Benin remain underrepresented in political positions. In the 2019 legislative elections, only 6% of National Assembly members were women. Similar to this trend, In the 2020 municipal elections, 4.3% of councilors were women and there were 4 women mayors out of 77 elected representatives. Benin has now deployed a new electoral code was deployed in 2021 which has reserved a minimum quota of 24 out of 109 seats for women, at the rate of one woman per electoral district.¹¹² At the communal counselor level, women had only 3.86 % representation in 2020.¹¹³

Social and cultural context

There is a continued problem of Gender Based violence experienced by women in Benin with 23.8% of women experiencing it at least once in their lifetime.¹¹⁴

There is a wide regional disparity in women's access to land across Benin. While women in central and southern Benin have more control over land, for example; 20.7% in Couffo, and 17.8 % in Plateau, women in other regions have significantly limited access like in Alibori, (10.9 %) & Atacora (12.2%).¹¹⁵

Health

Benin has a maternal mortality ratio (MMR) of 397 per 100,000 live births (as of 2017).¹¹⁶ As per the Benin Household Demographic and Health Survey analysis showed that while 89% of women reported going for at least one antenatal care (ANC) visit, only half of these women initiated ANC in the first trimester and completed 4 or more visits.¹¹⁷ This gap in the utilization rate of maternal health services, a study finds, is

¹¹⁰ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹¹¹ [World Bank. 2021. Gender Assessment in Benin : Women's Participation in Economic Opportunities and Decision-Making. World Bank, Washington, DC. © World Bank. Retrieved from <https://openknowledge.worldbank.org/handle/10986/38039>](#)

¹¹² [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹¹³ [World Bank. 2021. Gender Assessment in Benin : Women's Participation in Economic Opportunities and Decision-Making. World Bank, Washington, DC. © World Bank. Retrieved from <https://openknowledge.worldbank.org/handle/10986/38039>](#)

¹¹⁴ [UN Women, Global Database on Violence against Women](#)

¹¹⁵ [World Bank. 2021. Gender Assessment in Benin : Women's Participation in Economic Opportunities and Decision-Making. World Bank, Washington, DC. © World Bank. Retrieved from <https://openknowledge.worldbank.org/handle/10986/38039>](#)

¹¹⁶ [UNDP \(2022\) Gender Inequality Index](#)

¹¹⁷ [Benin Demographic and Household Survey \(2017-18\) Final Report \(French\)](#)

due to poor and/or disrespectful care received at the facility level, perception of care, and long wait times at facilities as well as lack of financial means among women to cover associated costs.¹¹⁸ Further, the contraceptive usage rate among women in the age group 15-49 is only 17% and there is an unmet family planning need of 24%. Additionally, only 34% of the demand for contraceptives is satisfied by modern methods.¹¹⁹

Education

The literacy rate among women in Benin is 31% compared to 54% for men¹²⁰ and 10% of females in primary school-going age are out of school.¹²¹ However, interestingly, at the tertiary level, Benin has one of the world's highest percentages of women graduating from Engineering and ICT courses at 54.6% and 55.1% respectively (as a percentage of total graduates in the sector).¹²² The country has a National Policy on Literacy and Adult Education which was adopted in 2001 and aimed at reducing the rate of illiteracy among men and women by half between 2001 and 2010¹²³ however World Bank data suggests that between 2002 and 2012, the literacy rate actually dropped by 1%.¹²⁴ While enrollment of girls is high at the primary level, even surpassing that of boys, there is an evident gender gap in secondary education enrollment which is 40.8% for men compared to 34.1% for women.¹²⁵ Benin has committed to a post-2015 Education Sector Plan (2018-2030) to promote access and retention in education for girls by waiving school fees at the secondary level for girls. The country has budgeted USD 6.4 million for the year 2019 for this fee waiver for girls. Further, Benin has budgeted USD 306,023 between 2018 and 2021 to provide scholarships for promoting girls' enrollment in agricultural and industrial education.¹²⁶

However, to improve the enrollment and retention rate of women at the secondary level, Benin needs to address structural gaps in its education system. The World Bank-supported Global Partnership for Education Project identified challenges that cripple Benin's education system currently including; high administrative burden among staff, slow-moving and highly political decision-making processes that interfere with managerial decisions, improper division of roles and poor coordination across Ministries, rapidly changing leadership at the institutional level and the problem of high-absenteeism of teaching staff.¹²⁷

For climate research education, as part of the programme for the West African Science Service Centre on Climate Change and Adapted Land Use (WASCAL) which is a partnership between ECOWAS Commission and the German government since 2012, MSc and Ph.D. programs in climate science at selected universities in Benin with staff exchanges between the West African institutions and their affiliates in Germany.¹²⁸

¹¹⁸ [Gryseels, C., Dossou, J. P., Vigan, A., Boyi Hounsou, C., Kanhonou, L., Benova, L., & Delvaux, T. \(2022\). Where and why do we lose women from the continuum of care in maternal health? A mixed-methods study in southern Benin. *Tropical Medicine & International Health*, 27\(3\), 236–243. <https://doi.org/10.1111/tmi.13729>](https://doi.org/10.1111/tmi.13729)

¹¹⁹ [United Nations Population Fund, World Population Dashboard, Benin](#)

¹²⁰ [World Bank, 2018](#)

¹²¹ [World Bank, 2018](#)

¹²² [UNESCO \(2021\). UNESCO Science Report: The race against time for smarter development](#)

¹²³ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹²⁴ [World Bank, 2018](#)

¹²⁵ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹²⁶ [UNESCO \(2021\) Global Education Monitoring Report, Benin](#)

¹²⁷ [The World Bank, 2019, Project Information Document Integrated Safeguards Data Sheet](#)

¹²⁸ [UNESCO \(2021\). UNESCO Science Report: The race against time for smarter development](#)

Women's economic participation and entrepreneurship

Benin has seen a growing Labour Force Participation among women in the past 2 decades. While the disparity between men's and women's labor force participation (LFPR) is not high overall with about 69.3% LFPR for women and 73.3% for men, the type of work conducted by women is largely informal. As per the Enquête Harmonisée Sur les Conditions de Vie des Ménages survey in Benin, 95.5% of women workers concentrated in the informal sector compared to 86% of men.¹²⁹ Women and young people constitute the majority of unemployed people, with 37.5% of 15-24 old being without a job, and 24.6% of them being women.

While the high share of female graduates in Engineering and ICT fields should create a thriving entrepreneurship environment in the country for women, as per data only 25.9% had female directors and 36.7% of businesses have women among their shareholders.¹³⁰ However, the country has an above-average estimated gross income per capita among women of 2,998\$, compared to other Sub-Saharan African nations.¹³¹ Women-led businesses are concentrated in service industries such as accommodation and food as well as retail and wholesale, which unfortunately have been most impacted due to COVID. Female-headed businesses in Benin remain unregistered with only 13% having certification compared to 21% of men-headed businesses having requisite registration.¹³²

Gender and energy in Benin

According to ERI-ESI 2018, 63.4% of households use charcoal compared to 22% for gas and 12.2% for wood, and energy use for cooking is predominantly biomass.¹³³ About 41% of households have access to electricity in Benin.¹³⁴ Only 35% of women have access to electricity, compared to 69% of men.¹³⁵

The Energy Ministry in Benin has prepared a gender profile for the energy sector in 2020 and a 2021-2025 National Action Plan for the Policy for Gender Mainstreaming and Energy Access in Benin and operationalized a Gender and Development Unit to ensure gender mainstreaming across the Ministry's projects.

ECREEE is implementing a project entitled: Promoting Agricultural Competitiveness through Solar Energy (PAC-SOLAR) in Benin with the aim to enhance women's access to renewable energy. As part of this project, women have also been trained and charged with the job of cleaning the solar panels regularly, and maintain solar PV installations.¹³⁶

However, overall, women in Benin are unable to access training and finance to learn high-skill trades in the energy sector and there continues to be a higher preference for employing and financing men in the energy

¹²⁹ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹³⁰ [World Bank. 2021. Gender Assessment in Benin : Women's Participation in Economic Opportunities and Decision-Making. World Bank, Washington, DC. © World Bank. Retrieved from <https://openknowledge.worldbank.org/handle/10986/38039>](#)

¹³¹ [UNDP \(2022\) Gender Development Index](#)

¹³² [International Trade Centre \(2020\) Promoting SME Competitiveness in Benin. COVID-19: An Inclusive Path Towards Resilience](#)

¹³³ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

¹³⁴ [International Renewable Energy Agency. \(2022\). Energy Profile, Benin](#)

¹³⁵ [Institut National de la Statistique et de l'Analyse Economique et AFRISTAT. 2019. Enquête Régionale Intégrée sur l'Emploi et le Secteur Informel, 2018. Cotonou, Bénin et Bamako, Mali : INSAE et AFRISTAT](#)

¹³⁶ [Renewable energy bolsters agricultural activities for women in rural communities in Benin. \(2022, April\). Global Network: Regional Sustainable Energy Centres.](#)

sector.¹³⁷ While Benin produces one of the world's highest shares of women graduates who are Engineers and ICT specialists, it is unfortunate that these women are not being absorbed in the clean energy sector.

¹³⁷ [African Development Bank Group and UN Women \(2021\) Republic of Benin, Country Gender Profile. Assessment of the Impacts of Covid-19 on Women and Girls](#)

Zambia

Overview of gender equality and gender gaps in the country

Zambia has a population of over 19 million,¹³⁸ of which 50.5% are women, and where over 55% of the population lives in rural areas – the majority of them being women.¹³⁹

Zambia has a comparatively equal development index when compared with other Sub-Saharan African countries, with a Gender Development Index of 0.965. The country is placed with ‘Group 2’ countries with an absolute deviation from gender parity between 2.5 percent and 5 percent. This is a result of the country achieving equality in SDG 4.4 of mean years of schooling with both girls and boys receiving 7.2 years of schooling.¹⁴⁰ Zambia has a Gender Inequality Index of 0.540 and is ranked 138, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.¹⁴¹

Indicator	Result	Source and date
Life expectancy at birth	63.9	UNDP (2022) Gender Development Index
Maternal mortality ratio	213	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	83%	World Bank, 2018
Women’s Labour Force Participation Rate (age 15 to 64)	71%	World Bank, 2019
Mean years of schooling[1]	7.2	UNDP (2022) Gender Development Index
Estimated gross income per capita	2,615	UNDP (2022) Gender Development Index

Legal status of women

Zambia has a favorable policy framework to advance gender equality in the country. The Constitution (Amendment) Act No. 2 of 2016 provides equal rights to women and men to freely participate in, determine and build a sustainable political, legal, economic and social order. The Constitution also underlines national values and principles which include equity, equality, and non-discrimination.¹⁴² Further, the country has also enacted the Gender Equity and Equality Act, 2015, to strengthen the legal framework for the elimination of all forms of discrimination against women and girls as committed under the Maputo Protocol; and put in

¹³⁸ [United Nations Population Fund, World Population Dashboard, Zambia](#)

¹³⁹ [World Bank 2021](#)

¹⁴⁰ [UNDP \(2022\) Gender Development Index](#)

¹⁴¹ [UNDP \(2022\) Gender Inequality Index](#)

¹⁴² [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

place affirmative action to eliminate all barriers which prevent women from participating meaningfully in all spheres of life and create a conducive environment for such participation. The Act also provides that “an employer shall not discriminate against a woman by dismissing or demoting 17 the woman on the basis of sex, marriage, disability, pregnancy or maternity leave or subject the woman to any other disadvantage in employment” further stipulating that women must be awarded maternity leave with pay without loss of employment, seniority or social allowances.¹⁴³ Progress has also been made with regard to women’s human rights with the revision of their penal code in 2005, 2010, and 2012 as per which punishments for sexual violence have been made more severe.¹⁴⁴

As a result of Zambia’s policy environment, the country has made strides across the board with respect to women’s health status, labor force participation, and literacy rate as is highlighted in the following sections.

Political participation

Women’s participation in political and economic spheres in Zambia is low, however, it has continuously risen. The proportion of parliamentary seats held by women increased from 14% in 2011 to 18.9% in 2018.¹⁴⁵ However, recently women’s political representation has seen a decline in the most recent August 2021 election, with women now holding 13% of parliamentary seats and 21% of cabinet positions compared to 17% and 40% of seats in the executive cabinet in 2016. This decline is a result of the recent limited skills and resources available to women candidates required for political mobilization and the inability of leading parties to fulfill mandatory quotas for the representation of women within their cadre. Further, the election apparatus in the country has also been unable to curtail violence during polls, which dissuades women candidates and also impedes voter turnout among women voters.¹⁴⁶

Social and cultural context

There are 73 ethnic groups in Zambia, categorized into three social systems which include matrilineal patrilineal, and bilateral social structures. However, overall, Zambian society is based on patriarchal values and is highly conservative. Zambia has a dual legal system consisting of civil law and customary law, with customary law favoring men over women with respect to land access and marriage.¹⁴⁷ However, data suggest that Zambia has a very low gender disparity in ownership of a house with 35% of women owning a house either alone or jointly and 38% of men having home ownership rights alone or jointly.¹⁴⁸

The country presents high rates of GBV, with more than one-third (36%) of women age 15-49 having experienced physical violence at least once since age 15, and 14% have experienced sexual violence.¹⁴⁹

Health

¹⁴³ [Republic of Zambia \(2015\) THE GENDER EQUITY AND EQUALITY ACT, 2015](#)

¹⁴⁴ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁴⁵ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁴⁶ [European Democracy Hub \(2021\). EU support for women’s political participation and leadership under the EU’s Gender Action Plan: A case study on Zambia](#)

¹⁴⁷ [Food and Agriculture Organization of the United Nations \(2018\). National gender profile of agriculture and rural livelihoods, Zambia](#)

¹⁴⁸ [Zambia Statistics Agency, Ministry of Health \(MOH\) Zambia, and ICF \(2019\). Zambia Demographic and Health Survey 2018. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, Ministry of Health, and ICF](#)

¹⁴⁹ [EU Gender Action Plan III, 2021-2025. Zambia Country Level Implementation Plan \(CLIP\)](#)

The MMR in the country is much lesser than in other Sub-Saharan African countries. Maternal mortality rates have been decreasing over the years, from 398 deaths per 100,000 live births in 2013-14 to 252 in 2018 and 84% of deliveries by women took place at health institutions. Further, 97% of women who delivered children in the 5 years preceding the survey received antenatal care (ANC) from a skilled provider.¹⁵⁰ The Zambian government under the Gender Equity and Equality Act, also protects and promotes women's sexual and reproductive health for girls, and reinforces women's right over family planning. The country has also added Comprehensive Sexuality Education to the school curriculum and there has been an increased focus on youth in sexual education and service provision through training programs for healthcare providers and peer educators in adolescent-friendly sexual reproductive health services across districts.¹⁵¹ However, there continues to be an unmet need for family planning among 20% of married women in the country.¹⁵²

Education

As per World Bank estimates, about 83% of adult women in the country are literate and this estimate has been steadily rising over the past 2 decades.¹⁵³ This is a result of the country's holistic Girls Education and Women's Empowerment and Livelihoods (GEWEL) project to support girls from vulnerable households with school requisites and fees to enable them to complete secondary education. The project aims to increase access to secondary education for 25,000 adolescent girls coming from extremely poor households.¹⁵⁴ However, higher education is very low in the country, with the DHS estimating that only about 6% of women and 8% of men attain higher education.¹⁵⁵ However, in an effort to increase women's access to tertiary education, the country has now reserved 30% of state-offered bursaries to support women. Further, to encourage women's admission in STEM fields at the tertiary level, the country has reduced the cut-off score for girls for entry into these courses.¹⁵⁶

Women's economic participation and entrepreneurship

As per ILO estimates, the Labour force participation rate for women is at 71%, which is rather high compared to other SAA countries. The 2020 Labour Force survey from the country points out a lower LFPR for women however the survey indicates that at the national level, females accounted for a higher percentage of the working-age population (15 years old or older) at 52.8% compared to males at 47.2%.¹⁵⁷ However, men constitute 62% of formal employment, while women only 38% and 73% of women contribute to unpaid family labor.¹⁵⁸ The largest proportion of women works in agriculture, followed by sales professions. Within the agricultural sector, women represent only 19% of formal employees while men make up 81% of formal employees in the Agri sector.¹⁵⁹ However, the government has extended social security

¹⁵⁰ [Zambia Statistics Agency, Ministry of Health \(MOH\) Zambia, and ICF \(2019\). Zambia Demographic and Health Survey 2018. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, Ministry of Health, and ICF](#)

¹⁵¹ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁵² [Zambia Statistics Agency, Ministry of Health \(MOH\) Zambia, and ICF \(2019\). Zambia Demographic and Health Survey 2018. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, Ministry of Health, and ICF](#)

¹⁵³ [World Bank, 2018](#)

¹⁵⁴ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁵⁵ [Zambia Statistics Agency, Ministry of Health \(MOH\) Zambia, and ICF \(2019\). Zambia Demographic and Health Survey 2018. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, Ministry of Health, and ICF](#)

¹⁵⁶ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁵⁷ [Ministry of Labour and Social Security \(2020\). Labour Force Survey Report.](#)

¹⁵⁸ [EU Gender Action Plan III, 2021-2025. Zambia Country Level Implementation Plan \(CLIP\)](#)

¹⁵⁹ [Ministry of Labour and Social Security \(2020\). Labour Force Survey Report.](#)

cover to the informal sector including occupational safety, health, and protection for women, the youth, and the disabled.¹⁶⁰

Within the agriculture sector, the country has put in place plans and measures to enhance women's participation, and productivity and to ensure food security. For instance, the Agricultural Development and Value Chain Enhancement (ADVANCE) Project targets the women and youth in 288 Chiefdoms of the country by providing tractors and tillers. Furthermore, the Government has launched the Climate Change Gender Action Plan (ccGAP) to mainstream gender in the country's climate change policy, including in funding and programming.¹⁶¹

Gender and energy in Zambia

Zambia have experienced a high rate of deforestation, posing an increasing challenge to women and girls who are primarily responsible for fuelwood collection and need to travel long distances to collect firewood for household energy. Estimates suggest that women in Zambia spend on average 800 hours a year in fuelwood collection.¹⁶² Overall, only 32.8% population of Zambia has access to electricity.¹⁶³ However, compared to other Southern African countries, Zambia has one of the highest shares of modern renewables in energy consumption at 27.6 in 2017%. However, this share has seen a slight decline from 29.4%.¹⁶⁴

The Ministry of Energy conducted a Gender Assessment of the Energy Sector in Zambia in 2018 and is in the process of developing a Gender Mainstreaming Strategy and Action Plan.¹⁶⁵ Zambia's Energy Regulatory Board (ERB) has also committed to participate in the Women in Energy Regulation Internship Program, organized by the National Association of Regulatory Utility Commissioners (NARUC) with funding support from the U.S. Agency for International Development (USAID) and Power Africa to provide education and training opportunities and foster a gender-equitable workforce in the energy sector.¹⁶⁶

¹⁶⁰ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁶¹ [The Ministry of Gender, Zambia and UN Women \(2019\). Republic of Zambia: Progress Report on the Implementation of the Beijing Declaration and Platform for Action \(1995\) and the Outcomes of the Twenty-Third Special Session of the General Assembly \(2000\)](#)

¹⁶² [Food and Agriculture Organization of the United Nations \(2018\). National gender profile of agriculture and rural livelihoods. Zambia](#)

¹⁶³ [Zambia Demographic and Health Survey 2018](#)

¹⁶⁴ [UNESCO \(2021\). UNESCO Science Report: The race against time for smarter development](#)

¹⁶⁵

¹⁶⁶ [Zambian Regulators Demonstrate Ongoing Commitment to Gender Equity in the Energy Sector by Hosting Naruc Women in Energy Regulation Internship Program. \(2020, November\). National Association of Regulatory Utility Commissioners. Retrieved from https://www.naruc.org/international/news/zambian-regulators-demonstrate-ongoing-commitment-to-gender-equity-in-the-energy-sector-by-hosting-naruc-women-in-energy-regulation-internship-program/.](#)

Mozambique

Overview of gender equality and gender gaps in the country

Mozambique is an East African nation that has a population of 33.1 million people, with a young and adolescent population of 34% between the age group of 10 to the age of 24 years. The fertility rate in the country stands at 4.6.¹⁶⁷

The Gender Development Index in Mozambique is 0.922 and the country is placed with 'Group 4' countries with a medium to low equality in HDI achievements between women and men (absolute deviation of 7.5–10 percent). This is primarily driven by a gender gap in expected and mean years of schooling of 2.4 years among females and 4.1 years among males.¹⁶⁸ Mozambique has a Gender Inequality Index of 0.537 and is ranked 136, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.¹⁶⁹

Indicator	Result	Source and date
Life expectancy at birth	62.4	UNDP (2022) Gender Development Index
Maternal mortality ratio	289	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	50%	World Bank, 2017
Women's Labour Force Participation Rate (age 15 to 64)	78%	World Bank, 2019
Mean years of schooling ¹⁷⁰	2.4	UNDP (2022) Gender Development Index
Estimated gross income per capita	1,096 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Mozambique has ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1997 and signed on the Beijing Declaration and Platform for Action in 1995. Further, the country signed The Declaration of Gender in Africa in 2005 and the SADC protocol about Gender and Development in 2008, and the Maputo Protocol.¹⁷¹ The country also has a conducive framework for women to have access to land and productive assets through the 1997 Land Law which enables women to register themselves for their land use and the 2005 Amended Commercial Code under which women can engage in business without permission by their husbands. Further, under its constitution, the country also formulated a Law on Domestic Violence in 2009 to protect women from abuse and violence.¹⁷² The country

¹⁶⁷ [United Nations Population Fund, World Population Dashboard, Mozambique](#)

¹⁶⁸ [UNDP \(2022\) Gender Development Index](#)

¹⁶⁹ [UNDP \(2022\) Gender Inequality Index](#)

¹⁷⁰ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

¹⁷¹ [Japan International Cooperation Agency \(2015\) Country Gender Profile: Mozambique](#)

¹⁷² [Japan International Cooperation Agency \(2015\) Country Gender Profile: Mozambique](#)

also has a Gender Policy which was approved in 2018 and its corresponding Implementation Strategy was operationalized in the National Action Plan for the Advancement of Women (PNAM 2018-2024). The PNAM “aims at re-enforcing the promotion of gender equality in all spheres of development of the country, adopting a transversal approach to gender dimension, becoming an instrument of inter-sectoral coordination, to ensure gender equality and empowerment of women”.¹⁷³ However, there is limited punishment rape within marriage under the country’s penal code (only 2 to 6 years) whereas it is 2 to 8 years outside of marriage. While the legal framework of the country creates a mechanism for gender-equal laws to be implemented, the process of enforcing laws is rife with challenges, especially in rural areas, and limited awareness among women of their rights under the Mozambique Constitution and associated laws.¹⁷⁴

Political participation

Mozambique has a comparatively high representation of women in politics and has a smaller gender gap, compared to even some developed countries, with 42.4% parliamentarians being women and 57.6% men.¹⁷⁵ The government has a policy mandating 30% participation of women in local councils and associations. However, the country is far from achieving the target of 50% representation as committed under the 2008 SADC Protocol on Gender and Development. Currently, the country is performing poorly in creating a conducive environment for women to participate in the leadership and decision-making bodies of political parties in the country. Even at the municipal level, women leaders have limited representation with only six women elected as Mayors and 8 as Presidents of the Municipal Assemblies, among 53 municipalities in 2018’s assembly elections.¹⁷⁶ However, under the Gender Policy and its Implementation Strategy, Area 2 on Governance the country has categorically tried to focus on introducing legislative measures to ensure the equitable inclusion of women in top positions in the electoral lists of the political parties and groups of citizens and in decision-making bodies. A study by USAID suggests that women (especially younger women) face a lot of party resistance when entering political life and public decision-making.¹⁷⁷

Social and cultural context

While legally, men and women are accorded equal rights, the Mozambiquan society is patriarchal and gender-unequal customs and social order continue to be reinforced across family, community, and institutional systems. 21% of women in the country have experienced intimate partner violence in their lifetime.¹⁷⁸ However, studies suggest that there is severe underreporting of violence due to the perception that it is a private matter, with survivors being unaware of formal mechanisms to report and seek help and most survivors reaching out to community leaders and families.¹⁷⁹

Health

Mozambique has a maternal mortality ratio of 289 per 100,000 live births which is lower than other Sub-Saharan African regions.¹⁸⁰ Further, women’s agency of their fertility is also reflected in the increasing use

¹⁷³ [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁷⁴ [USAID \(2019\). Gender Assessment For Usaid/Mozambique Country Development Cooperation Strategy](#)

¹⁷⁵ [World Economic Forum \(2022\) Global Gender Gap Report](#)

¹⁷⁶ [Muabisa, E. N., Mendes de Brito, M. O., & Muendane, E. \(2022, March 4\). The contribution of women to and gender issues in the 2021/2022 electoral reform process in Mozambique. Institute for Democracy and Electoral Assistance.](#)

¹⁷⁷ [USAID \(2019\). Gender Assessment For Usaid/Mozambique Country Development Cooperation Strategy](#)

¹⁷⁸ [UN Women, Global Database on Violence against Women](#)

¹⁷⁹ [Jethá, E., Keygnaert, J., Martins, E. *et al.* Domestic violence in Mozambique: from policy to practice. BMC Public Health 21, 772 \(2021\). <https://doi.org/10.1186/s12889-021-10820-x>](#)

¹⁸⁰ [United Nations Population Fund, World Population Dashboard, Mozambique](#)

of contraception from 16.5% to 26.0% between 2010 and 2020. Still, 22% of married women report an unmet need for Family Planning.¹⁸¹ Further, the presence of skilled birth attendants has also increased from 48% in 2010 to 73% in 2020 (UNDP 2010 and 2020 as cited in UN Women report). However, the prevalence of HIV and AIDS is much higher among women in Mozambique compared to men. Overall, 15.4% of women compared to 10.1% of men are HIV positive in the country. A UN Women report suggests that is a consequence of the limited decision-making power women in Mozambique have as compared to the African and global averages and traditional practices whereby women are coerced to have unprotected sexual contact with partners.¹⁸²

Education

The country's literacy rate for females aged 15 and above is 50%.¹⁸³ As per the Ministry of Education and human development's 2018 survey, girls represent around 48% of the pupils across levels of education. However, there are vast regional differences in educational attainment with gender parity decreasing as one looks at higher grades, across the central and northern provinces of the country (Mined 2018 as cited in UN Women report).¹⁸⁴ For girls, school attendance continues to be an issue often leading to high dropout rates for girls.¹⁸⁵ The issue of adolescent pregnancy is also prevalent in the country, causing low attendance rates among girls, with about 38% of adolescent girls giving birth in Mozambique, which is the highest among the Southern African Development Community.¹⁸⁶ Due to factors such as unintended pregnancy, high burden of domestic work, distance to school, safety concerns, and poor quality of education, there is a limited progression of women to secondary and university levels of education.¹⁸⁷ As a result, only 31% of girls transition to secondary school, and only 2% complete university.¹⁸⁸

The Government has also started providing partial or full scholarships for women to enroll in vocational and technical education courses.¹⁸⁹

Women's economic participation and entrepreneurship

As per the Gender Inequality Index, there is a negligible disparity in Labour force participation (LFPR) with women's LFPR at 77.7% and men's LFPR at 79%.¹⁹⁰ However, the gender gap is evident when one looks at the quality and intensity of employment, with women working predominantly in the informal and subsistence sectors as self-employed or unpaid workers and working lesser hours in productive employment compared to men.¹⁹¹ While 31% of men are unpaid workers, this figure is much higher, at 44% for women. Agriculture remains the main sector of employment for both men and women followed by sales

¹⁸¹ [United Nations Population Fund, World Population Dashboard, Mozambique](#)

¹⁸² [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁸³ [World Bank, 2017](#)

¹⁸⁴ [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁸⁵ [USAID \(2019\). Gender Assessment For Usaid/Mozambique Country Development Cooperation Strategy](#)

¹⁸⁶ [Jaén-Sánchez, N., González-Azpeitia, G., Saavedra-Santana, P., Saavedra-Sanjuán, E., Manguiza, A. A., Manwere, N., Carranza-Rodriguez, C., Pérez-Arellano, J. L., & Serra-Majem, L. \(2020\). Adolescent motherhood in Mozambique. Consequences for pregnant women and newborns. *PloS one*, 15\(6\), e0233985. <https://doi.org/10.1371/journal.pone.0233985>](#)

¹⁸⁷ [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁸⁸ [A supportive sisterhood creates education and leadership opportunities for girls in Mozambique. \(2022, May 25\). UNESCO.](#)

¹⁸⁹ [Japan International Cooperation Agency \(2015\) Country Gender Profile: Mozambique](#)

¹⁹⁰ [UNDP \(2022\) Gender Inequality Index](#)

¹⁹¹ [Gradin, C. and Tarp, F. \(2019\). Gender Inequality in Employment in Mozambique. *South African Journal of Economics*, 87: 180-199. <https://doi.org/10.1111/saje.12220>](#)

and services.¹⁹² Women in urban areas are also employed as housekeepers, informal vendors, and cleaners, and secretaries. There is also a high wage disparity in the country with women earning only 69% of men.¹⁹³

While agriculture is the predominant form of employment, smallholder farm ownership among women in Mozambique is three times lower than that of men.¹⁹⁴ However, the country has recently reformed mobile financing systems and created avenues for access to microfinance and group saving schemes among women. This has been imperative for women to invest in short-cycle activities such as short-cycle agriculture like horticulture, trade, and agro-processing.¹⁹⁵

Gender and energy in Mozambique

Currently 30.6% of the population in Mozambique has access to electricity.¹⁹⁶ About 80% of the energy use is from biomass¹⁹⁷ which places a disproportionate burden on women to collect firewood. To expand the country's electrification, Mozambique launched Mozambique's National Energy for All Programme in 2018 which targets full access to electricity by 2030.¹⁹⁸

The share of modern renewables in Mozambique's final energy consumption has significantly increased between 2014 and 2017 from 18.7% to 30%. Vast gas reserves have been discovered in the Rovuma Basin of Mozambique, which will enable the country to become one of the top 10 suppliers of liquefied natural gas. However, the power, gas, and mining sectors combined have not been able to generate much employment in the extractive industry, creating no more than 1.1% of jobs in 2018.¹⁹⁹

In the country, there are existing initiatives and organizations that focus on expanding women's representation and building the capacity of women who are engaged in the energy sector, For instance,

1. The [Mozambique Women Of Energy](#) is a platform for women in the energy sector who identify areas of action for gender equality in this sector
2. [BRILHO programme](#), an SNV - Netherlands Development Organisation program in partnership with TechnoServe aims to increase the number of women with access to modern energy solutions in off-grid communities and enhance the inclusion and empowerment of women in the sector as collaborators and leading businesses. They do this through technical training of companies on gender-forward company policies and practices

¹⁹² [USAID \(2019\). Gender Assessment For Usaid/Mozambique Country Development Cooperation Strategy](#)

¹⁹³ [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁹⁴ [CGAP \(2016\). National Survey and Segmentation of Smallholder Households in Mozambique Understanding Their Demand for Financial, Agricultural, and Digital Solutions](#)

¹⁹⁵ [UN Women \(n.d.\) Mozambique National Gender Statistics Assessment](#)

¹⁹⁶ [World Bank \(2020\)](#)

¹⁹⁷ [European Union Energy Initiative \(2013\). Biomass Energy Strategy \(BEST\) Mozambique](#)

¹⁹⁸ [UNESCO \(2021\). UNESCO Science Report: The race against time for smarter development](#)

¹⁹⁹ [UNESCO \(2021\). UNESCO Science Report: The race against time for smarter development](#)

Sierra Leone

Overview of gender equality and gender gaps in the country

Sierra Leone is a country in West Africa, on the Atlantic Ocean. The country has a population of 8.3 million people, with a young and adolescent population of 33% between the age group of 10 to the age of 24 years. The fertility rate in the country stands at 3.9 which is somewhat lower compared to some of the other West African nations.²⁰⁰

The Gender Development Index in Sierra Leone is 0.893 and the country is placed among 'Group 5' countries with a high absolute deviation from gender parity, of more than 10%. This is primarily driven by a gender gap in mean years of schooling; with 3.5 years among females and 5.8 years among males.²⁰¹ Sierra Leone has a Gender Inequality Index of 0.633 and is ranked 162, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.²⁰²

Indicator	Result	Source and date
Life expectancy at birth	61.4	UNDP (2022) Gender Development Index
Maternal mortality ratio	1,120	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	40%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	59%	World Bank, 2019
Mean years of schooling ²⁰³	3.5	UNDP (2022) Gender Development Index
Estimated gross income per capita	1,453 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Sierra Leone has ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1988 and the Convention on the Rights of the Child (CRC) in 1990.²⁰⁴ The country has also ratified the Beijing Platform for Action (1995); UN General Assembly Resolution 52/86 on Crime Prevention and Criminal Justice Measures to Eliminate Violence against Women (1998), The Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol) (2003). Sierra Leone has also committed to the Solemn Declaration on Gender Equality in Africa adopted by African Union Heads of State and Government in July 2004; UNSC Resolution 1820 (2008) which calls on states to end impunity for sexual violence and ensure that all victims have equal protection under the law. Under Article 15 of Sierra Leone's constitution the foundations for gender equality are laid; with the commitment that

²⁰⁰ [United Nations Population Fund, World Population Dashboard, Sierra Leone](#)

²⁰¹ [UNDP \(2022\) Gender Development Index](#)

²⁰² [UNDP \(2022\) Gender Inequality Index](#)

²⁰³ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

²⁰⁴ [United Nations Human Rights Treaty Bodies, Sierra Leone](#)

“..every person in Sierra Leone is entitled to the fundamental human rights and freedoms of the individual, that is to say, has the right, whatever his race, tribe, place of origin, political opinion, color, creed or sex..”²⁰⁵

The country has laid a solid legal foundation for gender equality with the recent Gender Equality and Women's Empowerment (GEWE) policy released in 2021, which is a consolidation of the country's existing National Policy on the Advancement of Women and the National Policy on Gender Mainstreaming adopted in 2000. Apart from this, the country has three other Gender Justice Acts; including the Child Rights Act; the Anti-Human Traffic Act; the Sexual Offences Act 2012 (amended in 2019). The GEWE has the following broad objectives; which it aims to meet through a multisectoral approach²⁰⁶

1. Achieve gender equity and equality in all sectors and enable women to take their rightful position in national development, political leadership, and decision-making;
2. Ensure satisfactory quality of life for women and girls and eliminate the feminization of poverty;
3. Eliminate all forms of GBV and discrimination

Taking cognizance of the need to educate and sensitize men, the country also released a Male Involvement Strategy for the Prevention of SGBV in Sierra Leone; aiming to encourage the participation of men and boys as change agents and allies in changing power relations to champion women's and girls' rights.²⁰⁷ However, a study has indicated that the donor-funded (UNFPA) objective and intended approach to male engagement in the country have not necessarily translated to project design and implementation by local organizations. And while the involvement of men towards maternal care is reaping dividends, the training aimed at empowering men to engage in women's empowerment, may in fact run a risk of reinforcing patriarchal structures that dictate that men are responsible for 'protecting women', taking the agency away from women.²⁰⁸

Political participation

In Sierra Leone, the position of women's political representation remains dismal with only 12.3% of women MPs being women.²⁰⁹ Under the Gender Equality and Women's Empowerment Policy, Sierra Leone's cabinet committed to ensuring women's active participation in political leadership and decision-making at all levels by setting aside 30% of elective and public offices for women.²¹⁰ To enforce compliance with this quota system; the country has also amended the Political Parties Act, to replace the Political Parties Registration Commission with the Political Parties Regulation Commission, enhancing their statutory powers to penalize parties that fail to comply with quota provisions.²¹¹ However, despite these measures, women candidates face multiple barriers that impede their participation in public political life including violent hostility, intimidation, limited fund availability, low literacy, and confidence.²¹² Further, customary

²⁰⁵ [The Ministry of Gender and Children's Affairs, Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²⁰⁶ [The Ministry of Gender and Children's Affairs, Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²⁰⁷ [The Ministry of Gender and Children's Affairs, Sierra Leone \(n.d\). National Male Involvement Strategy for the Prevention of Sexual and Gender-Based Violence in Sierra Leone](#)

²⁰⁸ [The Institute of Gender Research and Documentation \(2022\). The male engagement strategy in Sierra Leone: A feminist analysis](#)

²⁰⁹ [Inter-Parliamentary Union, Sierra Leone](#)

²¹⁰ [The Ministry of Gender and Children's Affairs, Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²¹¹ [M'Cormack-Hale, F., & Amoah, M. A. \(2022, December 1\). Sierra Leoneans welcome government efforts to address gender inequalities, call for more. Afro Barometer. Institute for Governance Reform \(IGR\) and Ghana Center for Democratic Development \(CDD-Ghana\). Retrieved January 24, 2023, from <https://www.afrobarometer.org/publication/ad580-sierra-leoneans-welcome-government-efforts-to-address-gender-inequalities-call-for-more/>.](#)

²¹² [UNDP. \(2022, November 24\). Women's voice matters: Supporting Inclusive Election in Sierra Leone \[web log\]. Retrieved January 22, 2023, from <https://www.undp.org/sierra-leone/stories/womens-voice-matters-supporting-inclusive-election-sierra-leone>](#)

laws are also in contrast to statutory laws especially at the local levels, with women not being recognized as Chieftains in the Northern Regions and in Kailahun and Kono Districts in the Eastern Region.²¹³

Social and cultural context

The presence of customary laws in Sierra Leone undermines women's position as owners of land tenure. Since the majority of ethnicities are patrilinear, women are not entitled to land. And while it is a criminal offense to evict widowed women from their land, there continues to be ignorance about the legal rights of women to land tenure and reported dispossession of land among widows or divorced women.²¹⁴ In Sierra Leone, 53% of women have experienced Physical and/or Sexual Intimate Partner Violence in their lifetime.²¹⁵

Gender-based violence remains a critical area of concern in Sierra Leone, with the President declaring a State of Emergency on rape and sexual violence in February 2019 which led to the amendment of the Sexual Offences Act of 2012 by Parliament in 2019.²¹⁶ As per the amended Sexual Offences Act, the maximum penalty for rape and sexual penetration of a child has been increased from fifteen years to life imprisonment; and provisions have been made for the introduction of prosecution for aggravated sexual assault under this Act.²¹⁷

Health

Sierra Leone has one of the world's highest maternal mortality ratios of 1,120 per 100,000 live births as per the UN database, just behind Chad and South Sudan.²¹⁸ The country's infrastructure withstood stresses under the country's civil war and Ebola epidemic and the public health infrastructure continues to face challenges of inadequate health practitioner staff, low skills among health workers, limited availability of medical equipment and supplies, as well as delayed referrals from lower level facilities. Further, while almost all pregnant women make 1 ANC visit, only half of these women undergo HIV testing during ANC visits.²¹⁹ Women's agency to decide their age of marriage and use contraception for family planning is also limited in Sierra Leone. 29.6% of girls are married before the age of 18, 21% of women aged 15-19 had a birth or were pregnant (in 2019) and 25% of married women report an unmet need of family planning, which has not changed between 2013 and 2019. Women also have poor nutritional status with 47% of women aged 15-49 being anemic.²²⁰

Education

Literacy rate among women aged 15 and above is only 40% in Sierra Leone. Regionally this is lower than other African countries and also much lower compared to the literacy rate among adult boys which stands at 55%, as per World Bank data.²²¹ The gap is even wider as per the country's Demographic and Health survey, with 42% literacy among women compared with 62% among men. Data from 2015 indicates that

²¹³ [The Ministry of Gender and Children's Affairs, Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²¹⁴ [S. B. Mabikke, R. A. Ahene, M. P. Rizzo, F. Romano. 2020. Innovative Solutions to Protect Women's Customary Land Rights in Sierra Leone. Land Tenure Journal no. 1-20. Rome. FAO.](#)

²¹⁵ [UN Women, Global Database on Violence against Women](#)

²¹⁶ [The Ministry of Gender and Children's Affairs, Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²¹⁷ [Sexual Offences \(Amendment\) Act, Vol. CXLX, No. 80 \(2019\)](#)

²¹⁸ [United Nations Population Fund, World Population Dashboard, Mozambique](#)

²¹⁹ [Ministry Of Health And Sanitation \(n.d\). Sierra Leone Rmncah Strategy 2017 - 2021](#)

²²⁰ [Statistics Sierra Leone \(Stats SL\) and ICF. 2020. Sierra Leone Demographic and Health Survey 2019. Freetown, Sierra Leone, and Rockville, Maryland, USA: Stats SL and ICF.](#)

²²¹ [World Bank, 2021](#)

for Grade 2 in lower secondary about 13.6% of girls dropped out of school compared to 10.6% of boys, attributed to harassment and abuse from teachers and male peers, forced marriages, lack of sexual health education, and inadequate support mechanisms.^{222,223} Further, the government had also imposed restrictions on pregnant girls from attending school curtailing the scope to complete their education which was only recently overturned in 2020.²²⁴

A UNESCO study cites that in 2017 almost a third of girls who dropped out of school did so because of pregnancy. The study however, also highlights an encouraging trend based on analysis of the country's tertiary institution census; the share of women enrolled in tertiary education has increased from 43% to 48% between 2017 to 2019, and the gender gap in access rate (number of students per 100,000 inhabitants) has halved from 350 to 159. Further, the study points out that a majority of students enrolled in Technical, Vocation Education and Training (TVET) are women (54%), and 39% of Science, Technology, Engineering, Agriculture, and Mathematics (STEAM) students are women. However, only one in five women take up STEAM studies in contrast with nearly one in three men, and enrollment to a variety of TVET courses is highly gendered with women concentrated in service sector training courses (like tailoring, catering, and hairdressing) compared to men in manufacturing and construction related training. To encourage STEM education, the Government has established a grant-in-aid policy for providing scholarships to all students studying these subjects.²²⁵

Women's economic participation and entrepreneurship

While World Bank estimates point to similar LFPR among men (58%) and women (59%) aged 15 to 64²²⁶, the country's DHS points to a disparity with men more likely to be employed than women (72% versus 69%).²²⁷ Although the trends around women's tertiary and technical education are encouraging, these don't necessarily translate to women's transition to gainful employment opportunities, where existing disparities and discrimination undermine young women's competitive position compared to men. In 2018 for instance, the Employment-to-Population Ratio for young women with tertiary education is 10 percentage points lower than young men's, and the gap widens to 12 percentage points in higher education. Tertiary educated women who are NEET (not in employment, education, or training) is 24% compared to 17% men. However, any form of vocational education significantly increases women's ability to start looking for work, much more effectively than men; with young female NEETs who received any formal vocational training relatively much less subject to idleness (defined by a state where they are not even looking for work) (58%) than their male counterparts (74%).²²⁸

Sectorally, women are majorly concentrated in agriculture (54%) and sales (37%) jobs, and are less likely, compared to men, to be in skilled manual work (4% versus 9%) or in managerial/technical/professional roles (3% versus 14%). Women's position as workers remains dismal, as an extremely high percentage of women in agricultural work (69%) are unpaid and 31% of women in non-agricultural work remain unpaid.²²⁹

²²² [The Ministry of Gender and Children's Affairs. Sierra Leone \(2021\). The Gender Equality and Women's Empowerment Policy](#)

²²³ [IEP-UNESCO Dakar \(2020\). Financial supportTechnical supportEducation Sector AnalysisAssessing the enabling environment for gender equality](#)

²²⁴ Calimoutou, E. (2020, September 17). Sierra Leone took a major first step towards keeping girls in school and inclusive education [web log]. Retrieved January 19, 2023, from <https://blogs.worldbank.org/nasikiliza/sierra-leone-took-major-first-step-towards-keeping-girls-school-and-inclusive-education>

²²⁵ [IEP-UNESCO Dakar \(2020\). Financial supportTechnical supportEducation Sector AnalysisAssessing the enabling environment for gender equality](#)

²²⁶ [World Bank, 2019](#)

²²⁷ [Statistics Sierra Leone \(Stats SL\) and ICF. 2020. Sierra Leone Demographic and Health Survey 2019. Freetown, Sierra Leone, and Rockville, Maryland, USA: Stats SL and ICF.](#)

²²⁸ [IEP-UNESCO Dakar \(2020\). Financial supportTechnical supportEducation Sector AnalysisAssessing the enabling environment for gender equality](#)

²²⁹ [Statistics Sierra Leone \(Stats SL\) and ICF. 2020. Sierra Leone Demographic and Health Survey 2019. Freetown, Sierra Leone, and Rockville, Maryland, USA: Stats SL and ICF.](#)

While there are no recent estimates of women entrepreneurs in Sierra Leone, an IFC study from 2015 estimated 2.1 million women entrepreneurs engaged in micro-enterprises. However, these women entrepreneurs face myriad barriers to accessing finance for their businesses and primarily access finance from personal sources such as gifts or personal savings or rely on informal community-based fund mobilization for petty cash, through the Self-Rotating Saving and Loan Scheme locally known as *osusu*.²³⁰

Gender and energy in Sierra Leone

Electrification rates in Sierra Leone are low compared to other countries in the region. Only a small portion of the population in Sierra Leone has access to electricity, estimated to be between 26.2% (World Bank, 2020)²³¹ and 23% (DHS, 2020). About 64.4% of cooking fuel comes from wood, and 32.6% is charcoal, putting women at a high risk of smoke inhalation and respiratory diseases.²³²

Sierra Leone is a part of ECOWAS and is implementing their policies for gender mainstreaming in Energy Access. Under ECOWAS, The Ministry of Energy is responsible for supervising the Barefoot Solar Engineering Training Centre, established to undertake rural electrification projects and to provide solar training to illiterate women.²³³ The country also has a Barefoot Women Solar Engineers Association of Sierra Leone, started by women trained at Barefoot college India on the installation and maintenance of solar PV systems, and installation of solar pumps. While these women are not formally educated, they have been able to play a critical role in last-mile access to energy for households in 32 towns and villages.²³⁴ The Barefoot women have also provided solar connections to 60 community clinics in northern Sierra Leone, which earlier lacked continuous electricity, putting lives at risk, especially making childbirth risky at these clinics. With solar panels, Barefoot women have been able to ensure cheap, clean, and stable connections thus making institutional deliveries at these facilities much safer for women.²³⁵ Between 2014 to 2017, 50 solar women technicians were trained under this program and solar PV systems were installed in 3,500 Households.²³⁶

Among off-grid energy providers operating in Sierra Leone, the women-led company Smiling Through Light founded by Mariamma Kamara is leading the way in deploying women to provide clean energy access through the distribution and sale of solar products, focusing on the last mile. The company has a 100% female sales agent force and 80% of the entire organization including leadership is made up of women. Smiling Through Light has trained 1000 individuals in Sierra Leone on clean energy access using a storytelling approach and provided solar lamps to 400 school children in off-grid communities under their 'One Child, One Lamp' campaign.²³⁷

²³⁰ [IFC \(2014\). National Study on Women's Access to Financing in Sierra Leone](#)

²³¹ [World Bank \(2020\)](#)

²³² [Statistics Sierra Leone \(Stats SL\) and ICF. 2020. Sierra Leone Demographic and Health Survey 2019. Freetown, Sierra Leone, and Rockville, Maryland, USA: Stats SL and ICF.](#)

²³³ [ECREE \(n.d\). Status of Gender Mainstreaming in Energy Access in Sierra Leone](#)

²³⁴ Burrell, L. (2021, March 5). The sky is the limit: Sierra Leone's 'barefoot' women solar engineers [web log]. Retrieved January 19, 2023, from <https://www.unido.org/stories/sky-limit-sierra-leones-barefoot-women-solar-engineers>.

²³⁵ [OneEarth \(n.d\). How women in Sierra Leone are powering health clinics with solar energy](#)

²³⁶ [Intellectap \(2019\). Mapping the Off-Grid Solar market in Sierra Leone](#)

²³⁷ [USAID \(n.d\) Women Champions in the Energy Sector](#)

Guinea

Overview of gender equality and gender gaps in the country

Guinea is a country in West Africa with vast nature reserves. The country has a population of 13.9 million people, with a young and adolescent population of 34% between the age group of 10 to the age of 24 years. The fertility rate in the country, at 4.4, is very high compared to some other countries in the West African region.²³⁸

The Gender Development Index in Guinea is 0.850 and the government is placed with 'Group 5' countries with a high absolute deviation from gender parity, of more than 10%. While the country does not have a very high difference between men's and women's income, the high deviation is potentially a result of high gender gap in mean years of schooling with 1.3 years among females and 3.2 years among males.²³⁹ Guinea has a Gender Inequality Index of 0.621 and is ranked 157, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.²⁴⁰

Indicator	Result	Source and date
Life expectancy at birth	60.1	UNDP (2022) Gender Development Index
Maternal mortality ratio	576	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	31%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	65%	World Bank, 2019
Mean years of schooling ²⁴¹	1.3	UNDP (2022) Gender Development Index
Estimated gross income per capita	2,320 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Guinea, similar to many other Sub-Saharan and West African countries recognizes customary law along with statutory laws. This, in some areas of social and cultural life, impedes women's ability to exercise their agency as equal citizens, especially in the inheritance of wealth and land, where while statutory law grants women 1/8th of wealth and equal rights to land, customary law prevents women from claiming these.²⁴² In the international arena, Guinea has committed to gender equality through ratification of various UN and regional conventions, such as; the Convention on the Elimination of all forms of Discrimination Against Women (CEDAW) in 1982, the African Charter on Human and Peoples' Rights on the Rights of Women to

²³⁸ [United Nations Population Fund, World Population Dashboard, Guinea](#)

²³⁹ [UNDP \(2022\) Gender Development Index](#)

²⁴⁰ [UNDP \(2022\) Gender Inequality Index](#)

²⁴¹ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

²⁴² [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

"ensure the elimination of all discrimination against women and ensure the protection of women's rights as stated in international declarations and conventions", the Draft Supplementary Act on Equality of Rights between Women and Men for Sustainable Development within the ECOWAS Region, the Draft ECOWAS Plan of Action on Gender and Trade, the Draft ECOWAS Gender and Migration Framework and Plan of Action; and the Common Gender Policy of the West African Economic and Monetary Union (WAEMU) in 2016.²⁴³

The Constitution of Guinea was amended in 2001 to put in place a regulatory environment that is non-discriminatory and supports gender equality.²⁴⁴ Further, in 2020, the constitution was amended to specify the age of marriage for girls at 18, and include "physical integrity" as part of their rights.²⁴⁵ However, various Civil Codes reinforce men's familial authority by recognizing them as the sole head of the household, with sole guardianship of children and Guinean women cannot transfer citizenship to children through marriage to men outside the country.²⁴⁶ However, certain limiting legal restrictions curtail women's full participation in the Guinean economy, such as women having to seek their husband's approval to pursue a trade or profession of their choice.²⁴⁷

Political participation

As per the Global Gender Gap report, there are 29.6% women in parliament in Guinea compared to 70.4% men.²⁴⁸ A recent report from 2021 by the country's first woman Economy and Finance Minister suggests that Guinea has close to 31% of ministerial portfolios being held by women as of 2021, which is about 9% higher than the average from the Economic Community of West African States (ECOWAS) countries and there has been a 57% increase in the number of women in politics between 2010 and 2021. The report however notes that while more women have joined public life, their representation in decision-making positions is still very low, with only 25 women hold holding directorial positions, compared to 67% having intermediary positions such as division and section heads.²⁴⁹ While Guinea has Gender Focal Points designated in all Ministries to undertake gender-responsive planning and budgeting, research suggests that GFPs (often women in low-level positions) do not have clear mandates and find it difficult to navigate and assert their authority in male-dominated ministerial networks.²⁵⁰

Further, many studies point to the ineffective civil registration system in Guinea, due to which women are unable to register for identification proofs, curtailing their ability to participate in public and political life.^{251,252} Further, mobility restrictions and safety concerns during political demonstrations, spousal permissions required, time poverty, and low education rates all limit women's ability to effectively contribute

²⁴³ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁴⁴ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁴⁵ [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

²⁴⁶ [OECD Development Centre \(n.d\). Social Institutions and Gender Index: Guinea](#)

²⁴⁷ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁴⁸ [World Economic Forum \(2022\) Global Gender Gap Report](#)

²⁴⁹ [Falémé Conseil \(2021\). Fc Succinct Sur La Diversité Genre: Tableau de bord](#)

²⁵⁰ [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

²⁵¹ [UN Women \(n.d\). Supporting Women's Empowerment And Gender Equality In Fragile States Research Brief: Guinea](#)

²⁵² [Centre Of Excellence for CRVS Systems \(n.d\). Knowledge Briefs on Gender and CRVS: Gender Equity in Guinea: How Social Norms Affect the Civil Registration of Women and Girls in West Africa](#)

to political life in Guinea.²⁵³ However, recently the country has adopted a significant legal mandate, which dictates that women must make up 50 percent of candidate lists for elective positions.²⁵⁴

Social and cultural context

Customary and religious practices in Guinea severely limit women's ability to realize their rights as citizens of Guinea. Within marriage and families, women's position is undermined under traditional and customary law. Child marriage is also rampant in the country with 46.5% of girls married off before the age of 18.²⁵⁵ Even though polygamy is prohibited under statutory law,²⁵⁶ among married women in age 15 to 49, 42% of women are in polygamous unions.²⁵⁷ Guinea also has one of the world's highest rates of female genital mutilation (FGM) with 94.5% of women in age 15-49 reporting that they have experienced FGM,²⁵⁸ traditionally regarded as an important socio-cultural rite of passage for initiation of girls into adulthood and marriage which is ceremoniously celebrated. On the other hand, patriarchal norms dictate that women must be chaste before marriage, and forced marriage (especially for widows) is common. At the community level, there is judicial inaction on FGM, leading to the continued practice of FGM. And while there have been increased awareness drives to educate people on legal sanctions imposed under the law on FGM, this has only led to people forgoing celebratory rituals to conduct FGM more privately.^{259,260}

Health

The state of reproductive agency among women is poor in Guinea. The high rate of FGM highlighted in the previous section is also coupled with inadequate poor maternal care women receive. While at 576, Guinea has a lower MMR than some other countries in the region, Guinea has a long way to go to reach the SDG target of less than 70 by 2030. Close to 28% of all female deaths in the country are maternal deaths.²⁶¹ This could be due to a variety of factors, such as high rates of anemia among women, 46% of women in the 15-49 age group are anemic, limited usage of and access to modern contraception, with 36% of unmarried sexually active women stating that their contraception needs are unmet and 22% of all women reporting unmet family planning needs.²⁶²

Education

Guinea has the lowest mean years of schooling among girls in the world at 1.3 years as per Gender Development Index.²⁶³ As per the country's DHS, there is a wide gender gap in education between young

²⁵³ [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

²⁵⁴ Guinea adopts Law on Parity, securing equal representation of women on electoral lists. (2019, May 10). *UN Women*. Retrieved January 20, 2023, from <https://www.unwomen.org/en/news/stories/2019/5/news-guinea-adopts-law-on-parity>.

²⁵⁵ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁵⁶ [OECD Development Centre \(n.d\). Social Institutions and Gender Index: Guinea](#)

²⁵⁷ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁵⁸ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁵⁹ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁶⁰ [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

²⁶¹ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁶² [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁶³ [UNDP \(2022\) Gender Development Index](#)

females and males with only 39% literate women in the age group, compared with 65% young men in the same age group.²⁶⁴ Further, as per world development indicators, the female-to-male enrollment ratio drops significantly as learners transition from primary to tertiary levels, and at the tertiary level, it stands at 0.45, which is lesser than the Sub-Saharan African region on average.²⁶⁵ As per the 2018 DHS, 63% male vs 58% female in primary education, and 32% male vs 25% female in secondary school. In rural areas, only 11% of girls attend secondary school vs 18% of boys.²⁶⁶

The country has rural second-chance education schools known as NAFA centers that cater to school dropouts or those that have never attended schools, supporting them to gain literacy and numeracy skills and as of 2019 there were 13,000 students enrolled in these centers.²⁶⁷

Women's economic participation and entrepreneurship

As per the country's DHS, the Labour force participation rate among women is 66% while among men it is 81%. Agriculture, animal husbandry, and fishing or hunting are the main sectors of employment for both men and women. Interestingly however, unlike a lot of other Sub-Saharan African nations, 64% of women in Guinea are self-employed in agriculture instead of working as family members on farms. However, only 29% of women working in agriculture were remunerated in cash.²⁶⁸ Women's agricultural work is 8 percentage points less productive than men's in Guinea due to disparity in land ownership and men being much more likely to use farm inputs such as fertilizer, irrigation, and high-quality seeds.²⁶⁹ Among women, there has been a steep decline in the proportion of those employed in agriculture, from 52% to 37%. There has been a transition of women from agriculture to sales professions, with 32% of women working in sales and services.²⁷⁰

Culturally, entrepreneurship is not seen as a respectable work avenue for women in Guinea.²⁷¹ Regionally, Guinea is much behind average in terms of female entrepreneurship with only 9% of firms having female founders, compared to 29% in the Sub-Saharan African region, as per the 2016 World Bank Enterprise Survey for Guinea (*conducted for formal enterprises, and excluding microenterprises, where women are largely concentrated*). Further, about 25% of enterprises in the country had no female employees.²⁷²

As per Global Findex women are behind men by 13% points in bank account ownership, with 24% of women having accounts, compared to 37% of men. Similarly, in terms of mobile money account holding, women are behind men with 17% of females having such accounts compared to 26% of men.²⁷³ While there has been significant growth of MFIs and fintech in the country, which could be a way to tackle gendered norms and empower women financially, tedious KYC processes, power connections in the hinterland, and weak

²⁶⁴ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁶⁵ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁶⁶ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁶⁷ [The World Bank \(2020\). Implementation Completion And Results Report On A Global Partnership For Education Grant To The Republic Of Guinea For The Guinea - Pooled-Fund For Basic Education](#)

²⁶⁸ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁶⁹ [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁷⁰ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁷¹ [Groggel, L., Sow, A. B., & Gnimassou, R.-M. A. \(2020\). \(rep.\). USAID/Guinea CDCS Gender Analysis Report. USAID/Guinea CDCS Gender Analysis Report.](#)

²⁷² [World Bank \(2019\). Guinea: The Economic Benefits of a Gender Inclusive Society. Washington, DC: World Bank Group](#)

²⁷³ [The World Bank \(2021\) The Global Findex Database](#)

IT infrastructure and interoperability between banks and fintech pose a threat to the evolution of mobile money in the country.²⁷⁴

Gender and energy in Guinea

Currently, only 44% of households in Guinea have access to electricity, and only 5% of the population has access to clean cooking technologies.²⁷⁵ In Guinea, cooking is predominantly done outside the home and 60% of households use wood for cooking.²⁷⁶

Among notable initiatives in the country, the Women's Fund for Gender Equality (FGE) developed a training program on renewable energy technologies to promote sustainable livelihoods for women and in partnership with a local grantee organization Partenariat - Recherches - Environnement - Médias (PREM), trained 300 women in Guinea's Tristao Islands how to plant, grow, harvest, and sell products made from moringa plants using solar polytunnel dryers. This resulted in 25,000 moringa trees being planted and greenhouse gas emissions being reduced by 40 tonnes as a result of solar drying technology, between 2013 to 2016.^{277, 278}

Currently, there is an ongoing Research Project on Energy Transition for Women's Economic Empowerment through the Horticultural Value Chain in Guinea, to understand and address mechanisms for use of solar irrigation systems in the horticultural value chain, for irrigation, processing, and preservation. The project which aims to improve the profitability and working time of rural women by 2025, is also building the capacity of women to use solar technologies and access finance. The program is being implemented by the *Initiative prospective agricole et rurale* (IPAR) in collaboration with the *Centre d'étude et de coopération internationale* (CECI) and is funded by the International Development Research Centre (IDRC).²⁷⁹

²⁷⁴ [African Center for Economic Transformation \(ACET\). 2019. Promoting Women and Youth Financial Inclusion for Entrepreneurship and Job Creation: Comparative Study of Selected Sub-Saharan African Countries: Guinea Country Policy Brief.](#)

²⁷⁵ [International Renewable Energy Agency. \(n.d\). Energy Profile, Guinea](#)

²⁷⁶ [Institut National de la Statistique \(INS\) et ICF. 2018. Enquête Démographique et de Santé en Guinée 2018. Conakry, Guinée, et Rockville, Maryland, USA : INS et ICF.](#)

²⁷⁷ [UN Women \(n.d\). Supporting Women's Empowerment And Gender Equality In Fragile States Research Brief: Guinea](#)

²⁷⁸ [The Borgen Project \(n.d\). Women's Empowerment In Guinea](#)

²⁷⁹ [Wansi, B. I. \(2022, June 20\). SENEGAL/GUINEA: Project to install solar irrigation systems \[web log\]. Retrieved January 31, 2023, from https://www.afrik21.africa/en/senegal-guinea-project-to-install-solar-irrigation-systems/.](#)

Burundi

Overview of gender equality and gender gaps in the country

Burundi is a country in East Africa, south of the Equator. The country has a population of 12.6 million people and one of Africa's highest population densities. Burundi has a young and adolescent population of 32% between the age group of 10 to 24. The fertility rate in the country stands at 5.1.²⁸⁰

The Gender Development Index in Burundi is 0.935, and the country is placed with 'Group 3' countries with medium equality in HDI achievements between women and men, with an absolute deviation between 5% and 7.5%. Although there is a gender gap in mean years of schooling, with 2.5 years among females and 3.9 years among males, there is a relatively low difference between women and men in the estimated gross national income per capita, compared to other East African countries (\$668 for women and \$797 for men).²⁸¹ Burundi has a Gender Inequality Index of 0.505 and is ranked 127; the index measures disparities between men and women in economic opportunities, reproductive health, and empowerment.²⁸²

Indicator	Result	Source and date
Life expectancy at birth	63.6	UNDP (2022) Gender Development Index
Maternal mortality ratio	548	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	68%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	82%	World Bank, 2019
Mean years of schooling ²⁸³	2.5	UNDP (2022) Gender Development Index
Estimated gross income per capita	668 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Several national and international policy instruments are in place in Burundi to advance gender equality. Burundi ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1992 and the Convention on the Rights of the Child (CRC) in 1990.²⁸⁴ As a member of the African Union (AU), the AU's Gender Equality and Women's Empowerment 2018-2028 (GEWE) strategy is part of Burundi's regional regulatory framework. The GEWE strategy focuses on four key pillars, namely:

- Maximizing opportunities, outcomes and E-TECH dividends
- Dignity, security and resilience
- Effective laws, policies, and institutions

²⁸⁰ [United Nations Population Fund, World Population Dashboard, Burundi](#)

²⁸¹ [UNDP \(2022\) Gender Development Index](#)

²⁸² [UNDP \(2022\) Gender Inequality Index](#)

²⁸³ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

²⁸⁴ [United Nations Human Rights Treaty Bodies, Burundi](#)

- Leadership, voice, and visibility²⁸⁵

The Burundian government also adopted the Optional Protocol to the Convention on the Rights of the Child on the Sale of Children, Child Prostitution, and Child Pornography in November 2007, reinforcing the legal protections from sexual violence and exploitation for women and girls.²⁸⁶

Within the sub-region, the Pact on Peace and Security in the Great Lakes region was adopted during the International Conference on the Great Lakes region in 2006, which established measures to address violence against women and children through its Protocol on the Prevention and Suppression of Sexual Violence Against Women and Children.²⁸⁷

Regarding the national legislation, the Constitution of the Republic of Burundi (2005, Art 13) recognizes gender equality stating that ‘All Burundians are born equal in merit and dignity. All citizens enjoy the same rights and are entitled to the same protection by the Law. No Burundian shall be excluded from the social, economic, or political life of the nation by reason of race, language, religion, sex, or ethnic origin’.²⁸⁸ The Public Administration Statute (2006) prohibits discrimination in recruitment, pay or dismissals based on sex, in line with the Labour Code.²⁸⁹ Recently (2020 – 2021), Burundi mandated equal remuneration for work of equal value.²⁹⁰

Political participation

The 2005 Constitution sets a quota of 30% to ensure the participation of women in decision-making bodies such as the National Assembly and government (Art. 129 and 164) to address women’s underrepresentation in public life.²⁹¹ This milestone was achieved through the leaders of women’s associations’ advocacy at the time of the Arusha Agreement.²⁹² ²⁹³ According to the 2022 Global Gender Gap report, women hold 38.2% of the seats in Parliament and 37.5% of the ministerial positions.²⁹⁴ Burundi had its first women Prime Ministry, Sylvie Kinig in 1994.

Social and cultural context

Burundi is characterized as a patriarchal society where women have very little decision-making power compared with men. These norms determine power dynamics between men and women from an early age and are reflected in the high rates of physical and economic exploitation of women.²⁹⁵ Women’s primary responsibility in Burundi is considered to be the care of the household, which results in carrying out a disproportionate amount of the burden of domestic work, leaving them with little time to accomplish anything

²⁸⁵ [African Union \(2018\). AU Strategy for Gender Equality & Women’s Empowerment 2018-2028](#)

²⁸⁶ [United Nations General Assembly \(2002\). Optional Protocol to the Convention on the Rights of the Child on the Sale of Children, Child Prostitution and Child Pornography](#)

²⁸⁷ [International Conference on the Great Lakes Region \(2006\). Pact on Security, Stability and Development in the Great Lakes Region](#)

²⁸⁸ [National Constitution of Burundi, 2005.](#)

²⁸⁹ Law No. 1-037 of 7 July 1993, Article 6. Cited in [UNDP \(2012\). Gender Equality and Women’s Empowerment in Public Administration. Burundi Case Study.](#)

²⁹⁰ [World Bank Group \(2022\). Women, Business and the Law](#)

²⁹¹ [National Constitution of Burundi, 2005.](#)

²⁹² The Arusha Peace and Reconciliation Agreement, signed in 2000 marks the end of 12 years of civil in Burundi.

²⁹³ [UNDP \(2012\). Gender Equality and Women’s Empowerment in Public Administration. Burundi Case Study.](#)

²⁹⁴ [World Economic Forum \(2022\) Global Gender Gap Report](#)

²⁹⁵ [USAID \(2021\). Analysis of cultural barriers to women’s economic empowerment in Burundi.](#)

else, including financial independence.²⁹⁶ The early marriage rate in Burundi is 9.9%, lower than in other countries in Sub-Saharan Africa.²⁹⁷

Women's rights over land are governed by customary law, which discriminates against women. In the context of a patriarchal culture such as the Burundian, "women do not inherit the land, they are only usufructuaries and their right to use it depends on the goodwill of their parents, brothers and sometimes their in-laws."²⁹⁸ This is particularly problematic for women because it leaves them unprotected in times of hardship or when they are abandoned or widowed.²⁹⁹

GBV is a pervasive problem in Burundi. Nearly close to half of the women in the country (46.7%) have suffered intimate partner physical and/or sexual violence.³⁰⁰ Domestic violence "is a mainstream expectation within family life and something which is seen as a necessity and, to a certain extent, as a good thing. It reflects the woman's status as being inferior".³⁰¹ Economic violence is also common in Burundi. It is exerted by prohibiting victims from accessing or using household assets, the crops harvested by them and, controlling their access to health care, employment, and excluding them from the household expenditures decision-making.³⁰²

Health

Women of reproductive age (15-49 years) often face barriers with respect to their sexual and reproductive health and rights: in 2017, 39.6% of women had their need for family planning satisfied with modern methods.³⁰³ The adolescent birth rate is 58.2 per 1,000 women aged 15-19 as of 2015, down from 84.5 per 1,000 in 2011. HIV/AIDS is also a problem that disproportionately affects women in Burundi. There are 85,000 people in Burundi living with HIV, from which 45,000 are adult women and 30,000 are adult men.³⁰⁴

Education

Gender norms in Burundi limit women's access to continued education, resulting in a higher rate of illiteracy in women.³⁰⁵ While 68% of women aged 15 and above are literate, 81% of their male counterparts are literate.³⁰⁶ Families educate girls emphasizing the productive and reproductive roles that women must assume. Girls are also expected to look after their younger siblings and help with domestic and agricultural work.³⁰⁷ This unequal division of labour in the household burdens young women, limiting their time to study, while young men have time for studying and gaining vocational skills. Girls are far more likely to be withdrawn from schools and retained at home. Some reasons for girls' premature withdrawal from school

²⁹⁶ [USAID \(2021\). Analysis of cultural barriers to women's economic empowerment in Burundi.](#)

²⁹⁷ [World Economic Forum \(2022\) Global Gender Gap Report](#)

²⁹⁸ [IGN FI \(2022\). Women and Land Rights in Burundi.](#)

²⁹⁹ [USAID \(2021\). Analysis of cultural barriers to women's economic empowerment in Burundi.](#)

³⁰⁰ [World Economic Forum \(2022\) Global Gender Gap Report](#)

³⁰¹ [CARE \(2017\). Norms and practices impeding gender equality in Burundian society.](#)

³⁰² [CARE \(2017\). Norms and practices impeding gender equality in Burundian society.](#)

³⁰³ [UN Women \(n.d.\). Country Fact sheet Burundi.](#)

³⁰⁴ [Frontline Aids Burundi](#)

³⁰⁵ [USAID \(2021\). Analysis of cultural barriers to women's economic empowerment in Burundi.](#)

³⁰⁶ [World Bank Data \(2021\)](#)

³⁰⁷ Courtois, A. C. (2016). Les femmes chefs de ménage à Bujumbura : marginalité, violences, et résilience. Cited in [USAID \(2021\). Analysis of cultural barriers to women's economic empowerment in Burundi.](#)

in Burundi include poverty, illness, fear of pregnancy, planned marriage or deeming education for a girl to be of no use, and early pregnancy.³⁰⁸

Women's economic participation and entrepreneurship

Women in Burundi play an active role in the agricultural sector, which provides 90% of the country's food production.³⁰⁹ They constitute 55% of the agricultural labour and carry out 70% of farm work. Despite women producing most of the food crops used for family consumption, they often do not have the decision-making power over the use of the harvest.³¹⁰

Women are highly involved in informal sectors, particularly in rural areas, which generally require low qualifications, rather than in the trade and commerce sector, which are better paid.³¹¹ Further, Burundian women are expected to carry out a disproportionate amount of the burden of household work, limiting their opportunities to find employment or start a business.

Regarding laws that promote or hinder women's entrepreneurship, there are no differences between the opportunities women and men have to start a business since both can sign a contract, register a business and open a bank account on unequal terms. However, Burundian law still does not prohibit discrimination in access to credit based on gender, hindering women's ability to pursue entrepreneurial activities.³¹²

Gender and energy in Burundi

Only 11.7% of the population in Burundi has access to electricity³¹³, a rate four times lower than the average in Sub-Saharan Africa of 48.2%,³¹⁴ and access to clean and modern energy in Burundi remains extremely low.³¹⁵ Almost all cooking relies on traditional fuels (99% of the households), and around 90% of rural households depend on conventional three-stone stoves.³¹⁶

Gender norms and gender dynamics play an important role in purchasing energy-related products and appliances within households. Women are often the primary users of energy appliances, and the use of fuels and technologies may directly impact their health. However, they may not be able to decide to buy cleaner energy technologies since many of them do not have control over their earnings and household purchases.³¹⁷

Some of the gender-focused programs implemented in the region include the Lumière program, implemented by local NGO FVS-Amade in partnership with UNICEF. Active since 2015, FVS-Amade works with Village Savings and Loan Associations (VSLAs) to serve the last mile consumers with clean energy

³⁰⁸ [USAID \(2021\). Analysis of cultural barriers to women's economic empowerment in Burundi.](#)

³⁰⁹ [UN Women Africa: Burundi.](#)

³¹⁰ [Iradukunda et. al., 2019. Understanding gender roles and practices in the household and on the farm: Implications for banana disease management innovation processes in Burundi.](#)

³¹¹ [Muchiri & Nzisabira \(2020\). Gender, poverty reduction and social work : a view from Burundi.](#)

³¹² [World Bank Group \(2022\). Women, Business and the Law](#)

³¹³ [World Bank Data \(2020\).](#)

³¹⁴ [World Bank Data \(2020\).](#)

³¹⁵ [DHS. \(2016-17\) Demographic and health data.](#)

³¹⁶ [Lighting Africa \(2020\). Burundi Market Assessment for OffGrid Solar and Improved Cooking Technologies for Households.](#)

³¹⁷ [DHS. \(2016-17\) Demographic and health data.](#)

products. UNICEF subsidizes around 60% to 70% of the purchase price, the purchase of solar lanterns by the NGO FVS Amade, which sells them either directly to households or to VSLAs.³¹⁸ By replacing battery lamps, kerosene, and candles with solar lanterns, the program aims to improve women's and children's health by preventing them from inhaling kerosene fumes.³¹⁹ In addition, the program also positively impacts women since VSLA model can help increase access to financial services among women.³²⁰

In 2020, the Republic of Burundi received a grant amounting to USD 100 million from the World Bank for the Solar Energy in Local Communities (SOLEIL) project. The project's Technical Assistance facility has a significant gender focus, highlighted below:³²¹

- Scholarship and Mentorship for women's capacity development in the energy sector: Establishment of a 100% scholarship program for 40 women studying STEM-related courses, supported by the General Director for Energy, who wish to join the energy workforce after graduation. Further, the program aims to provide leadership training and targeted mentorship programs for female talent in the sector
- Employment: Enabling 30% employment of women in overall jobs through mini-grid development by supporting private sector companies in mapping out relevant employment opportunities; such as during construction, and O&M (i.e. solar panel cleaning and small-scale gardening activities) and also upstream employment across mini-grid company staff footprint through local partners.
- Entrepreneurship: To foster female entrepreneurship in the sector, 20% of start-up grants are to be earmarked for female-headed or majority female-owned companies. Further, information sessions targeting female business associations (such as Association des Femmes d'Affaires au Burundi) would be hosted and business skills development support would be provided to women entrepreneurs in the sector
- Women customers: Differentiated subsidies will be provided to companies in order to target female consumers and increase the purchase of off-grid systems (asset ownership) by vulnerable households. Further, women's preferences and affordability constraints will be mapped at the household and enterprise level, followed by an assessment of the positive impact on women as a result of adopting off-grid products (in terms of time savings, drudgery reduction, and livelihoods).

³¹⁸ [Lighting Africa \(2020\). Burundi Market Assessment for OffGrid Solar and Improved Cooking Technologies for Households.](#)

³¹⁹ [UNICEF \(2018\). Impact Evaluation of the Lumière Project. Final Report.](#)

³²⁰ [CARE \(2019\). Gender Equality and Women's Empowerment Programme. GEWEP II 2016 – 2018 Burundi. Rapport de l'évaluation finale.](#)

³²¹ [World Bank \(2020\). Project Appraisal Document on a Proposed Grant in the amount of SDR 72.4 Million \(US \\$100 Million Equivalent\) to The Republic Of Burundi for a Solar Energy In Local Communities Project](#)

Togo

Overview of gender equality and gender gaps in the country

Togo is a country in western Africa. The country has a population of 8.7 million people, with a young and adolescent population of 32% between the age group of 10 to the age of 24 years. The fertility rate in the country stands at 4.1.³²²

The Gender Development Index in Togo is 0.849, and the country is placed in 'Group 5', countries with low equality in HDI achievements between women and men, with a high absolute deviation from gender parity of more than 10%. This is primarily driven by a gender gap in mean years of schooling, with 3.4 years among females and 6.8 years among males. Togo has a Gender Inequality Index of 0.580 and is ranked 149; the index measures disparities between men and women in economic opportunities, reproductive health, and empowerment.³²³

Indicator	Result	Source and date
Life expectancy at birth	62.4	UNDP (2022) Gender Development Index
Maternal mortality ratio	396	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	55%	World Bank, 2019
Women's Labour Force Participation Rate (age 15 to 64)	56%	World Bank, 2019
Mean years of schooling ³²⁴	3.4	UNDP (2022) Gender Development Index
Estimated gross income per capita	1885 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

In the international arena, Togo ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1983 but has not yet ratified the Optional Protocol on violence against women. The country also ratified the Convention on the Rights of the Child (CRC) in 1990,³²⁵ the Beijing Platform for Action in 1995,³²⁶ and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (the Maputo Protocol) in 2005.³²⁷ Togo incorporated the principle of gender equality in the Constitution in 1992, which states: "All human beings are equal in dignity and in right. The man and the woman are equal before the law. No one may be favoured or disadvantaged for the reason of their familial,

³²² [United Nations Population Fund \(2022\), World Population Dashboard, Togo](#)

³²³ [UNDP \(2022\) Gender Inequality Index](#)

³²⁴ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

³²⁵ [United Nations Human Rights Treaty Bodies, Togo](#)

³²⁶ [Ministry for Social Affairs, the Advancement of Women and the Protection of Children \(2004\). Evaluation of the Implementation of The Beijing Platform for Action](#)

³²⁷ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

ethnic, or regional origin, of their economic or social situation, of their political, religious, philosophical or other convictions” (Art. 2).³²⁸

Following the recommendations of CEDAW, Togo enacted the reforms in 2012 and 2014 to the Persons and Family Code, making significant progress towards gender equality. Before the reform, the Family Code restricted women's rights by not granting a woman an equal right to choose where to live, to get a job without her husband's permission, and to be named head of household.³²⁹ However, despite these progresses, the Women, Business and the Law (WBL) 2022 report notes a setback for Togo in terms of gender equality with the enactment of a new Labor Code in 2021, that no longer broadly prohibits the dismissal of pregnant workers leading to a decrease in its WBL score from 84.4 in 2021 to 81.9 in 2022.³³⁰

Political participation

Over the years, the participation of women in politics has increased, with women occupying high-level government positions, including the Prime Minister's Office, the National Assembly, and the Ministry of Defense.³³¹ However, women remain under-represented at the upper levels of power and decision-making: in the Executive, the Legislature, and the Judiciary and the control of information. Women hold only 18.7% of the National Assembly seats and 31% of the ministerial positions.³³² Togo currently has a women Prime Minister, Victoire Tomegah Dogbé, who has held the position since September 28, 2020, and is the first woman holding the position.

Social and cultural context

Despite the progress made in recent years in Togo on the legal status of women and their political participation, culture, religion, and strong traditions still allow discriminatory practices against women. While the practice of child marriage has declined in recent years, this practice still occurs in some regions of the country. According to the latest Global Gender Gap report, Togo has an early marriage rate of 13.8% of women.³³³ This practice is partly due to cultural factors such as the dowry that can be an economic motivation for low-income families to get girls married off early or for strengthening bonds between families.³³⁴ Both the Children's Code and the Personal and Family Code set the minimum age of marriage at 18 years, and the Law on Reproductive Health 2007 prohibits forced marriage (Art. 11, 2007).³³⁵

The Civil Law in Togo establishes that marriages can be monogamous or polygamous (Personal and Family Code, Art.42).³³⁶ Women still face discrimination regarding family laws in Togo, for instance, women have to wait 300 days to remarry, unlike men that do not have to wait. Further, there is no legislation specifically addressing domestic violence in Togo.³³⁷ Widows also face discrimination and violence in Togo. Despite existing specific protections under the civil law for widows that prohibit degrading mourning rites, such as tests to make a woman prove she didn't kill her husband, deny her access to food and any income-

³²⁸ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³²⁹ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³³⁰ [World Bank \(2022\). Women, Business and the Law](#)

³³¹ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³³² [World Economic Forum \(2022\) Global Gender Gap Report](#)

³³³ [World Economic Forum \(2022\) Global Gender Gap Report](#)

³³⁴ [France Office français de protection des réfugiés et apatrides \(OFPRA\) \(2016\). Togo: Les mariages forces.](#)

³³⁵ [OECD \(2019\). Social Institutions and Gender Index.](#)

³³⁶ [Droit civil, droit commercial, droit de la famille \(2012\).](#)

³³⁷ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

generating activity, and prohibit widows from any inheritance, unfortunately, these rituals persist in a majority of tribal, rural communities.³³⁸

In terms of inheritance, women are not entitled to inherit from their husbands or fathers under customary law in Togo. However, civil law provides both women and men equal rights to inheritance (Personal and Family Code), although it also stipulates that these rights may be waived if the spouses opt for customary rules in matters of succession (Art. 404).³³⁹ Further, civil law grants equal rights and access to land among women and men, but customary law often excludes women or limits their access to land.³⁴⁰ In fact, traditional practices relating to land ownership vary across ethnic groups. In some cases, women can only purchase land through a man, while in other areas, women are excluded from ownership altogether.³⁴¹

Gender-Based Violence (GBV) is another problem that hinders gender equality. In Togo, 22.1% of ever-partnered women have suffered intimate partner physical and/or sexual.³⁴² Some of the Togolese government's efforts to reduce the risk of GBV include the adoption of the *Politique Nationale de l'Equité et de l'Egalité de Genre du Togo* in 2011, a policy to establish an institutional, sociocultural, legal, and an economic environment favorable to the achievement of gender equity and equality in Togo; and to address forms of violence against women.³⁴³ FGM is also prohibited in the country since 1998 through Act No. 98-016, which is also considered a harmful practice and prohibited in the Law on Reproductive Health 2007 and the Child Code 2007.³⁴⁴

Despite the existence of a legal framework for FGM, there is still a lot of work to be done in terms of education and sensitization. A government study found that more than 20% of the population believes that it is acceptable and justifiable for a man to be violent against his wife.³⁴⁵

Health

Togo's national health policy, adopted in 1996 and revised in 1997 includes improving women's and children's access to primary health care by providing improved geographical coverage as one of its key objectives.³⁴⁶ Togo started taking into account the reproductive health component, and the gender dimension was mainstreamed in it, following the recommendations of the International Conference on Population and Development held in Cairo in 1994.³⁴⁷

Despite having equal rights to use contraception, women of reproductive age (15-49 years) in Togo face barriers to accessing sexual and reproductive health information and services. In 2017, only 29.6% of women made informed decisions regarding sexual relations and about 39.6% of women had their need for

³³⁸ [UN Trust Fund to End Violence against Women \(2017\). Ending harmful traditional widowhood practices in Togo](#)

³³⁹ [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁴⁰ [The Global Initiative \(2012\). Global Initiative for Economic, Social and Cultural Rights. Annual Report 2012.](#)

³⁴¹ Khan, Frida (2016). Islamic inheritance laws and their impact on rural women in Muslim societies: synthesis of studies from Asia and West Africa and emerging recommendations, International Land Coalition. ILC. Cited in [OECD n.d.\). Social Institutions and Gender Index.](#)

³⁴² [World Economic Forum \(2022\) Global Gender Gap Report](#)

³⁴³ [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³⁴⁴ [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁴⁵ Togo, Government of. 2014. Troisième Enquête Démographique et de Santé EDST-III. Togo 2014. Rapport Préliminaire Government of Togo. Cited in [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³⁴⁶ [Ministry for Social Affairs, the Advancement of Women and the Protection of Children \(2004\). Evaluation of the Implementation of The Beijing Platform for Action](#)

³⁴⁷ [Ministry for Social Affairs, the Advancement of Women and the Protection of Children \(2004\). Evaluation of the Implementation of The Beijing Platform for Action](#)

family planning satisfied with modern methods.³⁴⁸ Women in rural areas have less access to reproductive health services, and men partners can be against their wife's use of contraception. Abortion is legal in cases of rape or incest if the fetus is impaired or to save the pregnant woman's life.³⁴⁹

In recent years, Togo has increased its efforts to fight HIV/AIDS due to its rapid spread. In the country, 1.9% of the total population (% of the population ages 15-49) has HIV, of which 65% are women.³⁵⁰ As part of these efforts, the National AIDS Programme (PNLS) was reinforced in September 2001 with a permanent secretariat of a National Council to combat AIDS and other sexually transmitted infections. The Joint United Nations Programme on HIV/AIDS (UNAIDS) has supported Togo in its fight against HIV/AIDS.³⁵¹

Education

One of Togo's most significant challenges in terms of education is reducing the existing gender gaps. Only 55% of women aged 15 and above are literate, compared to 80% of men.³⁵² Many women reportedly drop out of school due to pregnancy, and there is a prioritizing of boys' education over girls.³⁵³ According to UNESCO, the lower secondary completion gap between boys and girls is 10.7, more significant than the gap in the Sub-Saharan Africa aggregate, which is 4.7.³⁵⁴

The Togolese government has made progress in reducing the gender gaps through the implementation of initiatives focused on girls' education, some of these include:

- Togo joined the Global Partnership for Education in 2010 and received support to implement multilevel interventions tackling school-level and institutional obstacles and harmful social norms affecting girls throughout their education to strengthen their school enrollment.³⁵⁵
 - Under this initiative, Togo received a 7 million COVID-19 grant from the World Bank, one of its key objectives included supporting girls with targeted back-to-school campaigns through sensitization programs on early marriage, gender-based violence, early pregnancies, and the importance of continuing to learn.³⁵⁶
- The country has also developed an education strategy for 2014 to 2025 to “develop a quality basic education to achieve universal primary education by 2022.”³⁵⁷

Women's economic participation and entrepreneurship

In Togo, women's participation in the labor force has remained roughly the same since 1990. In 2021, women's labor force participation rate was 55.5%, compared to 59.4% for men.³⁵⁸ However, women are

³⁴⁸ [UN Women \(n.d.\). Country Fact Sheet Togo.](#)

³⁴⁹ United Nations Development Programme (2010). World abortion policies, Cited in [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁵⁰ [World Bank Data \(2021\)](#)

³⁵¹ [Ministry for Social Affairs, the Advancement of Women and the Protection of Children \(2004\). Evaluation of the Implementation of The Beijing Platform for Action](#)

³⁵² [World Bank Data \(2019\)](#)

³⁵³ Emery, Robert E. (Editor) 2013 Cultural Sociology of Divorce: An Encyclopedia. Sage Publications. Committee on the Elimination of Discrimination against Women (CEDAW) (2011), Consideration of Reports Submitted by States Parties Under Article 18 of the Convention on the Elimination of All Forms of Discrimination against Women: Mali, Combined Sixth and Seventh Periodic Reports of States Parties, CEDAW/C/TGO/6-7, CEDAW, New York, NY. Cited in [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁵⁴ UNESCO Institute for Statistics (2021). [Cited in World Bank Gender Data Portal.](#)

³⁵⁵ [Global Partnership for Education \(2021\). Decision on program implementation grant to Togo](#)

³⁵⁶ [Global Partnership for Education](#)

³⁵⁷ [Republique Togolaise \(2014\). Plan Sectoriel de l'education Pse 2014-2025 Amelioration de l'acces, de l'equite Et de la Qualite de l'education au Togo.](#)

³⁵⁸ [World Bank \(2021\). Gender Data Portal](#)

primarily engaged in the informal sector, where 90% of workers are women, which directly impacts their ability to access the benefits of formal employment.³⁵⁹

In the past decades, Togo has made significant progress in supporting women's entrepreneurship and economic participation. The country has enacted reforms under all eight indicators in the WBL index (Mobility, Togo has made significant progress since 1970—the first year covered by the Women, Business, and the Law (WBL) index. The country has enacted reforms under all eight indicators in the index (Mobility, Workplace, Pay, Marriage, Parenthood, Entrepreneurship, Assets, and Pension). One of the most significant reforms was the enactment of the 2006 Labor Code that prohibited the dismissal of pregnant workers, mandated equal pay for the same work, and prohibited gender discrimination in employment.³⁶⁰ One of the latest setbacks regarding labor laws was enacting a new labor code that no longer broadly prohibits the dismissal of pregnant workers.^{361 362}

Under civil law in Togo, women and men are granted equal access and participation in all spheres of life, including economic life.³⁶³ This right allows women and men to open individual bank accounts and access formal financial services such as a line of credit or financial loan, regardless of marital status.³⁶⁴ But there are gender gaps in access to financial services and products because of barriers that women typically face, such as lack of collateral. Married under the marriage community of property regime are prohibited from providing such guarantees without their husbands' consent.³⁶⁵

Regarding leadership and entrepreneurship, according to the Global Gender Gap 2022 report, only 11.4% of the firms in Togo have female top managers, and 10.7% have a majority of women ownership. The same report highlights that Togo has the highest share of women in senior roles, with 70.1%. Further, the percentage of women business owners has increased in the last two decades, from 22% in 2014 to 32% in 2020.³⁶⁶

Women in Togo spend disproportionately more time on unpaid care work than men, which negatively impacts the time they can spend pursuing formal economic activities. As of 2016, women entrepreneurs in Togo spent 19.3 hours per week on care work compared to 8.8 for male entrepreneurs.³⁶⁷

Gender and energy in Togo

Only half (54%) of the Togolese population has access to electricity.³⁶⁸ The country has a relatively diversified energy mix, and more than 13% of its final energy consumption comes from renewable supplies of energy, mainly hydropower.³⁶⁹ In Togo, as in other countries of West Africa, women are the most vulnerable and affected by energy poverty and its effects. They are the most exposed to health risks related to the use of polluting fuels they are responsible for cooking. This burden also limits their participation in other activities, especially income-generating activities, causing a vicious circle by not having the means or

³⁵⁹ [World Economic Forum \(2022\) Global Gender Gap Report](#)

³⁶⁰ [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁶¹ [World Bank Group \(2021\). Women Business and the Law Togo.](#)

³⁶² [World Bank Group \(2022\). Women Business and the Law 2022 Report.](#)

³⁶³ Constitution of Togo, rev. 2007. Cited in [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁶⁴ [World Bank Group \(2015\) Women, Business and the Law 2016.](#)

³⁶⁵ [OECD \(n.d.\). Social Institutions and Gender Index.](#)

³⁶⁶ [World Bank. Entrepreneurship Database 2014-2022.](#)

³⁶⁷ World Bank. 2016. "IDA Togo—Systematic Country Diagnostic." Washington, DC: World Bank. Cited in [World Bank Group \(2022\). Reforming Discriminatory Laws to Empower Women in Togo. Global Indicators Briefs No. 12.](#)

³⁶⁸ [World Bank Data \(2020\).](#)

³⁶⁹ [IEA, Togo](#)

the decision-making to invest in clean energy products that can allow them to improve their living conditions and that of their family.³⁷⁰

Among notable initiatives in the sector, *Entrepreneurs du Monde* (EdM) is an initiative supported by the Economic Community of West African States (ECOWAS) and the government of France, created in 1998, that works with people living in fuel poverty by facilitating access to essential goods and services for their families. Based in Togo since 2012, EdM runs the Mivo Energie project that aims to combat global warming and deforestation in Togo by facilitating access to clean, modern, and affordable energy products for vulnerable populations, especially women.³⁷¹

Under the CIZO programme, the Government of Togo (GoT) has granted licenses to two companies to provide solar home systems in the country.³⁷² Bboxx is an off-grid solar energy provider, with operations across Sub-Saharan Africa, that manufactures and provides installation and maintenance services for solar home systems and cooking solutions through a Pay-As-You-Go (PAYG) model.³⁷³ Recently Bboxx secured an €11m loan from the Off-Grid Energy Access Fund (“OGEF”) to provide 1.5m Togolese people with clean energy and support smallholder farmers in Togo to access solar-powered irrigation.³⁷⁴ One of the key pillars of the company is Female Empowerment through the provision of affordable and safe alternatives to traditional fuels through an affordable PAYG model. The company is also known to collect sex-disaggregated data from customers and analyze their repayment rates.³⁷⁵ In other countries such as Nigeria, Bboxx is also working with local partners to train women entrepreneurs in building businesses and bringing clean energy to their communities, also leveraging local women's workforce across their distribution networks to reach target regions.³⁷⁶

³⁷⁰ [ECOWAS Centre for Renewable Energy and Energy Efficiency](#)

³⁷¹ [ECOWAS Centre for Renewable Energy and Energy Efficiency](#)

³⁷² [IFC / Lighting Global \(2021\) Off-Grid Solar Market Research for Togo](#)

³⁷³ [Bboxx \(2020\). Impact Report 2020](#)

³⁷⁴ [Bboxx. \(2022, November 30\). Bboxx EDF sign €11m deal with OGEF to accelerate electrification of an additional 1.5M people in Togo \[web log\]. Retrieved January 31, 2023, from <https://www.bboxx.com/news/bboxx-edf-sign-11m-deal-with-ogef-to-accelerate-electrification-of-an-additional-1-5m-people-in-togo/>.](#)

³⁷⁵ [Bboxx \(2020\). Impact Report 2020](#)

³⁷⁶ [Bboxx \(2022, June 28\). Bboxx partners with Solar Sister to accelerate clean energy access in Nigeria \[web log\] Retrieved January 31, 2023 from <https://www.bboxx.com/news/bboxx-partners-with-solar-sister-to-accelerate-clean-energy-access-in-nigeria/>](#)

Malawi

Overview of gender equality and gender gaps in the country

Malawi is a South-East African country bordering Mozambique, Zambia and Tanzania. The country has a total population of 20.2 million people, with a demographic dividend skewed toward the young and adolescent population (between the age group of 10 to the age of 24 years), who make up 34% of the population. The fertility rate in the country stands at 3.9 which is very high compared to regional averages.³⁷⁷

The Gender Development Index in Malawi is 0.968 and the country is placed among 'Group 2' countries with an absolute deviation from gender parity between 2.5 percent and 5 percent. The disparity in mean years of schooling for females (4.1 years) and males (4.7 years) is lower compared to other countries in the region.³⁷⁸ Malawi has a Gender Inequality Index of 0.554 and is ranked 142, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.³⁷⁹

Indicator	Result	Source and date
Life expectancy at birth	66.5	UNDP (2022) Gender Development Index
Maternal mortality ratio	349	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	64%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	74%	World Bank, 2019
Mean years of schooling ³⁸⁰	4.1	UNDP (2022) Gender Development Index
Estimated gross income per capita	1,232 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Malawi's constitution of 1994, upholds gender equality, prohibiting gender-based discrimination under Article 20 and accords women the same rights as men to enter contracts, maintain property, retain citizenship and custody of children. Further, the Constitution principally outlines gender considerations for national policy development and adoption of legislation to obtain gender equality through "(i) full participation of women in all spheres of Malawians society on the basis of equality with men; (ii) the implementation of the principles of non-discrimination and such other measures as may be required; and (iii) the implementation of policies to address social issues such as domestic violence, security of the person, lack of maternity benefits, economic exploitation and rights to property".³⁸¹ The country has also demonstrated its commitment to furthering gender equality by ratifying international treaties such as the

³⁷⁷ [United Nations Population Fund, World Population Dashboard, Malawi](#)

³⁷⁸ [UNDP \(2022\) Gender Development Index](#)

³⁷⁹ [UNDP \(2022\) Gender Inequality Index](#)

³⁸⁰ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

³⁸¹ [Constitution of Malawai from Malawai Gazette \(1994\). Retrived on Feburary, 3 from https://www.ilo.org/dyn/natlex/docs/ELECTRONIC/40069/90473/F1019822909/MWI40069.pdf](#)

Convention on the Elimination of all forms of Discrimination Against Women (CEDAW) in 1987 and the CRC - Convention on the Rights of the Child in 1991.³⁸² Further, Malawi is also a signatory to The Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (2005), The Beijing Declaration and Platform for Action (BPfA), and the Southern African Development Community (SADC) Protocol on Gender and Development.

The country has adopted the Gender Equality Act as a Law in 2013 with the aim to “promote gender equality, equal integration, influence, empowerment, dignity, and opportunities, for men and women in all functions of society, to prohibit and provide redress for sex discrimination, harmful practices and sexual harassment, to provide for public awareness on the promotion of gender equality, and to provide for connected matters” with a focus on Human Rights, Employment, Education, Sexual and Reproductive Health and Civic Awareness.³⁸³

Malawi's Ministry of Gender, Children, and Social Welfare consolidated a National Gender Policy in 2015, with the goal to reduce gender inequalities in priority areas including; education and training; health; agriculture, food security, and nutrition; natural resources, environment, and climate change management; economic development; governance and human rights. The following objectives have been identified by the policy:

1. To advocate for increased access, retention, and completion to quality education for girls and boys.
2. To ensure women, men, boys, and girls' sexual and reproductive health rights, and HIV AIDS status are improved.
3. To strengthen gender mainstreaming in all sectors of the economy.
4. To reduce poverty among women and other vulnerable groups (Orphans, widows, PLHA, persons with disabilities, and the elderly) through economic empowerment.
5. To promote women's participation in decision-making positions in both politics and public life. vi. To reduce gender-based violence.
6. To strengthen the capacity of the National Gender Machinery.

The policy commitment is complemented by a Monitoring and Evaluation Plan and has been developed in alignment with various other national policies such as the country's larger Growth and Development Strategy, Reproductive Health Policy, Education Policy, Agricultural and Environmental policies as well as the National Youth Policy.³⁸⁴

Political participation

Malawi boasts of electing the second African female president, Joyce Banda (from 2012- 2014). However, the country still lags behind in women's political participation.³⁸⁵ As per the Global Gender Gap report, 23% of the parliament is composed of women, compared to 71% of men. The gap widens for ministerial leadership, with 17.39% of women and 82.61% of men holding ministerial positions.³⁸⁶

³⁸² [United Nations Human Rights Treaty Bodies, Malawi](#)

³⁸³ [Republic of Malawi \(2013\) Gender Equality Act, Chapter 25:06](#)

³⁸⁴ [The Republic of Malawi \(2015\) National Gender Policy. Retrived February 3, 2023, from https://cepa.rmportal.net/Library/government-publications/National%20Gender%20Policy%202015.pdf](#)

³⁸⁵ [Tenthani, R. \(2012, April 10\). Joyce Banda: Malawi's first female president. BBC. Retrieved February 3, 2023, from https://www.bbc.com/news/world-africa-17662916](#)

³⁸⁶ [World Economic Forum \(2022\) Global Gender Gap Report](#)

Studies have suggested that Malawian voters view women as less credible leaders than men, expecting women to conform to gender norms, however, women are also held to a much higher standard than men parliamentarians. There is also a relegation of women to fixed roles within the party ranks, commonly, the 'women's wings' which do not have a clear mandate within political parties. Further, women candidates face discrimination, negative campaigning, and increasing expenses during campaigning, which causes high turnover among women candidates.³⁸⁷ The country is yet to take affirmative action on women's participation in politics, with no legislative quotas for women's representation.³⁸⁸

Social and cultural context

Malawi's social life is dictated by customary law which is patriarchal in tradition. While the country has areas with matrilinear communities, social norms across these still continue to favor men. Women also have limited access to productive resources like agricultural plots, and women-managed agricultural plots are also smaller than that of men (by 12%) with 25% low productivity as a result of limited access to farm inputs.³⁸⁹ The incidence of violence against women is very high in Malawi with 37.5 % of women having experienced physical and/or sexual intimate partner violence in their lifetime.³⁹⁰ This figure has continuously risen between 2004 to 2016 from 15% to 33%.³⁹¹ This continues even as the country's National Plan of Action to combat Gender-Based Violence (GBV) provides an overarching framework to prevent GBV and the Domestic Violence Act (2006) underlines the penalization of violent offenses against women.³⁹² The country also has one of the world's highest rates of child marriage, with 42% of girls married before the age of 18, and 9% being married below the age of 15.³⁹³ To tackle the pervasive issue of child marriage, the country has developed a National Strategy to Eliminate Child Marriage, which aims to reduce the prevalence of child marriage in Malawi by 20% by 2023. The implementation of the strategy is to be coordinated by a multi-sectoral task force and through the establishment of community child protection committees, community victim support units (CVSUs), Community-Based Child Care Centres (CBCCs) and Children's Corners (CCs).³⁹⁴

Health

³⁸⁷ O'Neill, T., Kanyongolo, N., Wales, J., & Mkandawire, M. W. (2016). (rep.). *Women and power Representation and influence in Malawi's parliament*. Overseas Development Institute. Retrieved February 3, 2023, from <https://cdn.odi.org/media/documents/10289.pdf>.

³⁸⁸ Amundsen, I., & Kayuni, H. (2016). (rep.). *Women in Politics in Malawi*. CHR Michelsen Institute. Retrieved February 3, 2023, from <https://www.cmi.no/publications/file/5923-women-in-politics-in-malawi.pdf>.

³⁸⁹ UN Women (2021). *Malawi National Gender Statistics Assessment*. Retrieved February 3, 2023, from <https://africa.unwomen.org/sites/default/files/Field%20Office%20Africa/Attachments/Publications/2021/12/Assessment%20of%20Malawi%20National%20Gender%20Statistics%20System%200311202103.pdf>

³⁹⁰ UN Women, *Global Database on Violence against Women*

³⁹¹ UN Women (2021). *Malawi National Gender Statistics Assessment*. Retrieved February 3, 2023, from <https://africa.unwomen.org/sites/default/files/Field%20Office%20Africa/Attachments/Publications/2021/12/Assessment%20of%20Malawi%20National%20Gender%20Statistics%20System%200311202103.pdf>

³⁹² World Bank. (2022) *Malawi Gender Assessment*. Retrieved February 3, 2023, from <https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf>

³⁹³ Makwemba, M., Chinsinga, B., Kantukule, C. T., Munthali, A., Woldegorgis, M., Haenni, S., & Lin, Q. (2019). (rep.). *Survey Report: Traditional Practices in Malawi*. UNICEF. Retrieved February 3, 2023, from <https://www.unicef.org/malawi/media/1546/file/Traditional%20Practices%20in%20Malawi-%20Survey%20Report.pdf>.

³⁹⁴ UNICEF (2020). *Budget Scoping on Programmes and Interventions to End Child Marriage in Malawi*

The maternal mortality rate in Malawi stands at 349 per 100,000 live births.³⁹⁵ The country has seen great progress in expanding women's access to institutional care, and as per the country's last DHS, Malawi has a very high rate of institutional deliveries, increasing significantly from 55% in 1992 to 91% in 2015-16. Although only half (51%) of women had at least 4 ANC check ups in their last pregnancy.³⁹⁶ Further, as per the country's fifth Integrated Household Survey (IHS5), 68% of women were assisted by nurses or midwives while 28% were assisted by doctors or clinical officers.³⁹⁷ There is also high access to family planning methods among women, with a lower unmet need reported among all women (13%) as compared to other countries in the region.³⁹⁸ A majority of married Malawian women users of family planning methods report making a joint decision regarding family planning methods along with their husbands. However, there has been no increase in the median age at first birth, since 1992, which is currently hovering at 18.9 years.³⁹⁹ The incidence of HIV among women in Malawi is high at 27.5%, and along with complications arising during pregnancy, they are the leading cause of death among girls of age 15-19 in Malawi.⁴⁰⁰

Education

Malawi has a universal primary education policy that is free for all.⁴⁰¹ As per the Integrated Household Survey, 62.6% of females are literate.⁴⁰² The high incidence of child marriage among Malawians also reflects attitudes toward girls' education. Among dropouts from secondary school, 32% of females mentioned marriage being the primary reason they had to drop out.⁴⁰³ There is a gender gap of 6.87 percent in the secondary education enrollment rate in Malawi as per the Global Gender Gap report (2022).⁴⁰⁴ Access to higher education beyond secondary schooling is limited in Malawi among both men and women (above the age of 15), with only 1.2% of women having tertiary education qualifications, and men having a somewhat higher rate of tertiary education at 2.7%.⁴⁰⁵

The country has developed a long-term National Educational sector Investment plan to improve enrollment and quality of education in the country and a particular objective to increase equity in secondary education with a particular focus on girls. To enhance girls' education attainment, the country has introduced affirmative action by mandating 50-50 selection criteria in secondary schools, construction of girls' hostels, provision of bursaries, and provision of health, nutrition, physical education, and sports. In Technical, Entrepreneurial, and Vocational Education and Training, a code of conduct has been introduced to combat gender-based violence (GBV).⁴⁰⁶

The Ministry of Labour, Youth, Sports, and Manpower Development (MoLYSMD) has also released The Malawi TEVET Policy, and the first priority area under this policy is to increase access and equity in Malawi's TEVET system and increase current levels of enrolment of female students and vulnerable students

³⁹⁵ [UNDP \(2022\) Gender Inequality Index](#)

³⁹⁶ [National Statistical Office \(NSO\) \[Malawi\] and ICF. 2017. Malawi Demographic and Health Survey 2015-16. Zomba, Malawi, and Rockville, Maryland, USA. NSO and ICF.](#)

³⁹⁷ [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

³⁹⁸ [United Nations Population Fund, World Population Dashboard, Malawi](#)

³⁹⁹ [National Statistical Office \(NSO\) \[Malawi\] and ICF. 2017. Malawi Demographic and Health Survey 2015-16. Zomba, Malawi, and Rockville, Maryland, USA. NSO and ICF.](#)

⁴⁰⁰ [UNFPA Malawi. \(2019\). Adolescent Pregnancy, Retrieved February 3, 2023, from https://malawi.unfpa.org/en/topics/adolescent-pregnancy-2](#)

⁴⁰¹ [World Bank. \(2022\) Malawi Gender Assessment. Retrieved February 3, 2023, from https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf](#)

⁴⁰² [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

⁴⁰³ [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

⁴⁰⁴ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴⁰⁵ [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

⁴⁰⁶ [Ministry of Education, The Republic of Malawi \(2020\). National Education Sector Investment Plan 2020 – 2030](#)

in TEVET training from less than 30% to 60%. However, a study by UNESCO found that issues of sexual harassment, lack of female teachers and role models at TEVET centers, untrained teachers, and poor sanitation and accommodation infrastructure can cause a lack of success and non-completion among women learners. The study also found that there is a stereotypically gendered dimension to enrollment in these courses, with women opting for administration and tailoring programs and men enrolling in courses like construction.⁴⁰⁷

Women's economic participation and entrepreneurship

While Malawi has a comparatively high Female Labour Force Participation in the sub-region, their conditions of work are far from ideal. As per the country's Fifth Integrated Household Survey (2019-20), 91.8% of women in Malawi between the ages 15-64 are engaged in income-generating activities. The highest proportion of women (81.7%) in Malawi have engaged in agriculture or fishing activities and wage work is the least common with only 6.6% of women engaged in it. Wage work is conducted by men more commonly (10.6%), and casual work employs a larger proportion of working women (58%) than men (47%).⁴⁰⁸ Unpaid work is also more common among women, with the 2015-16 DHS revealing that 59% of women are not paid for their work compared to 26% of men.⁴⁰⁹ There is also a gender pay gap in Malawi, with women earning only 64 cents per dollar paid to men.⁴¹⁰ The Integrated Household Survey also finds that across the board, women wage earners work fewer hours, and spend more time on non-market activities such as household chores.⁴¹¹

As per the Global Gender Gap report, only 14.2% of firms in the country have female top managers, and only 11.4% of firms have female majority ownership.⁴¹² While women entrepreneurship is proportionately high in Malawi, they are relegated to non-IT sectors and access to resources and inputs are key barriers limiting their productivity. As per IHS 5, the most common form of non-farm enterprise in Malawi is in the Wholesale and retail trade; Accommodation, and food service sectors. A higher percentage of enterprises in this sector is operated by female-headed households (71%) compared to men (58%). On the other hand, in the Transportation and storage; Information and communication sector, men hold a higher proportion of enterprises (10.8%) compared to women (4.5%).⁴¹³ A World Bank analysis also found that women business managers' sales are about 65% less than those of male managers, however, the difference reduces to 45% when controlled for household and enterprise-level factors which indicated that return on resources is the major reason for the gender gap. Further, the analysis found that while men are more likely to use saved-up agricultural income as start-up capital, women rely on gifts from friends or family.⁴¹⁴

Gender and energy in Malawi

As per World Bank Data, a very small minority of households in the country have access to electricity (14.9%) as of 2020.⁴¹⁵ As per the country's IHS, 79% of the population use firewood as the main source of

⁴⁰⁷ [UNESCO \(2018\). Making technical and vocational training work for women and vulnerable students in Malawi: improving access and success.](https://unesdoc.unesco.org/ark:/48223/pf0000266142) Retrieved February 4, 2023, from <https://unesdoc.unesco.org/ark:/48223/pf0000266142>

⁴⁰⁸ [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

⁴⁰⁹ [National Statistical Office \(NSO\) \[Malawi\] and ICF. 2017. Malawi Demographic and Health Survey 2015-16. Zomba, Malawi, and Rockville, Maryland, USA. NSO and ICF.](#)

⁴¹⁰ [World Bank. \(2022\) Malawi Gender Assessment. Retrieved February 3, 2023, from https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf](https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf)

⁴¹¹ *ibid*

⁴¹² [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴¹³ [National Statistical Office, The Republic of Malawi \(2020\). The Fifth Integrated Household Survey \(IHS5\) 2020 Report](#)

⁴¹⁴ [World Bank. \(2022\) Malawi Gender Assessment. Retrieved February 3, 2023, from https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf](https://documents1.worldbank.org/curated/en/099315004212212759/pdf/P176395040da430500beb7042a0d5846ee6.pdf)

⁴¹⁵ [World Bank \(2020\). Access to electricity \(% of population\) - Malawi](#)

fuel for cooking, followed by charcoal (18.5%) and electricity (1.2%). For lighting, 11.2% of households use electricity, and a high majority, 79.7% use torches with batteries.⁴¹⁶

In Malawi, women have been at the forefront of various private sector and policy advocacy initiatives to expand energy access. Breaking the Barriers program being implemented by Christian Aid is empowering women in Karonga district, Malawi, to advocate for increased rural electrification and increased job opportunities and incomes for women. In 2019, women from Breaking the Barriers engaged with the Malawian parliamentary women's caucus to address energy poverty in the country, and in September of 2019, Malawi passed a tax amendment legislation to remove import duty on solar products.⁴¹⁷

FINCOOP is a financial cooperative that provides affordable financial services, serving micro, small and medium entrepreneurs, farmers, and salaried employees. The company's Green Loan Programme targets low-income households across Malawi using multiple channels such as Village Savings and Loan Associations (VSLAs), civil services, and technology partners. FINCOOP has tailored its payment terms to fit women's requirements through the deployment of payment methods such as Pay As You Go models, cash, and group loans, and by offering longer repayment terms. From 2018 to 2020, more than 1212 female-headed households were able to access solar products through this program.⁴¹⁸

⁴¹⁶ National Statistical Office, The Republic of Malawi (2020). *The Fifth Integrated Household Survey (IHS5) 2020 Report*

⁴¹⁷ Christian Aid. (2020, January 16). *Women in Malawi fight energy poverty*. *European Union*. Retrieved January 4, 2023, from <https://www.christianaid.org.uk/news/women-malawi-fight-energy-poverty>.

⁴¹⁸ UK Aid, AECF (2021). *The impact of financing business models on access to solar home systems and women's financial inclusion: Insights from FINCOOP Savings and Credit Cooperative in Malawi and MoneyMart Finance in Zimbabwe*. Retrieved January 4, 2023, from <https://www.aecfafrica.org/wp-content/uploads/2021/12/AECF-Womens-financial-inclusion-Summary-Final-Draft-1.pdf>

Lesotho

Overview of gender equality and gender gaps in the country

Lesotho is a landlocked country in Southern Africa. The country has a total population of 2.2 million people, with a young and adolescent population (in the age group 10-24) of 29%. The fertility rate in the country stands at 3 per woman, which is lower compared to other countries in the continent.⁴¹⁹

The Gender Development Index in Lesotho is 0.985 and the country is placed among 'Group 1' countries with high equality in HDI achievements between women and men (an absolute deviation of less than 2.5 percent). Lesotho is the only country among those categorized as 'low Human Development Indexed countries' that falls in Group 1 of GDI and has higher mean years of schooling for females (6.6 years) than males (6.0 years).⁴²⁰ Lesotho has a Gender Inequality Index of 0.557 and is ranked 144, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.

⁴²¹

Indicator	Result	Source and date
Life expectancy at birth	55.9	UNDP (2022) Gender Development Index
Maternal mortality ratio	544	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	89%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	63%	World Bank, 2019
Mean years of schooling ⁴²²	6.6	UNDP (2022) Gender Development Index
Estimated gross income per capita	2,107 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Lesotho achieved democracy in 1993 and the country's constitution's Bill of Rights enshrines non-discrimination and equality with respect to legislation, governance, sustainable development considerations, and democratic institutions of the State.⁴²³ The country, like many others in the region, has a pluralistic legal system, recognizing customary law, Roman-Dutch Law (Common/Civil Law) along with Statutory Law. While the Statutory law under the Constitution forms the supreme law of the land and provides equal rights to men and women, customary and common laws often relegate women to minority status. Internationally and regionally, Lesotho is a signatory to key conventions upholding women's rights including The Beijing Declaration and its Platform for Action (in 1995), and the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1995. To empower women, and

⁴¹⁹ [United Nations Population Fund, World Population Dashboard, Lesotho](#)

⁴²⁰ [UNDP \(2022\) Gender Development Index](#)

⁴²¹ [UNDP \(2022\) Gender Inequality Index](#)

⁴²² Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

⁴²³ [The Kingdom of Lesotho \(n.d.\). Lesotho Government Beijing +25 Review Report. Retrieved February 6, 2023, from https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf](#)

mitigate violence against them, Lesotho has also put in place legal frameworks such as The Legal Capacity of Married Persons Act, which entitles both men and women to the same rights and obligations regarding their marital duties, Land Act, and Companies Act which reinforces women's rights to own property and control productive assets.⁴²⁴

The country has a strong gender focus as part of its National Strategy Development Plan, and aims to mainstream gender across the 4 focus areas of the Plan including inclusive and sustainable economic growth and job creation, human capital strengthening, enabling infrastructure building as well as governance and accountability system strengthening.⁴²⁵

The country also has an ambitious Gender and Development Policy outlining 12 gender objectives to be implemented through a rights-based, multi-sectoral approach. These objectives range from providing legal protection, political representation, and quality education access for women, to addressing issues related to gender and climate change as well as gender-based violence, providing food security, health, and sanitation access, and aiding women's involvement in peacebuilding and media & ICT. Lesotho has also charted out a robust mechanism for the implementation of this policy with cross-cutting strategies including effective research, coordination, capacity building of the governing machinery, and Gender Responsive Budgeting and financing.⁴²⁶

The World Bank's Women, Business, and Law index score Lesotho 78.1 out of 100. The country scores particularly low in the 'Parenthood' indicator since the country is yet to institute legislation mandating maternity leave for women. Further, the country does not have any provisions for criminal prosecution or civil penalties against sexual harassment faced by women in the workplace.⁴²⁷

Political participation

With a strongly articulated policy objective around gender and governance, Lesotho demonstrates a high commitment to ensuring women's political participation. However, women still remain underrepresented in both national and local level political positions, with only 26.4% of women in the National Assembly and 24.2% in the Senate.⁴²⁸ The Global Gender Gap report also highlights a gap between women and men's representation in ministerial positions with women holding only 22% of positions.⁴²⁹

Lesotho has introduced affirmative action with a 30% quota for women at the local council level and a 'zebra-list' model for the 40 proportional representation (PR) seats in the National Assembly where women must be listed alternatively with men, in submitted PR lists, thus granting 40 out of 120 seats in the National Assembly to women. However, regardless of these actions, there has been a steady decline in the political representation of women, falling from 58% to 40% between 2005 and 2017 in local councils, and from 27% in 2012 to 25% in 2015 in the National Assembly.⁴³⁰

⁴²⁴ [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf)

⁴²⁵ [The Kingdom of Lesotho \(n.d.\). Lesotho Government Beijing +25 Review Report. Retrieved February 6, 2023, from https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf](https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf)

⁴²⁶ [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf)

⁴²⁷ [World Bank Group \(2021\). Women, Business and the Law. Lesotho](#)

⁴²⁸ [Inter-Parliamentary Union, Lesotho](#)

⁴²⁹ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴³⁰ [Nyane, H., & Rakolobe, M. \(2021\). Women's representation in Lesotho's legislative bodies. A politico-legal analysis of the effectiveness of electoral gender quotas. *Journal of African Elections*, 20\(2\), 81–101. https://doi.org/10.20940/jae/2021/v20i2a5](https://doi.org/10.20940/jae/2021/v20i2a5)

Social and cultural context

The progressive nature of Lesotho's institutional framework to address gender inequalities is evident in key legal Amendments made by the country presented in the previous section, however, the country is still patriarchal, and Law of Inheritance Act 26 of 1873 has not reversed the Basotho customary law where the male child becomes the principal heir of his father's property.⁴³¹ Further, Gender Based Violence and Child Marriage are still pervasive in the country. As per the country's Multiple Indicator Cluster Survey (MICS) from 2018, 19.4% of girls between the ages 20-49 were married before the age of 18 and 2.1% were married before the age of 15.⁴³² To reduce child marriages in the country, Lesotho launched a nationwide campaign on Child Marriage in 2016, to raise awareness of the impact of early marriages on girls, reinforcing the need to eradicate it.⁴³³ The Gender and Development Policy (2018-2030) has committed to zero tolerance towards all forms of violence against women and the country's Sexual Offence Act (2003) also recognizes marital rape as an offense.⁴³⁴ Further, the country is also in the process of developing Counter Domestic Violence legislation that criminalizes different types of violence, with the support of UNFPA.⁴³⁵ Although even within this framework, some reports indicate that as many as 86% of women experience GBV in their lifetime.⁴³⁶

Health

Lesotho has faced multiple catastrophic droughts including the El Nino, which have led to crop failures, food shortages, violence, malnutrition, and displacement.⁴³⁷ These climate shocks often have a disproportionately adverse impact on women's health as they disrupt care received by women. Data suggests that while the country expanded access to essential healthcare for women, quality provision of care remains a challenge. The maternal mortality rate in Lesotho stands at 544 per 100,000 live births.⁴³⁸ While the rate of institutional deliveries in the country is high (89.4%) and so is the rate of antenatal care (ANC) received by pregnant women, with 91% of women receiving at least ANC check ups at least once.⁴³⁹ The country still has a high rate of maternal deaths due to inadequate, inappropriate care received or negligence at the health facility level.⁴⁴⁰ Similarly, with respect to reproductive and sexual healthcare, compared to other countries in the region, Lesotho has a low rate of unmet need for family planning, with

⁴³¹ Ntho, M. N., Tale, L. M., Thabane, K., Kaaba, T., Makhotla, M., Konstabile, T., Mapetja, S., & Motsamai, S. (2015). (rep.). *The Status of Women in Lesotho*. Lesotho Council of Non-Governmental Organisations. European Union. Retrieved February 6, 2023, from <http://www.lcn.org.ls/Resource/The%20Status%20of%20Women%20in%20Lesotho.pdf>.

⁴³² Bureau of Statistics (2019). *Lesotho Multiple Indicator Cluster Survey 2018: Survey Findings Report*. Maseru, Lesotho: Bureau of Statistics

⁴³³ The Kingdom of Lesotho (n.d.). *Lesotho Government Beijing +25 Review Report*. Retrieved February 6, 2023, from <https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf>

⁴³⁴ Ntho, M. N., Tale, L. M., Thabane, K., Kaaba, T., Makhotla, M., Konstabile, T., Mapetja, S., & Motsamai, S. (2015). (rep.). *The Status of Women in Lesotho*. Lesotho Council of Non-Governmental Organisations. European Union. Retrieved February 6, 2023, from <http://www.lcn.org.ls/Resource/The%20Status%20of%20Women%20in%20Lesotho.pdf>.

⁴³⁵ The Kingdom of Lesotho (n.d.). *Lesotho Government Beijing +25 Review Report*. Retrieved February 6, 2023, from <https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf>

⁴³⁶ Chipatiso, L. M., Machisa, M., Nyambo, V., & Chiramba, K. (2014). (rep.). *The Gender Based Violence Indicators Study Lesotho*. Gender Links. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/imported/articles/attachments/20068_final_gbv_ind_lesotho.pdf.

⁴³⁷ The Kingdom of Lesotho (2020). *Gender and Development Policy 2018-2030*. Retrieved February 6, 2023, from <https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf>

⁴³⁸ UNDP (2022) *Gender Inequality Index*

⁴³⁹ Bureau of Statistics (2019). *Lesotho Multiple Indicator Cluster Survey 2018: Survey Findings Report*. Maseru, Lesotho: Bureau of Statistics

⁴⁴⁰ UNFPA. (2021, June 28). *Maternal Death Review shows that the loss of many women's lives could been avoided if action was taken* [web log]. Retrieved February 7, 2023, from <https://lesotho.unfpa.org/en/news/maternal-death-review-shows-loss-many-womens-lives-could-been-avoided-if-action-was-taken>

only 11% of married women reporting unmet needs.⁴⁴¹ However, cervical cancer is the most common cause of death among women aged 40-49, due to late detection and treatment initiation.⁴⁴²

Lesotho also has one of the world's highest prevalence rates of HIV among women with 30.4% of women infected with HIV compared to 20.8% of men. This is due to unequal power relations as well as GBV, with a significantly higher prevalence in women who had been forced into sexual relationships (39.3%).⁴⁴³ In 2018, the country also set an ambitious target to end HIV/AIDS by 2030 under their National HIV & AIDS Strategic Plan 2018/19 – 2022/23 which focuses on innovation in interventions and delivery models, multi-sectoral coordination, and adoption of results-based planning language and accountability frameworks. The strategy has a strengthened focus on gender, with objectives around the reduction of Sexual and Gender-Based Violence, along with a reduction in gender inequalities and reducing cultural and normative barriers for women.⁴⁴⁴

Education

Lesotho has an overall high literacy rate among women compared with other countries in the region, and has a higher literacy rate among women than men. As per the country's 2018 MICS, 92% of women are literate with 56.2% having attended secondary education and 13.2% of women having attended higher education. Comparatively, only 76% of men are literate, with only 43% having attended secondary education and 12.2% having attended higher education.⁴⁴⁵ However, at the tertiary education level, there is a gendered divide in STEM education, with the proportion of female students to male students being 0.4% in the faculty of Agriculture, Science, and Technology at the National University of Lesotho.⁴⁴⁶

The country has a progressive education policy, under the Education Act (2010) which provides for free and compulsory primary education for girls and boys. The policy also underlines that girls who become pregnant while in school must be integrated back into schooling post after the delivery of their child.⁴⁴⁷ The country's Gender and Development policy also recognizes the gendered dimension to choice of disciplines, with girls dominating 'soft disciplines', such as humanities, education, law, social sciences, crafts, and nutrition while the 'hard disciplines' such as sciences' have more male students. To correct this imbalance, the policy categorically commits to establishing mechanisms "to encourage women, girls, and other vulnerable groups to study Science, Technology, Engineering, and Mathematics (STEM) subjects".⁴⁴⁸ The Technical, Vocational Education and Training courses in Lesotho are government-sponsored to enable girls to complete these trainings.⁴⁴⁹

⁴⁴¹ [United Nations Population Fund, World Population Dashboard, Lesotho](#)

⁴⁴² [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](#)

⁴⁴³ [Ministry of Health, Lesotho, Centers for Disease Control and Prevention \(CDC\), and ICAP at Columbia University. \(2019\). Lesotho Population-based HIV Impact Assessment \(LePHIA\) 2016-2017: Final Report. Maseru, Lesotho, Atlanta, Georgia, and New York, USA. Retrieved February 7, 2023, from https://phia.icap.columbia.edu/wp-content/uploads/2019/09/LePHIA_FinalReport_Web.pdf](#)

⁴⁴⁴ [Kingdom of Lesotho \(2018\). National HIV & AIDS Strategic Plan 2018/19 – 2022/23. Retrieved February 7, 2023, from http://nac.org.ls/wp-content/uploads/2020/02/National-HIV-Strategic-Plan-2018-2023-Aug-30.18.pdf](#)

⁴⁴⁵ [Bureau of Statistics \(2019\). Lesotho Multiple Indicator Cluster Survey 2018: Survey Findings Report. Maseru, Lesotho: Bureau of Statistics](#)

⁴⁴⁶ [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](#)

⁴⁴⁷ [Gouvernement gazette, The Kingdom of Lesotho \(2010\). Education Act \(2010\). Retrieved February 7, 2023, from https://planipolis.iiep.unesco.org/sites/default/files/ressources/lesotho_education_act_2010.pdf](#)

⁴⁴⁸ [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](#)

⁴⁴⁹ [The Kingdom of Lesotho \(n.d.\). Lesotho Government Beijing +25 Review Report. Retrieved February 6, 2023, from https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/CSW/64/National-reviews/Lesotho.pdf](#)

Women's economic participation and entrepreneurship

As per Lesotho's 2019 Labour Force Survey, the country has a female labor force participation rate (LFPR) of 45.3%, which is lower than other countries in the region and about 10 percentage points lower than LFPR of men (55%). The survey also highlights that most women are engaged in household-related activities as employers for undifferentiated goods and services-producing activities of the households for their own usage (25%) followed by the manufacturing sector (18%). Comparatively a high percentage of men were engaged in agriculture, forestry, and fishery-related activities (17.4%) and construction (16%). Further, the mean earnings for women are much lower than men, i.e. Maloti 3,377.22 and Malori 4,672.92 respectively.⁴⁵⁰

As per the Global Gender Gap report 2022, there is also a wide gender gap with regards to the representation of women as legislators, senior officials, and managers, with only 33.7% women and 66.2% men as well as professional and technical workers, where women make up 39% workers.⁴⁵¹

To encourage women's participation in the labour force, Lesotho underlines very ambitious policy goals under the Gender and Development policy, including (but not limited to) the expansion of women's control over productive resources, including land, information, technology, safe water, energy and finance, harmonization of legislation and policy to provide better protection and working conditions for women, putting in place public procurement quotas to source directly from women businesses, as well as advocating a 30% quota for women in decision making positions in the private sector.⁴⁵²

The country also has a Science and Technology policy from 2008, which aims to achieve high participation of women in Science and technology-related work, including teaching, learning research, and development. Further, the policy places special focus on women entrepreneurs by providing them access to training, technical assistance, marketing, and export development support, as well as increased access to technology-based incubation.⁴⁵³ However, the successes of this policy are yet to be evaluated.

Gender and energy in Lesotho

In Lesotho, the majority of households do not have access to electricity. As per the MICS, only 43.8% of people have access.⁴⁵⁴ The Department of Energy along with the UNDP, have collaborated on a gender mainstreaming framework under the Sustainable Energy 4 All project with the following objectives:⁴⁵⁵

1. To build an inclusive energy sector with institutions free of discrimination and inequality
2. To build the capacity of women entrepreneurs to participate in the energy sector supply value chain
3. To facilitate the integration of gender during the implementation of electrification projects.

The energy policy (2015-2025) recognises gender mainstreaming as one of its guiding principles and states that gender equality is an integral part of the formulation and implementation of energy programmes and activities.⁴⁵⁶

⁴⁵⁰ [Beaureau of Statistics. Kingdom of Lesotho \(2019\). 2019 Labour Force Survey \(LFS\) Report. Retrieved February 7, 2023, from https://www.bos.gov.ls/New%20Folder/Copy%20of%20Demography/2019_Lesotho_LFS_Report.pdf](https://www.bos.gov.ls/New%20Folder/Copy%20of%20Demography/2019_Lesotho_LFS_Report.pdf)

⁴⁵¹ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴⁵² [The Kingdom of Lesotho \(2020\). Gender and Development Policy 2018-2030. Retrieved February 6, 2023, from https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf](https://genderlinks.org.za/wp-content/uploads/2020/03/Gender-and-Development-Policy-2018-2030.pdf)

⁴⁵³ [UNESCO \(2018\). Lesotho Science & Technology Policy 2006-2011. Retrieved February 7, 2023, from https://unesdoc.unesco.org/ark:/48223/pf0000158512](https://unesdoc.unesco.org/ark:/48223/pf0000158512)

⁴⁵⁴ [Bureau of Statistics \(2019\). Lesotho Multiple Indicator Cluster Survey 2018: Survey Findings Report. Maseru, Lesotho: Bureau of Statistics](#)

⁴⁵⁵ [Kingdom of Lesotho, UNDP \(2020\). Gender Mainstreaming Strategy for the Energy Sector, Lesotho 2020-2024. Retrieved February 7, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/ls/undp-lesotho-gender--energy-sector-report-2020.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/ls/undp-lesotho-gender--energy-sector-report-2020.pdf)

⁴⁵⁶ [Kingdom of Lesotho \(2011\). Lesotho Energy Policy \(2015-2025\). Retrieved February 15, 2023, from https://nul-erc.s3.amazonaws.com/public/documents/reports/lesotho-energy-policy-2015-2025-1532182755.pdf](https://nul-erc.s3.amazonaws.com/public/documents/reports/lesotho-energy-policy-2015-2025-1532182755.pdf)

A recent report found that within the Department of Energy, women represent a very small minority of the workforce (5%) overall. Further, in The LEC Transmission and Distribution Unit, only women were found to represent only 3.5% of staff in technical roles. However, in the private sector, there seems to be a high representation of women (44%) in senior management positions, as most of the private companies are family businesses with couples representing Founding Directors. The overall representation of women in private sector energy companies is very high compared to regional and global averages (42%), however these women are reportedly concentrated in administrative roles rather than technical or managerial roles.⁴⁵⁷

Among notable organisations active in the energy sector, Women In Sustainable Energy Lesotho (WISE-L) is an organisation that advocates for and promotes a gender balance in the energy sector as well as promotes entrepreneurship amongst women in this sector.⁴⁵⁸

⁴⁵⁷ [Kingdom of Lesotho, UNDP \(2020\). Gender Mainstreaming Strategy for the Energy Sector, Lesotho 2020-2024. Retrieved February 7, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/ls/undp-lesotho-gender--energy-sector-report-2020.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/ls/undp-lesotho-gender--energy-sector-report-2020.pdf)

⁴⁵⁸ [Lesotho: Energy. Wise-L Celebrate Women's Month. \(2022, August 30\). APO Group: Africa Newsroom. Retrieved February 7, 2023, from https://www.africa-newsroom.com/press/lesotho-energy-wisel-celebrate-womens-month?lang=en.](https://www.africa-newsroom.com/press/lesotho-energy-wisel-celebrate-womens-month?lang=en)

Burkina Faso

Overview of gender equality and gender gaps in the country

Burkina Faso is a landlocked country in West Africa. It has a population of 22.1 million people, with a young and adolescent population of 33% between the age group of 10 to the age of 24 years. The fertility rate in the country stands at 4.9.⁴⁵⁹

The Gender Development Index in Burkina Faso is 0.903, and the country is placed with 'Group 4' countries with medium to low equality in HDI achievements between women and men (absolute deviation of 7.5 to 10 percent). This is partly due to the country's significant gender gap in women's and men's estimated gross national income per capita, which is \$921.⁴⁶⁰ Burkina Faso has a Gender Inequality Index of 0.621 and is ranked 157, which measures disparities between men and women in economic opportunities, reproductive health, and empowerment.⁴⁶¹

Indicator	Result	Source and date
Life expectancy at birth	61	UNDP (2022) Gender Development Index
Maternal mortality ratio	320	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	38%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	60%	World Bank, 2019
Mean years of schooling ⁴⁶²	1.6	UNDP (2022) Gender Development Index
Estimated gross income per capita	1659 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

At the international level, Burkina Faso ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) in 1987, including the Optional Protocol on violence against women, and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (the Maputo Protocol) in 2006.⁴⁶³ The country also ratified the Convention on the Rights of the Child (CRC) in 1990,⁴⁶⁴ and signed the Beijing Platform for Action Conference recommendations and conclusions and committed to implementing them.⁴⁶⁵

⁴⁵⁹ [United Nations Population Fund. World Population Dashboard. Burkina Faso](#)

⁴⁶⁰ [UNDP \(2022\) Gender Development Index](#)

⁴⁶¹ [UNDP \(2022\) Gender Inequality Index](#)

⁴⁶² Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

⁴⁶³ [OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁴⁶⁴ [United Nations Human Rights Treaty Bodies. Burkina Faso](#)

⁴⁶⁵ [Burkina Faso Ministry for the Advancement Of Women \(2004\). Reply to the Questionnaire To Governments On Implementation Of The Beijing Platform For Action \(1995\) And The Outcome Of The Twenty-Third Special Session Of The General Assembly \(2000\).](#)

At the national level, the 1991 Constitution (Art. 1) prohibits any discrimination on the basis of sex, stating that all citizens are equal.⁴⁶⁶ In 1997 was created a Minister dedicated to women and gender issues (currently the Ministry of Women, National Solidarity, Family and Humanitarian Action). Burkina Faso adopted a National Gender Policy in 2009, which specific objectives are:

- Promote equal rights and equal opportunities in terms of access to and control over basic social services.
- Promote participatory economic development, access, and distribution of more equitable resources and incomes.
- Develop equal participation of men and women in decision-making spheres at all levels.
- Promote the institutionalization of gender in all areas.
- Promote a dynamic partnership for gender and development.
- Develop information and awareness-raising mechanisms aimed at all actors for a change in behavior and mentality in favor of equity and equality in relations between men and women.⁴⁶⁷

In terms of laws and regulations that affect women's economic opportunity in Burkina Faso, the country got a score of 82.5 out of 100 in the Women Business and the Law's 2022 index. This score is higher than the regional average observed across Sub-Saharan Africa (71.5).⁴⁶⁸ However, when it comes to restrictions on freedom of mobility, men have had the preference to choose where the family will reside (*Code des Personnes et de la Famille*, Art. 294). In addition, the current legislation affects women's employment and income by not having laws that enforce equal pay for work of equal value and paid parental leave. The existence of a law that prohibits women from working in an industrial job in the same way as men (*Code des Personnes et de la Famille*, Arts. 246 and 393) also limits women's opportunities.

Political participation

Despite the quotas of 30% of women for legislative and local elections, implemented in 2009 to promote women's representation in the political sphere,⁴⁶⁹ women's representation in the Assemblée Nationale is still low, with women holding only 17% of the seats. The original Law on Quotas from 2009 established financial sanctions (losing 50% of public funding) for non-compliance with the 30% quota, but these were removed in 2020. However, if a party reaches or exceeds the 30 percent quota, it receives additional funding as prescribed by the regulations related to public funding of political parties (Law on Quotas, Article 5 & 6).⁴⁷⁰ Burkina Faso has one of the lowest scores in the Global Gender Gap's Political Empowerment subindex, just before Nigeria, and it ranks 138 out of 146 globally.⁴⁷¹

Social and cultural context

Discriminatory practices towards women in Burkina Faso still prevail, due to the variations in social relationships between men and women in the different ethnic groups, as each one has its customs and culture, as well as the social and economic conditions of women in urban and rural areas, and women are generally regarded as having a lower status than men.⁴⁷² Often, society and the authorities disregard the

⁴⁶⁶ [Burkina Faso Ministry for the Advancement Of Women \(2004\). Reply to the Questionnaire To Governments On Implementation Of The Beijing Platform For Action \(1995\) And The Outcome Of The Twenty-Third Special Session Of The General Assembly \(2000\).](#)

⁴⁶⁷ [Burkina Faso Ministère de la Promotion de la Femme \(2009\). Document de la Politique Nationale Genre Du Burkina Faso.](#)

⁴⁶⁸ [World Bank Group \(2021\). Women Business and the Law. Burkina Faso.](#)

⁴⁶⁹ [OECD SIGI \(2018\). Étude Pays Sigi-Burkina Faso.](#)

⁴⁷⁰ [International Institute for Democracy and Electoral Assistance \(2023\). Gender Quotas Database, Burkina Faso.](#)

⁴⁷¹ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴⁷² [Japan International Cooperation Agency \(2013\). Country Gender Profile: Burkina Faso.](#)

Family and Penal Codes favoring the customary law, which perpetuates practices that discriminate against women.⁴⁷³

Despite the Burkinabe criminal and civil law establishing that forced marriage is illegal,⁴⁷⁴ early marriage is still prevalent in Burkina Faso. Half of the women aged 20–24 (51.6%) were married or in a union before age 18.⁴⁷⁵ This is in part because many families force their daughters into marriage due to poverty. Dowry motivates families to arrange marriages for their daughters as soon as they reach puberty.⁴⁷⁶ Polygamy is legal under the Family Code.⁴⁷⁷ A large part of the population is married under a polygamous regime (i.e. 38% of unions).⁴⁷⁸ Levirate is forbidden under the Criminal Code (Art. 376). However, under customary law, widows are obliged to marry a brother of their deceased husband.⁴⁷⁹

Regarding inheritance, widows and girls have equal legal rights to inherit property under articles 742-744 of the Family Code. However, the customary law grants no right of inheritance to widows or minor children, so they cannot inherit land either.⁴⁸⁰ Although civil law does not discriminate against women, women's access to land, particularly concerning rural land, has always been challenging.⁴⁸¹ Only 32% of married women have access to land, and in most cases, they share it with others. Further, only 8.2% of women own land alone.⁴⁸² The poor access of women to land is a recurring problem due to customary laws, in which the land is generally the family's property and only reserves rights of use to the woman, and the fact that inheritance is the primary means of accessing land.⁴⁸³

In Burkina Faso, the percentage of women who have experienced intimate partner violence (11.5) is less than the world average of 27%.⁴⁸⁴ However, gender-based violence (GBV) continues to manifest in many ways, such as female genital mutilation (FGM), which is still common in Burkina Faso, despite its ban in 1996. 76% of Burkinabe women have had the procedure done.⁴⁸⁵ The country is one of the few in the region to have prosecuted perpetrators under legislation banning FGM (Penal Code Art. 513-7, 513-8, 513-9).⁴⁸⁶

Health

While legislation is in place that grants the right to women's reproductive health, more than half of the Burkinabe women (54%), are excluded from the decision-making process relating to their own health.⁴⁸⁷ The national family planning policy, adopted in 2004, guarantees access to contraception to all women. However, women face challenges accessing contraception mainly due to the disapproval of husbands and problems with access, information, or cost.⁴⁸⁸ There is a higher incidence of HIV/AIDS among women than men because they are still highly exposed to infection due to practices such as early or forced marriage,

⁴⁷³ [OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁴⁷⁴ [OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁴⁷⁵ [UN Women \(n/d\) Country Sheet, Burkina Faso.](#)

⁴⁷⁶ CEDAW (2009) 'Consideration of reports submitted by States parties under article 18 of the Convention on the Elimination of All Forms of Discrimination against Women, [Cited in OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁴⁷⁷ [Burkina Faso: Code des personnes et de la famille \(1989\)](#)

⁴⁷⁸ [OECD SIGI \(2018\). Étude Pays Sigi-Burkina Faso.](#)

⁴⁷⁹ [Code et lois au Burkina, Burkina Faso](#)

⁴⁸⁰ UN Economic Commission for Africa (UNECA) (2009a) 'Questionnaire à l'intention des pays en prevision de l'examen et de l'évaluation de la mise en oeuvre du programme d'action de Beijing quinze ans après son adoption (Beijing + 15). [Cited in OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁴⁸¹ [Rapport National Beijing+25](#)

⁴⁸² Enquêtes démographiques et de Santé (EDS) (2010). [Cited in Rapport National Beijing+25](#)

⁴⁸³ [Rapport National Beijing+25](#)

⁴⁸⁴ [The World Bank, Gender data portal \(2010\). Burkina Faso.](#)

⁴⁸⁵ [Stat compiler, the DHS Program \(2010\). Burkina Faso.](#)

⁴⁸⁶ [Rapport National Beijing+25](#)

⁴⁸⁷ [OECD SIGI \(2018\). Étude Pays Sigi-Burkina Faso.](#)

⁴⁸⁸ [OECD SIGI \(2018\). Étude Pays Sigi-Burkina Faso.](#)

the unwillingness of their partners to wear condoms, levirate marriage, and polygamy.⁴⁸⁹ It is estimated that there are 51,000 women aged 15 and over living with HIV compared to 31,000 men.⁴⁹⁰

Social norms tend to exclude women from the family planning decision-making process. Although most of the people think that the decisions must be made by both partners jointly; between a quarter and a third of the people say that women do not have to take part in the use of contraceptives, birth spacing, and the number of children since this is the husband's decision.⁴⁹¹ Induced abortion is legally restricted in Burkina Faso. It is estimated that 23 women abort per 1000 women ages 15–49 in 2020.⁴⁹² Most of the time, they are performed using non-recommended methods by untrained providers, causing a greater risk of complications for women.⁴⁹³

Education

In terms of education, the literacy rate is higher in men 49.2% compared to 31% in women.⁴⁹⁴ The primary completion rate is higher in girls (57.3%) than in boys (67.7%), but this situation is reversed in secondary, where the completion rate for girls is 18.4% and 21.6% for boys, and this gap is even wider in rural areas.⁴⁹⁵ Only 20.59% of women graduate with STEM degrees in Burkina Faso compared to 79.41% of men.⁴⁹⁶

The Directorate for the Promotion of Girls' Education (DPEF) was established within the Ministry of Basic Education and Literacy (MEBA) to promote activities to reduce gender-based inequalities. The DPEF also leads activities to raise awareness to keep girls in school.⁴⁹⁷

Women's economic participation and entrepreneurship

60% of Burkinabe women (% of women ages 15-64) participate in the labor force participation compared to 76% of men (% of men ages 15-64).⁴⁹⁸ Women's participation in the informal sector is high in Burkina Faso, with 95% of the women working in the informal sector.⁴⁹⁹

Regarding entrepreneurship, there are no legal restrictions for women. Under civil law, women can access credits, sign contracts, register a business, and open a bank account like men.⁵⁰⁰ However, formal financial institutions consider women as "high-risk" applicants as they often do not have financial or material security.⁵⁰¹ This situation puts women at a disadvantage regarding access to bank loans, which can hinder women's entrepreneurship. While there are no legal restrictions on Burkinabe women's access to bank

⁴⁸⁹ [Burkina Faso Ministry for the Advancement Of Women \(2004\). Reply to the Questionnaire To Governments On Implementation Of The Beijing](#)

⁴⁹⁰ [UNAIDS \(2021\)](#)

⁴⁹¹ [OECD SIGI \(2018\). Étude Pays Sigi-Burkina Faso.](#)

⁴⁹² [Performance Monitoring for Action \(2022\). PMA Abortion Survey Results, Burkina Faso, December 2020 – March 2021.](#)

⁴⁹³ [Performance Monitoring for Action \(2022\). PMA Abortion Survey Results, Burkina Faso, December 2020 – March 2021.](#)

⁴⁹⁴ [UNESCO \(2018\).](#)

⁴⁹⁵ [Ministère de l'éducation Nationale, de l'alphabetisation et de la Promotion des Langues Nationales \(2022\). Rapport de suivi annuel 2021 du Programme de développement stratégique de l'éducation de base](#)

⁴⁹⁶ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁴⁹⁷ [UNESCO \(2007\). Burkina Faso Country case study](#)

⁴⁹⁸ [The World Bank \(2019\)](#)

⁴⁹⁹ [World Economic Forum \(2022\) Global Gender Gap Report](#)

⁵⁰⁰ [World Bank Group \(2021\). Women Business and the Law, Burkina Faso.](#)

⁵⁰¹ CEDAW (2009) 'Consideration of reports submitted by States parties under article 18 of the Convention on the Elimination of All Forms of Discrimination against Women, [Cited in OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

loans, the government reports that some entrepreneurial activities are closed to rural women due to social and cultural barriers.⁵⁰²

Gender and energy in Burkina Faso

Burkina Faso's energy access rate is 19%, one of the lowest in the West African sub-region.⁵⁰³ The expansion of off-grid electrification faces critical barriers, including inadequate policy and regulatory frameworks, lack of access to finance, lack of local technical capacity, and high cost of new connections, among others.⁵⁰⁴ These barriers may make it more difficult for women to access off-grid electrification due to gender gaps in education, access to finance, land tenure, and cultural norms that prevail in the country.

The Africa Minigrids Program supported by UNDP and financed by the GEF, seeks to demonstrate productive uses of electricity to reduce costs, increase the viability and enable mini-grid development at scale. The primary targets and actors of the pilot projects are small-scale farmers (women and youth, mostly) who will benefit directly from the electricity supplied by the solar mini-grids. The project started in 2022 and it is planned to end in 2026.⁵⁰⁵

Among other initiatives, the GCF-funded Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries⁵⁰⁶ program is implemented by the International Fund for Agricultural Development (IFAD) and local organizations from the participant countries, including Burkina Faso. The program seeks to provide concessional loans to foster the best adaptation and mitigation practices for green businesses within the selected agricultural value chains in each target country. Operated by local, national agricultural banks (LNABs) – the Agricultural Bank of Burkina Faso– the facility will offer special lines of financing for green business projects prepared by farmer's organizations (FOs), women and youth organizations, cooperatives, and MSMEs (including agribusiness dealers and solar operators). Besides providing loans, the program includes a Technical Assistance Facility (TAF) aimed at addressing capacity, knowledge, and policy gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks, and clients (FOs, MSMEs, and cooperatives), with a special emphasis on women and youth.⁵⁰⁷

⁵⁰² CEDAW (2009) 'Consideration of reports submitted by States parties under article 18 of the Convention on the Elimination of All Forms of Discrimination against Women. [Cited in OECD \(2014\). Social Institutions and Gender Index, Burkina Faso.](#)

⁵⁰³ [World Bank Data \(2020\).](#)

⁵⁰⁴ Off-Grid Solar Market Assessment Burkina Faso Report" in the framework of the Regional Offgrid Electrification Project (ROGEP) cited in [UNDP & Government of Burkina Faso \(2021\) project document: National Child Project under the GEF Africa Minigrids Program – Burkina Faso](#)

⁵⁰⁵ Off-Grid Solar Market Assessment Burkina Faso Report" in the framework of the Regional Offgrid Electrification Project (ROGEP) cited in [UNDP & Government of Burkina Faso \(2021\) project document: National Child Project under the GEF Africa Minigrids Program – Burkina Faso](#)

⁵⁰⁶ The countries participating in this program are Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan.

⁵⁰⁷ [The Inclusive Green Financing Initiative \(IGREENFIN I\): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall \(GGW\) countries](#)

Guinea-Bissau

Overview of gender equality and gender gaps in the country

Guinea-Bissau is a coastal country in West Africa. has a population of 2.1 million people, with a young and adolescent population of 32% between the age group of 10 to the age of 24 years. The fertility rate in the country stands at 4.2.⁵⁰⁸

The Gender Development Index in Guinea-Bissau is 0.867, and the country is placed in 'Group 5', countries with low equality in HDI achievements between women and men, with a high absolute deviation from gender parity of more than 10%. The country particularly has a wide gender gap in mean years of schooling, with 2.4 years among girls and 4.9 years among men. Guinea-Bissau has a Gender Inequality Index of 0.627 and is ranked 159; the index measures disparities between men and women in economic opportunities, reproductive health, and empowerment.⁵⁰⁹

Indicator	Result	Source and date
Life expectancy at birth	61.8	UNDP (2022) Gender Development Index
Maternal mortality ratio	667	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	40%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	67%	World Bank, 2019
Mean years of schooling ⁵¹⁰	2.4	UNDP (2022) Gender Development Index
Estimated gross income per capita	1561 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Internationally and regionally, Guinea-Bissau has made several commitments to equal rights for women. The country ratified the Optional Protocol to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1981, as well as the Protocol to the Charter of Human Rights and Peoples on the Rights of Women in Africa.⁵¹¹ Further, the country's "Social Protection Framework Law", Law 4/2007, defines the conditions for the protection of women workers in the formal market, including maternity leave provisions. However, It does not contain specific protection regulations for the informal labor market.⁵¹²

⁵⁰⁸ [United Nations Population Fund, World Population Dashboard, Guinea-Bissau](#)

⁵⁰⁹ [UNDP \(2022\) Gender Inequality Index](#)

⁵¹⁰ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

⁵¹¹ [Coalition Of Organizations Defense Of Children's Rights In Guinea-Bissau \(2019\). Civil Society Report Of The Implementation Of Girl's Rights In Guinea-Bissau Upr - Gaps Implementation Of Girls Rights In Guinea-Bissau](#)

⁵¹² [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](#)

Legislation exists in the country to protect women from violence, however, the implementation of these has been weak with pervasive violence against women, as is examined in the next section. The Law on Domestic Violence (Law Nbr. 6/2014 of 4 February) aims to combat cases of physical, psychological, property, and sexual violence. The country also has a Law to combat Female Genital Mutilation (Law Nbr. 14/2011 of 06 June) which aims to prevent and combat FGM, the practice is still prevalent, as expanded in the next section.⁵¹³ While the country's Land Law guarantees the right to private use of land without discrimination, the Constitution does not have any specific provisions to protect women's right to use land, and their land tenure, as a result, is often insecure. Men continue to control and manage agricultural and household land, while women live under the looming threat of having their land taken away.⁵¹⁴

As per the Women, Business and Law index, the country has a score of 42.5 out of 100, with the poorest performance in the workplace, pay, and entrepreneurship indicators, under Lei No. 2 of 1986, Art. 155(4) of the country's legislation, no provisions were found to grant women equal rights to an industrial job, same as men, or equal pay. Under the country's Civil Code, Art. 1680, women do not have the same right to register a business or open a bank account. The country also does not have any legislation, criminal penalties, or civil remedies for sexual harassment in employment, nor does it have maternity or paternity benefits in place for working women.⁵¹⁵

The country had developed a National Policy for Gender Equality and Equity, aimed at addressing gender inequalities and combating discriminatory practices, however, this policy has not been adopted into implementation yet.⁵¹⁶ With technical support from UNDP, the country's National People's Assembly is currently in the process of revising the constitution with the integration of a gender lens.⁵¹⁷

Social and cultural context

Guinea-Bissau is ethnically diverse with numerous sub-groups, with all these having a patriarchal social structure (except the *Bijagó* group of the Bijagos Archipelago) and customary law aligned with traditional gender roles.⁵¹⁸ Social life among these ethnic groups is characteristically male-dominated, under the culture of *matchundaadi*. The *Matchundaadi* culture is based on hierarchical control that men are guaranteed over women, placing them in decision-making roles, and institutionalizing exalted men above the law.⁵¹⁹ Polygamy is a common form of marriage, with 39.2% of women in the age group 15-49 being in polygamous unions,⁵²⁰ with men exerting their dominance over multiple women under such unions, where men must prove their masculinity by producing more children.⁵²¹ Gender-based violence, including Female Genital Mutilation or "*Fanadu di mulher*" is a critical aspect of the *matchundaadi* culture, and is considered

⁵¹³ [Coalition Of Organizations Defense Of Children's Rights In Guinea-Bissau \(2019\). Civil Society Report Of The Implementation Of Girl's Rights In Guinea-Bissau Upr - Gaps Implementation Of Girls Rights In Guinea-Bissau](#)

⁵¹⁴ [Silva, Y. N. \(2021\). \(rep.\). Guinea-Bissau: Where land rights are not secure for women. African Arguments . Retrieved February 11, 2023, from https://africanarguments.org/2021/10/guinea-bissau-where-land-rights-are-not-secure-for-women/.](#)

⁵¹⁵ [World Bank Group \(2021\). Women, Business and the Law. Guinea-Bissau](#)

⁵¹⁶ [Mabyea, K. E., & Salla, F. O. M. \(2021\). \(rep.\). Final External Evaluation Of The Project "Supporting Women And Youth's Political Participation For Peace And Development In Guinea-Bissau". UN Women. Retrieved February 11, 2023, from https://www.un.org/peacebuilding/sites/www.un.org.peacebuilding/files/documents/pbf_irf-211_evaluation_pbf_project_gbis_wypp_final.pdf.](#)

⁵¹⁷ [UNDP. \(2021, July 1\). A gender lens in the revision of the Constitution of Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/guinea-bissau/news/gender-lens-revision-constitution-guinea-bissau.](#)

⁵¹⁸ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](#)

⁵¹⁹ [Moreira, J. \(2017\). A Cultura di Matchundadi na Guiné-Bissau: Género, Violências e Instabilidade Política. Lisbon: ISCTE.](#)

⁵²⁰ [Ministério da Economia e Finanças, Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019, Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵²¹ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](#)

a rite of passage for girls' marriage.⁵²² Circumcision is also associated with a higher bride price, and uncircumcised women face the threat of social exclusion.⁵²³ As per the country's 6th Multiple Indicator Cluster Survey (MICS) 2020, 52% of women in the country (in the age group 15-49) have reportedly undergone some form of Female Genital Mutilation and 29.7% of young girls (aged 0-14) have been subjected to FGM.⁵²⁴

The country's legal age for marriage was raised to 18 years, however, there is no penalization for perpetrators articulated in the country's law.⁵²⁵ As per the MICS, 29.9% of women in the age group 20-49 were married off before they reached the age of 18.⁵²⁶ While official data is limited, the country is also affected by human trafficking, with the trafficking of women for sexual exploitation and of children (*Talibés*) who are entrusted to those pretending to be religious leaders (*Marabouts*) on the pretext of religious education.⁵²⁷

Political participation

Guinea-Bissau has experienced political instability for decades. The political system is semi-presidential. The country took affirmative action in 2018, by introducing legislation demonstrating its commitment to uplifting women's position in political decision-making. The country's Parity Law establishes that there must be a minimum of 36% of women in party lists for elective positions in the National Parliament and in local municipal elections.⁵²⁸ The law also underlines sanctions in case of non-compliance. In 2019, only 8 of 21 parties that participated in the election complied with the Parity Law.⁵²⁹ election saw an increase in the number of women members of parliament, from 7,8% (1999) to 13,7% however, the country's government has indicated their commitment women's political participation by conferring 11 of the 31 ministerial portfolios to women and a total of 35% female participation in the Executive of Guinea-Bissau.⁵³⁰

Health

Years of political instability and unrest have impacted women's well-being in Guinea-Bissau. The country's deep-rooted customs that legitimize violence against women, forced marriages, and early initiation into reproductive labor, have adversely impacted women's health. As per the MICS (2020), 27% of women in the age group 20-24 reported that they had their first pregnancy before the age of 18. Further, women have to travel far to access health facilities, which combined with poor road infrastructure makes the situation

⁵²² [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵²³ Baldé, F. (2018). Mutilação Genital Feminina, Uma Questão de Direitos Humanos Das Mulheres Na Guiné-Bissau. In C. Carvalho, R. Falcão, & M. Patrício (Eds.), Corte/Mutilação Genital Feminina: Respostas Institucionais Integradas (pp. 97–106).

⁵²⁴ [Ministério da Economia e Finanças. Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019. Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵²⁵ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵²⁶ [Ministério da Economia e Finanças. Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019. Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵²⁷ [United Nations Guinea Bissau \(2020\). UN Common Country Analysis \(CCA\) Retrieved February 11, 2023, from https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_\(CCA\).pdf](https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_(CCA).pdf)

⁵²⁸ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵²⁹ [Noronha, N. de. \(2019, March 8\). Eleições: Mulheres guineenses "desapontadas" com partidos. DW. Retrieved February 11, 2023, from https://www.dw.com/pt-002/dia-da-mulher-mulheres-guineenses-desapontadas-com-partidos/a-47822069.](https://www.dw.com/pt-002/dia-da-mulher-mulheres-guineenses-desapontadas-com-partidos/a-47822069)

⁵³⁰ [Dansó, I. \(2019, July 5\). Alta participação feminina no novo Governo guineense. DW. Retrieved February 11, 2023, from https://www.dw.com/pt-002/novo-governo-guineense-tem-alto-%C3%AAdndice-de-participa%C3%A7%C3%A3o-feminina/a-49492419.](https://www.dw.com/pt-002/novo-governo-guineense-tem-alto-%C3%AAdndice-de-participa%C3%A7%C3%A3o-feminina/a-49492419)

precarious for women.⁵³¹ Close to a majority of women in the country had home births (48%) and only 50.4% of women had assisted births at health facilities.⁵³² The country has a high maternal mortality rate of 667 deaths per 100,000 live births, which is one of the highest in the world.⁵³³ Further, women's access to contraception is limited, and 21% of women aged 15-49 who are currently married or in a union have an unmet need for family planning.⁵³⁴

The country also has West Africa's second-highest burden of HIV among the reproductive population, with 5.3% of the reproductive-age population living with HIV, and women making up 70% of these cases.⁵³⁵

Education

In a country, where about half of the households spent more than 65% of their incomes on food, education becomes a non-essential expense.⁵³⁶ While the country's Basic Law of the Educational System (2011) despite many gaps in its implementation, guarantees universal, free, and compulsory education up to the 6th year,⁵³⁷ there is a long way to go in achieving gender parity in education at the secondary and above levels. As per the country's MICS (2020) literacy rate among girls (aged 15-49) is 32.6%, and Guinea-Bissau has a twenty percentage point literacy gap between boys and girls. While the difference at the primary schooling level is not that stark, at the secondary school level there are only 11.7% females compared to 20.5% boys. Guinea-Bissau has only a small percentage of boys and girls who attain technical and higher education, with 2% of women having a technical education and 4% of men with technical education, and 2.1% of women with a university education compared to 4.8% of men.⁵³⁸

The country has an education sector plan (2017- 2025) that promotes schooling and encourages the retention of women in the first two basic cycles of education.⁵³⁹ However, quality of education and infrastructure is a looming issue in the country, with most schools, not having basics such as instructor's manuals, and other teaching materials; and the students not having access to pencils, paper, schoolbooks, etc.⁵⁴⁰

Women's economic participation and entrepreneurship

⁵³¹ [UNFPA \(2018, August 14\) Relief for women facing treacherous journey to medical care in Guinea-Bissau. Retrieved February 11, 2023, from https://www.unfpa.org/news/relief-women-facing-treacherous-journey-medical-care-guinea-bissau](https://www.unfpa.org/news/relief-women-facing-treacherous-journey-medical-care-guinea-bissau)

⁵³² [Ministério da Economia e Finanças, Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019, Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵³³ [UNDP \(2022\) Gender Inequality Index](#)

⁵³⁴ [Ministério da Economia e Finanças, Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019, Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵³⁵ [UNICEF \(n.d\). Children and AIDS. Keeping children and adolescents free from HIV-AIDS.Guinea Bissau Retrieved February 11, 2023, from https://www.unicef.org/guineabissau/children-and-aids#:~:text=Challenges,highest%20rate%20in%20west%20africa](https://www.unicef.org/guineabissau/children-and-aids#:~:text=Challenges,highest%20rate%20in%20west%20africa)

⁵³⁶ [United Nations Guinea Bissau \(2020\). UN Common Country Analysis \(CCA\) Retrieved February 11, 2023, from https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_\(CCA\).pdf](https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_(CCA).pdf)

⁵³⁷ [UNESCO \(n.d\). Education Profiles, Guines Bissau, Inclusion. Retrieved February 11, 2023, from https://education-profiles.org/sub-saharan-africa/guinea-bissau/-inclusion](https://education-profiles.org/sub-saharan-africa/guinea-bissau/-inclusion)

⁵³⁸ [Ministério da Economia e Finanças, Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019, Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵³⁹ [Ministère de l'Education Nationale, République da Guiné-Bissau \(2017\). Programme Sectoriel de l'Education de la Guinée Bissau \(2017 -2025\). Retrieved February 11, 2023, from https://planipolis.iiep.unesco.org/sites/default/files/ressources/guinea-bissau-esp-2017-2025.pdf](https://planipolis.iiep.unesco.org/sites/default/files/ressources/guinea-bissau-esp-2017-2025.pdf)

⁵⁴⁰ [United Nations Guinea Bissau \(2020\). UN Common Country Analysis \(CCA\) Retrieved February 11, 2023, from https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_\(CCA\).pdf](https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_(CCA).pdf)

Guinea-Bissau's economy is predominantly based around agriculture, with cashew farming being the primary source of income and employment for most of the country's poor.⁵⁴¹ As per the country's Integrated Regional Survey on Employment and Informal Sector (2019), 96% of women in Guinea-Bissau work in the informal sector. Women in Guinea-Bissau are relegated to informal work and they compose the vast majority of self-employed people at home; street vendors and domestic workers.⁵⁴² As per the ERI-ESI survey, 51.9% of women are in vulnerable employment as part-time, family workers, or self-employed, compared with 33.4% of men. On the other hand, women are almost non-existent in high productivity informal non-agrarian sector work such as civil construction work, transport activities, larger commercial establishments, and far distance and regional trade.⁵⁴³ 'Idleness' among disillusioned youth is also higher among women, with 31.8% of women in the age group 15-24, not in the education system or in employment, compared to 18.8% of men. Further, only 9.7% of women have formal remunerated work in the agriculture sector, while 27.5% of men have salaried work in the formal agriculture sector.⁵⁴⁴

Gender and energy in Guinea-Bissau

Currently, access to electricity among Guinea-Bissau's population is low. As per the MICS (2020), 18% of households have electricity access through a connection to a public grid and 17.7% of households have electricity through a non-public network connection.⁵⁴⁵ Due to limited access to modern energy sources, women have to spend a lot of time collecting wood and exposing themselves to harmful biomass fumes. Notably, women are engaged in various biomass-intensive processing industries including charcoal production, fish smoking, and street-food vending, and hence are important agents of change in the shift towards cleaner energy use in the country.⁵⁴⁶

The country has released a National Action Plan for Gender Mainstreaming in Access to Energy and The Guinea-Bissau Ministry of Natural Resources and Energy developed a National Action Plan (NAP), with a specific focus on mainstreaming gender into national development plans. The NAP aims to create an institutional framework for gender mainstreaming in access to energy followed by an intervention framework. The plan highlights the need to train women and young people and raise awareness around gender mainstreaming in energy, to enable a diverse and inclusive labor market in the energy sector. The NAP also aims to periodically measure the progress of its implementation and assess effectiveness.⁵⁴⁷

UNDP is currently implementing a project aimed at promoting better access to modern energy services through sustainable mini-grids and low-carbon bioenergy technologies among Guinea-Bissau's forest-dependent communities. As part of the project learnings, it was found that there is tremendous scope for clean off-grid renewable energy to create entrepreneurship opportunities for women, especially in the food processing and fishing sector (which is the second largest subsistence activity in Guinea-Bissau) and as

⁵⁴¹ [United Nations Guinea Bissau \(2020\). UN Common Country Analysis \(CCA\) Retrieved February 11, 2023, from https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_\(CCA\).pdf](https://minio.dev.devqube.io/uninfo-production-main/6a71e449-ab2f-4348-a456-7bc702ffec53_Guinea-Bissau_UN_Common_Country_Analysis_(CCA).pdf)

⁵⁴² [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵⁴³ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵⁴⁴ [Instituto Nacional de Estatística e AFRISTAT. 2019. Inquérito Regional Integrado sobre Emprego e Setor Informal. 2017/ 2018: Relatório final. Bissau, Guiné-Bissau e Bamako, Mali : INE e AFRISTAT.](#)

⁵⁴⁵ [Ministério da Economia e Finanças. Direção Geral do Plano/Instituto Nacional de Estatística \(INE\). 2020. Inquérito aos Indicadores Múltiplos \(MICS6\) 2018-2019. Relatório Final. Bissau, Guiné-Bissau: Ministério da Economia e Finanças e Direção Geral do Plano/ Instituto Nacional de Estatística \(INE\)](#)

⁵⁴⁶ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵⁴⁷ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

resellers/traders of energy products. Within the fishing sector, it was found that due to the limited availability of wood fuel, the fresh catch is often thrown away by women in coastal communities. There is thus an opportunity for women to be embedded in the fishing cold storage supply chain; particularly cooling facility/service provision, ice production, and fish-vending.⁵⁴⁸

African Development Bank Group's Country Strategy Paper (CSP) for Guinea Bissau covering 2022-2026 (aligned with the "*Hora Tchiga*") intends to focus their support on two infrastructure sectors—transport and energy. The cross-cutting theme of the country strategy that will become an integral part of the lending and non-lending operations in Guinea-Bissau is, climate, gender, and governance.⁵⁴⁹

Niger

Overview of gender equality and gender gaps in the country

Niger is a West African country with vast desert terrain. The country has a population of 26.1 million people, with a young and adolescent population of 33% between the age group of 10 to the age of 24 years.⁵⁵⁰ The country has one of the world's highest fertility rates of 6.6 per woman.⁵⁵¹

The Gender Development Index in Niger is 0.835, and the country is placed in 'Group 5' countries with low equality in HDI achievements between women and men, with a high absolute deviation from gender parity of more than 10%. The country has a high gender gap in mean years of schooling, with 1.7 years among girls and 2.8 years among boys, and in gross national income per capita. Niger has a Gender Inequality Index of 0.611 and is ranked 153; the index measures disparities between men and women in economic opportunities, reproductive health, and empowerment.⁵⁵²

Indicator	Result	Source and date
Life expectancy at birth	62.8	UNDP (2022) Gender Development Index
Maternal mortality ratio	509	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	29%	World Bank, 2021
Women's Labour Force Participation Rate (age 15 to 64)	62%	World Bank, 2019
Mean years of schooling ⁵⁵³	1.7	UNDP (2022) Gender Development Index

⁵⁴⁸ [UNDP \(2021\). UNDP Guinea-Bissau Gender Analysis Guinea-Bissau. Retrieved February 11, 2023, from https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf](https://www.undp.org/sites/g/files/zskgke326/files/migration/gw/Gender-Analysis-Final-01.03.2021.pdf)

⁵⁴⁹ [African Development Bank Group \(September, 2022\). Guinea Bissau - Country Strategy Paper 2022-2026. Retrieved February 11, 2023, from https://www.afdb.org/en/documents/guinea-bissau-country-strategy-paper-2022-2026](https://www.afdb.org/en/documents/guinea-bissau-country-strategy-paper-2022-2026)

⁵⁵⁰ [United Nations Population Fund. World Population Dashboard, Niger](#)

⁵⁵¹ [Charbit, Y., & Becker, S. \(2021, June 17\). Is a demographic disaster unavoidable in Niger? Retrieved March 1, 2023, from https://ideas4development.org/en/demographic-growth-niger/](https://ideas4development.org/en/demographic-growth-niger/)

⁵⁵² [UNDP \(2022\) Gender Inequality Index](#)

⁵⁵³ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

Estimated gross income per capita	936 (PPP \$)	UNDP (2022) Gender Development Index
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Legal status of women

The Constitution of Niger which was adopted in 2010 upholds equal rights for men and women and guarantees the same rights and duties for women and men in Articles 8 and 10. Further, Article 22 explicitly mandates the State to take measures to combat gender violence and take measures to ensure the full participation of women in public institutions, and in national development, through the country's policies. The country's quota law defines minimum representation percentages for women in public affairs and the commercial and labor codes prohibit gender-based discrimination in hiring, work allocation, wages, and promotion practices.⁵⁵⁴

However, Niger is one of the few countries in the region, which has not yet ratified the African Charter on Human and Peoples' Rights on the Rights of Women.⁵⁵⁵ The country ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) in 1999, however, Niger has expressed reservations about its implementation.^{556,557}

Further, the country's legal and regulatory system accords supremacy to customary law over statutory law. For instance, the Civil Code, denotes men as heads of the household with marital authority over wives, the Family Code recognized customary systems around matrimony and property, and the Rural Code fails to provide promote and protect women's ownership rights over land.^{558,559}

The country has articulated its gender commitment in some key country policies and strategy documents including the Sustainable Development and Inclusive Growth Strategy (SDDCI, Niger 2035), the National Gender Policy, the National Women's Economic Empowerment Strategy, National GBV Prevention and Response Strategy, and the Gender and Islam Strategy.⁵⁶⁰ The country's Gender Policy has adopted 4 key lenses to reach its equality goals including; (i) improvement of the socio-cultural environment in relation to demography, peace, and security; (ii) strengthening of the institutional and legal framework to fight against gender-based violence and to enable equitable participation of men and women in the management of power; (iii) economic empowerment and inclusive growth in relation to sustainable management of the environment, disaster risk management, migration, and humanitarian emergencies; (iv) strengthening of institutional mechanisms and organizational frameworks for coordination, monitoring-evaluation, and partnership. The country also has a National Observatory for Gender Promotion under the

⁵⁵⁴ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁵⁵ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁵⁶ [UN Human rights treaty bodies, Ratification status for Niger](#)

⁵⁵⁷ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁵⁸ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁵⁹ [Green Climate Fund \(2021\). Gender Assessment for FP176: Hydro-agricultural development with smart agriculture practices resilient to climate change in Niger. Retrieved March 1, 202 from https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf](https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf)

⁵⁶⁰ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

Prime Minister's office with a mandate to monitor, and promote gender equity and equality.⁵⁶¹ Apart from this, the country also has gender cells and focal points within sectoral ministries, which are responsible for gender mainstreaming in policy-making and programmatic decision-making at ministerial and institutional levels, however, these cells are mostly non-operational.⁵⁶²

Social and cultural context

The prevalence of Gender Based Violence in Niger is at 28.4%.⁵⁶³ Women displaced in some regions of the country, due to extremist Islamic groups' armed incursions face a high threat of violence in resettlement or refugee camps. Niger revised the Penal Code in 2003 to make provisions for certain forms of violence against women, particularly sexual harassment, female genital mutilation, slavery, pimping, incitement to debauchery, and rape however the country is yet to take holistic, adequate, and concrete steps to prevent, punish and manage cases of gender-based violence (GBV) and child marriage.⁵⁶⁴ Niger also has the world's highest incidence of child marriage (76%), with close to 4.8 million child brides of which 2 million were married before the age of 15. To combat this, In 2021, the government issued a decree institutionalizing communal and village committees as formal child protection committees.⁵⁶⁵ This further impacts girls' ability to remain in school and complete their education. Polygamy is widely practiced in the country, and high fertility is often a tool for women to command the respect of their husbands.⁵⁶⁶

Political participation

Niger has implemented quotas to enable the effective participation of women in decision-making by passing Law No. 2000-008 in 2000, and further revising the quotas in 2019 up to 25% for elective positions and 30% for appointive positions.⁵⁶⁷ As per the latest Inter-Parliamentary Union data, in Niger's National Assembly, 30.72% of MPs are women, as per their December 2020 election.⁵⁶⁸ In the central administration, out of 1,764 decision-making positions only 26% i.e. 466 are held by women, and at the deconcentrated level (regional and departmental services of the Ministries), out of a total of 1,007 decision-making positions, 8% are held by women. At the decentralized level, there are 17 women councilors out of a total of 104 councilors in City Councils however there are no female City Council Chairperson. In Regional Councils, 34 women were elected out of a total of 260 councilors (13%).⁵⁶⁹

There is a wide gender gap in Niger's ministerial representation, with only 15.79% women ministers compared to 84.21% men.⁵⁷⁰ The country has supported the establishment of the Niger Chapter of the

⁵⁶¹ [Green Climate Fund \(2021\). Gender Assessment for FP176: Hydro-agricultural development with smart agriculture practices resilient to climate change in Niger. Retrieved March 1, 202 from https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf](https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf)

⁵⁶² [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁶³ [Green Climate Fund \(2021\). Gender Assessment for FP176: Hydro-agricultural development with smart agriculture practices resilient to climate change in Niger. Retrieved March 1, 202 from https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf](https://www.greenclimate.fund/sites/default/files/document/fp176-gender-assessment.pdf)

⁵⁶⁴ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁶⁵ [UNICEF \(2021\). Country Profile 2021 For Every Child: UNFPA-UNICEF Global Programme To End Child Marriage. Retrieved 1 March, 2023, from https://www.unicef.org/media/128371/file/Country-profile-Niger-2021.pdf](https://www.unicef.org/media/128371/file/Country-profile-Niger-2021.pdf)

⁵⁶⁶ [CARE \(2021\). Gender analysis on sexual and reproductive health: Niger - Maradi \(July 2021\). Retrieved 1 March, 2023, from https://reliefweb.int/report/niger/gender-analysis-sexual-and-reproductive-health-niger-maradi-july-2021](https://reliefweb.int/report/niger/gender-analysis-sexual-and-reproductive-health-niger-maradi-july-2021)

⁵⁶⁷ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁶⁸ [Inter-Parliamentary Union \(2020\). Niger](#)

⁵⁶⁹ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁷⁰ [World Economic Forum \(2022\) Global Gender Gap Report](#)

African Women Ministers and Parliamentarians Network (REFAMP) which is working towards the promotion of female leadership in politics.⁵⁷¹

Health

The high fertility rate in Niger has adverse impacts on women's health and well-being. Niger has a maternal mortality rate of 509 as per the Gender Inequality Index.⁵⁷² The country also has very poor coverage of ANC and PNC checkups, with only 27.8% of women having 4 or more ANC visits in 2016-18 and 39.4% of women receiving post-natal care. In 2017 institutional births were also limited (38.2%) and only 36.8% of births were attended by skilled health professionals. Anemia among women is also highly prevalent with 46% of women being anemic.⁵⁷³ The uptake of contraception in Niger is low due to religious beliefs under Islamic law and women's fears about seeking reproductive health services from male healthcare workers.⁵⁷⁴ Further, 18% of married women have an unmet need for family planning and only 50% of their demand is met with modern contraceptive methods.⁵⁷⁵

Education

As per the World Bank data, Niger has a very low female literacy rate of 29%.⁵⁷⁶ Further, girls in rural areas have low enrollment rates in schools. In 2016-17, the Gross Enrollment Rate for girls in primary school was 72% and only 59% of girls completed schooling in rural areas. This further declined at the secondary schooling and higher education level with only 31% and 2% of girls enrolling respectively.⁵⁷⁷ Women face access constraints due to inadequate financial resources available for their education, as well as early marriage. Within schools, poor infrastructure, gender-based violence, and sanitation further exacerbate these constraints for girls. To promote equality and gender parity in education, the country has developed a National Strategy for Accelerating Girls and Women's Education and Training (SNAEFFE). Under this policy, through presidential decrees, the government in Niger has been working towards setting up literacy and 'second chance' education programs for girls who have dropped out of schools by providing assistance to parents, food rations, scholarships, and deliberation of a decentralization policy for secondary education so that community schools can be set up and attended by girls and women.⁵⁷⁸ Considering early marriage and early pregnancy often leads to girls dropping out of schools, integrating women back into the schooling system is a key challenge being faced, with these girls often ostracised by their communities for falling pregnant. In 2019, the government of Niger adopted an order directing schools to allow married and/or pregnant girls to return back to schools, continue their studies, and return after they have delivered/have given birth.⁵⁷⁹

For promoting vocational and technical education, the country has developed a Gender Strategy in Vocational and Technical Education and established 103 community-level training centers and 11

⁵⁷¹ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁷² [UNDP \(2022\) Gender Inequality Index](#)

⁵⁷³ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁷⁴ [WHO \(n.d\). Situational analysis to scale up and sustain postpartum family planning services in Niger Retrieved 1 March, 2023, from https://cdn.who.int/media/docs/default-source/reproductive-health/contraception-family-planning/niger-fp.pdf?sfvrsn=a4ddb73_4&download=true](https://cdn.who.int/media/docs/default-source/reproductive-health/contraception-family-planning/niger-fp.pdf?sfvrsn=a4ddb73_4&download=true)

⁵⁷⁵ [United Nations Population Fund, World Population Dashboard, Niger](#)

⁵⁷⁶ [World Bank, 2021](#)

⁵⁷⁷ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁷⁸ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁷⁹ [Machizi, K. K., Dijbo, M. M. \(2022, August 30\). Interview: New Niger Order Protects Girls' Rights to Education. Human Rights Watch . Retrieved March 1, 2023, from https://www.hrw.org/news/2022/08/30/interview-new-niger-order-protects-girls-rights-education.](https://www.hrw.org/news/2022/08/30/interview-new-niger-order-protects-girls-rights-education)

vocational and technical training centers, and 8 high schools. Further, for adolescent girls a 6-month vocational training program is being piloted.⁵⁸⁰

Women's economic participation and entrepreneurship

Female labor force participation rate (LFPR) in Niger stands at 61.73%, and the country has a gender disparity of 22.5 percentage points between men and women, with LFPR of 84.27% among men.⁵⁸¹ The high fertility rate, constricting societal norms, and women's roles in managing their households are some reasons they are unable to participate effectively in the labor force. As per the country's Employment survey from 2016, women work an average of 28 hours a week compared to men who work 43 hours. Agriculture is the dominant sector, and most women work as informal agricultural workers. World Bank estimates that about 91% of women are in vulnerable employment as of 2017.⁵⁸² However, there is a wide disparity in agricultural productivity of plots managed by women compared to men, with 19% less production per hectare, which is due to land ownership issues, limited access to and issues in supervising hired farm labour, and the quantity and quality of fertilizer used.⁵⁸³ In formal employment, public officers have a lower representation of women civil servants, especially as senior staff, while at the junior level, there are 73% of women in the Ministry of Public Service.⁵⁸⁴ A high proportion of women youth (77.1%) are also 'idle' and not in education, employment, or training.⁵⁸⁵ Within wage employment, there is an estimated 29% gender gap in earnings between men and women in similar roles, due to disparity in educational attainment and concentration of women in low-productivity sectors.

Entrepreneurship in the informal sector is a crucial phenomenon among women in Niger, and 23% of women aged 16-64 own an enterprise in the country. However, there is a vast gender gap of 61% in profits accrued by women-owned businesses versus men-owned. This is due to factors such as the limited availability of labor to help run the enterprise, literacy, and numeracy gaps, and informality of enterprise registration. Textile manufacturing followed by wholesale retail trade and motor vehicle repairs are two main sectors with the high disparity in profits, 82% and 68% respectively.⁵⁸⁶

Gender and energy in Niger

Currently, there are only 16.6% of households in Niger with access to electricity. The main source of fuel in the country is wood. This causes a lot of time burden on women who are primarily the ones responsible for firewood collection. Butane gas is another alternative source of energy in the country, however, the use is expensive for poorer Nigeriens.⁵⁸⁷ In urban 38.2% of women-headed households have access to grid

⁵⁸⁰ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁸¹ [World Economic Forum \(2022\) Global Gender Gap Report](https://www.weforum.org/publications/global-gender-gap-report)

⁵⁸² [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁸³ [McGee, K., & Yetna, P. B. \(2015\). \(working paper\). Gender Differentials and Agricultural Productivity in Niger. The World Bank. Retrieved March 1, 2023, from https://documents1.worldbank.org/curated/en/159921468289531733/pdf/Gender-differentials-and-agricultural-productivity-in-Niger.pdf](https://documents1.worldbank.org/curated/en/159921468289531733/pdf/Gender-differentials-and-agricultural-productivity-in-Niger.pdf)

⁵⁸⁴ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁸⁵ [The World Bank. Niger Gender Landscape](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

⁵⁸⁶ [World Bank Group . \(2019\). \(rep.\). Economic Impacts Of Gender Inequality In Niger. Retrieved March 1, 2023, from https://documents1.worldbank.org/curated/en/541361576613122122/pdf/Economic-Impacts-of-Gender-Inequality-in-Niger.pdf](https://documents1.worldbank.org/curated/en/541361576613122122/pdf/Economic-Impacts-of-Gender-Inequality-in-Niger.pdf)

⁵⁸⁷ [African Development Bank Group \(2020\). Niger - Country Gender Profile - 2020. Retrieved 1 March, 2023, from https://www.afdb.org/en/documents/niger-country-gender-profile-2020](https://www.afdb.org/en/documents/niger-country-gender-profile-2020)

electricity compared to 46.1% of male-headed households. Women-headed households in both rural and urban areas have lower access to off-grid solutions.⁵⁸⁸

⁵⁸⁸ [Brutinel, M., Nabourema, F., Angelou, N., Koo, B. B., Portale, E., & Rysankova, D. \(2020\). \(rep.\). Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework Niger. World Bank Group . Retrieved March 1, 2023, from https://documents1.worldbank.org/curated/en/929061603687020211/pdf/Niger-Beyond-Connections-Energy-Access-Diagnostic-Report-Based-on-the-Multi-Tier-Framework.pdf.](https://documents1.worldbank.org/curated/en/929061603687020211/pdf/Niger-Beyond-Connections-Energy-Access-Diagnostic-Report-Based-on-the-Multi-Tier-Framework.pdf)

Chad

Overview of gender equality and gender gaps in the country

Chad is a Central African country, with a population of around 17 million people – 49.8% of which are women – an annual population growth of 3.2%, with 48% of the total population under the age of 15.

The Gender Development Index for Chad is 0.77. According to the GDI scoring, Chad belongs to the group of countries with the lowest human development (countries with low equality in HDI achievements between women and men and absolute deviation from gender parity of more than 10 percent). On the Gender Inequality Index (GII) which measures disparities between men and women in the areas of economic opportunities, reproductive health, and empowerment, the country is ranked among the highest in the world (165).

Chad also scores low on human development, with women being in worse conditions compared to men (0.339 HDI for women versus men 0.441), and Chad ranks 190th out of 191 countries on the Human Development Index (2019).

Indicator	Result	Source and date
Life expectancy at birth	54.3	UNDP (2022) Gender Development Index
Maternal mortality ratio	1,140	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	18	World Bank, 2016
Women's Labour Force Participation Rate (age 15 to 64)	64%	World Bank, 2019
Mean years of schooling ⁵⁸⁹	1.5	UNDP (2022) Gender Development Index
Estimated gross income per capita	965 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

The Republic of Chad is a member of the United Nations and the African Union. It has ratified a number of UN Human Rights Conventions and thus has made binding international commitments to adhere to the standards laid down in these universal human rights documents. All inhabitants of Chad may turn to the UN Human Rights Committee through procedure 1503, to the Special Rapporteurs for violations of specific human rights or to ECOSOC for women's rights violations.⁵⁹⁰ Women in Chad also have full voting rights. However, despite improvements over the years in the legal framework, there are still important challenges in reflecting women's legal rights in practice. Chad has a strong constitutional framework for equality which

⁵⁸⁹ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

⁵⁹⁰ [ClaimingHumanRights.org Claiming Human Rights - in Chad](#)

includes the State's obligation to eliminate all forms of discrimination against women, and has made some progress towards achieving women's right to equality. For example, Chad has taken the necessary legislative measures to ban female genital mutilation and child marriage.⁵⁹¹ Unfortunately, enforcement of these laws is inadequate. Women and girls continue to face the prospect of lives of gender-based violence.

There are signs of positive steps. In 2022, the Ministry of National Education and Civic Promotion (MENPC) of the Republic of Chad, a member of the Gender at the Center Initiative (GCI), sought international support to provide technical capacity-building towards integration of gender equality issues into education policies. In addition, the Government of Chad is preparing its Partnership Pact, a sector-wide document that will articulate a priority gender-sensitive reform that a handful of donors and non-governmental organizations will support over the next five years.⁵⁹²

Political participation

In Chad, women represent more than 50% of the total population and are potentially a decisive electorate. A law on representative parity exists, though there is currently less than a 50% representation of women in seats of national and local governments.⁵⁹³ As in other countries in the continent, gender norms impact women's political participation, and women's groups within the country have called on the government to guarantee parity in nominative and elective functions in the Republic of Chad and at regional level.⁵⁹⁴ Despite the challenges, 32.3% of seats in Chadian Parliament are held by women, which is higher than the sub-Saharan Africa average of 25.7%.⁵⁹⁵

Social and cultural context

The social and cultural context of women is shaped by factors such as religion, traditional beliefs, and societal norms. Despite legal protections, women in Chad often face discrimination in various domains, including education, employment, and inheritance rights. Harmful practices like child marriage and female genital mutilation/cutting (FGM/C) persist, particularly in rural areas, despite being prohibited by law. Religion, predominantly Islam, significantly influences the social and cultural context of women's rights in Chad. Conservative interpretations and cultural traditions may curtail women's autonomy in areas such as marriage, divorce, and inheritance. In addition, Chad is a country where land ownership is often governed by customary laws and practices that may vary among different ethnic groups and regions, and women's access to and ownership of land can be influenced by social, cultural, and legal factors.

Traditional gender roles and norms also play a significant role, with women often expected to conform to traditional roles as wives, mothers, and caregivers, constraining their opportunities for education and employment. Women in Chad have made progress in advocating for their rights, with active women's rights organizations and civil society groups working towards gender equality and women's empowerment. The

⁵⁹¹ UN OHCHR Chad (2017): UN experts say poverty and culture cannot justify discrimination against women <https://www.ohchr.org/en/press-releases/2017/12/chad-un-experts-say-poverty-and-culture-cannot-justify-discrimination>

⁵⁹² UNESCO Office for Africa (2022) <https://dakar.iiep.unesco.org/en/news/chad-commits-mainstreaming-gender-equality-public-education-policies>

⁵⁹³ Africa News (2021) Chad: Women hold decisive vote in upcoming presidential election <https://www.africanews.com/2021/04/09/chad-women-hold-decisive-vote-in-upcoming-presidential-election/>

⁵⁹⁴ International Federation for Human Rights (2022) <https://www.fidh.org/en/region/Africa/chad/chad-encourage-women-s-contribution-to-the-political-transition>

⁵⁹⁵ [UNDP Gender Inequality Index, 2022](#)

Chadian government has also taken steps to promote gender equality through policy reforms and legal protections, although challenges persist in implementation and enforcement.

Health

A high maternal mortality rate is a major health issue in Chad. 1,140 women die per 100,000 live births, which is more than double the sub-Saharan African average, although it has improved from 1,420 in 2000.⁵⁹⁶ The high maternal mortality rate is caused by several factors such as lack of access to medical treatment, early marriage, and high birth rates. Chad also has a high adolescent fertility rate of 138.3 births per 1,000 women aged 15-19. This is higher than the sub-Saharan average of 100.9.⁵⁹⁷

Throughout the country, the average age at first birth is 18.1 years. Furthermore, 24% of women aged 20-24 were married before their fifteenth birthday, and 60% of women aged 20-24 were married before their eighteenth birthday,⁵⁹⁸ compared with 8% of men in the same age group.

Chadian women suffer from poorer health compared to men, with only 31% of pregnant women receiving adequate prenatal healthcare,⁵⁹⁹ and 34% of women being subjected to female genital mutilation.⁶⁰⁰ Moreover, women of reproductive age (15-49 years) often face barriers with respect to their sexual and reproductive health and rights: despite progress, in 2019, only 17.5% of women had their need for family planning satisfied with modern methods,⁶⁰¹ and 30% of married women had an unmet need for contraception.⁶⁰² Furthermore, severe legal restrictions on abortion lead women to have abortions performed clandestinely, usually in unhealthy conditions.⁶⁰³

Education

The education situation for girls in Chad faces numerous challenges, including limited access to quality education, low enrollment rates, high dropout rates, early marriage and pregnancy, gender-based violence, cultural and social norms, and poverty. These challenges often result in lower levels of educational attainment for girls compared to boys, which can have long-term consequences for their empowerment, well-being, and economic opportunities.

The average Chadian woman will benefit from 1.5 years of formal schooling compared to 3.7 years for men.⁶⁰⁴ Furthermore, only 8% of women have attended secondary school compared to 24% of men.⁶⁰⁵ Adult literacy in Chad is also lower among women than among men, with 18.2% of women being literate and 35.4% of men being literate. The gap in adult literacy between men and women is 17.2, which is larger than the gap of the sub-Saharan Africa aggregate of 13.⁶⁰⁶

⁵⁹⁶ [UNDP Gender Inequality Index, 2022](#)

⁵⁹⁷ [UNDP Gender Inequality Index, 2022](#)

⁵⁹⁸ [World Bank Gender Data Portal Chad Profile, n.d.](#)

⁵⁹⁹ [World Bank Chad Gender Landscape, 2021](#)

⁶⁰⁰ [28 Too Many Chad Profile, n.d.](#)

⁶⁰¹ [UN Women Chad Country Profile, n.d.](#)

⁶⁰² [World Bank Chad Gender Landscape, 2021](#)

⁶⁰³ [OECD Dev Social Institutions & Gender Index, 2019](#)

⁶⁰⁴ [UNDP Gender Development Index, 2022](#)

⁶⁰⁵ [UNDP Gender Inequality Index, 2022](#)

⁶⁰⁶ [World Bank Gender Data Portal Chad Profile, n.d.](#)

Traditionally, education for women and girls is less valued and households often prioritize education for boys compared to girls. However, the Government of Chad has committed to mainstreaming gender equality in its public education policies, which is an encouraging step in promoting education for girls and women in Chad.⁶⁰⁷

Women's economic participation and entrepreneurship

The labor force participation rate among females is 48.4% and among males is 70.8%, giving Chad a higher labor force participation gap than the rest of sub-Saharan Africa. Factors such as early marriage, early pregnancy, cultural norms that prioritize male education, and lack of access to quality education and vocational training may hinder women's ability to enter the workforce. Furthermore, social norms and traditional gender roles in Chad often dictate that women primarily engage in household and caregiving responsibilities, while men are expected to be the breadwinners. This can limit women's ability to participate in economic activities outside of their homes, and women may face discrimination, biases, and stereotypes in the workplace or as entrepreneurs.

In Chad, 75% of employed people work in agriculture,⁶⁰⁸ which includes subsistence farming, fishing, and animal husbandry. Women play a critical role in food production and contribute to the rural economy by cultivating crops, raising livestock, and engaging in other agricultural activities to sustain their households and communities. They also participate in the informal sector, which includes jobs in informal markets, petty trade, and small-scale enterprises. Despite their important contribution, 98.8% of female workers are in vulnerable employment, compared to 86.6% of men, both of which are higher than the sub-Saharan African average. These are workers who are the least likely to have formal work arrangements, social protection, and safety nets to guard against economic shocks; thus they are more likely to fall into poverty, posing challenges to women's economic well-being and empowerment.⁶⁰⁹ As such, 41% of the employed female population lives below the poverty line compared to 39% of men.⁶¹⁰

There are also gender gaps in the formal sector. It is estimated that only 1% of female workers are salaried workers, compared to 13% of male workers,⁶¹¹ and women constitute less than 18% of professional and technical workers.⁶¹² There is also an entrepreneurship gap, with only 9% of firms having majority female ownership and only 12% of firms with female top managers.⁶¹³ Furthermore, female entrepreneurs may face challenges such as limited access to finance, markets, and business development services due to gender-based discrimination.

The principle of non-discrimination on the ground of sex, age or nationality prevails within Chad's Labour Code (1996) with regards to employment, notably recruitment, hiring, terms and conditions, promotions, training, assignments and termination. The Labour Code also guarantees paid maternity leave of 14 weeks, during which mothers are entitled to the totality of their wages, with a further 6 months of unpaid parental leave. However, women are mostly involved in the informal sector and precarious jobs and thus remain excluded from the legal protection of the formal sector. Furthermore, even women within the formal sector face gender bias. The Labour Code prohibits women from entering certain professions which could

⁶⁰⁷ [UNESCO Chad commits to mainstreaming gender equality in public education policies, 2022](#)

⁶⁰⁸ [World Bank Data Bank, n.d.](#)

⁶⁰⁹ [World Bank Gender Data Portal Chad Profile, n.d.](#)

⁶¹⁰ [UN Women Chad Country Profile, n.d.](#)

⁶¹¹ [World Bank Chad Gender Landscape, 2021](#)

⁶¹² [World Economic Forum Global Gender Gap Report, 2022](#)

⁶¹³ [World Economic Forum Global Gender Gap Report, 2022](#)

endanger their health, safety or morality, and the Civil Code stipulates that a woman can choose a profession on the condition that her husband agrees.⁶¹⁴

Women's economic empowerment is also threatened through challenges in accessing land, financial services, credit, and agricultural extension services, which can limit their productivity and income generation potential. Chad's Civil Code stipulates that a woman can open a bank account at a formal institution for the funds given by her husband. However, the husband must be notified and must authorize withdrawals from the account.⁶¹⁵ Women's access to credit is also limited due to discrimination and stereotypes, their lack of collateral and lack of information regarding credit. Additionally, it is reported that in some parts of Chad, women must seek the approval of their husband if they wish to take a bank loan to open a business. Even though Law on the Limitation of Land Rights guarantees the right to everyone to own, use, make decisions and use as collateral, customary law and practices often prioritize male ownership and inheritance of land. Most women don't own land and usually work on the land of their father or husband, and they generate income from those lands, it is usually given back to their father or husband.⁶¹⁶

Gender and energy in Chad

Only 11% of the population have access to electricity in Chad. Women in Chad, particularly those in rural areas, often face additional challenges in accessing energy services due to factors such as poverty, social norms, lack of infrastructure, and limited access to financial resources. These challenges can disproportionately impact women's ability to access and benefit from modern energy services, such as electricity for lighting, cooking, and productive uses. Sex-disaggregated data on access to energy as well as women's participation in the sector is absent.

In 2022, the World Bank approved the \$295M Chad Energy Access Scale Up Project (PAAET) which aims to increase access to electricity and clean cooking solutions via expansion of the main power grid and mini-grids, standalone solar systems, deployment of improved stoves, and natural resource management.⁶¹⁷ The project will include targeted interventions to improve access to electricity for female-headed households, including in refugee camps, and female-led enterprises, promote women entrepreneurship, and build awareness relevant to women and girls around energy services.⁶¹⁸

⁶¹⁴ [OECD Dev Social Institutions & Gender Index, 2019](#)

⁶¹⁵ [OECD Dev Social Institutions & Gender Index, 2019](#)

⁶¹⁶ [OECD Dev Social Institutions & Gender Index, 2019](#)

⁶¹⁷ [World Bank Chad Scales Up Its Access to Energy, 2022](#)

⁶¹⁸ [World Bank Project Information Document, 2020](#)

Somalia

Overview of gender equality and gender gaps in the country

Somalia is an East African country, with a population of around 17 million people – 49.9% of which are women – an annual population growth of 3.1%, with 47% of the total population under the age of 15.⁶¹⁹

There is a scarcity of socioeconomic data in Somalia, including on gender, because of years of political instability and civil war. This has led to lack of institutions and systems needed to collect, analyze, and disseminate reliable data, limited capacity of the Government to undertake comprehensive data collection, an inability for researchers and organizations to access certain regions for data collection, and significant displacement and migration of Somali people which complicates data collection and tracking of social and economic indicators. As such, Somalia does not have a Human Development Index, Gender Inequality Index nor a Gender Development Index.⁶²⁰

It's likely that Somalia would belong to the group with the lowest human development group for all three indices, and that gender inequality and deviation from gender parity is high.

Indicator	Result	Source and date
Life expectancy at birth	57.4	UNDP (2022) Gender Development Index
Maternal mortality ratio	829	UNDP (2022) Gender Inequality Index
Literacy rate (female aged 15 and above)	n.a	World Bank
Women's Labour Force Participation Rate (age 15 to 64)	23%	World Bank, 2019
Mean years of schooling ⁶²¹	n.a	UNDP (2022) Gender Development Index
Estimated gross income per capita	545 (PPP \$)	UNDP (2022) Gender Development Index

Legal status of women

Women in Somalia face difficulties in accessing justice due to cultural restrictions, differing interpretations of judicial regimes, and the absence of women within the judicial system. As of 2021, there are only two female judges out of a total of 295 and 14 female prosecutors out of a total of 65, highlighting the need for gender parity in the judicial system.⁶²²

⁶¹⁹ [World Bank Data Bank, 2021](#)

⁶²⁰ [FAO National gender profile of agriculture and rural livelihoods – Somalia, 2021](#)

⁶²¹ Average number of years of education received by people ages 25 and older, converted from educational attainment levels using official durations of each level.

⁶²² [UNWomen.Org - Where we are Eastern & Southern Africa \(2022\)](#)

Somalia has a Ministry of Women and Human Rights Development, mandated to advance the promotion and protection of gender equality and human rights, including the rights of women, children and other vulnerable groups. Increased awareness of the rights of women and girls has been driven by this ministry alongside international rights groups. The Federal Government of Somalia launched the Somali National Action Plan (NAP) in September 2022.⁶²³ Its purpose is to lay out and guide implementation of the Somali Women's Charter and UN Security Council Resolution 1325 (UNSCR 1325). The inclusion and participation of Somali women in peacebuilding and decision-making at all levels is the primary goal of the NAP, where these ideals are recognized as a crucial foundation for effective governance, long-term peace, and development.

Despite progress in the formal recognition of women's legal status, the United Nations has reported that cultural factors and customary law can be more accepting of inequities or victimization of women and girls.⁶²⁴ For example, while no laws limit the participation of women or members of historically marginalized groups in the political process (and they do participate), cultural factors have limited their participation.⁶²⁵

There is some hope the National Action Plan will be able to have a broad impact on women's lived experience in Somalia, including their legal standing. Conflict-related sexual and gender-based violence, access to justice, inclusion in constitutional review processes, state building and peacebuilding, national reconciliation and transitional justice mechanisms, prevention of violent extremism, as well as climate change, humanitarian emergencies, and COVID-19, are all identified as relevant themes/action areas for women's wellbeing and equity, and are to be addressed by the national action plan.⁶²⁶

Political participation

In Somalia, women account for 50.15% of the population.⁶²⁷ In 2016, Somalia reached a milestone with nearly a quarter of parliamentary seats occupied by women. In January 2021, a quota was announced for a third of parliamentary seats to be reserved for female lawmakers. There has since been limited progress in female representation in the national legislative bodies - the 329 seats in Somalia's lower and upper houses of parliament - despite the quota. After the parliamentary elections in 2021-2022, women's representation fell to 20% in the House of the People, which is lower than the 2016 figure of 24%.⁶²⁸ There were some gains made with 26% seats for women in the Upper House in 11th Parliament, in comparison to 24% in 10th Parliament. Given the clan system at the foundation of Somalia's politics, which is male-dominated, Somali women largely remain excluded from key political and decision-making processes.⁶²⁹

Social and cultural context

Traditional customs, religious doctrines and societal expectations are among the various factors that contribute to the social and cultural environment that Somali women experience. Women and girls across the country face significant challenges, as gender inequality is pervasive. Somalia ranks as the fourth-lowest country in the world for gender equality, with high rates of maternal and infant mortality, as well as

⁶²³ [UNWomen.Org Somalia Launches National Action Plan \(2022\)](https://africa.unwomen.org/en/stories/news/2023/01/somalia-launches-national-action-plan-on-unscr-13-for-women-and-security)<https://africa.unwomen.org/en/stories/news/2023/01/somalia-launches-national-action-plan-on-unscr-13-for-women-and-security>

⁶²⁴ [Somalia Human Rights Report \(2022\)](#)

⁶²⁵ [Somalia Human Rights Report \(2022\)](#)

⁶²⁶ [UNWomen.Org Somalia Launches National Action Plan \(2022\)](#)

⁶²⁷ [Electoral Institute for Sustainable Democracy in Africa - Somalia Situation Review \(2022\)](#)

⁶²⁸ [UNWomen.Org - Where we are Eastern & Southern Africa \(2022\)](#)

⁶²⁹ [UNWomen.Org - Where we are Eastern & Southern Africa \(2022\)](#)

prevalence of early marriage.⁶³⁰ Stereotypes about women in the workplace, clan-based identity politics that center men, and deep-rooted gender-based typecasts are among the factors behind gender inequity and injustice. For example, barriers to education systematically exclude women from attaining skills, which has implications for female participation in the economy. Certain social norms negatively impact women: for example, gender based violence through female genital mutilation, which can have serious physiological and psychological impacts throughout a woman's life, and the expectation to marry early.⁶³¹

Despite the introduction of the Sexual Offences Bill (SOB) in 2018, which would provide greater protection for victims of sexual violence and help end the culture of impunity relating to GBV, women in Somalia remain under-represented within and excluded from key political and decision-making processes. Efforts from government bodies such as the Ministry of Women and Human Rights Development display some awareness of the inequities and challenges existing in the sociocultural context. However, amid ongoing fragility in terms of the security situation within Somalia, with numerous attacks carried out by militant groups in both urban and rural areas, and with GBV being a feature of this conflict,⁶³² there are even greater challenges for the government to adequately provide for the security, protection, social, environmental, and economic needs of the female population.

Health

A high maternal mortality rate is a major health issue in Somalia. 829 women die per 100,000 live births, which is higher than the sub-Saharan African average.⁶³³ The high maternal mortality rate is caused by several factors such as limited access to healthcare, poor healthcare infrastructure, socio-cultural factors like early childbearing, and ongoing conflict.

Somali women suffer from worse health outcomes than men and face significant barriers to reproductive health, with only 24% of pregnant women receiving adequate prenatal healthcare, and 37% of married women with unmet need for contraception.⁶³⁴ Furthermore, 99% of women in girls and Somalia have undergone female genital mutilation,⁶³⁵ which can lead to significant health issues throughout life, including complications in childbirth which can lead to maternal mortality. What's more, abortion in Somalia has been criminalized, except to save the life of the mother,⁶³⁶ and a cesarean is performed only if the husband and/or the mother-in-law give permission.⁶³⁷ Women are more malnourished than men and are less food secure.⁶³⁸

Education

The education situation for girls in Somalia is characterized by various challenges and disparities, such as long distances to school, safety concerns, and lack of female teachers. Although there have been some improvements in recent years, girls' education in the country continues to face obstacles. Socio-cultural norms in Somalia often favor boys' education over girls', leading to gender disparities in access to

⁶³⁰ [UNWomen.Org - Where we are Eastern & Southern Africa \(2022\)](#)

⁶³¹ [World Bank Blog \(2022\)](#)

⁶³² [Somalia Human Rights Report \(2022\)](#)

⁶³³ [UNDP Gender Inequality Index, 2022](#)

⁶³⁴ [World Bank Somalia Gender Landscape, 2021](#)

⁶³⁵ [UNICEF FGM Somalia, 2021](#)

⁶³⁶ [OECD Dev Social Institutions and Gender Index, 2019](#)

⁶³⁷ [FAO National gender profile of agriculture and rural livelihoods – Somalia, 2021](#)

⁶³⁸ Ibid.

education. The deeply rooted cultural and religious beliefs, coupled with gender stereotypes, contribute to the marginalization of girls in the education system. There is limited data on school enrolment and attendance in Somalia, although it's estimated that more than three million children in Somalia are out of school.⁶³⁹ There is also limited data on literacy rates in Somalia, although they are likely to be particularly low amongst rural women.⁶⁴⁰

Early marriage and childbearing also prevent girls in Somalia from attaining an equal education to boys. Somalia has a high adolescent fertility rate of 118.0 births per 1,000 women aged 15-19.⁶⁴¹ This is also higher than the sub-Saharan average of 100.9, although the rate of adolescent fertility has decreased since 2010. Furthermore, 16.7% of women were married before the age of 15,⁶⁴² and 35.% were married before the age of 18.⁶⁴³

Women's economic participation and entrepreneurship

The labor force participation rate among females is 21.2% and among males is 46.9%, giving Somalia a higher labor force participation gap than the rest of sub-Saharan Africa.⁶⁴⁴ An overwhelming majority of the women are self-employed and work in agriculture-related activities. Very few women take up economic activity that records high profits through exports and imports. For example, in the livestock and fishing export industries, women are hardly represented. Female farmers also have less access to agricultural resources, capital and information than their male counterparts.⁶⁴⁵

96.5% of women work in vulnerable employment in Somalia compared to 88.8% of men, both of which are higher than the sub-Saharan African average.⁶⁴⁶ These are workers who are the least likely to have formal work arrangements, social protection, and safety nets to guard against economic shocks; thus they are more likely to fall into poverty, posing challenges to women's economic well-being and empowerment.

There are also gender gaps in the formal sector. Only 3% of women in the labor force are salaried workers, compared with 10% of men.⁶⁴⁷ Furthermore, women represented only 27.7% of those employed in senior and middle management in 2019, which falls in the second quintile of all countries for which there are data.⁶⁴⁸

Women are also less likely to be able to access capital. In 2014, 33.7% of women and 43.6% of men in Somalia had an account with a financial institution or with a mobile-money-service provider, both of which are lower than the sub-Saharan average.⁶⁴⁹ The few institutions that provide financial services have demanding conditions and procedures to access funding, such as strict requirements for collateral securities (for example, land title), which few rural women own as compared to men.⁶⁵⁰

⁶³⁹ [UNICEF Somalia Education, n.d.](#)

⁶⁴⁰ [FAO National gender profile of agriculture and rural livelihoods – Somalia, 2021](#)

⁶⁴¹ [UNDP Gender Inequality Index, 2022](#)

⁶⁴² [World Bank Gender Data Portal, 2020](#)

⁶⁴³ [UN Women Somalia Fact Sheet, n.d.](#)

⁶⁴⁴ [World Bank Gender Data Portal Somalia Profile, n.d.](#)

⁶⁴⁵ [FAO National gender profile of agriculture and rural livelihoods – Somalia, 2021](#)

⁶⁴⁶ [World Bank Gender Data Portal Somalia Profile, n.d.](#)

⁶⁴⁷ [World Bank Somalia Gender Landscape, 2021](#)

⁶⁴⁸ [World Bank Gender Data Portal Somalia Profile, n.d.](#)

⁶⁴⁹ [World Bank Gender Data Portal Somalia Profile, n.d.](#)

⁶⁵⁰ [FAO National gender profile of agriculture and rural livelihoods – Somalia, 2021](#)

Gender and energy in Somalia

Energy access in Somalia is limited, particularly in rural areas, and women are disproportionately affected by the lack of reliable and affordable energy sources. National energy access in Somalia is 18% and less than 5% in rural areas.⁶⁵¹ The lack of electricity affects women in various ways, such as limiting their access to information, communication, and education, as well as constraining their economic opportunities. Sex-disaggregated data on access to energy as well as women's participation in the sector is absent.

Several programs and initiatives, led by international organizations, non-governmental organizations, and the Somali government, aim to improve women's access to energy in Somalia. These initiatives focus on promoting renewable energy technologies, building capacity, and fostering gender equality in the energy sector. For instance, the Sustainable Charcoal Reduction and Alternative Livelihoods (SCREAL) project aims to reduce dependency on charcoal by promoting sustainable energy alternatives, including fuel-efficient stoves, and creating alternative livelihood opportunities, particularly for women and youth.⁶⁵² The Africa Minigrids Program (AMP) aims to bring new development opportunities to rural communities and to increase access to electricity for 67K people, half of whom will be women.⁶⁵³

⁶⁵¹ [IEA WEO, 2020](#)

⁶⁵² [UNDP Somalia Projects, n.d.](#)

⁶⁵³ [UNDP Sustainable Energy Hub, 2023](#)

Conclusions and recommendation for the Gender Action Plan

This assessment provides an overview of the status of gender equality in the sixteen H2R countries. The H2R initiative has the opportunity to address some of the underlying causes of gender inequality in the countries by enhancing access for women to clean energy technologies and job opportunities in the sector. In particular, the following objectives are relevant:

H2R's Gender Objectives:

1. Elevate women in the OGS sector as customers, employees and entrepreneurs
2. Do no harm from a gender perspective
3. Provide poorest and most vulnerable women and Women Headed Households with clean energy
4. Invest in women-led businesses and businesses with gender-forward business practices
5. Positively impact organizational culture for women employees through the influencing portfolio companies to adopt and/or expand gender business practices

The H2R Gender Assessment informs the development of the Gender Action Plan through the identification of key barriers and entry points for the H2R initiative. In particular, the country-level assessments identified barriers for engaging women as employees and leaders in these contexts, with prevalent negative gender norms, incomplete education attainment, and limited access and control of assets influencing women's participation in the economy and in productive activities. Elevating women as customers, employees, and business leaders in the OGS will require a comprehensive approach both within the investment process and in the allocation of capital.

H2R's Gender Action Plan will achieve its objectives through 4 components. The first is to apply a **gender lens across its investment processes** to mitigate gender biases and identify hidden opportunities. This involves integrating gender considerations at each step of the decision-making process to produce a portfolio of investments positioned to meet H2R's gender objectives. Second, H2R will offer **post-investment gender-focused TA to investees** to incentivize companies to adopt, expand, and/or implement gender-forward business practices. Post-investment TA support will focus on the development and implementation of company Gender Action Plans (GAP) and on providing training to level set any gaps in gender awareness at the investee level. Third, H2R will work to **build opportunities for women in the broader ecosystem** through a dedicated training and employment facility to drive employment access and readiness in OGS markets. Finally, H2R will catalyze OGS ecosystems in H2R markets by crowding in a variety of stakeholders and sharing lessons. H2R will **invest in insights, data aggregation, and OGS ecosystem development** to build a robust community of actors working together towards common goals.

These components are further detailed in the GAP.

H2R Gender Action Plan

Impact Statement: If the appropriate investment capital and TA is provided to OGS companies, then they will be able to deliver clean energy access to women customers and women-headed households and to increase diversity in their workforce, increasing companies' growth opportunities and contributing to women economic empowerment because they will be adequately incentivized and capacitated to operate in 'Hardest-to-Reach' (H2R) contexts.

Outcome statement:

- Access to clean and decentralized energy to 34,047,623 women (47% of 72,441,750 total beneficiaries)
- Sustainable and inclusive growth of OGS businesses
- Reduction in 5.1 m tCO₂e to preserve and improve women's health
- Increased resilience of women and girls via improved health, safety and wellbeing
- Reduced vulnerability of women

Output statement:

- Funding and technical assistance lead to increased capacity of OGS companies to serve low-income women customers and to hire and retain women employees, as well as creating gender inclusive workplaces.

H2R's Gender Objectives:

1. Elevate women in the OGS sector as customers, employees and entrepreneurs
2. Do no harm from a gender perspective
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4. Invest in women-led businesses and businesses with gender-forward business practices
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Activities	Baseline	Targets	Indicators	Timeline	Responsibility	Indicative budget
1. Apply a gender lens to its investment decision-making						
1.1 Identify women-led or women-owned businesses eligible for funding ¹	n/a	Track number (annually)	# of women owned/led companies investees	Year 1-10	Gender expert at H2R (Acumen)	Staff time
1.2 Perform assessments of how companies are currently operating (before investment) that serves as a baseline for monitoring process in gender equity.	n/a	100% of company beneficiaries (annually)	# of gender inclusion self-assessment surveys	Year 1-10	Implementation supported by gender consultants, Nexus Platform	\$150,000
1.3 Identify companies that are 2X Challenge compliant and support portfolio growth in this area	n/a	30% of company beneficiaries (final)	# of companies that are 2X Challenge compliant	Year 1-10	Implementation supported by gender consultants, with the oversight of Gender Expert at H2R	Staff time
1.4 Require companies to develop a Gender Action Plan within 90 days of signing loan agreements	n/a	100% of company beneficiaries (annually)	# of GAPs developed	Year 1-10	Deployed by investee companies Reviewed by Gender Expert at H2R	USD 8,000 for each company USD 456,000 for 57 companies (TA budget) ²
1.5 Require companies to collect and report sex-disaggregated customer data	To be defined at due	100% of companies	# of companies that report sex-	Year 1-10	Deployed by investee companies Reviewed by Gender	USD 190,000 (TA budget)

¹ Women-owned: 51% of more owned by woman/women; women-led: 51% of women in leadership positions (C-suite)

² For assumptions used to calculate the TA budget and breakdown by activities, please see Gender Lens Investing Strategy.

	diligence phase	beneficiaries (annually)	disaggregated sales data		Expert at H2R	
2. Offer post-transaction gender-focused technical assistance to investees						
2.1 Support the development and implementation of gender inclusive business practices by offering gender TA support to investees to meet minimum requirements and to implement their GAP	n/a	57 companies (final)	# of companies participating in TA activities	Year 1-10	Implementation by gender consultants with the oversight of Gender Expert at H2R	USD 456,000 (TA budget)
2.2 Provide job opportunities for women and increase diversity in the workforce across all levels of investee companies	33% women employees (global benchmark) ³	30% women employees across different levels of companies' workforce (to be confirmed once companies baseline assessments are done) 33% of jobs	% of women employees in investee companies # of jobs created for women	Year 1-10	Deployed by investee companies. Reviewed by gender expert at H2R Acumen	

³ Please refer to the gender assessment for sources used to calculate the benchmark

		created for women (final)				
3. Support dedicated training and employment opportunities for women in off-grid solar						
3.1 Strengthening women's participation in OGS value chains through dedicated training for technicians and assemblers of OGS products	0	200 women technicians trained (final)	# of women that participates in training activities	Year 2-10	Implemented by partner organisation, with the oversight of Gender Expert at H2R	USD 200,000
3.2 Support companies to identify, recruit and train women in commission-based roles through partnerships with community-based organisations	0	1-2 companies	# of participating companies		Implemented by partner organisation, with the oversight of Gender Expert at H2R	USD 100,000
3.3 Support towards hiring women as change agents/ambassadors for awareness generation in the communities	0	75 women ambassadors trained (final)	# of women trained	Year 2-10	Implemented by partner organisation, with the oversight of Gender Expert at H2R	USD 100,000
3.4 Showcase opportunities for employment by supporting training institutions and companies to create opportunities for young women (e.g. internships, leadership mentoring, open days for STEM students)	0	3 (final)	# of companies participating in showcasing training opportunities	Year 2-10	Deployed by investee companies with the support of gender consultant.	USD 100,000

3.5 Support female customer awareness of OGS products, climate resilience, and benefits of electrification via company-level marketing campaigns	0	5 (final)	# of companies with marketing campaigns	Year 2-10	Deployed by investee companies with the support of gender consultant.	USD 100,000
4. Insights, data aggregation and OGS ecosystem development						
4.1 Gather sex-disaggregated statistics on beneficiaries and employees and share lessons with business-supporting organisations, ecosystem players, investors and companies filling a much needed data gap and in informing future programming and investments.	0	2 (final)	# of knowledge products developed (publications or webinars)	Year 6-10	Gender Expert at H2R	Staff time
4.2 Establish inter-portfolio partnerships and convene ecosystem players in the sector (gender lens investors, women associations, industry aggregators etc.) to support scale of companies and their impact	0	1 (final)	# of events organized	Year 8-10	Gender Expert at H2R	\$50,000
4.3 Annual stakeholder engagement meetings with NDAs, civil society organizations, private institutions, and women's organizations. H2R reporting on project activities including gender activities, and seeks to	0	10	# of events organized	Year 1-10	Gender Expert at H2R and AE	Staff time

learn from others.						
Total budget for GAP implementation						USD 2,041,336